

The Complete 2008 Honda Fit Book

*Everything a new driver and a worried parent need to know to
drive, own, fix, and even race a 2008 Honda Fit.*

DISCLAIMER

This book is not affiliated with Honda. Working on cars can injure you. Follow your owner's manual when it disagrees with this book.

HOW TO USE THIS BOOK

This book is about one specific car: the **2008 Honda Fit, US base trim, first-generation chassis code GD3**. Honda sold a fancier Fit Sport trim that year too, and a completely different second-generation Fit started in 2009. Neither of those is this book. If your Fit is a Sport trim, or a 2009 model or newer, some of what is here still applies, but a lot of it does not. Check Chapter 2 to see exactly what your base trim does and does not have, and do not assume a part number, a torque value, or a step in this book is correct for any Fit other than the 2007–2008 GD3 base car.

This book covers both transmissions Honda offered on the 2008 Fit: the 5-speed manual and the 5-speed automatic. Wherever the two work differently, the book says so and walks through both, usually in side-by-side sections. If a chapter does not mention a transmission difference, the information applies to both.

How the sourcing works. Every fact, number, and procedure in this book was checked against a real source before it was written down: Honda's own owner's manual, government safety databases, published repair manuals, or (when nothing better existed) the Honda Fit owner community. Each sentence that carries a specific claim has an invisible tag behind it in the file this book was built from, a short code like **F-WP01-003** for a fact or **P-WP11-8** for a repair procedure. You will never see these codes on the printed page; they are hidden markers, not footnote numbers. What they mean for you is this: nothing in this book was made up. If you ever want to check exactly where a claim came from, the codes tie every sentence back to a real, dated source.

Two phrases in this book flag information that deserves a little extra caution:

- **"Owners report that..."** This phrase means the information came from the Honda Fit owner community, not from Honda or a government agency. It is only used when at least two separate owners or forum sources said the same thing independently. It is usually reliable, but it has not been confirmed by Honda's own paperwork, so treat it as a strong lead rather than a guarantee.
- **UNVERIFIED.** When this book could not find a trustworthy source for a specific number or claim, even after a real search, it says so plainly instead of guessing. You will see a note like "We could not confirm this for the 2008 Fit. Check your owner's manual, a dealer, or a repair shop before you rely on this." Guessing about a car is how people get hurt or waste money; this book would rather tell you it does not know.

How the book is organized. The book has ten parts. Part I introduces the car and helps you confirm exactly which Fit you own. Part II covers the basics: keys, seats, starting the car, and every control and gauge. Part III teaches you to actually drive it, in good conditions and bad. Part IV covers the paperwork side of ownership. Part V is the maintenance you can do yourself, from an oil change on up. Part VI teaches you to diagnose a problem before you pay anyone to look at it. Part VII walks through bigger repairs at home. Part VIII

is for readers who want to modify the car, and Part IX is for readers who want to take it to an autocross lot or a track day. Part X, the appendices, gathers the book's reference tables (specs, torque values, fuse maps, part numbers, a glossary, and more) into one place you can flip to quickly instead of hunting through a chapter.

Read the parts in order if this is your first car. Jump straight to the part you need if it is not.

CONTENTS

Part I: Meet Your Fit	9
Chapter 1: What This Car Is	10
Chapter 2: Identify Your Exact Car	13
Chapter 3: Specifications at a Glance	17
Part II: The Absolute Basics.....	23
Chapter 4: Keys, locks, and getting in	24
Chapter 5: Sitting right	29
Chapter 6: Starting and stopping the car.....	34
Chapter 7: The dashboard explained.....	39
Chapter 8: Controls You Will Use Every Day	47
Chapter 9: The Magic Seat and Cargo Tricks	55
Chapter 10: Fuel.....	62
Part III: Driving It	66
Chapter 11: First Drive for a New Driver	67
Chapter 12: Driving a Manual Transmission from Zero.....	71
Chapter 13: Parking, Reversing, and Tight Spaces.....	76
Chapter 14: Highway, Rain, Snow, Heat, and Night	78
Chapter 15: Getting Good Fuel Economy	81
Chapter 16: When Something Goes Wrong on the Road	84
Part IV: Owning It (the paperwork)	94
Chapter 17: Title, Registration, Insurance, Inspection, and Emissions Basics	95
Chapter 18: Recalls and Service Bulletins for This Exact Car, and How to Check for New Ones.....	98

Chapter 19: Documents to Keep in the Glovebox and at Home.....	105
Chapter 20: Buying a Used 2008 Fit: Inspection Checklist and Known Trouble Spots	107
Part V: Maintenance	113
Chapter 21: The maintenance schedule	114
Chapter 22: Tools and Supplies for a Starter Garage.....	120
Chapter 23: Safety First — Jacking, Jack Stands, Wheel Chocks, Hot Parts, Fluids	125
Chapter 24: Oil and Filter Change	129
Chapter 25: Air Filter and Cabin Filter.....	133
Chapter 26: Tires — Pressure, Rotation, Wear, Replacement, the Spare	137
Chapter 27: Brakes — Inspection, Pads, Rotors, Fluid	142
Chapter 28: Battery: Testing, Cleaning, Replacing, Jump Starting	149
Chapter 29: Spark Plugs	155
Chapter 30: Coolant, Transmission Fluid, Brake Fluid, Power Steering, and Washer Fluid	159
Chapter 31: Belts, Hoses, and the Timing Chain.....	168
Chapter 32: Lights and Bulbs, Wipers, Fuses	172
Chapter 33: Keeping It Clean Inside and Out.....	180
Part VI: Diagnosing Problems	184
Chapter 34: How to Describe a Problem	185
Chapter 35: The Check Engine Light and OBD-II Codes	190
Chapter 36: Won't Start, Won't Crank, or Cranks But Won't Start.....	199
Chapter 37: Noises, Vibrations, Smells, and Leaks: Lookup Tables	204
Chapter 38: Known Weak Spots on the 2007-2008 Fit	219
Chapter 39: When to Go to a Shop, and How to Not Get Ripped Off.....	226

Part VII: Repairs You Can Do at Home	230
Chapter 40: Brake job, full	231
Chapter 41: Suspension: struts, shocks, sway bar links, bushings	236
Chapter 42: Cooling system: thermostat, radiator, water pump	243
Chapter 43: Starter, alternator, ignition switch	250
Chapter 44: Exhaust and O2 sensors	256
Chapter 45: Interior fixes: door handles, window regulators, seat repair	262
Chapter 46: Body: dents, rust, paint touch-up, headlight restoration	269
Part VIII: Souping It Up	275
Chapter 47: Modding Philosophy — Goals, Budget, and the Law	276
Chapter 48: Stage 0 — Get Your Baseline Right First	281
Chapter 49: Tires and Wheels — Sizes, Offset, and Weight	285
Chapter 50: Suspension — Springs, Coilovers, Sway Bars, and Alignment..	289
Chapter 51: Brakes for Mods — Pads, Fluid, Lines, Rotors, and Cooling.....	298
Chapter 52: Intake, Header, and Exhaust.....	303
Chapter 53: Engine Management	307
Chapter 54: Power Adders	312
Chapter 55: Engine Swaps	317
Chapter 56: Weight reduction and aerodynamics	322
Chapter 57: Seats, harnesses, roll bars.....	327
Part IX: Getting Ready for the Track	332
Chapter 58: Autocross.....	333
Chapter 59: HPDE / track days.....	337
Chapter 60: Track-day prep checklist	341
Chapter 61: Driver skills.....	344
Chapter 62: After the track.....	347

Part X: Appendices 350

Appendix A: Full specifications 351

Appendix B: Fluid capacities and types 359

Appendix C: Torque specifications..... 365

Appendix D: Fuse and relay maps..... 370

Appendix E: Tire and wheel fitment table 377

Appendix F: Part numbers for common maintenance items..... 380

Appendix G: Glossary 388

Appendix H: Resources 408

Appendix I: Recall and TSB list 412

PART I

Meet Your Fit

CHAPTER 1

What This Car Is

What you will learn

- Why Honda built the Fit, and what kind of car it is.
- Why the year and the generation of your Fit matter.
- What a chassis code is, and which one your car has.
- Where your 2008 Fit was built, and which trims Honda sold in the US that year.

The short version

Your car is a 2008 Honda Fit, base trim. Honda built it at its plant in Suzuka, Japan. In the US, Honda sold only two versions of the 2008 Fit: the base Fit and the Fit Sport. Your car belongs to the first generation of the Fit, a car with the internal code GD3. It is not the same car as the Fit that came after it.

The full explanation

A small car built to hold a lot

The Honda Fit is a small hatchback, a car with a rear door that swings up instead of a trunk lid. Honda built the Fit around a 1.5-liter engine, a size measurement that tells you how much air and fuel the engine's cylinders can hold. Later chapters cover the engine, the size of the car, and how far a tank of gas takes you. For now, the key point is this: the Fit was built to be small on the outside but roomy on the inside. It was also built to use less fuel than a bigger car.

Honda's engine for this car carries the code L15A1. An engine code is a short label a manufacturer uses to identify one specific engine design, separate from any other engine that company builds. You will see this code on parts boxes and repair information, so it is worth remembering.

Why the generation of your Fit matters

A chassis code is a short label Honda uses inside the company to identify one specific version of a car's structure and parts. Your 2008 Fit carries the chassis code GD3. Every part built for this car, from body panels to engine mounts, was designed around that GD3 structure.

The safety-testing group IIHS states that the Fit "was introduced in the 2007 model year" in the US. That means the GD3 Fit was sold new in the US only for the 2007 and 2008 model years. A model year is the year Honda assigns to a specific version of a car. It does not always match the calendar year the car was actually built or sold.

Honda later replaced this car with a new design. That newer Fit used a different chassis code and different parts throughout, even though it kept the same name. This book covers only the first-generation, 2008, GD3 Fit. If your car is a Fit built for model year 2009 or later, the two cars share a name, but they do not share a design. Do not use this book's specific numbers, part information, or procedures for that later car. Whenever this book gives you a number, a part, or a step, it is speaking about the 2008 GD3 Fit only.

Base trim and Fit Sport

Honda's 2008 US lineup for the Fit had exactly two trims, a trim being one specific equipment level of a car that a manufacturer sells under one name. Those two trims were the base Fit and the Fit Sport. This book is written for the base Fit. Chapter 2 walks through exactly what the base trim has, and what it does not have compared with the Fit Sport.

Where your car was built

Honda built the 2008 Fit at its plant in Suzuka, in Japan's Mie Prefecture. Every 2008 US-market Fit, base and Sport alike, came from that same plant.

Quick check

1. What chassis code does a 2008 base Fit have?
2. In what city and country was your 2008 Fit built?
3. How many US trims did Honda sell for the 2008 Fit, and what were they called?
4. Why does it matter that your Fit is a first-generation car, not the generation that came after it?

CHAPTER 2

Identify Your Exact Car

What you will learn

- How to read your car's VIN, the Vehicle Identification Number that names your exact car.
- Where to find the metal plates and stickers that identify your Fit.
- Exactly what your base Fit has, and what it does not have compared with a Fit Sport.
- What Honda knows, and does not know for sure, about your car's paint color code.

The short version

Your car's VIN is a 17-character code that identifies your exact Fit. You can find that VIN on a plate at the top of your dashboard, on a label inside the driver's doorjamb, and stamped into the engine compartment. A base Fit and a Fit Sport look similar, but the base trim is missing several features the Sport has, such as alloy wheels, fog lights, and cruise control.

The full explanation

Reading your VIN

A VIN, short for Vehicle Identification Number, is a 17-character code that no two cars in the world share. Reading a VIN means breaking that code into groups of characters, where each group tells you something different about the car.

For your 2008 base Fit, here is what the groups mean:

VIN positions	What they tell you	Value for your car
1-3	WMI, the World Manufacturer Identifier	JHM = Honda Motor Co., Ltd., Japan, passenger car
4-6	Platform or chassis code	GD3, the first-generation Fit structure
10	Model year code	The digit "8" marks a 2008 model
11	Assembly plant code	"S" marks the Suzuka, Japan plant
12-17	Serial number	A running production number unique to your car

If you want to decode your own real VIN with full confidence, use the official decoder at the NHTSA website. NHTSA stands for the National Highway Traffic Safety Administration. Hand-counting the characters yourself is a helpful way to understand the pattern. But the government tool checks the whole code, including a check digit that confirms the VIN is valid.

Where to find the VIN on your car

Honda puts your VIN in three places. The easiest one to check is a metal plate fastened to the top of the dashboard. You can see this plate by looking through the windshield from outside, on the driver's side. The VIN also appears on the certification label inside the driver's doorjamb, printed both as text and as a barcode. It is also stamped into the engine compartment's bulkhead, the wall that separates the engine bay from the passenger cabin.

The sticker that tells you your tire information

That same doorjamb area holds a second important label. This one lists the tire sizes and cold tire pressures Honda recommends for your car. "Cold" means the tires have not been driven on recently, since driving heats up the air inside a tire and raises its pressure reading. Check this label, not a number you remember or a tire shop's default setting, when you set your tire pressure.

What base got, and what base did not get

Honda sold your 2008 Fit as one of two US trims: the plain "Fit," which this book calls base, and the "Fit Sport." Both trims came standard with air conditioning, power windows, power door locks, and power mirrors. Both also came with the 60/40 split "Magic Seat" rear seat system, which lets you fold or flip the rear seats into several different cargo configurations. The base Fit's factory tires were sized P175/65R14, mounted on 14-inch steel wheels with full wheel covers; the Fit Sport instead used a larger, wider tire, P195/55R15, on alloy wheels.

Compared with the Fit Sport, the base Fit did not come with any of the following:

- Paddle shifters mounted on the steering wheel, available only on Sport models with the automatic transmission.
- An underbody aero kit and a rear roofline spoiler.
- Fog lights.
- A remote keyless entry system with a security system.
- Cruise control.
- 15-inch aluminum alloy wheels.
- The upgraded 200-watt, six-speaker audio system with an auxiliary audio jack for plugging in an outside music player; base cars instead came with a four-speaker AM/FM/CD system.

One research note worth passing along: a Canadian-market Honda specification page describes a similar "base" trim as including an auxiliary audio jack, which conflicts with the US information above. This may reflect a real difference between the US and Canadian versions of the same trim, but it could not be confirmed either way. If your factory radio has an auxiliary jack and everything else here matches your car, that is not necessarily a mistake in this book.

Paint colors

Honda's own announcement for the 2008 Fit states that two colors were new that year: Tidewater Blue Metallic and Blackberry Pearl. A parts and pricing website lists seven total exterior colors for the 2008 Fit. They are:

- Milano Red
- Storm Silver Metallic
- Tidewater Blue Metallic
- Vivid Blue Pearl
- Blackberry Pearl
- Blaze Orange Metallic
- Nighthawk Black Pearl

A paint code is a short factory code that identifies one specific paint color, and body shops use it to mix a matching paint. Some paint codes were found for these colors, but they came from a general Honda color reference that was not confirmed as accurate specifically for the 2008 Fit. Do not order paint by a code from this book. Instead, check the paint code sticker on your own driver's doorjamb, the same area that holds your VIN and tire information, and match that code exactly.

The emissions label

We could not confirm this for the 2008 Fit. Check your owner's manual.

Quick check

1. What does "VIN" stand for, and how many characters does it have?
2. Where are the three places Honda put the VIN on your car?
3. Name three features the Fit Sport had that the base Fit did not.
4. Where should you check for your car's real, confirmed paint code, instead of trusting a general list?

CHAPTER 3

Specifications at a Glance

What you will learn

- The exact engine numbers for your 2008 base Fit.
- The size and weight of your car, and how much it can carry.
- How many miles per gallon the EPA (Environmental Protection Agency) rated your Fit for.
- Your car's original price, and how it did in crash testing.

The short version

Your 2008 base Fit uses a 1.5-liter engine that makes 109 horsepower. It is a small car, under 158 inches long, that still holds over 21 cubic feet of cargo with the rear seats up. Honda sold it new for \$13,950 with a manual transmission or \$14,750 with an automatic.

The full explanation

Engine

Horsepower is a measurement of how much work an engine can do over time. Torque is a measurement of an engine's twisting force, the pulling power you feel when you accelerate. RPM stands for revolutions per minute, or how many full turns the engine makes every minute. Compression ratio compares the space inside a cylinder before and after the piston squeezes the air-fuel mixture; a higher number means a tighter squeeze. Bore and stroke describe the width of each cylinder and the distance its piston travels. Cc stands for cubic centimeters, a way of measuring the total volume of all the cylinders together. Redline is the highest RPM the engine is designed to reach safely.

Spec	Value
Engine code	L15A1
Displacement	1.5 L (1,497 cc)
Horsepower	109 hp @ 5,800 rpm
Torque	105 lb-ft @ 4,800 rpm
Compression ratio	10.4:1
Bore x stroke	73.0 mm x 89.4 mm
Required fuel	Regular unleaded gasoline
Redline	6,500 rpm

Honda's own specification sheet names the required fuel as "Regular Unleaded" without giving an exact minimum octane number, a number that measures a gasoline's resistance to knocking. Standard US regular-grade gasoline is commonly labeled 87 octane. But no source specific to this car states that number, so this book does not claim it as confirmed for your Fit.

Transmission

A gear ratio is the number of times the engine turns for every one turn of the wheels in a given gear. A higher ratio gives more pulling power at low speed, and a lower ratio saves fuel at speed. Your base Fit could be ordered with a 5-speed manual transmission, which was standard, or an optional 5-speed automatic transmission.

Gear	Manual ratio	Automatic ratio
1st	3.462	2.996
2nd	1.870	1.679
3rd	1.321	1.067
4th	0.970	0.756
5th	0.757	0.550
Reverse	3.231	1.957
Final drive	4.29	4.56

Weight and size

Curb weight is how much the car weighs by itself, full of fuel and fluids, but with no passengers or cargo aboard. Wheelbase is the distance between the center of the front wheel and the center of the rear wheel. Ground clearance is the gap between the lowest part of the car and the ground.

Spec	Value
Curb weight, manual	2,432 lb
Curb weight, automatic	2,514 lb
Length	157.4 in
Width	66.2 in
Height	60.0 in
Wheelbase	96.5 in
Ground clearance	We could not confirm this for the 2008 Fit. Check your owner's manual.

Cargo and fuel

Cargo volume is the amount of space, measured in cubic feet, available for luggage and other items.

Spec	Value
Cargo volume, rear seats up	21.3 cu ft
Cargo volume, rear seats folded down	56.8 cu ft
Fuel tank capacity	10.8 gal (41 L)

A few other sources gave different numbers for cargo volume with the seats folded down, ranging from about 42 to about 59 cubic feet. This likely happened because different companies measure cargo space using different methods. This book uses the number from Honda's own 2008 US specification sheet, 56.8 cubic feet, as the most trustworthy figure for your car.

Fuel economy

The EPA rates every new car's fuel economy in miles per gallon, written as MPG, so shoppers can compare cars on a level basis. City MPG estimates driving in stop-and-go traffic; highway MPG estimates steady-speed driving.

Transmission	City MPG	Highway MPG
5-speed manual	28	34
5-speed automatic	27	34

Tires and wheels

Your base Fit's factory tire size is P175/65R14 81S, mounted on a 14-inch steel wheel, 5.5 inches wide, with a full wheel cover.

Crash test ratings

NHTSA is the National Highway Traffic Safety Administration, and IIHS is the Insurance Institute for Highway Safety. They are the two US groups that crash-test new cars. Both publish star or letter ratings so buyers can compare safety.

NHTSA rated the 2008 Fit this way, on a 1-to-5-star scale where 5 stars is the best score:

Test	Rating
Frontal crash, driver side	5 stars
Frontal crash, passenger side	5 stars
Side crash, driver side	5 stars
Side crash, passenger side	3 stars
Rollover	4 stars (13.3% rollover risk)

Rollover risk is NHTSA's estimate of how likely a car is to roll over in a single-vehicle crash; a lower percentage is safer. NHTSA did not publish a single combined overall star rating for the 2008 Fit, because NHTSA did not start using one combined score until testing for the 2011 model year.

IIHS graded the same car on a scale of Good, Acceptable, Marginal, and Poor:

Test	Rating
Moderate overlap front crash	Good
Side crash	Good
Head restraints and seats (whiplash protection)	Poor

IIHS states that these results apply to both the 2007 and 2008 model years, since the two years share the same structure. IIHS had not yet created its small overlap front crash test when your Fit was built, so no small overlap rating exists for this car.

Original price

MSRP stands for Manufacturer's Suggested Retail Price, the price a manufacturer recommends a dealer charge before any negotiation, taxes, or fees.

Transmission	MSRP
5-speed manual	\$13,950
5-speed automatic	\$14,750

These prices exclude tax, license, registration, and a \$670 destination charge, the fee a manufacturer adds to cover shipping the car to the dealer. Actual dealer prices could vary.

Quick check

- 1. How many horsepower does your Fit's engine make, and at what RPM?
- 2. What is the difference between curb weight and cargo volume?

3. Which transmission gets better city fuel economy, the manual or the automatic?
4. What was your Fit's original MSRP with a manual transmission, and what does MSRP stand for?
5. Why does this book not give you a confirmed ground clearance number?

PART II

The Absolute Basics

CHAPTER 4

Keys, locks, and getting in

What you will learn

- How the key and the anti-theft chip inside it work
- Why your Fit does not have a remote fob, and how to lock and unlock it instead
- How to use the child safety lock on the back doors
- How to find the fuel door and the tailgate (the rear hatch door) release
- How to open the hood safely, step by step

The short version

Your base 2008 Fit uses a plain metal key, not a fob, and that key has a hidden chip that talks to the car's anti-theft system. You lock and unlock the car with the key, a small tab on each door, or a power switch on the driver's door. The hood, fuel door, and tailgate (the rear hatch, since the Fit is a hatchback and has no separate trunk) each open a different way. This chapter walks you through all of them.

The full explanation

The key and the immobilizer

Your car came with two metal keys and a small tag with a key number on it, which you use to order a new key later. Each key has a chip built into it. This chip talks to your car's immobilizer, a system that stops the engine from starting unless it senses the right chip. The key has no battery, but a damaged chip still stops the car from starting.

Some Fit models come with a remote transmitter, a small fob you press to lock or unlock the car from a distance. Your base Fit does not have one. That feature is built into the Sport trim only. On your car, you lock and unlock every door using the key or the switches inside the car.

Locking and unlocking the doors

Every door has a small tab on top. Push the tab in to lock that door, and pull it out to unlock it. Your Fit also has power door locks, which use an electric switch instead of your hand at each door. On the driver's door, a master switch locks every door and the tailgate at once. Press the top of the switch to lock them, and press the bottom to unlock them. Turning the key in the driver's door lock also works. The first turn locks or unlocks only the driver's door. The second turn locks or unlocks the rest of the car. One safety note: the master switch will not work while the driver's door is open with the key in the ignition. The ignition is the slot where you turn the key to start the engine.

The child safety lock

Each rear door has a small lock lever near its rear edge. Set that lever to LOCK, and the rear door cannot open from inside, no matter how the regular door lock tab is set. The door still opens normally from outside using the outer handle. Use this lock any time a child rides in the back seat.

The fuel door

Your base Fit's fuel door has no lever or cable inside the car. To open it, walk around to the side of the car and pull on the notched edge of the fuel door itself. The fuel cap, the small cap that seals the gas tank opening, stays attached to the car by a short tether. This means you cannot drop it or lose it while refueling.

The tailgate

The Fit is a hatchback, so its rear cargo door is called the tailgate, not a trunk. It opens only from the outside: pull its handle, then lift it up. It has no separate button or lever inside the car. Instead, it locks and unlocks together with the driver's door whenever you use the key, the door lock tab, or the master power-lock switch. On your base trim, you can also lock or unlock the tailgate directly with the key.

Step by step: Open the hood

Safety warnings

- Always park on level, flat ground and set the parking brake before you open the hood.
- Always let the engine cool before you open the hood if you recently finished driving.
- Do not open the hood if you see steam or spray coming from underneath it. Wait until it stops.
- Always check that the support rod, the metal rod that holds the hood open, is properly seated in its hole before you let go of the hood.

Steps

1. Park on level ground, turn off the engine, and set the parking brake.
2. Reach under the lower-left corner of the dashboard, near your left knee, and pull the hood release handle.
3. The hood will pop up slightly but stay held by a second, hidden latch.
4. Walk to the front of the car and slide your fingers under the hood's front edge, near the center.
5. Feel for the second latch handle and push it to release the hood.
6. Lift the hood the rest of the way open.
7. Pull the metal support rod free from its clip under the hood.

8. Fit the tip of the support rod into the matching hole on the underside of the hood.
9. Let go of the hood slowly to make sure the rod is holding its weight.

To close the hood

1. Lift the hood slightly to free the support rod from its hole, then clip the rod back in place.
2. Lower the hood to about one foot above the front fenders, the body panels over the front wheels, then let it drop the rest of the way.
3. Push down lightly on the hood to confirm it caught and is fully latched.

How you know it worked: The hood stays open on its own, held only by the rod, with no need for you to hold it up. When closed, it sits flush and does not move when you push down on it.

What can go wrong

- You pull only the interior release and never find the second hood latch, so the hood will not lift.
- You let go of an open hood before the support rod is properly seated in its hole.
- You slam the hood shut from too low, instead of letting it drop the last foot on its own.

Quick check

1. What part inside your key allows the car to start, even though the key has no battery?
2. Does your base 2008 Fit come with a remote fob to lock and unlock the doors? Why or why not?
3. How do you open the fuel door on your Fit?

4. What is the correct name for the Fit's rear cargo door, and how many ways can you unlock it?
5. Name one safety warning you must follow before you open the hood.

CHAPTER 5

Sitting right

What you will learn

- How to slide and recline the driver's seat correctly
- How to tilt the steering wheel and adjust the head restraints (headrests)
- How your power mirrors work, and why you set them before you drive
- How to fasten your seat belt so it protects you correctly
- The full setup routine to run every time you get in, before you start driving

The short version

Before you drive, you slide your seat, tilt the steering wheel, set your head restraints, aim your mirrors, and fasten your seat belt correctly. Your base Fit has power mirrors and a tilt-only steering wheel, so you use switches and a lever instead of pushing on things by hand. Doing all of this before you move keeps you safe, because these adjustments protect you in a crash and help you see the road.

The full explanation

The seat

The driver's seat slides forward and backward using a lever under the front edge of the seat cushion. Lift the lever, slide the seat to where you want it, then let go and rock the seat to confirm it caught. A second lever on the side of the seat-back, the part of the seat behind your back, tilts that backrest angle. Pull the lever out, lean the seat-back to the angle you want, and release the lever to lock it. The rear outer seat-backs recline too, using a release lever on

the outside edge of each one. The owner's manual does not give an exact travel distance or a number of notches for either seat adjustment. Reclining your seat-back too far is dangerous, because a reclined seat-back stops your seat belt from holding you properly in a crash.

The steering wheel

Your base Fit's steering wheel tilts up and down, but it does not telescope, meaning it does not slide toward or away from you. To adjust it, push the lever under the steering column, the part of the steering assembly under the wheel, all the way down to unlock it. Move the wheel to an angle where you can still see all the gauges, the dials in front of you, then push the lever back up to lock it.

The head restraints

Every seating position has a head restraint, the padded part above the seat-back that supports your head. The two front head restraints slide up and down. Pull straight up to raise one, or press its release button sideways while you push it down to lower it. Always use both hands, and never do this while driving. Line up the middle of the back of your head with the center of the restraint. The rear outer head restraints do not adjust for height; they only lift out for cleaning after you press a release button.

The mirrors

Your base Fit has power mirrors on the outside, not manual ones you push by hand. With the ignition on, use a selector switch to pick the driver's or the passenger's mirror, then use a second switch to tilt that mirror up, down, left, or right. Return the selector switch to its center, off position when you finish, so the setting locks in place. The inside rear-view mirror also has a day and night flip tab that cuts glare from headlights behind you at night.

The seat belt

To buckle up, push the metal latch plate into the buckle and tug on it to confirm it caught. Press the red button on the buckle when you want to release it. For a correct fit, pull the lap portion down low across your hip bones first. Then pull the shoulder portion up to remove slack, so it crosses the middle of your chest and shoulder, not your neck or arm. If the belt still crosses your neck or arm, change the shoulder anchor height instead. Each front shoulder belt anchor, the fixed point where the belt attaches near your shoulder, slides up and down a track with four height positions. Squeeze the two small buttons on the anchor to unlock it and slide it. This height adjuster exists only on the front seats, not the rear ones.

Step by step: Adjust your seat and mirrors before driving

Safety warnings

- Always make every seat, head restraint, mirror, and steering wheel adjustment before you start driving, never while the vehicle is moving.
- Always sit as far back from the steering wheel as you safely can while still reaching the controls. Sitting too close raises your injury risk if the front airbag inflates.
- Do not recline the seat-back too far. A reclined seat-back stops the seat belt from holding you properly in a crash.
- Always confirm the seat and seat-back are locked in position by rocking them before you drive.

Steps

1. Sit in the seat and pull up the lever under the front edge of the seat cushion.
2. Slide the seat to a position where you can fully press the pedals, then release the lever.
3. Rock the seat forward and back to make sure it is locked in place.

4. Pull the lever on the side of the seat-back and lean to a comfortable, upright angle.
5. Release the lever to lock the seat-back, then rock it to confirm it caught.
6. Pull straight up on each front head restraint, or press its side button and push down, until it is level with the middle of your head.
7. Push the lever under the steering column down to unlock the wheel.
8. Move the wheel up or down until you can see all the gauges clearly, then push the lever back up to lock it.
9. Try moving the wheel to confirm it is locked in place.
10. Flip the tab under the inside mirror to day mode, then tilt it for a clear view out the back window.
11. Turn the ignition to the ON position and move the mirror selector switch to the driver's side.
12. Press the edges of the mirror adjustment switch until that mirror shows a clear view, then repeat for the passenger's side mirror.
13. Return the mirror selector switch to its center, off position so the setting is not changed by accident.
14. Fasten your seat belt, then pull the lap portion low and snug and the shoulder portion across your chest.
15. If the belt crosses your neck or arm, squeeze the two buttons on the shoulder anchor and slide it to a better height.

How you know it worked: You can fully press the pedals and see every gauge. Both outside mirrors show a clear view without you leaning, and the shoulder belt rests across the middle of your chest, not your neck.

What can go wrong

- You adjust the mirrors, seat, or steering wheel after you already started driving.
- You leave the seat too close to the steering wheel.

- You forget to lock the seat-back angle lever.
- You leave the mirror selector switch turned on instead of centering it, which can let the setting drift.

Quick check

1. What could happen in a crash if you recline your seat-back too far?
2. What do you push under the steering column to unlock its tilt adjustment?
3. Why does your base Fit use switches, not your hand, to aim the outside mirrors?
4. How many height positions does the front shoulder belt anchor have?
5. Name two adjustments you must always make before driving, never while driving.

CHAPTER 6

Starting and stopping the car

What you will learn

- The four positions of the ignition switch, the slot where you turn the key to start the car
- How to start a manual transmission car, step by step
- How to start an automatic transmission car, step by step
- What the immobilizer light means and what to do if it blinks
- How to release the shift lock on an automatic if the shifter will not move out of Park

The short version

Before you turn the key, you set the parking brake and check that the shifter is in the right spot. On a manual, you press the clutch pedal all the way down first. On an automatic, you press the brake pedal and confirm the shift lever is in Park. Once the engine starts, ease off the accelerator, the gas pedal, so the engine does not race.

The full explanation

The ignition switch

Your Fit's ignition switch has four positions. LOCK is where you insert or remove the key; on an automatic, the shift lever must also be in Park to turn the key to LOCK. ACCESSORY runs the audio system and the accessory power socket. ON is the normal driving position, and switching from

ACCESSORY to ON lights up several dashboard indicators briefly as a test. START is spring-loaded and used only to start the engine; it springs back to ON when you let go.

A reminder beeper sounds if you open the driver's door while the key is left in LOCK or ACCESSORY, and removing the key stops it. On an automatic, the shift lever must be in Park before you can remove the key. Never remove the key while driving. Doing so locks the steering wheel and can cause you to lose control of the car.

Before you start driving, the owner's manual's own pre-drive checklist tells you to check the gauges and indicators on the instrument panel. The instrument panel is the panel of dials and lights in front of you. The same checklist also covers your mirrors, seat position, seat belts, and closed doors.

Safety warning (general knowledge, true of any gasoline engine, not a Honda-specific fact): Never start or run the engine in a closed garage or other enclosed space. A running engine produces carbon monoxide, a gas you cannot see or smell that can kill you in minutes. Always start the car outdoors or with the garage door fully open.

Starting a manual transmission Fit

1. Apply the parking brake.
2. In cold weather, turn off electrical accessories to reduce battery drain.
3. Press the clutch pedal all the way to the floor. The START position will not work unless the clutch pedal is fully pressed.
4. Without touching the accelerator, turn the key to START.
5. Do not hold START for more than 15 seconds at a time. If the engine does not start, wait at least 10 seconds before you try again.
6. Once the engine starts, ease off the accelerator so it does not race.

If the engine still does not start after 15 seconds, or it starts and stalls right away, repeat the START step with the accelerator pressed about halfway down. If it still will not start, the engine may be flooded, meaning too much fuel has reached it. Press the accelerator all the way down and hold it there while starting; if that does not work, go back to the halfway-down step. Cold weather and altitudes above 8,000 feet (2,400 m) make starting harder.

Starting an automatic transmission Fit

1. Apply the parking brake.
2. In cold weather, turn off electrical accessories to reduce battery drain.
3. Make sure the shift lever is in Park, and press the brake pedal.
4. Without touching the accelerator, turn the key to START.
5. Do not hold START for more than 15 seconds at a time. If the engine does not start, wait at least 10 seconds before you try again.
6. Once the engine starts, ease off the accelerator so it does not race.

The flooding and hard-starting notes above apply the same way on an automatic.

The automatic transmission's shift positions

Your base automatic Fit has these shift lever positions: P (Park), R (Reverse), N (Neutral), and D (Drive, which uses gears one through five on its own). It also has D3, which is like D but locks out 4th and 5th gear, useful for engine braking downhill or in stop-and-go traffic. Position 2 locks 2nd gear, for climbing power, engine braking, or slippery starts, and position 1 locks 1st gear. An "S" position and paddle shifters exist only on the Sport trim, not on your base car.

Releasing the shift lock

Normally, you move the automatic shift lever out of Park by pressing the brake pedal and pressing the lever's release button. If that does not work, your Fit has a shift lock release slot, a hidden covered slot in the shift lever console, next to the lever's release button. To use it, set the parking brake and remove the key from the ignition. Protect the notch of the slot's cover with a cloth, then carefully pry the cover off with a small flat-tip screwdriver or a metal fingernail file. Insert the key into the slot, push down on the key while you press the shift lever's release button, and move the lever out of Park to Neutral. Remove the key from the slot, reinstall the cover with its notch on the left side, press the brake pedal, and restart the engine. Needing this procedure means your car is developing a problem, so have it checked by a dealer.

The manual transmission's shift pattern

Your 5-speed manual is synchronized in all forward gears, meaning the gears mesh smoothly without extra effort from you. It has a mechanical lockout that prevents shifting directly from 5th gear into reverse. Always press the clutch pedal all the way down when you shift. Do not rest your foot on the clutch pedal when you are not shifting, since that wears the clutch faster. Come to a complete stop before you shift into reverse; shifting into reverse while moving can damage the transmission. Pause briefly with the clutch down, or pass through a forward gear for a moment, before you engage reverse, to stop the gears from grinding.

The immobilizer light

The key symbol on your dashboard is the immobilizer system indicator. At key-ON, it should light briefly, then go off, confirming the car recognized your key's chip. If it blinks instead, the system did not recognize your key. Turn the key to LOCK, remove it, reinsert it, and turn to ON again. Another key or metal object near the ignition switch can also cause this, so keep spare keys and fobs away from the switch while starting. If it keeps blinking, call the dealer. The light also blinks a few times when you turn the key from ON

down to ACCESSORY or LOCK; that is normal, not a fault. Never alter the immobilizer system or add devices to it, and always take your key with you when you leave the car.

Quick check

1. Name the four positions of the ignition switch, in order.
2. What must you do with your foot before turning the key to START on a manual? What about on an automatic?
3. What should you do if the engine has not started after holding START for 15 seconds?
4. Where is the shift lock release slot, and when would you need to use it?
5. If the immobilizer light blinks instead of going off, what are your first three steps?

CHAPTER 7

The dashboard explained

What you will learn

- What every warning light on your base 2008 Fit's dashboard means
- What to do for each light, and whether it is safe to keep driving
- Which lights described in the owner's manual you will never see on your base car
- What each gauge and readout on your instrument panel, the panel of dials and lights in front of you, shows you

The short version

Your instrument panel has 20 warning lights and messages that apply to your base Fit, and this chapter lists every one. Some lights only give you information, some tell you to schedule a repair soon, and a few tell you to pull over right away. Learn this table before your first drive, because it is the page you will come back to most.

The full explanation

How many lights does your Fit have

The owner's manual describes 26 named indicators across US and Canadian Fit models. Six of those do not apply to a US base Fit, leaving 20 that you may actually see. The immobilizer system indicator, the key-shaped light that confirms your key's anti-theft chip was recognized, is covered in Chapter 6, since it comes up during starting.

The lights you will never see

Your base Fit does not have six indicators that the manual describes, because they belong to other trims or other countries. These are the fog light indicator (Sport only) and the security system indicator (Sport only, since base has no remote entry system). They also include the cruise main and cruise control indicators, found only on Sport and Canadian LX models, since base has no cruise control. The last two are the daytime running lights indicator and the washer level indicator, both Canadian models only. If your base car ever shows one of these six, something aftermarket, meaning added after the factory build, is probably involved.

The warning light table

Light	What it means	What you do	Safe to keep driving?
Malfunction Indicator Lamp (MIL), also called the check engine light — an engine-block icon	An emission-control part may have a problem, or the fuel cap is loose or missing	Check the fuel cap first and tighten it until it clicks. If the light stays on or comes back, see a dealer soon	Yes for now, but do not ignore it. Driving with it on can damage emissions parts and the engine
Low Oil Pressure Indicator — an oil can icon	Engine oil pressure has dropped very low or is lost	Pull off the road right away and shut off the engine. Check the oil level, add oil if it is low, then watch the light when you restart	No. Stop the car and shut off the engine as soon as it is safe
Charging System Indicator — a battery icon	The battery is not being charged while the engine runs	Turn off accessories you do not need, keep the engine running, and drive to a service station for help	Only for a short distance to get help. The battery will drain further if you stop the engine
Brake System Indicator (red) — an exclamation mark in a circle	Either the parking brake is not fully released, or the brake fluid is low or there is a brake problem	If the parking brake is on, release it fully. If the light stays on, check the brake fluid; if the pedal feels wrong, pull over and call for a tow	Yes if it is only the parking-brake reminder. No if the brake pedal feels different than normal
Seat Belt Reminder Indicator — a person wearing a belt	The driver or front passenger seat belt is not fastened	Fasten your seat belt, and make sure your passenger does too	Yes, but always buckle up before you drive
Supplemental Restraint System (SRS) Indicator — a person in front of an airbag shape	A possible problem with the airbags, seat belt tensioners, or related sensors	Have a dealer inspect the car as soon as possible	Yes, but the airbags may not work correctly in a crash until it is checked
Side Airbag Off Indicator — reads	The front passenger's side	No action needed unless it seems un-	Yes, this light is informational only

Light	What it means	What you do	Safe to keep driving?
"SIDE AIRBAG OFF"	airbag has automatically shut off	expected; if so, have it checked	
Anti-lock Brake System (ABS) Indicator — reads "ABS" in a circle	A problem with the anti-lock function, the system that keeps your wheels from locking up under hard braking	Have a dealer inspect the car soon	Yes. Normal braking still works, but drive extra carefully, especially on wet or icy roads
Electric Power Steering (EPS) Indicator — a steering wheel icon	A problem in the electric power steering system, which helps you turn the wheel with less effort	Stop in a safe place, turn off the engine, then restart it to reset the system. If it returns, have it checked	The manual says to stop and shut off the engine right away. Treat this as a pull-over light
Low Fuel Indicator — a fuel pump icon	About 1.7 US gallons (6.3 L) of fuel remain in the tank	Refuel soon	Yes, but plan to refuel before it runs out
Door and Tailgate Open Indicator — an open-door icon	A door or the tailgate is not closed tightly	Close all doors and the tailgate firmly	Yes, but close it first for safety
High Temperature Indicator — a red thermometer icon	Engine coolant, the liquid that keeps your engine from overheating, is running hot	If it blinks, slow down. If it stays on, pull over safely and shut off the engine	No if the light stays on. Pull over and shut off the engine to avoid engine damage
Low Temperature Indicator — a blue thermometer icon	Normal when the engine is cold. A problem if it stays on once the engine has warmed up	If it stays on after the engine warms up, have a dealer check it	Yes, this is normal at startup in cold weather
Turn Signal Indicators — left and right green arrows	Shows an active turn signal. A fast blink usually means a burned-out bulb	Replace the bulb soon if it blinks fast or does not blink at all	Yes, but fix a burned-out bulb soon so other drivers can see your signal
Hazard Warning (four-way flashers)	The hazard button was pressed, for use when you are parked in a dangerous spot	Use it when stopped in traffic danger or broken down	Yes, this is a safety feature

Light	What it means	What you do	Safe to keep driving?
— both arrows flashing together	or your car is disabled		
Low Tire Pressure Indicator — a tire cross-section with an exclamation mark	One or more tires is significantly under-inflated	Pull over when safe, check your tires, and inflate or replace the low one with the compact spare	Drive slowly and briefly to a safe stop. If two or more tires are low, call for a tow
Tire Pressure Monitoring System (TPMS) Indicator — reads "TPMS"	The tire-pressure monitoring system itself has a problem and can no longer watch your tire pressure	Check tire pressure by hand at every fill-up, and have a dealer check the system	Yes, but you lose the automatic low-tire warning until it is fixed
Lights On Indicator	Your exterior lights are switched on	Turn off exterior lights when you park, or they will drain the battery	Yes, this light is informational only
High Beam Indicator — a blue headlight icon	Your high-beam headlights are on	Switch to low beams when other traffic is oncoming	Yes, this light is informational only
Maintenance Minder Indicator — a wrench icon	It is time for scheduled maintenance	Check the information display for which service is due, and schedule it	Yes, but do not ignore it for long
"D" Indicator (automatic transmission only)	Shows the shift lever is in Drive. A flashing "D" while driving can mean a transmission problem	If it flashes, avoid hard acceleration and have the transmission checked soon	Yes for a short trip, but get it checked, especially if the check engine light is also on
"CHECK FUEL CAP" message (on the information display, not a separate light)	The gas cap is loose or missing	Turn off the engine, and check and retighten the cap until it clicks once	Yes, but the message can take several days of driving to clear even after you fix the cap

The gauges

Your instrument panel shows a tachometer, a gauge that measures how fast your engine is spinning, along with a speedometer and a fuel gauge. It also has an odometer, which shows total distance driven, a trip meter with A and B settings you can reset separately, and a select/reset knob. An information display shows text messages like the Maintenance Minder and "CHECK FUEL CAP." Your Fit has no separate needle gauge for coolant temperature. Instead, the High and Low Temperature indicator lights tell you about coolant temperature.

Press the select/reset knob to switch between Trip A and Trip B, and press and hold it until the display reads "0.0" to reset one. The odometer shows total miles driven; tampering with it to change the reading is illegal under US federal law.

The fuel gauge shows approximately how much fuel is in the tank, and it may read slightly more or less than the actual amount. The needle drops to empty after you turn off the ignition and comes back up once the engine runs again. Do not drive with an extremely low fuel level. Running out of fuel can make the engine misfire and can damage the catalytic converter, a part of the exhaust system that reduces pollution.

The tachometer has a red zone, the danger zone the engine speed limiter protects against on upshifts and the reference point for safe engine braking on downshifts. The 2008 Fit's factory redline, the highest safe engine speed, is 6,500 rpm. We could not confirm the normal idle RPM for the 2008 Fit. Check your owner's manual.

Your speedometer is a standard analog needle gauge, and US models read in mph.

Quick check

1. If your Low Oil Pressure light comes on, what should you do first, and is it safe to keep driving?
2. Name two lights that are informational only, meaning you do not need to change your driving.
3. Which light tells you your car is close to running out of fuel, and about how much fuel is left when it comes on?
4. Why does your base Fit's dashboard never show a cruise control indicator?
5. What is the difference between your engine's redline and its normal idle speed, and which one could this book confirm?

CHAPTER 8

Controls You Will Use Every Day

What you will learn

- How to work the lights, turn signals, hazards, and horn
- How to run the wipers, washers, and both defrosters
- How the heating and air conditioning (HVAC, or heating, ventilation, and air conditioning) controls work
- How the power windows, door locks, and child locks work
- How the radio works, and why the base Fit has no aux jack (a port for plugging in a phone or music player)

The short version

Most controls sit on the two steering-column levers or on the dashboard near the gauges. The left lever runs your lights and turn signals. The right lever runs your wipers and washers. The radio plays AM, FM, and CDs, but the US base Fit has no aux jack, so you cannot plug in a phone or MP3 player directly .

The full explanation

Horn

Press the center pad of the steering wheel to sound the horn .

Headlights

The light switch sits on the left steering-column lever, the same lever you use for turn signals . Turn it one position on to light the parking lights, taillights, instrument panel, side markers, and rear license plate . Turn it to the next position to turn on the headlights . A "lights on" indicator on the instrument panel reminds you the lights are on. If you leave the lights on and turn the ignition to ACCESSORY or LOCK, that indicator stays lit, and a chime sounds when you open the driver's door .

High beams

Push the left lever forward to turn on the high beams; a high-beam indicator lights up . Pull the lever back to return to low beams . To flash your high beams without leaving them on, pull the lever back lightly and let go; the high beams stay on only while you hold the lever .

Turn signals

The same left lever signals your turns. Push it down to signal a left turn and up to signal a right turn . The lever returns to center on its own once the turn finishes, or when you release it . For a lane change, push the lever lightly and hold it; it will not latch, so it stops signaling when you let go .

Hazard warning lights

A dedicated hazard button flashes all four turn signals and both dashboard turn-signal indicators at once . It sits near the instrument panel, grouped with the brightness knob and the rear defogger button .

Windshield wipers and washer

The right steering-column lever runs your wipers . It has six positions. MIST runs the wipers at high speed only while you hold the lever up. OFF stops them. INT (intermittent) sweeps the wipers every few seconds. LO runs them at a steady low speed. HI runs them at a steady high speed. Pulling the lever

toward you sprays washer fluid; the wipers run at low speed and make one more sweep after you let go . Two-speed, intermittent front wipers come standard on the base Fit, not only on the Sport trim .

Rear wiper and washer

A separate rotary switch runs the rear wiper and washer . OFF parks the wiper. Turning the switch clockwise turns the rear wiper on. Holding it past ON also sprays the rear washer while the wiper runs. Turning the switch counterclockwise sprays the rear washer alone, without running the wiper . Both washers share one fluid tank; there is no separate rear tank . The rear wiper and washer come standard on base trim .

Defrosting the front windshield

Turn the fan on; high speed clears the glass fastest. Slide the fresh air/recirculation lever to fresh air mode . Turn the mode control dial to the windshield-defroster setting . This turns on the air conditioner automatically, since dry air clears the glass faster; you cannot switch the A/C off in this mode . Set the temperature so the air feels warm. Closing the round dashboard corner vents sends more air to the defroster vents . Once the glass clears, switch back to fresh air mode so the windows do not fog again .

Defrosting the rear window

Press the rear window defogger button on the dash to turn its heating grid on and off; the button has its own indicator light . This button includes a built-in shutoff timer, so it turns itself off after a set time rather than running forever . Never scrape the inside of the rear glass up and down; always wipe it side to side . The defogger's heating wires run along the inside of the glass, and scraping can damage them .

Heating and air conditioning (HVAC)

Four main controls sit across the dash, left to right . The fan control dial raises or lowers airflow. The temperature control dial makes the air warmer as you turn it clockwise. The fresh air/recirculation lever picks where the air comes from. The mode control dial picks which vents the air flows from: dashboard vents, floor vents, a mix of the two, the windshield defroster, or a floor-and-defroster mix . A small amount of air always flows from the dashboard corner vents, no matter which mode you pick . A separate A/C button turns the air conditioning compressor on and off, with its own indicator light . Air conditioning comes standard on every 2008 US Fit, including base trim .

The fresh air/recirculation lever controls where intake air comes from . Slide it to fresh air, and outside air enters through intakes at the base of the windshield; keep that area clear of leaves and debris . Slide it to recirculation, and the system reuses air already inside the cabin . Leave the system in fresh air mode most of the time, since recirculation can fog your windows, especially with the A/C off . Switch to recirculation only for short stretches through dust or smoke, then switch back .

Power windows

The ignition must be in the ON position for any window to move . Push a window switch down and hold it to lower that window; release it to stop partway . Pull back and hold to raise a window . The driver's window has an AUTO feature: push its switch firmly down past the first light push, then let go, and the window lowers all the way by itself . Pulling back briefly while it lowers stops it partway . A MAIN switch on the driver's door disables the passenger window switches when pushed in; the driver's own window keeps working . Push the MAIN switch again to turn the passenger switches back on . Keep the MAIN switch off, disabling the passenger windows, whenever children ride in the car, so they cannot operate a window and hurt themselves .

Always check that everyone is clear before closing any window; a closing power window can seriously injure a hand or fingers . Power windows, including auto-down on the driver's side, come standard on base trim .

Door locks

Each door has a lock tab on top; push it in to lock and pull it out to unlock . You can also lock or unlock the front doors with the ignition key . On the driver's door, pushing the top of the master power-lock switch locks all doors and the tailgate together; pushing the bottom unlocks them all . Each passenger door's own tab locks or unlocks only that door . Turning the key once in the driver's door from outside unlocks only that door; turning it again unlocks the rest . To lock a passenger door from outside, push in its tab and close the door . To lock the driver's door from outside, take the key out, push in the tab or press the top of the master switch, then close the door . A lockout-prevention feature exists too. With the driver's door open and the key still in the ignition, the master switch will not work, and pushing in the driver's tab does not lock the car . This design stops you from locking your only key inside by accident .

Child safety locks

Each rear door has a childproof lock lever near its edge . Set it to LOCK, and that rear door cannot open from the inside handle at all, no matter how the regular lock tab is set . It opens only from outside . To release it, pull the lock tab up and open the door from outside .

Dome light

The ceiling light has a three-position switch: ON, Door Activated, and OFF . In Door Activated mode, the light turns on when you open any door, or unlock the doors with the key . It fades out over about 30 seconds after every door closes tightly . It also turns on when you remove the key from the ignition,

and fades the same way if you do not then open a door . If a door stays open with no key in the ignition, the light shuts off completely after 3 minutes to protect the battery .

12-volt power outlet

The accessory power socket sits in the dash's center pocket, directly above the aux jack pocket; pull up its rubber cover to use it . It powers 12-volt DC (direct current, the steady, one-direction electricity a car battery supplies) accessories rated 120 watts or less, about 10 amps; it will not power a heating element like an old cigarette lighter . The ignition must be in ACCESSORY or ON for it to have power; it goes dead in LOCK . This outlet comes standard on base trim .

Radio and audio

The base Fit comes with a 160-watt AM/FM/CD system and 4 speakers . With the ignition in ACCESSORY or ON, turn the system on with the PWR/VOL knob or the AM/FM button, and adjust volume on the same knob . Change bands with the AM/FM button . Five ways exist to find a station. TUNE steps through frequencies one at a time, and SEEK searches automatically for the next strong signal . SCAN samples every strong station for about 10 seconds each . Six preset buttons store one AM station and up to two FM stations each, for 6 AM and 12 FM presets total . AUTO SELECT finds and stores strong local stations, which helps on a road trip far from home . The system also plays CDs and raises volume on its own as road noise increases with speed .

Base trim has no aux jack. This is an important, real limit of the car. An aux jack lets you plug a phone or music player into the stereo with a cable. Two official Honda sources confirm the aux jack ships only on the US Fit Sport and on Canadian models, never on the US base Fit . Compared to the Sport's 200-watt, 6-speaker system, the base system also lacks MP3/WMA playback and a 5-mode equalizer (a tool for adjusting different sound frequencies) .

Speed-sensitive volume, however, works on both trims . The stereo gives you no direct way to plug a phone in. If you want to play music from a phone in a base Fit, you will need a separate solution, such as an FM transmitter.

To set the clock on base trim's radio, press and hold the CLOCK button until you hear a beep; the displayed time starts blinking . Press the H button repeatedly to set the hour, then the M button repeatedly to set the minutes . Press CLOCK again to lock in the time . To round to the nearest hour instead, press CLOCK until it beeps, then press the R button .

Steering wheel

The steering wheel tilts up and down; this feature comes standard on base trim . Push the lever under the steering column all the way down to unlock the wheel . Move the wheel up or down while keeping the gauges visible, then push the lever back up to lock it in place . The wheel does not telescope, meaning it does not slide closer to or farther from you .

Step by step: Defrost a foggy windshield

1. Turn the fan on; pick high speed to clear the glass faster .
2. Slide the fresh air/recirculation lever to fresh air mode .
3. Turn the mode control dial to the windshield-defroster setting .
4. Set the temperature dial so the air feels warm.
5. Close the round dashboard corner vents to send more air to the defroster vents .
6. Once the windshield clears, switch back to fresh air mode .

What can go wrong

- Closing a power window on someone's fingers can cause serious injury; always check that hands and fingers are clear first .

- Leaving the passenger window switches active with a child in the car lets that child operate a window and get hurt; use the MAIN switch to disable them .
- Scraping the inside of the rear window up and down can damage the defogger's heating wires; always wipe side to side .
- Expecting an aux jack on a base-trim stereo will leave you stuck; the US base Fit does not have one .
- Leaving a door open with the key in the ignition can drain your battery through the dome light and chimes. The light itself shuts off after 3 minutes to help protect the battery .

Quick check

1. Which lever controls your turn signals, and which direction do you push it for a left turn?
2. Name two of the six positions on the windshield wiper lever.
3. Does the base 2008 Fit have an aux jack for a phone or music player?
4. What should you do before closing a power window, and why?
5. How do you stop a child in the back seat from opening a rear door from the inside?

CHAPTER 9

The Magic Seat and Cargo Tricks

What you will learn

- What the Magic Seat is and why the Fit's cargo space is unusual for such a small car
- How to fold the rear seat flat for cargo (Utility Mode)
- How to flip the rear seat cushions up for tall items (Tall Mode)
- How to make one long flat surface for lumber or surfboards (Long Mode)
- How to recline the front seats flat to rest (Refresh Mode)

The short version

The Magic Seat is Honda's name for the Fit's 60/40-split rear seat, split so each side folds on its own . It can fold four different ways beyond normal seating, called Utility, Tall, Long, and Refresh mode . Each mode takes a few minutes and no tools, but each one also has real safety rules you must follow before you start.

The full explanation

Honda built the Magic Seat as a 60/40-split rear seat, meaning the two sides split unevenly . This lets you fold one side and leave a passenger seated on the other . Honda describes five total seating and cargo setups: the standard five-passenger mode, plus refresh mode, tall object mode, long object mode, and utility mode . This chapter covers the four non-standard modes as step-by-step procedures.

With every seat up in standard mode, the Fit holds 90.1 cubic feet of passenger space and 21.3 cubic feet of cargo space behind the back seat . Fold the rear seat-backs down flat for Utility Mode, and cargo space grows to 56.8 cubic feet .

Two basic seat motions make up all four modes. First, a rear seat cushion can fold up and lock against the seat-back, which is the base move behind Tall Mode . Second, a rear seat-back can fold down flat, which is the base move behind Utility Mode and part of Long Mode . Long Mode and Refresh Mode add more steps on top of these two basic motions.

Step by step

Utility Mode: fold the rear seat down flat for cargo

Utility Mode gives you a flat load floor from the tailgate forward, for boxes, groceries, or bikes .

Safety warnings, before you start: - Always remove any items from the rear seat before you fold it. - Always make sure nothing sits on the floor before you fold the seat-back down. - Always keep cargo below the bottom of the windows, so it does not block the side curtain airbags (airbags built into the roof rail above the windows). - Always tie down loose cargo so it cannot fly forward during hard braking. - Do not put heavy items directly on a folded seat-back. - Do not let a passenger sit on a rear seat you have folded down; return each side fully upright, locked, and belted first.

Steps: 1. Decide which side of the 60/40 rear seat you are folding: driver's side, passenger's side, or both . 2. Remove everything from that rear seat. 3. Move the matching front seat as far forward as it will go . 4. Lock the rear outer head restraint (the padded support behind your head) on that side in its lowest position . 5. If you are folding the driver's side, also set the rear-center head restraint to its lowest position . 6. If you are folding the driver's side, pull the latch plate to release the center seat belt from its anchor, and let the belt

retract into the ceiling holder . Store the buckle there too . 7. Pull up the release lever on the outside of that seat-back . 8. Fold the seat-back forward until it lies flat . 9. Repeat steps 2 through 8 on the other side if you need the full-width flat floor. 10. Load your cargo, keeping it below window height and tied down.

How you know it worked: The seat-back lies flat and level, forming a flat load floor from the tailgate forward. It does not spring back up when you let go.

Tall Mode: flip the rear cushions up for tall items

Tall Mode opens floor space in front of the seat-back, for items too tall to fit under a normal seat .

Safety warnings, before you start: - Always remove any items from the rear seat before you fold the cushion up. - Always make sure the cushion locks firmly against the seat-back before you load anything against it. - Do not stack items higher than the top of the seat-backs; they can block your rear view and be thrown around the car in a crash. - Do not place cargo where it blocks your view through the rear window.

Steps: 1. Remove everything from the rear seat cushion you are flipping up. 2. Pull up on the rear seat cushion . 3. Fold the seat leg down as the cushion comes up . 4. Push the seat cushion firmly against the seat-back until it locks in place . 5. Repeat for the other side if you want both cushions up. 6. Load your tall item into the open floor space in front of the seat-back. 7. To return to normal seating later, make sure the floor under the seat is empty first . 8. Pull the seat leg up fully . 9. Slowly push the seat cushion back down while holding it with both hands . 10. Set the seat leg into its floor guide; a latch appears when it seats correctly .

How you know it worked: The seat cushion stays locked upright against the seat-back on its own. It does not fall back down when you let go.

Long Mode: carry long items like lumber or surfboards

Owners report that Long Mode combines the Tall and Utility motions to create one flat path from the tailgate to the dashboard . We could not confirm an official Honda step list for this exact combined mode on the 2008 GD3; check your owner's manual before loading anything long or heavy .

Safety warnings, before you start: - Always keep at least one working seat belt and seating position free for every person riding in the car. Do not use more seats than are actually available once you reconfigure them. - Always tie down long cargo. An unsecured long item can fly forward with great force during hard braking, which can seriously injure or kill someone in the front seat. - Do not let cargo extend into the space where an airbag needs to open. - Do not drive with cargo blocking your view through the rear window or a side mirror. - We could not fully confirm these exact steps against the 2008 GD3 owner's manual. Follow your own owner's manual's cargo-loading warnings if they differ from this book .

Steps (community-reported): 1. Fold the passenger-side rear seat cushion up, as in Tall Mode steps 1 through 4, or fold that seat-back down flat, as in Utility Mode . Pick whichever gives you the flattest path to your item . 2. Move the front passenger seat as far forward as it will go . 3. Recline the front passenger seat-back forward until it rests against the folded rear seat, forming one flat surface from the tailgate to the dashboard on that side . 4. Leave the driver's-side rear seat upright if you still need to carry a rear passenger . 5. Load the long item, such as lumber, a surfboard, or a ladder, along the flattened path. 6. Tie the item down using the cargo area's tie-down anchors so it cannot slide or shift.

How you know it worked: The item lies flat and secured along one continuous surface from the tailgate to the dashboard. It does not shift when you push on it by hand.

Refresh Mode: recline the front seats flat to rest

Refresh Mode reclines the front seats flat, level with the rear cushions, for resting when the car is parked . This is not a driving position.

Safety warnings, before you start: - Always park the car, set the parking brake, and turn off the engine before you recline the seats this far. - Always reinstall both front head restraints and lock both seats securely upright again before anyone drives or rides in the car. - Do not drive the car, or let anyone ride in a moving car, with the front seats in this reclined position. Seat belts are not designed to protect a reclined occupant (a person lying back in the seat). - Do not let a child use this mode unsupervised; the fold-down lever and mechanism could pinch fingers.

Steps: 1. Park the car, set the parking brake, and turn off the engine. 2. Recline the rear seat-backs as far back as they will go . 3. Remove both front head restraints and store them somewhere secure so they do not roll around . 4. Using the seat slide lever, move both front seat cushions as far forward as they will go . 5. Pull up the seat-back angle adjustment lever on each front seat . 6. Pivot each front seat-back backward until it is level with the rear seat cushions, creating one large cushioned area . 7. To return to the normal driving position, hold each seat-back with your hand first. 8. Pivot each seat-back back upright, controlling the motion so it does not come up too fast. 9. Reinstall both head restraints . 10. Lock both seats securely before anyone drives or rides in the car.

How you know it worked: The front seat-backs lie flat and level with the rear seat cushions, forming one continuous cushioned surface. Each seat-back stays down on its own without you holding it.

What can go wrong

Utility Mode

- Forgetting to move the front seat forward first stops the folded seat-back from lying flat.
- Leaving the head restraint up blocks the seat-back from folding flush.
- Skipping the center seat belt's anchor release on the driver's side can strain or damage that belt.
- Not checking the passenger airbag-off indicator after you restore the seat can leave you unsure if the front passenger's weight sensor still reads correctly.

Tall Mode

- Not pushing the cushion firmly enough against the seat-back means it is not truly locked, and it can fall during driving.
- Forgetting to fold the seat leg down can leave the mechanism loose.
- Loading an item so tall that it blocks the rear window can also interfere with the seat-backs later, when you want to return the seat to normal.

Long Mode

- Forgetting to move the front passenger seat forward before reclining it leaves a gap instead of a flat surface.
- Not tying the item down lets it slide during braking or turns.
- Loading an item so long that it sticks out past the rear bumper without a proper reflective flag or marker is unsafe, and may be illegal in some states.

Refresh Mode

- Forgetting to remove the head restraints first stops the seat-back from lying fully flat.

- Letting go of the seat-back too quickly when you return it upright is risky; it can spring up faster than expected, so control it by hand.
- Forgetting to relock the seats and reinstall the head restraints before driving again leaves the car unsafe for the next trip.

Quick check

1. What are the names of the four non-standard Magic Seat modes?
2. Before you fold the driver's-side rear seat-back down, what must you do with the center seat belt?
3. Why should you never stack cargo higher than the tops of the seat-backs in Tall Mode?
4. Why is Refresh Mode only safe when the car is parked with the engine off?
5. Why does this book label the Long Mode steps as "community-reported" instead of manual-confirmed?

CHAPTER 10

Fuel

What you will learn

- What fuel grade your Fit needs
- How big the fuel tank is, and roughly how far it will take you
- What the low-fuel light means, and what happens if you ignore it
- How to fill the tank correctly and safely
- What the "CHECK FUEL CAP" message means and how to clear it

The short version

Your 2008 Fit needs 87 AKI (pump octane) regular unleaded gasoline or higher . The tank holds 10.8 US gallons (41 liters) . Fill it outdoors, away from any flame or spark, and never try to add fuel again after the pump nozzle clicks off .

The full explanation

Fuel grade

Use 87 AKI (pump octane, the number posted on the gas station pump) unleaded gasoline or higher . Using a lower-octane fuel can cause a heavy, persistent metallic knocking sound in the engine, and that knock can damage the engine over time . Honda also recommends gasoline with detergent additives (chemicals that help keep the engine clean) . Where you can find it, choose gasoline without manganese-based additives such as MMT . Those additives can trigger the malfunction indicator lamp, the dashboard warning light also called the "check engine" light . Your Fit is designed for gasoline

blended with up to 10% ethanol or up to 15% MTBE (a fuel additive; short for methyl tertiary-butyl ether) by volume, but never gasoline containing methanol .

Tank size and range

The fuel tank holds 10.8 US gallons, or 41 liters . The owner's manual and this book do not give an exact miles-per-gallon figure in this chapter's sources. Your real-world range will depend on how and where you drive, so use your fuel gauge, not a mental estimate, to judge how far you can go.

The low-fuel light

The low fuel indicator (a dashboard light) turns on when about 1.7 US gallons (6.3 liters) remain in the tank, before the needle reaches E . A small extra reserve remains even once the needle reaches E, but treat that reserve as a safety margin, not fuel to drive on normally . Running the tank out of fuel can make the engine misfire and can damage the catalytic converter (a part that cleans your exhaust) . Avoid driving with an extremely low fuel level .

The fuel door and cap

The fuel fill door opens from outside the car; pull its notched edge to open it, since this trim has no separate release lever inside the car . The fuel fill cap stays tethered to the filler neck, so you cannot drop or lose it . You may hear a hissing sound as you remove the cap; that is normal tank pressure escaping . The cap's tether fits into a slit on the fuel fill door, so it can hang there while you refuel . Screw the cap back on until it clicks at least once; an improperly tightened cap can trigger the malfunction indicator lamp and a "CHECK FUEL CAP" message .

The "CHECK FUEL CAP" message

If the fuel fill cap is loose or missing, a "CHECK FUEL CAP" message scrolls across the instrument panel's information display the next time you start the engine . You can press the trip-meter select/reset knob to clear the message

from the display early . It comes back every time you restart the engine until you fix the cap . If the system keeps detecting a loose or missing cap, the malfunction indicator lamp also turns on . To fix it, turn the engine off, check that the cap is installed, and tighten it until it clicks at least once . The malfunction indicator lamp will not turn off right away; it can take several days of normal driving after you fix the cap . If it stays on longer than that, have a dealer inspect the car .

Step by step: Refuel the car

Safety warnings, before you start: - Gasoline is highly flammable and explosive. You can be burned or seriously injured while handling it . - Always turn off the engine before you start fueling . - Always keep heat, sparks, and open flame away from the fuel fill area . - Always handle fuel outdoors only, never in an enclosed space . - Always wipe up any spilled fuel right away . - Do not try to "top off" the tank once the pump nozzle clicks off on its own; leave room for the fuel to expand as its temperature changes .

Steps: 1. Park with the driver's side of the car closest to the fuel pump. 2. Turn off the engine . 3. From outside the car, open the fuel fill door by pulling its notched edge . 4. Remove the fuel fill cap slowly; a hissing sound as tank pressure escapes is normal . 5. Fit the cap's tether into the slit on the fuel fill door, so the cap hangs there while you fuel . 6. Insert the pump nozzle and select 87 AKI regular unleaded fuel or higher . 7. Fill until the nozzle clicks off automatically, then stop; do not try to add more . 8. If the nozzle clicks off repeatedly before the tank feels full, try a different pump; this can mean a problem with the fuel vapor recovery system . A dealer visit may help if it keeps happening . 9. Screw the fuel fill cap back on until it clicks at least once . 10. Close the fuel fill door.

How you know it worked: The pump handles cleanly, the cap clicks at least once when you tighten it, and the fuel gauge needle moves up. No "CHECK FUEL CAP" message or malfunction indicator lamp appears on your next drive.

What can go wrong

- Not tightening the fuel cap until it clicks can trigger a "CHECK FUEL CAP" message and the malfunction indicator lamp . This can take several days of driving to clear, even after you fix the cap .
- Trying to "top off" the tank past the automatic click-off risks a spill, and leaves no room for the fuel to expand .
- Forgetting to close the fuel fill door before you drive away leaves it open and exposed.
- Driving with the fuel gauge near empty risks a misfire and can damage the catalytic converter .

Quick check

1. What octane rating does your 2008 Fit need?
2. About how many gallons remain in the tank when the low fuel indicator first turns on?
3. Why should you never try to "top off" the tank after the pump clicks off?
4. What does a "CHECK FUEL CAP" message mean, and how do you fix the underlying problem?
5. Why must you turn off the engine before you start fueling?

PART III

Driving It

CHAPTER 11

First Drive for a New Driver

What you will learn

- What is different about starting off in a manual Fit versus an automatic Fit
- The gear positions on the base-trim automatic transmission, and what each one does
- A short first-drive routine for each transmission type
- Where to find the full manual-transmission practice routine in this book

The short version

A manual Fit needs your left foot on the clutch pedal before it will move. An automatic Fit only needs you to pick the right gear lever position and press the gas. Both transmissions reward a slow, gentle first drive in an empty parking lot.

The full explanation

This chapter gives you a first look at driving your Fit, split by transmission type. If you do not know which one your car has, look at the floor. A clutch pedal on the far left, next to the brake and gas, means you have a manual. Only two pedals means you have an automatic.

If your Fit has a manual transmission

A manual transmission asks you to pick every gear yourself using a lever and a clutch pedal. The clutch is the pedal that disconnects the engine from the wheels so you can change gears without grinding them.

The Fit's manual gearbox uses a standard five-speed "H" shift pattern. First and third gear sit in the top row of the pattern, and second, fourth, and reverse sit in the bottom row, with reverse next to fourth and fifth. The gearbox is synchronized in every forward gear, which means the gears are built to mesh smoothly as you shift.

This chapter's first-drive routine for a manual is short on purpose. Chapter 12 walks through the full sequence, step by step, with the exact clutch and gas technique to use, so read that chapter before your first real practice session.

If your Fit has an automatic transmission

The base-trim automatic Fit has a full shift gate, meaning it offers more lever positions than only "Drive." The positions are Park, Reverse, Neutral, Drive, a position marked "D" with a small "3" (called D3), Second, and First. The Sport-model Fit has a shorter gate with paddle shifters instead, but the base trim you are learning on has all seven positions.

Here is what each position after Park, Reverse, and Neutral actually does:

- **Drive (D):** your normal, everyday driving position. The transmission picks a suitable gear on its own, from first through fifth, based on your speed and how hard you are pressing the gas. When the engine is cold, it may shift up at a higher engine speed than usual, to help the engine warm up faster.
- **D3:** works like Drive, but the transmission stays in first, second, or third gear and will not shift into fourth or fifth. Honda intends this position to give you engine braking on a steep downhill, and to stop the transmission from constantly shifting between third and fourth in stop-and-go traffic. Engine braking means using the drag of the engine, instead of only the brakes, to slow the car down.
- **Second (2):** locks the transmission in second gear, so it will not drop to first when you stop. The manual recommends this position for climbing a grade with more power, and for extra engine braking on a steep downhill.

It also helps for starting out on a slippery surface or in deep snow, and for reducing wheel spin.

- **First (1):** locks the transmission in first gear. By moving the lever among First, Second, D3, and Drive as conditions call for it, you can shift the automatic somewhat like a manual, only without a clutch pedal.

For an ordinary first drive, you will spend almost all of your time in Drive. Save Second, First, and D3 for hills, snow, or heavy traffic, and remember to move the lever back to Drive once you no longer need them.

Note for parents and new drivers: this chapter's driving sources cover the gear lever positions themselves. They do not cover the exact steps for starting the engine and pulling out of a parking space on an automatic. Chapter 6 of this book, on starting and stopping the car, covers the engine-start sequence in full.

Step by step

Manual transmission — short version. Full safety warnings and the full 11-step routine are in Chapter 12. In brief: fasten your seatbelt and adjust your seat and mirrors, then press the clutch fully to the floor and start the engine. Shift into first gear, then slowly release the clutch while pressing the gas gently until the car moves. Go to Chapter 12 before you try this for the first time.

Automatic transmission — first drive. 1. Fasten your seatbelt and adjust your seat and mirrors before you shift out of Park. 2. Press the brake pedal, and move the shift lever from Park to Drive. 3. Release the parking brake. 4. Slowly release the brake pedal, then press the gas pedal gently. 5. Let the transmission pick its own gears as your speed changes; you do not need to shift it yourself in Drive. 6. If you are climbing a long, steep hill and want more power, or descending a steep hill and want more engine braking, move the lever to D3. 7. Move the lever back to Drive once you are back on a normal road.

What can go wrong

- **Manual:** letting the clutch pedal up too fast causes the engine to stall, which means it shuts off because it cannot keep running against the load. Chapter 12 covers this in detail.
- **Automatic — forgetting to shift back to Drive:** if you leave the lever in Second, First, or D3 after a hill or snow, the transmission stays stuck there. It will not use all five gears until you move the lever back. Get in the habit of checking the lever position once conditions are back to normal.
- **Automatic — confusing D3 with Drive:** D3 feels almost identical to Drive at low speed, since both use first through third gear in that range. The difference only shows up above the speed where Drive would normally shift into fourth or fifth.

Quick check

1. How can you tell, by looking at the pedals alone, whether your Fit is a manual or an automatic?
2. Name the seven lever positions on the base-trim automatic Fit's shift gate.
3. What is the difference between Drive and D3?
4. Which position would you choose to help start out on deep snow or a slippery surface?
5. Where in this book should you go to learn the full manual-transmission first-drive routine?

CHAPTER 12

Driving a Manual Transmission from Zero

What you will learn

- How the clutch pedal works and why letting it up too fast stalls the engine
- The Fit's shift pattern, and how to get into reverse without grinding the gears
- The recommended speeds for shifting up, for the best fuel economy
- A full first-drive routine, plus a routine for starting off on a hill

The short version

The clutch pedal disconnects the engine from the wheels so you can change gears. Let it up slowly and smoothly, with a little gas, and the car moves off without stalling. Practice in an empty lot before you try traffic or hills.

The full explanation

A manual transmission gives you full control of which gear the engine is in, using a gear lever and a clutch pedal. The clutch pedal is the leftmost of the three pedals. Pressing it all the way down disconnects the engine from the wheels, so you can move the gear lever without grinding the gears.

The shift pattern

The Fit's manual gearbox uses a standard five-speed "H" pattern. First and third gear are in the top row of the pattern; second, fourth, and reverse are in the bottom row, with reverse next to fourth and fifth. Every forward gear is synchronized, meaning the gearbox is built to let the gears mesh smoothly as you shift.

Getting into reverse without grinding

The transmission has a built-in lockout that stops you from shifting straight from fifth gear into reverse. To shift into reverse without the gears grinding, come to a full stop and push the clutch pedal all the way down. Then either pause a few seconds before moving the lever into reverse, or shift briefly into a forward gear first. Either action gives the input shaft, the spinning part inside the gearbox that reverse gear must line up with, time to stop turning before reverse engages.

Shifting for good fuel economy

The owner's manual gives recommended shift points for normal acceleration. Shift from first to second around 15 mph (24 km/h), and second to third around 27 mph (43 km/h). Shift from third to fourth around 39 mph (62 km/h), and fourth to fifth around 53 mph (85 km/h). As a general rule, drive in the highest gear that still lets the engine run and speed up smoothly. That gives you the best fuel economy and the lowest emissions.

Starting off on a hill

Honda's owner's manual for this generation of Fit does not describe a dedicated hill-start procedure. This car also does not have a Hill Start Assist feature, which is a system that appears on some later Honda models. The manual's page on parking does cover parking on a hill. Turn your wheels, use

first gear if you are facing uphill or reverse if you are facing downhill, and set the parking brake. It does not, however, describe how to pull away from a stop on an uphill grade.

Because of that gap, the hill-start routine below is a general three-pedal technique common to manual-transmission cars, built from this car's own documented clutch and parking-brake behavior. It is good general practice, not a Honda-published Fit procedure.

Step by step

Drive a manual transmission for the first time

Safety warnings, before you touch a pedal: - Always practice first in an empty parking lot or a very quiet street, not in traffic. - Always fasten your seatbelt and adjust your seat and mirrors before you start the engine. - Always press the clutch pedal all the way to the floor before turning the key to start the engine; the starter will not engage otherwise. - Do not rev the engine hard or ride the clutch pedal partly pressed. This wears the clutch out faster and can damage it. - Always come to a full stop before shifting into reverse. - Do not try to shift straight from fifth gear into reverse. The transmission has a lockout and needs a short pause, or a pass through a forward gear, first.

Steps: 1. Adjust your seat, mirrors, and seatbelt before starting the engine. 2. Push the clutch pedal all the way to the floor. The starter only works with the clutch down. 3. Turn the key to start the engine, then shift the lever into first gear. 4. Keep the brake pedal or parking brake applied while you slowly let the clutch pedal rise. 5. Feel for the "friction point," the moment the engine note changes and the car wants to creep forward. 6. Gently press the gas pedal while you keep easing the clutch pedal up. 7. Once the car is moving, finish releasing the clutch pedal and keep your foot off it. 8. Shift up around 15 mph into second, 27 mph into third, 39 mph into fourth, and 53 mph into fifth. 9. To shift up, press the clutch fully down, move the lever to the next gear, then

ease the clutch back up. 10. To slow down, press the clutch down fully right before the engine would stall, then brake normally. 11. Once stopped, shift to neutral and take your foot off the clutch until you are ready to move again.

How you know it worked: you can pull away smoothly and shift up and down through the gears. You can also come to a full stop without the engine stalling or the car jerking.

Start on a hill (manual transmission)

Safety warnings, before you try this: - Always practice this first in an empty parking lot or a quiet street with a gentle slope, not on a busy or steep hill. - Always keep your right foot ready to move to the brake pedal in case the car starts to roll backward. - Do not practice on a hill with another car stopped close behind you until you are confident you will not roll back. - Always turn on your hazard lights if you stall on a hill and are blocking traffic, then try again. - Do not rest your foot on the clutch pedal once you are moving; this wears the clutch out faster.

Steps: 1. Come to a full stop on the hill using the brake pedal. 2. Pull the parking brake up firmly with your right hand. 3. Push the clutch pedal all the way to the floor. 4. Shift the lever into first gear. 5. Move your right foot from the brake pedal to the gas pedal. 6. Press the gas gently while slowly letting the clutch pedal rise, until you feel the engine pull against the car. 7. When you feel the car want to move forward, release the parking brake fully. 8. Keep easing the clutch pedal up while giving a little more gas, so the engine does not bog down or stall. 9. Once the car is moving smoothly forward, finish releasing the clutch pedal all the way.

How you know it worked: the car moves forward up the hill smoothly, does not roll backward at any point, and the engine does not stall.

What can go wrong

- **Letting the clutch pedal up too fast.** This stalls the engine, meaning it shuts off because it cannot keep running against the sudden load.
- **Riding the clutch.** Holding the clutch pedal partway up while driving wears the clutch out faster.
- **Giving too little gas while releasing the clutch.** This also stalls the engine, especially on a hill.
- **Releasing the parking brake before the clutch has caught, on a hill start.** This lets the car roll backward.
- **Forgetting to fully release the parking brake once moving.** This can overheat the rear brakes.
- **Looking down at the shift lever instead of watching the road.** Learn the shift pattern by feel.
- **Trying to shift straight from fifth gear to reverse.** Pause first, or pass through a forward gear, instead.

Quick check

1. What does the clutch pedal do, and where is it located?
2. What are the recommended speeds for shifting from first to second, and from fourth to fifth, for good fuel economy?
3. Why can you not shift straight from fifth gear into reverse, and what two things can you do instead?
4. What should your right foot be ready to do during a hill start, in case the car rolls backward?
5. Does the 2008 Fit have a Hill Start Assist feature?

CHAPTER 13

Parking, Reversing, and Tight Spaces

What you will learn

- Where reverse gear sits in the manual shift pattern, and how to get into it smoothly
- What the owner's manual says about parking on a hill
- Why this chapter is shorter than you might expect, and where to look for more

The short version

Reverse sits next to fourth and fifth gear in the Fit's shift pattern, and the transmission has a lockout so you cannot grind into it from fifth by mistake. On a hill, turn your wheels, leave the car in gear, and set the parking brake. Beyond that, the sources for this book do not cover detailed parking or reversing technique for this specific car.

The full explanation

Reverse gear, on a manual Fit

In the Fit's five-speed "H" shift pattern, reverse sits in the bottom row of the pattern, next to fourth gear. The transmission has a lockout that prevents you from shifting straight from fifth gear into reverse. To get into reverse smoothly, come to a full stop and push the clutch pedal all the way down. Then either pause a few seconds or shift briefly into a forward gear before moving the lever into reverse. Either action gives the gearbox's spinning input

shaft time to stop turning before reverse engages. On an automatic Fit, reverse is one of the seven positions on the base-trim shift gate, alongside Park, Neutral, Drive, D3, Second, and First.

Parking on a hill

The owner's manual's page on parking covers what to do when you park on a hill. Turn your front wheels toward the curb or the edge of the road. Leave the transmission in first gear if you are facing uphill, or in reverse gear if you are facing downhill, and set the parking brake. This gives the parked car a second way to resist rolling, in addition to the parking brake, if it were to slip.

What this chapter does not cover, and why

The sources used for this book do not include specific technique for parallel parking or backing into a space. They also do not cover using mirrors while reversing, or judging distances in tight spaces, for the 2008 Fit. We could not confirm Honda-specific guidance on these points. Rather than invent tips that are not in the source material, this book points you to general safe-driving practice instead. Go slowly, and check your mirrors and look over your shoulder before and during any reversing move. If you are unsure a space is tight, get out and look before you try it. A driving instructor or your state's driver's manual is a good place to learn parking and reversing technique, step by step.

Quick check

1. Where does reverse gear sit in the Fit's manual shift pattern?
2. What must you do with the clutch pedal before shifting into reverse, and why?
3. According to the owner's manual, what gear should you leave the car in when parked facing uphill? Facing downhill?
4. Why does this chapter not include detailed parking or reversing tips?

CHAPTER 14

Highway, Rain, Snow, Heat, and Night

What you will learn

- Why driving fast on the highway costs you fuel economy
- What kind of tires the owner's manual recommends for winter driving
- The rules for using tire chains on a Fit
- Which weather and lighting conditions this chapter's sources do not cover

The short version

Speed above about 45 to 50 mph costs you fuel economy fast, because wind resistance rises sharply. For winter, "M+S" or all-season tires cover most conditions, and real snow tires or approved cable-type chains, on the front only, help further. The sources for this book do not cover rain, extreme heat, or night driving for this specific car.

The full explanation

Highway speed

Aerodynamic drag, the resistance of the air pushing back against the car, rises sharply above 45 mph (75 km/h). The EPA's general driving guidance says gas mileage usually drops quickly above 50 mph. Each 5 mph you drive over 50 mph acts like paying about \$0.29 more per gallon, based on EPA's national average fuel price at the time that figure was published. This EPA guidance is written for gasoline cars in general, not for the Fit specifically, but it applies to any gasoline car including the Fit. Roof racks, carriers, and trailers add extra drag on top of that.

Winter tires and traction

Tires marked "M+S," which stands for mud and snow, or marked "All Season" on the sidewall, are suitable for most winter driving. For the best traction in snow or ice, the owner's manual recommends installing dedicated snow tires or tire chains, and notes these may be legally required in some places. If you mount snow tires, they must be radial tires of the same size and load range as your originals. They must go on all four wheels, not only the drive wheels. Dry-road traction may be lower than with your original tires. Ask your tire dealer for a maximum speed recommendation for the snow tires you choose.

Tire chains

Tire chains are allowed on the Fit, but with limits specific to this car's tight tire clearance. Mount chains only on the front tires, never the rear. Only SAE Class "S" cable-type traction devices with rubber chain tensioners are allowed. The manual says not to use metal link-style chains. Install chains only when conditions or local law require them, and follow the chain maker's instructions. Mount them as tight as possible and drive slowly. Check that they are not contacting the brake lines or suspension, and stop right away if you hear contact. Remove the chains as soon as the road is clear.

Rain, heat, and night driving

This book's research did not find Honda-specific guidance for the 2008 Fit on driving in rain, in extreme heat, or at night. We could not confirm this for the 2008 Fit. Check your owner's manual, and until you find more specific guidance, use general safe-driving practice instead. Slow down in rain, since wet roads reduce traction. Watch your temperature gauge and coolant level more closely in extreme heat. Chapter 16 covers overheating in detail. Use your headlights and reduce speed at night, when your view of the road is shorter.

Quick check

1. About how fast can you drive before aerodynamic drag starts costing you noticeably more fuel?
2. What two words does "M+S" stand for on a tire sidewall?
3. Which wheels should carry tire chains on a Fit, and what type of chain does the manual allow?
4. Does this chapter's source material cover night driving for the 2008 Fit? What should you do instead?

CHAPTER 15

Getting Good Fuel Economy

What you will learn

- The shift points that give the best fuel economy on a manual Fit
- What the owner's manual and the EPA say hurts and helps your fuel economy
- Why the break-in period matters for your car's long-term health
- Why you should not tow a trailer with this car

The short version

Drive smoothly, keep your tires inflated, keep up with maintenance, and stay off the gas pedal in stop-and-go traffic. On a manual, shift up early. Above about 45 to 50 mph, wind resistance starts eating into your mileage fast.

The full explanation

Shifting for economy, on a manual

The owner's manual's recommended shift points, for normal acceleration, are: first to second around 15 mph (24 km/h), and second to third around 27 mph (43 km/h). Third to fourth comes around 39 mph (62 km/h), and fourth to fifth around 53 mph (85 km/h). The general rule is to drive in the highest gear that still lets the engine run and speed up smoothly.

What the owner's manual says hurts and helps your fuel economy

Factors that lower fuel economy, according to the owner's manual, include aggressive driving, meaning hard acceleration or hard braking, and excess idling and stop-and-go acceleration and braking. Running the engine cold, driving with a heavy load or the air conditioning on, and under-inflated tires also hurt it. Tips to improve it include keeping up with maintenance, using the correct oil viscosity, and keeping your tires properly inflated. Avoid excess weight in the car, and keep the underside clean of snow or mud buildup. Drive moderately, meaning you avoid rapid acceleration, hard cornering, and hard braking, and observe the speed limit. Aerodynamic drag, the resistance of the air pushing back against the car, rises sharply above 45 mph (75 km/h). Roof racks, carriers, and trailers add extra drag. On a manual transmission, use the highest gear that still lets the engine run smoothly.

What the EPA says, in general

The EPA's general driving-habit guidance applies to any gasoline car, including the Fit. It says aggressive driving, meaning speeding and hard acceleration or braking, can cut your gas mileage by roughly 15 to 30 percent at highway speeds. In stop-and-go traffic, aggressive driving can cut mileage by 10 to 40 percent. Gas mileage usually drops quickly above 50 mph. The EPA estimates that each 5 mph you drive over 50 mph acts like paying about \$0.29 more per gallon. That figure is tied to national average fuel prices at the time it was published, and is not specific to the Fit. This is general EPA guidance for gasoline cars, not text written specifically about the Fit.

The break-in period

Honda asks you to pay extra attention to your driving style for the vehicle's first 600 miles (1,000 km). This is called the break-in period. During this time, avoid full-throttle starts and rapid acceleration for the whole 600 miles, and avoid hard braking for the first 200 miles (300 km). Do not change the engine oil early. Wait until the regular scheduled maintenance interval. The same

advice applies again after an engine overhaul or exchange, or after the brakes are replaced. Driving gently during break-in protects the engine and brakes, which pays off in reliability and economy for the life of the car.

Towing

The owner's manual states that this vehicle is not built to tow a trailer at all, and that doing so can void the vehicle's warranties. No numeric trailer-towing weight rating is given anywhere in the manual, because the manual's position is that trailer towing should not be done with this car. A trailer adds weight and drag, and the manual already lists both as things that hurt fuel economy. So towing one would work against everything else in this chapter.

Quick check

1. At what speed does the owner's manual recommend shifting from first to second gear, on a manual Fit?
2. Name three things the owner's manual says will hurt your fuel economy.
3. How long is the Fit's break-in period, and what should you avoid doing during the first part of it?
4. Is towing a trailer recommended for this car? What does the owner's manual say about it?

CHAPTER 16

When Something Goes Wrong on the Road

What you will learn

- What each red or important warning light means, and what to do about it
- How to change a flat tire safely, step by step
- How to jump-start a dead battery, step by step
- How to handle an overheating engine, step by step
- The rules for towing this car, and what the front towing hook is really for
- Why this chapter cannot give you a Honda-specific accident checklist, and what to do instead

The short version

Pull over safely, turn on your hazard lights, and read the warning light before you decide what to do next. A flashing or steady red brake light, a low oil pressure light, or a stuck-on high temperature light all mean stop and deal with it, not keep driving. Keep the compact spare's speed limit and the flat-tow distance and speed limits in mind, since both are lower than normal driving speed.

The full explanation

Warning lights you may see on the road

The owner's manual only gives an explicit color label, "(Red)," to one indicator: the Parking Brake and Brake System Indicator. It comes on whenever you turn the ignition to ON, as a parking-brake reminder, and stays

on until the parking brake is fully released. A beeper also sounds if you drive with the parking brake partly applied. If it stays on after you release the parking brake, or comes on while you are driving, something is wrong. The brake fluid level is probably low, or there is a brake system problem. Press the brake pedal lightly to check if it feels normal. If it feels normal, check the brake fluid level at your next stop. If the pedal does not feel normal, meaning it goes down much farther or needs much harder pressure, slow down by downshifting and pull over immediately. Do not keep driving; have the vehicle towed and repaired. If the ABS indicator, the light for the anti-lock brake system, comes on together with this brake indicator, have the vehicle inspected by a dealer right away.

The Low Oil Pressure Indicator should never come on while the engine is running. If it flashes or stays on, oil pressure has dropped very low. Pull over safely, shut off the engine, and turn on your hazard lights. Let the vehicle sit a minute, then open the hood and check the oil level, adding oil to the full mark on the dipstick if needed. Then restart the engine and watch the indicator. If it does not go out within 10 seconds, shut the engine off again; there is a mechanical problem that needs repair before you keep driving. This indicator is commonly described as a red, oil-can-shaped light, but that color could not be confirmed for the 2008 Fit specifically.

The Charging System Indicator means, if it comes on brightly while the engine is running, that the battery is not being charged. Immediately turn off all electrical accessories, and avoid using power windows and similar loads. Keep the engine running, since starting it again would drain the battery further, and go to a service station or garage for help. Owners report that this light is a red battery-shaped symbol, though the owner's manual text itself does not name a color for it.

The Malfunction Indicator Lamp, also called the check-engine light, means one of the engine's emissions-control systems may have a problem. The car may feel unchanged, but fuel economy can drop and emissions can rise. Continued driving may cause serious damage, so have the vehicle checked by

a dealer as soon as possible if it comes on repeatedly. If it comes on right after you refuel, check for a loose or missing fuel cap first. The manual may also show a "CHECK FUEL CAP" message. Tightening the cap can take several days of driving to clear the light. This indicator does not demand pulling over right away, the way the brake, oil-pressure, or overheating indicators do, but the manual still says not to keep driving on it indefinitely. We could not confirm this indicator's color for the 2008 Fit specifically; the manual does not label it at all. Do not assume the check-engine light is a red light. Based on the manual's own wording, only the brake system indicator is confirmed red; the check-engine light is never given a color in the manual.

Hazard lights and reflectors

The hazard warning button, also called the four-way flashers, is on the instrument panel. Pressing it flashes all four exterior turn signals and both instrument-panel turn indicators at the same time. Use the hazard lights if you need to park in a dangerous spot near heavy traffic, or if your vehicle is disabled. Every roadside procedure in this chapter, flat tire, overheating, and low oil pressure, tells you to turn on the hazard lights as an early step.

We could not confirm this for the 2008 Fit. A full read of the owner's manual found no mention anywhere of a warning triangle, reflective triangle, road flare, or roadside reflector. It is not listed as standard equipment or as a recommendation. Check your owner's manual, and consider carrying a reflective triangle or flares as extra roadside protection even though the Fit's manual does not require or mention one.

Towing this car, and the towing hook

Honda names two approved ways to tow this vehicle for ordinary, non-emergency situations. Flat-bed equipment, where the whole car is loaded onto the back of a truck, is what the manual calls the best way to transport the vehicle. Wheel-lift equipment, where the tow truck's two arms lift the front

tires off the ground while the rear tires stay down, is called an acceptable way to tow the vehicle. The manual says to call a professional towing service, and never to tow the car with only a rope or chain, since that is very dangerous.

The owner's manual does not offer flat towing, meaning all four wheels rolling on the ground, as a supported method for either transmission. It only allows front-wheels-on-the-ground towing as a damage-only contingency, when flatbed or wheel-lift equipment cannot be used. This method is capped at 50 miles (80 km) and 35 mph (55 km/h) maximum. For a manual transmission, release the parking brake, shift to neutral, and leave the ignition switch in the ACCESSORY position so the steering wheel does not lock. For an automatic transmission, release the parking brake, start the engine, shift to D, and hold for 5 seconds. Then shift to N, let the engine idle for 3 minutes, and turn it off. Leave the ignition switch in ACCESSORY. If you cannot shift the transmission or cannot start the engine, the vehicle must be transported with the front wheels off the ground instead. The front-wheels-down method does not apply in that case. Do not lift the vehicle, or tow it, using the bumpers. They cannot support the vehicle's weight and will be seriously damaged. Always confirm the steering wheel turns freely, with the ignition in ACCESSORY, before towing begins.

The car also carries a single detachable towing hook and its extension tool, stored in the tool kit in the cargo area. It mounts into a bolt hole behind a cover panel in the front bumper. The manual frames this hook as a "get unstuck" tool. It is meant for freeing the car from sand, mud, or snow over very short distances, or for a towing service to use for straight, flat-ground pulling only. It is not for loading the car onto a flatbed, and must not be used as a cargo tie-down. Do not tow at an angle with it, and call a towing service for anything beyond a very short pull.

Accidents

We could not confirm this for the 2008 Fit. None of this book's research sources describe a Honda-specific accident checklist, meaning a list of what to do right after a collision. If you are ever in an accident, use general safe practice instead. Check for injuries first, and move to a safe spot off the road if the car can still move. Turn on your hazard lights, call emergency services if anyone is hurt, and exchange insurance and contact information with anyone else involved. Consult your state's driver's handbook or local authorities for the exact steps required where you live, since accident-reporting rules vary by state.

Step by step

Change a flat tire

Safety warnings, before you touch the jack: - Always stop far away from traffic lanes, on firm, level, non-slippery ground, before you start. - Always turn on the hazard warning lights before you get out of the car. - Always get every passenger out of the car before you change the tire. - Do not get under the car at any time. It can roll off the jack and crush you. - Always follow the jacking steps in order. Do not skip the jacking-point check. - Always handle wheel nuts and the brake hub carefully; they can be hot from driving. - Never exceed 50 mph (80 km/h) on the compact spare tire. It is a temporary tire only.

Steps: 1. Pull well off the road, onto firm, level, non-slippery ground, away from traffic. 2. Turn on the hazard warning lights. 3. Put the transmission in Park (automatic) or reverse (manual). Set the parking brake. 4. Turn the ignition off. Have everyone get out of the car and stand away from traffic. 5. Open the tailgate and lift the cargo floor lid using its strap. Hook the strap to the head restraint leg to hold it open. 6. Take the tool kit out of the center of the spare tire. 7. Turn the jack's end bracket counterclockwise to loosen it, then lift the jack straight out. 8. Unscrew the wing bolt and lift the spare tire out of its well. 9. Loosen each wheel nut on the flat tire one half turn with the

wheel nut wrench. Do not remove them yet. 10. On base-trim cars, pry off the wheel cover using the flat tip of the extension. 11. Find the jacking point nearest the flat tire. Look for the mark molded into the body underside. 12. Set the jack under that jacking point. Turn the end bracket clockwise until the jack meets the point. 13. Check that the jacking point's tab is resting inside the jack's notch before you raise the car. 14. Use the extension with the wheel nut wrench to raise the car until the flat tire is off the ground. 15. Remove the loosened wheel nuts, then pull the flat tire off. Set it down with the outside face up. 16. Wipe dirt off the wheel hub and the spare's mounting surface with a clean cloth. 17. Put the spare tire onto the hub. Thread the wheel nuts on finger-tight in a crisscross pattern. 18. Lower the car to the ground with the jack, then remove the jack. 19. Tighten the wheel nuts the rest of the way in the same crisscross pattern, to 80 lbf·ft (108 N·m). 20. Place the flat tire face down in the spare tire well and secure it with the wing bolt. 21. Store the jack and place the tool kit back in the center of the flat tire. 22. Store the wheel cover or center cap in the cargo area so it does not get scratched. 23. Unhook the strap, lower the cargo floor lid, and close the tailgate. 24. Drive to a service shop soon to have the wheel nut torque checked and the flat tire repaired.

How you know it worked: the spare tire sits flat on the ground, all wheel nuts are snug and torqued, and the car sits level. It drives straight with no pulling, wobble, or noise below 50 mph.

Jump start the car

Safety warnings, before you connect a cable: - Always keep sparks, open flames, and smoking materials away from the battery. It can explode. - Do not jump-start a battery with slushy or frozen electrolyte, the liquid inside the battery. Let it thaw first. - Do not connect the second jumper cable to the dead battery's negative terminal or to any other engine part. Use the grounding point, called "the stay" in the manual, instead. - Always turn off all electrical accessories on both vehicles before connecting cables. - Do not try to push- or

pull-start a car with an automatic transmission. It will not work. - Always keep loose jumper cable ends away from each other and from metal until fully disconnected.

Steps: 1. Open the hood and check the physical condition of your battery before doing anything else. 2. In cold weather, check that the battery's electrolyte is not slushy or frozen. 3. Turn off the heater, A/C, audio system, lights, and every other electrical accessory. 4. Put the transmission in neutral (manual) or Park (automatic), and set the parking brake. 5. If you are using another vehicle, have someone start it and run it at a fast idle. 6. Connect one jumper cable to the positive (+) terminal on your dead battery. 7. Connect the other end of that cable to the positive (+) terminal on the booster battery. 8. Connect the second jumper cable to the negative (-) terminal on the booster battery. 9. Connect the other end of the second cable to the grounding stay on your car, not the battery. 10. Start your engine. If cranking is slow, recheck that every cable clamp has a solid metal contact. 11. Once your car is running, disconnect the negative cable from your car first, then from the booster battery. 12. Disconnect the positive cable from your car, then from the booster battery. 13. Keep the loose cable ends apart and away from any metal until both cables are fully removed.

How you know it worked: the engine starts and keeps running on its own once both cables are disconnected, and the charging system indicator is off.

The car is overheating

Safety warnings, before you open the hood: - Do not open the hood if you see steam or spray coming from under it. Wait until it stops. - Steam and spray from an overheated engine can scald you badly. Stay clear until things cool. - Always let the engine and radiator cool down before removing the radiator cap. - Do not remove the radiator cap while the engine is hot. Hot coolant, the liquid that keeps the engine's temperature down, can spray out and scald you. - Do not keep driving with the high temperature indicator on. It can cause serious engine damage.

Steps: 1. Safely pull to the side of the road, away from traffic. 2. Shift to neutral (manual) or Park (automatic), and set the parking brake. 3. Turn off all accessories and turn on the hazard warning lights. 4. If you see steam or spray from under the hood, turn off the engine right away. 5. Wait until no more steam or spray is visible before you open the hood. 6. If there is no steam or spray, you may leave the engine running and watch the indicator. 7. If the indicator goes off within a few minutes, wait for it, then continue driving. 8. If the indicator stays on, turn off the engine and let it sit. 9. Look for an obvious coolant leak, such as a split hose. Everything is still very hot. 10. If you find a leak, do not drive again until it is repaired. 11. If you see no leak, check the coolant level in the radiator reserve tank. 12. Add coolant to the reserve tank if the level is below the MIN mark. 13. If the reserve tank was empty, let the engine and radiator cool fully before going further. 14. Using gloves or a heavy cloth, turn the radiator cap counterclockwise to the first stop only. 15. Let the pressure release fully, then push down on the cap and turn it off. 16. Start the engine and set the temperature dial to maximum heat. 17. Add coolant to the radiator up to the base of the filler neck. 18. If you have no coolant mix on hand, plain water is an acceptable emergency substitute. 19. Put the radiator cap back on tightly, then run the engine and watch the indicator. 20. If the indicator blinks or comes on again, stop driving. The engine needs repair. 21. If the temperature stays normal, top the reserve tank to MAX and close its cap tightly. 22. Have the cooling system drained and refilled with the correct coolant mixture soon.

How you know it worked: the high temperature indicator stays off during normal driving, the reserve tank sits between MIN and MAX, and you see no new coolant leaks.

What can go wrong

- **Flat tire — loosening wheel nuts after the car is already jacked up.** The wheel spins in place instead of loosening. Loosen them a half turn first, while the tire is still on the ground.

- **Flat tire — setting the jack on bodywork instead of the marked jacking point.** Check for the molded mark before you raise the car.
- **Flat tire — driving the spare above 50 mph, or for a long distance, instead of repairing the flat promptly.** The compact spare is a temporary tire only.
- **Flat tire — skipping the shop visit to have wheel nut torque re-checked.** The manual calls for this after any fresh tightening.
- **Jump start — connecting the last cable to the dead battery's negative terminal instead of a ground point.** This raises the risk of sparks near battery gases.
- **Jump start — leaving accessories on while connecting cables, or trying to push-start an automatic.** Neither works safely.
- **Jump start — letting stray cable ends touch metal while still connected on one end.** Keep the ends apart until both cables are fully off.
- **Overheating — opening the hood while steam is still visible, or removing the radiator cap on a hot engine.** Both can cause serious scalding.
- **Overheating — continuing to drive after the indicator stays on, instead of pulling over.** This can cause serious engine damage.
- **Overheating — forgetting to have the correct coolant mixture put back in, after adding plain water in an emergency.** Plain water is a stop-gap, not a permanent fix.
- **Towing — using the front towing hook for anything beyond a short, straight, flat-ground pull.** It is a "get unstuck" tool, not a general towing point, and must not be used to load the car onto a flatbed.
- **Towing an automatic without the D-then-N cycle.** Skipping the 5-second Drive step, the 3-minute idle in Neutral, and letting the engine cool before shifting can damage the transmission.

Quick check

1. What color does the owner's manual explicitly confirm for a warning light, and which light is it?
2. What should you do if the brake pedal does not feel normal when you press it?
3. Name the maximum speed limit for the compact spare tire, and the distance and speed limit for front-wheels-on-the-ground towing.
4. In what order do you disconnect jumper cables once the engine is running?
5. Why should you never open the hood if you see steam coming from it?
6. Does this book have a Honda-specific accident checklist for the 2008 Fit? What should you do if you are in an accident?

PART IV

Owning It (the paperwork)

CHAPTER 17

Title, Registration, Insurance, Inspection, and Emissions Basics

What you will learn

- What a **title** is, and why you should never leave it in the glovebox.
- What paperwork you should get when you buy from a dealer.
- What paperwork you should get when you buy from a private person.
- What the federal **Buyers Guide** sticker must tell you at a used car lot.
- Why registration, insurance, and inspection rules are different in every state, and where to check yours.

The short version

A title is the legal paper that proves who owns a car, and it is the most important paper you will ever get with it. When you buy a car, you should also get a bill of sale and a written record of the mileage. Registration, insurance, and inspection rules are set by each state, so always check your state DMV's website for the exact steps.

The full explanation

Buying from a dealer

A **title** is the legal document that proves who owns a car . If you buy a used 2008 Fit from a licensed dealer, you should receive five things at the sale . First, the title, or proof the dealer can transfer it to you . Second, a **bill of sale**, which is the dealer's own sales contract . It shows the price, the taxes, and the fees . Third, a **Buyers Guide** window sticker, which becomes your copy once

you buy the car . Fourth, an **odometer disclosure statement**, which is the mileage reading, signed by the seller . Federal law requires this statement for most cars under 20 model years old at the time of sale . Fifth, a temporary plate, or a finished application to title and register the car . This lets you drive it home legally while you wait for permanent plates .

Buying from a private person

Buying from a private seller works a little differently . You should still get the title, but the seller must sign it over to you . The seller's name on the title must match their name exactly . If it does not match, you will need extra paperwork. A chain of title is a paper trail showing each past owner in order. A power of attorney is a signed paper letting one person sign for another . You should also get a bill of sale with both people's names, addresses, and signatures . It should list the car's year, make, and **VIN** . A VIN is the 17-character code that identifies one specific vehicle; you will use it again in Chapter 18 . You need a written record of the mileage at the time of sale too . Some states also want a release-of-liability form . This form tells the DMV that the seller is no longer responsible for the car after the sale .

Why this book cannot give you exact state rules

Every state runs its own registration, insurance, and inspection system . Some states require a safety inspection or an emissions test . Some states require neither one . Some states want a notarized signature on the title, and some do not . Because these rules do not match from state to state, and because they can change, this book cannot list every one correctly and keep it current . Check your state DMV's website for the fees, deadlines, and inspection rules where you live .

The Buyers Guide sticker

The Federal Trade Commission's Used Car Rule forces dealers to post a Buyers Guide sticker on every used car . Private sellers do not have to post one . The dealer must put the sticker in the window before you are even

allowed to inspect the car for buying . The sticker must say whether the dealer offers a warranty . If it does, the sticker must state how long the warranty lasts, what share of repair costs the dealer pays, and which systems it covers . It must list the car's 14 major systems and the kinds of defects that can happen in them . It must suggest asking the dealer if you can get the car inspected by an independent mechanic first . It must warn you not to trust spoken promises that are not written down . Since a 2018 update, it must also recommend getting a vehicle history report . The dealer must hand you this same sticker as your copy when you buy the car .

Quick check

1. What is a title, and why should you never leave it in the glovebox?
2. Name three of the five things you should get when buying from a dealer.
3. What must the Buyers Guide sticker tell you about a warranty?
4. Why can't this book list your state's exact registration and inspection rules?
5. Where should you go to find your state's exact rules?

CHAPTER 18

Recalls and Service Bulletins for This Exact Car, and How to Check for New Ones

What you will learn

- What a **recall** is, and what a **campaign number** means.
- What a **TSB (Technical Service Bulletin)** is, and how it differs from a recall.
- All 8 confirmed NHTSA recalls for the 2007-2008 Honda Fit, in one table.
- Two recalls that sound like they apply to your car but do not.
- The top kinds of problems other 2008 Fit owners have reported to the government.
- How to check your exact car's VIN for open recalls, right now.

The short version

A recall is a free, safety-related repair that the maker must offer because a part can fail in a dangerous way. Your 2008 Fit has 8 confirmed recalls, and you should check every one against your car's own VIN. You can check for free at nhtsa.gov/recalls using your car's 17-character VIN. Do this today, and check again every so often, not only once when you buy the car.

The full explanation

What a recall is

A **recall** is a safety fix that a car maker must offer for free, because a part on the car can fail in a way that could hurt someone . The National Highway Traffic Safety Administration, or **NHTSA**, is the US government agency that tracks recalls . Every recall gets a **campaign number**, which is a code that identifies that exact recall . A **TSB**, or Technical Service Bulletin, is different from a recall . A TSB is a repair instruction the maker sends to dealers for a known problem that is not considered a safety defect . A TSB is not free unless it is still covered by your warranty .

The 8 confirmed recalls for your 2008 Fit

NHTSA's own database lists 8 recall campaigns for the 2007-2008 Honda Fit . Before you drive a used 2008 Fit, or even your own car if you have not checked lately, confirm every one of these shows as completed, not open .

Campaign number	Date recall began	Component	Remedy
10V033000	March 19, 2010	Power window master switch (driver's door) — water can enter through an open window and damage the switch, which can overheat or smoke	Dealer inspects the switch and installs a water cover. Dealer replaces the switch if it is already damaged. Free
13V260000	Parts ready October 18, 2013	Power window master switch (driver's door) — a repeat of the same defect, because the 2010 fix did not fully solve it	Dealer replaces the power window master switch with an updated part. Free
10V624000	January 24, 2011	Headlight low-beam wiring harness and combination switch — a bad connection can make low-beam headlights stop working	Dealer inspects the low-beam terminal and repairs it as needed. Free
07V549000	December 12, 2007	Frontal airbag Occupant Detection System (ODS) wire harness — road salt tracked onto the floor can corrode this wire under the carpet. This one only applies if your car was ever sold or registered in one of 21 listed salt-belt states, or Washington, DC	Dealer inspects the wire harness and replaces it if needed. Free
16V344000	June 27, 2016	Passenger frontal airbag inflator, made by a supplier named Takata — this covers both original and replacement inflat-	Dealer replaces the passenger airbag inflator. Free

Campaign number	Date recall began	Component	Remedy
		ors, and it can rupture and throw metal fragments	
17V029000	March 2017	Passenger frontal airbag inflator — a later expansion of the same Takata defect as 16V344000 above	Dealer replaces the front passenger airbag inflator. Free
19V501000	On or about August 12, 2019	Passenger frontal airbag inflator — this one only applies if the inflator had already been swapped once under an earlier Takata campaign	Dealer replaces the passenger airbag inflator again. Free
20V770000	NHTSA notified December 10, 2020	Drive shaft — a coating defect can let road salt corrode the shaft until it breaks. This one only applies to manual-transmission 2008 Fits	Dealer inspects the drive shaft and replaces it if necessary. Free

The ODS wire harness fix (07V549000) uses two related sensor systems, called ODS and OPDS . Together they detect a child seat or an out-of-position rider in the front passenger seat, so the airbag can be turned off for that seat when it should be . **Takata** was a company that made a type of airbag inflator that can explode over time as its propellant breaks down . That is why three separate campaigns above deal with a Takata passenger inflator . **Salt-belt states** are states that salt their roads heavily in winter, which speeds up rust on the underside of a car .

Two look-alike recalls that do NOT apply to your car

It is easy to search for "Honda Fit recall" and find a campaign that looks like a match but is not . The 2009 Fit is a different car underneath a similar name . Campaign 18V268000 covers a passenger airbag installation error, but its own list of affected years is 2007 and 2009 through 2013 — 2008 is not on that list . Campaign 19V182000 covers a driver-side airbag inflator, but it names only the 2007 Fit, not the 2008 Fit . Your car's passenger-side inflator is still covered, through the three campaigns in the table above . This is why you should always check your own car's VIN, instead of only searching by year and model .

One service bulletin worth knowing about

Electric power steering (EPS) is the electric motor that helps you turn the steering wheel . Honda Service Bulletin 14-058 extended the EPS warranty to 10 years or 150,000 miles, whichever came first . It applies to 2007-2008 Fits with a stored trouble code called a **DTC**, which stands for diagnostic trouble code . The specific codes are DTC 32-09 or DTC 61-04 . The symptom is steering that feels heavier than normal, especially when the car is not moving . By 2026 a typical 2008 Fit is past both limits of that warranty, so this repair is now an out-of-pocket cost, not a free warranty claim . A second bulletin, 13-021, covers a rocker arm oil pressure switch . It can trigger the **MIL**, or Malfunction Indicator Lamp, which is the dashboard light most people call the check engine light . It applies to 2007-2011 Fits and is fixed by replacing the switch and its seal .

What other 2008 Fit owners have reported to NHTSA

NHTSA keeps a public database of consumer complaints . As of this research, it held 217 total complaint records for the 2008 Fit . A single complaint can be tagged with more than one category, so the counts below add up to more than 217 . Here are the top categories, in order, with the number of complaints in each: air bags, 57 . Structure, 43 . Power train, 24 . Engine, 23 . Electrical

system, 18 . Exterior lighting, 17 . Vehicle speed control, 14 . Brake-related complaints combined, 22 . Air bags being the top category matches the recall list above . Three of your car's eight recalls involve airbags .

How to check your own car's VIN, right now

Go to **[nhtsa.gov/recalls](https://www.nhtsa.gov/recalls)** and use the "Search by YMM or VIN" option . Type in your car's full 17-character VIN and click Search . The VIN is printed on a small plate at the lower left corner of the windshield, visible from outside the car . It is also on your registration card and often your insurance card . NHTSA also lets you search by license plate and state instead . If your car has no open recalls, the tool says "0 unrepai red recalls associated with this VIN" . A search by year, make, and model instead of VIN gives you general recall and complaint information, not your specific car's repair status . Note that this VIN search shows unrepai red-recall status "for certain manufacturers," so it may not show every kind of record for every maker . American Honda's own owner site, owners.honda.com, also offers a VIN-based recall check, and Honda's customer service line, 1-800-999-1009, is another way to ask about recall or repair status . Using both the NHTSA tool and Honda's own site is a good habit, since a brand-new campaign can show up on one before the other . The exact page behavior on Honda's own site was not independently confirmed this pass, so treat it as a helpful backup, not your only check .

Quick check

1. What is the difference between a recall and a TSB?
2. Name any three of your car's 8 recall campaign numbers and what part they fix.
3. Why doesn't campaign 19V182000 apply to your 2008 Fit, even though it covers a Fit?
4. Where on the car do you find the VIN, and what website do you type it into?

5. What is Takata, and why does it show up in three different recall campaigns?

CHAPTER 19

Documents to Keep in the Glovebox and at Home

What you will learn

- Which three or four papers belong in your car's glovebox.
- Which papers belong at home instead, and why.
- Why the title should never travel in the car.
- How keeping good records now helps you later, including when you sell the car.

The short version

Keep your registration, your insurance card, and your inspection sticker in the glovebox, because you may need to show them at any time. Keep your title, your bill of sale, and your maintenance records at home instead, in a safe place. Never keep the title in the car, because a stolen title makes a stolen car much easier to resell.

The full explanation

What belongs in the glovebox

Keep your current **registration** in the glovebox . Registration is the state's record that your car is legally allowed on the road . Keep your proof of insurance card there too . If your state issues a safety or emissions inspection sticker or certificate, keep the current one in the car as well . Some owners also keep a copy of the Buyers Guide sticker from when they bought the car, described in Chapter 17 . A card listing an emergency contact, plus any

medical conditions or allergies, is also worth keeping in the glovebox . Do **not** keep the title in the glovebox . A stolen title makes a stolen car much easier for a thief to resell .

What belongs at home

Keep your **title** at home, in a safe place . The title is the single most important paper for this car, and it is not something you need on a daily drive . Keep your original bill of sale and purchase paperwork with it . Keep your Buyers Guide sticker copy from the sale too . Keep the owner's manual at home, or write down where the digital copy is saved . Keep your maintenance and repair receipts together . They matter for warranty claims, and they help prove the car's service history if you ever sell it . Keep any recall notice letters you receive from Honda or NHTSA, covered in Chapter 18 . Keep your auto insurance policy documents at home too, separate from the insurance card in the car .

Why this matters later

Good records at home make it much easier to answer "what wears by then" questions if you ever sell the car . Chapter 20 covers this in detail for the 2008 Fit specifically .

Quick check

1. Name three documents that belong in the glovebox.
2. Name three documents that belong at home instead.
3. Why should the title never stay in the car?
4. Why are maintenance receipts worth keeping, even for routine work?

CHAPTER 20

Buying a Used 2008 Fit: Inspection Checklist and Known Trouble Spots

What you will learn

- What a general **pre-purchase inspection (PPI)** covers on any used car.
- How to fold the 8 recalls from Chapter 18 into your buying checklist.
- The one service bulletin worth asking about before you buy.
- Eight trouble spots owners report on first-generation Fits, and how to check each one on a test drive.
- What a used 2008 Fit tends to cost in today's market, in a clear price table.
- Why two of the pricing sources disagree, and what a typical odometer reading looks like at this car's age.

The short version

Before you buy a used 2008 Fit, run its VIN through the recall check from Chapter 18 and get a mechanic's inspection. Owners report eight specific wear spots on this car, so check each one on the test drive. Expect a wide price range depending on the source and the car's condition. Expect the odometer to show anywhere from about 100,000 to 250,000 or more miles.

The full explanation

A general pre-purchase inspection

A **pre-purchase inspection**, or **PPI**, is a check of a used car before you buy it, done by you or by a paid independent mechanic . A general PPI covers five main areas on any car . First, the exterior and body: look for crash damage,

mismatched paint, rust, and signs of flood damage . Also check whether the car sits level, and check for cracked or chipped glass . Second, tires and brakes: check tread depth, the tire's age code on the sidewall, uneven wear, and the condition of the pads, rotors, and calipers . Third, the engine bay and the underside: check for fluid leaks, worn belts and hoses, and rust on the frame . A puddle under a parked car is worth investigating, and a transmission leak is especially expensive to ignore . Fourth, the interior and electronics: test the heat and air conditioning, and check the instrument cluster and warning lights . Confirm every gauge and light actually works . Fifth, a test drive that checks braking, steering feel, suspension, and acceleration . Also ask the mechanic to scan the car's computer for stored **diagnostic trouble codes (DTCs)** . A code can be cleared right before a sale, only to come back after you have driven the car for a while . A professional PPI from an independent shop commonly costs about \$100 to \$200 .

Confirm every recall is closed

Run the car's VIN through the recall check described in Chapter 18 before you buy . Confirm all 8 recalls from that chapter's table show as completed, not open . Pay special attention to two of them . The drive shaft recall (20V770000) only applies if the car has a manual transmission . The ODS wire harness recall (07V549000) only applies if the car was ever sold or registered in a salt-belt state . Do not be alarmed if a general web search turns up campaign 19V182000, since that recall names only the 2007 Fit, not your 2008 Fit . Trust what the VIN search itself shows for your specific car, not a general search .

Ask about the steering bulletin too

Ask the seller whether the check engine light has ever come on with a steering-related code . Also ask whether the steering has ever felt unusually heavy, especially when the car is stopped . This relates to Honda bulletin 14-058, covered in Chapter 18, which once extended the steering system's

warranty to 10 years or 150,000 miles . By now that warranty window has passed on any 2008 Fit, so a repair would come out of your own pocket . Still, knowing the car's history helps you judge its condition .

Eight trouble spots owners report on first-generation Fits

These eight items come from real owner reports on first-generation Fit forums . They are not from Honda or the government, so they are labeled as owner reports rather than confirmed defects .

1. Owners report that motor mounts wear out over time, causing a clunk when shifting into or out of gear, or vibration at idle .
2. Owners report that the timing chain tensioner can develop a ticking or rattling noise under load, though a healthy engine on this car is naturally a little "ticky" too .
3. The clutch master and slave cylinder are the small hydraulic parts that move the clutch when you press the pedal . On manual-transmission cars, owners report that one of these can suddenly feel spongy or stop working, sometimes with no visible fluid leak .
4. On automatic-transmission cars, owners report a shudder or hesitation holding a steady 55 to 65 miles per hour . More than one part can cause this, including worn spark plugs or ignition coils, not only the transmission .
5. Owners report air conditioning problems in two forms: a compressor that rattles when it engages, and cooling that starts strong but does not stay cold . Also check the condenser, which sits ahead of the radiator, for rock-strike damage .
6. Owners report a metallic clunk or knock from the front suspension over bumps, most often traced to worn sway-bar end links or strut top-mount bushings .

7. Owners report a dashboard or glovebox rattle, most often from a loose panel under the windshield or a part behind the factory radio . It tends to get worse in cold weather .
8. Owners report unexplained battery drain . Ask if the car has ever gone dead after sitting for a few days . A healthy 2008 Fit should draw roughly 15 to 17 **milliamps (mA)** of power with everything off . A milliamp (mA) is a small unit of electrical current .

Owners also report a specific rust spot worth checking, especially on a car that has spent winters in a state that salts its roads . Look where the rear quarter panel meets the rear bumper cover . Also check the seam where a plastic cover hides the joint between the quarter panel, the rocker panel, and the underside of the car . This pocket collects grit and moisture, and it is reported as the first place rust shows up on this car . Left unchecked, it can progress to holes in the driver's rear wheel well .

One test drive, eight checks

You can check most of these trouble spots on a single test drive . Shift firmly into and out of gear and listen for a clunk, which points to worn motor mounts . Accelerate under moderate load and listen for a new ticking or rattling beyond the engine's normal light idle sound . On a manual car, push the clutch pedal fully and feel for sponginess or a change in pedal travel . On an automatic car, hold a steady 55 to 65 miles per hour and feel for shudder or hesitation . Run the air conditioning on max for several minutes and confirm it stays cold the whole time, not only for the first minute . Drive over a few bumps and rough pavement, listening for suspension clunks and dashboard rattles .

What a used 2008 Fit costs in today's market

Different pricing sources measure different things, so their numbers do not match, and you should not read them as contradicting each other .

Source	What it measures	Base-trim range
Kelley Blue Book (KBB)	Dealer retail price for the base "Hatchback 4D" trim	\$4,575 to \$5,850
KBB	Estimated trade-in value (what a dealer might offer you for your own car)	About \$2,550 to \$2,825
KBB	Estimated private-party value (selling car to car, not through a dealer)	About \$3,925 to \$5,125
Edmunds	Appraisal-tool trade-in value	About \$1,022
Edmunds	Appraisal-tool private-party value	About \$1,446
CarGurus	Average price on cars actually listed for sale right now	About \$6,521, with real listings ranging from roughly \$2,600 to over \$15,000 depending on the city

Note: KBB's own figures for 2026 show the Sport trim pricing lower than the base "Hatchback 4D" trim . This book uses the Hatchback 4D numbers above for base-trim buying guidance, not the Sport numbers .

KBB and Edmunds are **appraisal-tool estimates**, meaning they assume an average, typical-mileage, typical-condition car, and both run lower than real asking prices . CarGurus reflects **actual asking prices** on real listings . That is why its number runs higher, and why it varies so much by city, mileage, and condition . This price gap between the two kinds of sources is a known, logged disagreement between them, not a mistake in one source or the other . A fair way to read all this: expect a rough floor around \$1,000 to \$3,000 for a high-mileage, rough-condition example . Expect a realistic asking price of roughly \$3,000 to \$10,000 or more for a clean, lower-mileage example . Use these numbers as a starting point, since used-car prices move over time and by region .

What the odometer is likely to show

At the commonly cited average of about 12,000 miles driven per year, an 18-year-old car would show roughly 216,000 miles by simple math . The US Federal Highway Administration reports drivers actually average 13,400 to 14,300 miles per year . That would put the same math closer to 241,000 to 257,000 miles . Real 2008 Fit listings show much wider variation than either estimate. Two examples showed 206,169 miles and 128,493 miles . Many owners drive less than the national average, especially as a car gets older . Treat any reading from roughly 100,000 to 250,000 or more miles as plausible and normal for a car this age .

Routine maintenance, such as oil, filters, brakes, and tires, is covered elsewhere in this book . Beyond that, expect at least some of the eight trouble spots above to show wear, or to have already been fixed, on any real 100,000-plus-mile example . Clearcoat is the clear protective top layer over the paint. Also check the paint on horizontal surfaces like the roof and hood for clearcoat peeling, another item owners report on this car . Use what you find as a talking point for negotiating the price . Do not treat it as an automatic reason to walk away from an otherwise well cared for car .

Quick check

1. Name the five or six areas a general pre-purchase inspection covers.
2. Which two recalls from Chapter 18 only apply under certain conditions, and what are those conditions?
3. Name three of the eight owner-reported trouble spots, and how you would check each one on a test drive.
4. Why do KBB's and CarGurus' prices for the same car look so different?
5. What odometer range should you consider normal for an 18-year-old Fit, and why is it so wide?

PART V

Maintenance

CHAPTER 21

The maintenance schedule

What you will learn

- What the Maintenance Minder is, and why your Fit has no printed mileage chart for service
- How the Maintenance Minder decides when your car needs service
- What the codes "Item A," "Item B," and the numbered sub items mean
- Five fixed rules that apply no matter what the Maintenance Minder says
- What you, the owner, should check on your own, and how often
- An honest note on cost, since this chapter's sources do not cover shop prices

The short version

Your 2008 Fit does not use a printed table of miles for oil changes and other service. Instead, a computer called the Maintenance Minder watches how you drive and shows a code on the dashboard display when something is due. A handful of items still follow a fixed schedule no matter what that display says, and this chapter lists every one of them.

The full explanation

What the Maintenance Minder is

The Maintenance Minder is a computer system built into your Fit. It tracks your driving and tells you when service is due, instead of relying on a fixed mileage chart. Many older Honda models, and some other cars, print a table with two columns: one for normal driving and one for severe driving, such as

short trips or dusty roads. The driver has to decide which column applies. Your Fit's owner's manual has no such table. The Maintenance Minder computer factors driving severity into its own math, so you do not have to guess which column you fall into.

The system tracks something called oil life. Oil life is a percentage the computer calculates from your driving conditions and how many times your engine has turned over. As oil life drops, the display eventually shows a maintenance code telling you it is time for service. Chapter 7 shows you exactly what this looks like on your instrument panel and covers every other dashboard light and icon.

Reading the codes: Item A, Item B, and sub items

When the Maintenance Minder decides service is due, it shows one main code and, sometimes, one or more numbered sub items.

Main Item A means it is time to replace your engine oil. Main Item B means more. It calls for a new engine oil and filter, plus a shop inspection. That inspection covers your front and rear brakes, your parking brake adjustment, and your tie rod ends, steering gear box, and boots. It also covers your suspension parts, driveshaft boots, and brake hoses and lines. Last, it covers all your fluid levels, your exhaust system, and your fuel lines.

Alongside Item A or B, a sub item number can appear for other jobs the Maintenance Minder tracks. These include rotating your tires, replacing your engine air filter, and replacing your cabin air filter, the filter that cleans the air coming through your heater and air conditioner. They also include inspecting and adjusting your drive belt, replacing your transmission fluid, replacing your spark plugs, and replacing your engine coolant, the liquid that keeps your engine from overheating. None of these sub items has a fixed mileage number printed in your owner's manual. The Maintenance Minder decides the timing for each one based on how you drive.

One more item rides along with services A, B, 1, 2, or 3: your valve clearance. Valve clearance is the small gap inside the engine that lets the valves open and close correctly. It gets checked at those visits, but it only gets adjusted if the engine sounds noisy. There is no separate mileage number for this adjustment. It is tied to whichever service visit comes up next.

Five fixed rules that beat the Maintenance Minder

Most service on your Fit runs on the Maintenance Minder's own timing. But five items follow a fixed rule that applies no matter what the dashboard display says. These are the only true "fixed schedule" items in the whole book for this car, and every new driver and every parent should know all five.

1. **Brake fluid, every 3 years.** Change it every 3 years regardless of any message on the display.
2. **Idle speed, every 160,000 miles (256,000 km).** Have it inspected at this mileage, independent of the Maintenance Minder.
3. **Engine oil, at least once a year.** If a full 12 months pass since your last reset without the display asking for an oil change, change the oil anyway.
4. **Air filter and cabin filter, every 15,000 miles (24,000 km) if conditions are dusty or sooty.** If you drive in dusty conditions, replace your engine air filter at this interval instead of waiting for the display. If you drive mainly in urban areas with heavy soot from industry or diesel traffic, replace your cabin filter at this same interval.
5. **Automatic transmission fluid, on two special conditions.** Say you drive an automatic-transmission Fit regularly at low speed in mountains. At 60,000 miles (100,000 km), have the fluid changed. Repeat that change every 30,000 miles (48,000 km) from then on. Say instead that you tow a Fit behind a motorhome. Change the automatic transmission fluid every 2 years, or every 30,000 miles (48,000 km) — whichever happens first. Both of these rules apply to the automatic transmission only.

Outside of these five, your owner's manual names no other severe-condition trigger with a fixed number. If you are shopping for parts or fluid, note that the Fit's automatic transmission fluid, Honda ATF-Z1, has since been discontinued industry-wide. Ask your parts store for its replacement, Honda ATF DW-1, if ATF-Z1 is not on the shelf.

What you check yourself, and how often

Between service visits, a few checks are yours to do, and none of them need a mechanic.

- Engine oil level and engine coolant level: check both every time you stop for fuel.
- Automatic transmission fluid level and brake fluid level: check monthly.
- Tire pressure and tread: check monthly.
- Headlights, parking lights, taillights, high-mount brake light, and license plate lights: check monthly that they all work.
- Battery condition: check monthly using the test indicator window on top of the battery.

A note on cost

This chapter's sources are Honda's own owner's manual and parts-fitment pages. None of them give a dollar figure for what any of this costs, whether you do it yourself or pay a shop. We could not confirm DIY-versus-shop costs for the maintenance schedule from this work package's sources. Check current prices with a local shop or parts store, or look for cost information in Chapters 22 through 33, which cover each maintenance job on its own.

Step by step: Making sense of your maintenance display

1. Look at your information display for an oil life percentage or a maintenance code.
2. If the code is Item A, plan to change your engine oil.

3. If the code is Item B, plan to change your engine oil and filter, and have the longer inspection list checked too.
4. Check for any sub item number next to the main code, and match it to the job list above.
5. Even with no code showing, change your oil once a year, change your brake fluid every 3 years, and check idle speed at 160,000 miles.
6. If you drive in dusty or sooty conditions, or tow, or drive slowly in mountains with an automatic, apply the matching fixed rule above instead of waiting on the display.

What can go wrong

Because most of your Fit's maintenance runs on a computer instead of a printed chart, it is easy to assume the dashboard will catch everything. It will not. The five fixed rules in this chapter run on their own clock, separate from the Maintenance Minder, and the display will not necessarily remind you about them. A parent or new driver who only reacts to dashboard codes could miss a 3-year brake fluid change or a dusty-conditions air filter swap for years. Write these five rules down, or set your own calendar reminders, instead of relying on the display alone.

Quick check

1. What does the Maintenance Minder do differently from an older car's printed service chart?
2. What is the difference between Main Item A and Main Item B?
3. Name all five fixed rules that apply no matter what the Maintenance Minder display shows.
4. How often should you check your tire pressure and your automatic transmission fluid level on your own?

5. Why can't this chapter tell you exactly how much a maintenance visit will cost?

CHAPTER 22

Tools and Supplies for a Starter Garage

What you will learn

- Every tool this book's maintenance chapters actually call for
- What each tool does, in plain words, the first time it comes up
- Which tools you need for almost every job, and which ones are for bigger jobs only
- A sensible order to buy tools in, so you are not buying everything at once
- Safety gear and shop supplies to keep on hand besides tools

The short version

This chapter lists only the tools named in this book's own maintenance procedures, nothing extra. You do not need a full mechanic's toolbox to start. A handful of tools cover most jobs, and a few more only matter once you tackle brakes or oil changes.

The full explanation

Turning and tightening tools

Most jobs in this book start with turning a bolt. A **ratchet** is a handle that turns a socket, a metal cap that fits over a bolt head, in one direction without you lifting it off between turns. A **breaker bar** is a plain, longer handle with no ratcheting action, used when a bolt is stuck and needs extra force. You need a ratchet or breaker bar with sockets sized to your car's bolts for the oil drain plug and for many other jobs.

A **torque wrench** is a tool that tells you exactly how hard you are tightening a bolt, by feel or by a dial. It measures in pound-feet (lbf·ft) or newton-meters (N·m). You need one that reaches at least 80 lbf·ft (108 N·m), the torque for your wheel nuts. The same torque wrench should also reach down to 29 lbf·ft (39 N·m), the torque for your oil drain bolt. Guessing at torque by feel alone risks a leak or a stripped bolt, so this tool is worth buying early.

An **oil filter wrench** is a tool shaped to grip the round oil filter so you can twist it loose. The owner's manual itself calls this "a special wrench, available from your dealer." You also need a **lug wrench**, or a socket that fits your wheel nuts plus a breaker bar, any time you take a wheel off. A **flat-tip (flathead) screwdriver** and a **Phillips screwdriver** cover the clips and screws on your air filter housing.

Lifting and support tools

A **floor jack** is a tool on wheels that lifts one corner or one end of the car using a hand pump or lever. Buy one rated for at least your car's weight. The small scissor jack that lives in your trunk is only rated for a roadside flat tire, not for working underneath the car.

Jack stands are metal supports you slide under the car once it is lifted, so the car's weight rests on solid steel instead of the jack. Buy at least two, and ideally four, each rated for at least your car's weight. Chapter 23 covers how to use a jack and jack stands safely, and that chapter's warnings apply every time you use these tools.

Wheel chocks are wedges you place against a tire that stays on the ground, so the car cannot roll while another corner is lifted. You need at least two.

Tire and brake tools

A **tire pressure gauge** measures the air pressure inside a tire, in pounds per square inch (psi). The owner's manual recommends using your own gauge every time, instead of a different shop gauge each visit, so your readings stay consistent. You will also want a portable inflator, or access to a gas station air pump, to add air.

A **flashlight** lets you see the brake pad through the wheel spokes without removing the wheel; a ruler or a dedicated pad-thickness gauge is a helpful, optional extra.

Once you move on to replacing brake pads, a **C-clamp**, or a dedicated brake piston tool, pushes the caliper piston back into its housing to make room for new pads. Rear drum brakes need their own set of tools. Use large C-clamps or brake-spring pliers to compress the brake shoes, plus a drum brake spring and hold-down tool kit to handle the springs and pins safely. A **wire brush** and **brake-parts cleaner**, a solvent spray made for cleaning brake dust, are needed for both front pad and rear drum jobs.

Bleeding your brakes means pushing old fluid and trapped air out of the brake lines. For that job, you need a wrench sized to the small **bleeder screw** on each brake. You also need clear plastic tubing that fits snugly over it, and a clean catch bottle for the old fluid. A turkey baster or a fluid syringe helps remove old fluid from the reservoir before you start.

Safety gear and shop supplies

Nitrile gloves and safety glasses protect your hands and eyes any time you work with oil, fresh or used. Keep a stack of shop rags on hand for wiping up drips and cleaning parts. You will also want a sealed container to carry used oil to a recycling center, since used oil cannot go in the trash or on the ground. For brake work, keep high-temperature brake grease on hand for pad backing

and slide pins. For the parking-brake lever pivot on drum brakes, use a specific product called Molykote 44MA, or an equivalent high-temperature brake grease.

Chapters 24 through 27 name the exact oil, filters, pads, shoes, and fluid your car needs. This chapter only covers the tools that install them.

Step by step: Building your toolkit in a sensible order

You do not need every tool on day one. Buy in this order, based on which jobs come first in this book.

1. Buy a tire pressure gauge first. It is cheap, and Chapter 26 asks you to check your tires often.
2. Add nitrile gloves, safety glasses, and shop rags. Every job after this uses at least one of them.
3. Add a ratchet or breaker bar with a socket set, a lug wrench, and a flashlight. These cover inspection jobs and light repairs.
4. Before your first oil change, add a torque wrench, an oil filter wrench, a drain pan, and a funnel.
5. Before you lift the car for any reason, add a floor jack, jack stands, and wheel chocks, and read Chapter 23 first.
6. Before a brake job, add the C-clamp or piston tool, wire brush, brake-parts cleaner, and, for rear drums, the spring and hold-down tool kit.
7. Before bleeding brakes, add the bleeder wrench, clear tubing, a catch bottle, and a turkey baster or syringe.

What can go wrong

Several procedure cards behind this book flag one bolt size or another as unconfirmed, telling you to check your own car before buying a socket rather than guess. Buying the wrong socket size wastes a trip to the store, so measure

or check the bolt on your own car first when a size is not given. A torque wrench is not optional if a job names a torque number. Tightening by feel alone can leave a bolt too loose, which can fail later, or too tight, which can strip threads or crack a part. Do not swap a floor jack alone for jack stands. The floor jack lifts the car, but only the jack stands are meant to hold its weight while you work underneath.

Quick check

1. What is the difference between a ratchet and a breaker bar?
2. Why does this book recommend a torque wrench instead of tightening bolts by feel?
3. Name two tools you need before you can safely lift the car and work underneath it.
4. Why should you use your own tire pressure gauge every time instead of a different gauge at each stop?
5. What should you do if a procedure card does not give you an exact bolt size?

CHAPTER 23

Safety First — Jacking, Jack Stands, Wheel Chocks, Hot Parts, Fluids

What you will learn

- The one rule that matters most in this whole chapter: never trust a jack alone
- How to jack up your Fit and support it safely on jack stands
- Why hot parts under the hood and under the car can burn you
- How to handle engine oil, brake fluid, and other fluids without hurting yourself or your car
- What can go wrong when someone skips a safety step

The short version

A jack lifts your car, but a jack can slip or roll off its point without warning. Jack stands, wedges you place under the car once it is up, are what actually holds the car's weight while you work underneath it. Every job in Chapters 24 through 27 that lifts the car depends on you doing this one step correctly.

The full explanation

Why a jack alone is not enough

A **jack** is a tool that lifts one corner or end of a car. A **jack stand** is a separate metal support you slide underneath once the car is up, so its weight rests on solid steel instead of the jack. The owner's manual itself warns that a car can

easily roll off a jack and seriously injure anyone underneath it. That is why every maintenance job in this book that needs the car off the ground also needs jack stands, not the jack by itself.

Your Fit's trunk holds a small scissor jack, meant only for changing a flat tire on the side of the road. A **floor jack** is a separate, larger tool on wheels, rated to hold much more weight, and it is what you use for jack-stand work in the garage. Buy a floor jack and jack stands that are each rated for at least your car's weight. Do this before you start any job in Chapters 24 through 27 that needs the car raised.

Where to lift, and where never to lift

Your owner's manual shows a molded mark on the underbody near each wheel. This is the lift point built for the small scissor jack. For jack-stand work, there are also reinforced seams along each rocker panel (the body panel below the doors) and along the front subframe, set well back from the bumper. These reinforced seam points come from community reports, not from the owner's manual itself, so check them against your own car and a second source before you rely on them.

Never set a jack or a jack stand under the floor pan, the exhaust, a plastic under-cover, or a random point on the frame. Any of these can bend, crush, or puncture under the car's weight.

Hot parts, fluids, and other everyday hazards

Beyond jacking, a few hazards come up again and again in Chapters 24 through 27. An engine, its exhaust, and its brakes all get hot enough to burn skin, so let them cool before you touch anything nearby. Used and fresh engine oil can irritate your skin and eyes, so wear gloves and eye protection whenever you handle it. Brake fluid splashes easily, damages paint on contact, and hurts eyes, so eye protection matters there too. Brake dust should never be blown around with compressed air or brushed dry; use a brake-parts cleaner spray instead, and never breathe it in.

Used oil is hazardous to the environment. Never pour it in the trash or on the ground; take it to a recycling center in a sealed container.

Step by step: Jack up the car and set it on jack stands

Safety warnings, before you touch the jack: - Do not ever get any part of your body under a car supported only by a jack. The car can roll off a jack and seriously injure anyone underneath. - Always work on a flat, level, hard surface. Never jack up a car on a slope, on soft ground, or on an incline. - Always set the parking brake and chock the wheels that will stay on the ground before you start jacking. - Always use the vehicle's marked or reinforced jacking points, never the floor pan, exhaust, plastic under-covers, or a random point on the frame. - Do not exceed a jack's or a jack stand's rated weight capacity. Check the rating sticker before you buy or borrow one. - Always give the car a firm shake test on the stands before working underneath, to confirm it will not slip.

Steps: 1. Park on a flat, level, solid surface and set the parking brake. 2. Chock the wheels that will remain on the ground. 3. Locate the lift point: the owner's manual shows a molded mark on the underbody near each wheel, for the factory scissor jack. 4. For jack-stand work, also find the reinforced pinch-weld seams on each rocker panel and the front subframe, well back from the bumper. 5. Treat step 4's points with caution: they are community-reported, not from the owner's manual, so double-check against your own car and a second reference. 6. Position the floor jack under the chosen point and raise the car slowly. 7. Stop as soon as the tire lifts a couple of inches and check that the jack is still centered on the point. 8. Continue raising to the height you need. 9. Slide a jack stand under a solid point near the jack and adjust it to the right height. 10. Slowly lower the floor jack until the car's weight rests on the jack stand. 11. Repeat steps 6 through 10 for the other end or side, if you are lifting the whole car. 12. Give the car a firm push and shake test before you get near or under it. 13. Leave the floor jack nearby as a backup, but do not

rely on it to hold weight once the stands are set. 14. When done, reverse the process: raise slightly with the floor jack, remove the stands, then lower slowly and fully.

How you know it worked: the car sits solidly on the stands with no rocking when pushed. All four (or two) contact points look evenly loaded. The wheels you did not lift stay chocked and on the ground the whole time.

What can go wrong

The most common mistake is placing a jack or a jack stand on the floor pan, a plastic cover, or the exhaust instead of a reinforced point. That mistake can bend the body or crush the part. The second most common, and most dangerous, mistake is working under a car held up by the floor jack alone, with no stands underneath. Jacking on a slope or soft ground can let the car shift, or let the jack sink into the ground while you work. Forgetting to chock the wheels that stay on the ground can let the whole car roll while one corner is in the air.

Quick check

1. Why is a jack alone not enough to support the car while you work underneath it?
2. What is the difference between the trunk's scissor jack and a floor jack?
3. Name two places you should never set a jack or a jack stand.
4. What should you do before you fully trust the car's weight to the jack stands?
5. Why should you never blow brake dust around with compressed air?

CHAPTER 24

Oil and Filter Change

What you will learn

- Why your Fit needs an oil and filter change, and what oil actually does
- The exact oil type and amount your Fit needs
- The full step-by-step process, from warming the engine to disposing of the old oil
- The one torque number that matters most in this job
- Common mistakes that cause leaks or engine damage

The short version

Engine oil lubricates the moving parts inside your engine and carries away heat and dirt. Your Fit needs 3.8 US quarts (3.6 L) of 5W-20 oil and a new filter each time you change it. This is rated as a job that needs the car raised on jack stands, so read Chapter 23 first if you have not already.

The full explanation

Why oil matters, and what your Fit needs

Oil coats the metal parts inside your engine so they can move against each other without wearing out. Over time, oil picks up heat, dirt, and tiny bits of metal, and it needs to be drained out and replaced.

Your owner's manual calls for API Premium-grade 5W-20 detergent oil, carrying the API Certification Seal marked "For Gasoline Engines." Honda's own motor oil is the preferred choice, though a synthetic oil that meets the same seal and viscosity is acceptable too. No oil additive is needed or recommended.

A full oil change, with a new filter, takes 3.8 US quarts (3.6 L) of oil. Without a filter change, it takes slightly less, 3.6 US quarts (3.4 L). After a full engine rebuild, when the engine is completely dry, it takes 4.4 US quarts (4.2 L). You will not need that last figure for a routine oil change. It is here so you recognize it if you ever see it on a parts receipt or in a shop's paperwork.

When to change it

Your Fit uses a computer called the Maintenance Minder instead of a printed mileage chart, as Chapter 21 explains in full. Even so, one fixed rule always applies: if a full 12 months pass without the display asking for an oil change, change the oil anyway.

The one number to memorize: drain bolt torque

The bolt at the bottom of your oil pan that lets old oil drain out is called the drain bolt. Tighten it to 29 lbf·ft (39 N·m, 4.0 kgf·m) with a torque wrench, no more and no less. Honda gives no specific torque number for the oil filter itself. Instead, install a new filter according to the instructions printed on it. This usually means turning it by hand until it is snug against the engine, plus a small extra turn.

Step by step: Change the engine oil and filter

Safety warnings, before you start: - Always let the engine run first, then shut it off. The manual says to run the engine to normal temperature before draining so the oil flows and carries out contaminants. - Always let the engine and exhaust cool before you touch anything underneath the car, once you shut it off. Hot oil and hot metal can burn you. - Do not get under a vehicle held up

only by a jack. Use jack stands, as Chapter 23 explains. - Always wear eye protection and gloves. Used oil and fresh oil can irritate skin and eyes. - Do not dump used oil in the trash or on the ground. It is hazardous to the environment. - Always clean up any spilled oil right away. Spilled oil can damage engine parts and creates a slip hazard.

Steps: 1. Warm up the engine by running it, then shut it off and set the parking brake. 2. Raise the front of the car and support it on jack stands, following Chapter 23. 3. Open the hood and remove the engine oil fill cap so air can enter as oil drains. 4. Slide the drain pan under the drain bolt at the bottom of the engine. 5. Loosen and remove the drain bolt and its washer with your socket. 6. Let all the oil drain fully into the pan; this can take several minutes. 7. Wipe the drain area, then reinstall the drain bolt with a brand-new washer. 8. Tighten the drain bolt to 29 lbf·ft (39 N·m) with a torque wrench. 9. Move the drain pan under the oil filter and remove the old filter with the filter wrench. 10. Check that the old filter's rubber gasket did not stick to the engine; remove it if it did. 11. Wipe a thin film of new oil on the new filter's gasket, then thread it on by hand until snug. 12. Lower the car back to the ground and remove the jack stands. 13. Pour in 3.8 US quarts (3.6 L) of the recommended 5W-20 oil through the funnel. 14. Reinstall the oil fill cap and tighten it securely. 15. Start the engine and watch the oil pressure light; it should go out within 5 seconds. 16. Let the engine run a few minutes, then shut it off and check the drain bolt and filter for leaks. 17. Wait several more minutes, then check the level on the dipstick, the thin metal rod you pull out to see the oil level, and top up if needed. 18. Seal the used oil in a container and take it to a recycling center.

How you know it worked: the oil pressure warning light goes out within 5 seconds of starting the engine. There are no drips under the car after it has sat overnight. The dipstick reads between the marks.

What can go wrong

Overfilling the engine past the dipstick's upper mark can damage the engine; the manual warns against this directly. Forgetting the new drain-bolt washer, called a crush washer, causes a slow leak. Over-tightening the oil filter by hand past snug can make it very hard to remove next time and can crush its gasket. Not fully re-torquing the drain bolt, or over-torquing it, can strip the pan's threads. And working under a car held up by only a jack, instead of jack stands, is the most dangerous mistake on this list.

Quick check

1. What kind and amount of oil does your Fit need for a full oil change with a new filter?
2. What torque should you use on the oil drain bolt, and with what tool?
3. Why should you run the engine before draining the oil, and let it cool before you go underneath?
4. What is a crush washer, and what happens if you forget to replace it?
5. What should the oil pressure light do within 5 seconds of starting the engine, and what does that tell you?

Air Filter and Cabin Filter

What you will learn

- What the engine air filter and the cabin air filter each do, and why they are different
- The exact part numbers Honda used for your Fit
- When to replace each filter
- How to replace both filters, step by step, with no jacking or fluids involved
- Common mistakes that reduce airflow instead of improving it

The short version

Your Fit has two separate air filters. The engine air filter cleans air before it reaches the engine, and the cabin air filter cleans air coming through your heater and air conditioner. Both jobs are simple, need no tools beyond a screwdriver, and take well under 20 minutes combined.

The full explanation

Two filters, two jobs

The **engine air filter** sits in a black plastic housing in the engine bay and cleans the air the engine breathes before that air reaches the fuel injection system. Its part number is Honda 17220-PWA-505, and it fits the 2007-2008 Fit; this number replaced an older part, 17220-PWA-J10.

The **cabin air filter**, also called the dust and pollen filter, cleans the air that comes through your heater and air conditioner vents before it reaches you. Its part number is Honda 80291-SAA-J01, and it fits the 2007-2008 Fit; this number has since been superseded by 80291-SAA-316. This filter is fitted on models with air conditioning, per the owner's manual.

When to replace each one

Both filters follow the Maintenance Minder's own timing under normal driving, with no fixed mileage number printed in the owner's manual. Two exceptions apply. If you drive mostly in dusty conditions, replace the engine air filter every 15,000 miles (24,000 km). If you drive mostly in urban areas with heavy soot from industry or diesel traffic, replace the cabin filter every 15,000 miles (24,000 km). Replace it sooner if airflow from your heater or air conditioner feels weaker than usual.

Step by step

Replace the engine air filter

Safety warnings, before you start: - Always let the engine cool before opening the hood if it has just been running. Parts near the filter box can be hot. - Do not start the engine with the air filter housing open or the element removed.

Steps: 1. Open the hood and find the air filter housing, a black plastic box on the driver's side of the engine bay. 2. Release the clips holding the two halves of the housing together (there are usually four). 3. Lift the top half of the housing up and out of the way. 4. Note which way the old filter element sits before you remove it. 5. Lift out the old, dirty filter element. 6. Wipe out any dust or leaves from inside the housing with a dry rag. 7. Place the new filter element in the same orientation as the old one. 8. Close the top half of the housing and press down until every clip snaps shut. 9. Check that all clips are fully closed before starting the engine.

How you know it worked: the housing halves sit flush together, every clip is snapped closed, and the engine starts and idles normally.

Replace the cabin (dust and pollen) filter

Safety warnings, before you start: - Do not force the glove box past its stops. You can break the hinge or the stop tabs.

Steps: 1. Open the glove box fully. 2. Squeeze both sides of the glove box inward to release the stop tabs. 3. Let the glove box swing down out of the way. 4. Find the filter tray in the housing behind the glove box opening. 5. Pinch the two clips on the tray's left and right sides and pull the tray out. 6. Slide the old filter out of the tray. 7. Look at the arrow printed on the new filter's frame. 8. Slide the new filter into the tray with the arrow pointing down. 9. Push the tray back into the housing until both clips click. 10. Swing the glove box back up and press it closed.

How you know it worked: the glove box closes and latches normally, and air from the vents feels stronger than before if the old filter was clogged.

What can go wrong

Forcing a stiff clip on the engine air filter housing can crack the plastic. Installing the new engine air filter upside down or backward, or leaving one clip unlatched, lets unfiltered air leak past the filter instead of through it. On the cabin filter, installing it with the arrow pointing the wrong way can reduce airflow instead of improving it. Not seating both tray clips lets the tray rattle or fall loose later.

Quick check

1. What is the difference between the engine air filter and the cabin air filter?
2. What is the part number for your Fit's engine air filter, and what older number did it replace?

3. Under what driving condition should you replace the engine air filter every 15,000 miles instead of waiting on the Maintenance Minder?
4. Which way should the arrow point when you install a new cabin filter?
5. What can happen if you force a stiff clip on the air filter housing instead of releasing it properly?

CHAPTER 26

Tires — Pressure, Rotation, Wear, Replacement, the Spare

What you will learn

- Your Fit's correct tire pressure, tire size, and wheel size
- How to check and set tire pressure the right way
- How to rotate your tires safely, and the torque your wheel nuts need
- What to look for when checking tires for wear
- Where to find the flat-tire and spare-tire change procedure in this book

The short version

Your Fit's base trim uses 32 psi (220 kPa) in every tire, front and rear, and the compact spare needs 60 psi (420 kPa). Checking pressure takes about 10 minutes and needs only a gauge. Rotating all four tires needs jack stands and a torque wrench, and takes about 45 minutes.

The full explanation

Your Fit's tire and wheel numbers

Your base-trim Fit's factory tire size is P175/65R14 81S, mounted on a 14-inch by 5.5J steel wheel with a full wheel cover. The wheel bolts on a 4x100 mm pattern, meaning four lug nuts arranged around a circle 100 mm across. The wheel's center bore, the hole in the middle that centers the wheel on the hub, measures 56.1 mm. Offset means how far the wheel's mounting face sits from its centerline. The factory wheel's offset is ET45, within a reported range of 43 to 47 mm for this tire and wheel combination.

Recommended pressure

Both base and Sport trims use the same cold tire pressure: 32 psi (220 kPa, 2.2 kgf/cm²), front and rear. The compact spare tire needs a higher pressure, 60 psi (420 kPa, 4.2 kgf/cm²), since it is a smaller, temporary tire. "Cold" means the tire has been parked for at least 3 hours, or driven less than 1 mile (1.6 km). A hot tire reads 4 to 6 psi higher than its true pressure, and this can fool you into underfilling it.

Rotating your tires

Rotating tires means moving each wheel to a new position on the car, so all four tires wear more evenly. This is a Maintenance Minder sub item, with no fixed mileage number printed in the owner's manual; the Minder decides the timing. Whichever pattern you use, torque every wheel nut to 80 lbf·ft (108 N·m) once the car is back on the ground.

Checking for wear

While you check pressure or rotate tires, look each tire over for bulges, cuts, or objects stuck in the tread. These sources do not give a specific tread-depth number for deciding when a tire needs replacing. We could not confirm a tread-wear replacement number for the 2008 Fit from this chapter's sources. Check your tire's own wear indicators, small raised bars molded into the tread, or ask a tire shop.

The spare tire

Your Fit carries a compact spare tire for roadside use. Chapter 16 covers the full step-by-step procedure for changing a flat tire and using that spare, including the jacking steps, so this chapter does not repeat it. Come back to Chapter 16 if you have a flat on the road; use this chapter for routine pressure checks and rotations instead.

Step by step

Check and set your tire pressure

Safety warnings, before you start: - Always check pressure when the tires are cold: parked 3+ hours, or driven less than 1 mile (1.6 km). Hot tires read 4-6 psi high and will fool you into underfilling them. - Do not let air out of a hot tire to try to match the cold-pressure number; you will underinflate it. - Always check the spare tire too, not just the four road tires.

Steps: 1. Find the recommended pressures on the driver's doorjamb sticker, or use 32 psi front and rear as a cold-pressure fallback. 2. Remove the valve cap from the first tire and set it somewhere you won't lose it. 3. Press the tire gauge firmly onto the valve stem until the hiss stops and read the number. 4. Compare the reading to the recommended pressure for that tire. 5. If the pressure is low, add air in short bursts, checking often, until it matches the number. 6. If the pressure is too high, press the small pin inside the valve to release a little air, then recheck. 7. Reinstall the valve cap. 8. Repeat steps 2 through 7 for the other three tires. 9. Check the compact spare the same way; it should be 60 psi (420 kPa). 10. While you are down there, look each tire over for bulges, cuts, or objects stuck in the tread.

How you know it worked: every tire, including the spare, reads within about 1 psi of its recommended cold pressure, and no tire shows a bulge, cut, or embedded object.

Rotate the tires

Safety warnings, before you start: - Do not work under a car held up only by a jack. Use four jack stands. - Always loosen every wheel nut a bit before you raise that corner off the ground. - Always chock the wheels that stay on the ground before you start jacking. - Do not fully tighten wheel nuts with the wheel hanging free; snug them, lower the car, then torque them on the ground.

Steps: 1. Park on level, solid ground and set the parking brake. 2. Chock the wheels you are not lifting. 3. Loosen each wheel nut about a half turn while all four tires are still on the ground. 4. Raise one corner and support it on a jack stand; remove that wheel. 5. Repeat step 4 for the other three corners so all four wheels are off. 6. Move each wheel to its new position, front-to-back and side-to-side per the rotation chart, unless your tires are directional. A **directional tire** has a tread pattern designed to roll only one way, marked with an arrow on the sidewall. 7. If your tires are marked directional, only swap front-to-back on the same side, never side-to-side. 8. Put each wheel back on its new hub and thread the nuts on by hand. 9. Snug the nuts in a crisscross (star) pattern with the wrench, one corner at a time. 10. Lower the car until the tire touches the ground but still has some jack support. 11. Torque each wheel nut to 80 lbf·ft (108 N·m) in the crisscross pattern. 12. Lower the car fully and remove the jack stands. 13. Check and set the pressure in all four tires once they are all back on the ground.

How you know it worked: every wheel nut is torqued to 80 lbf·ft. No wheel wobbles when you drive at low speed. The tires are now in their new rotation positions.

What can go wrong

Checking pressure right after driving gives a falsely high reading. Using a different gauge each time, without noticing it reads differently from your own gauge, causes confusion over time. Forgetting the spare tire is easy, and it slowly loses pressure over months even if it is never used. Ignoring a low-tire-pressure warning light and skipping the physical check anyway defeats the point of having the light.

During rotation, trying to fully torque a wheel nut while the wheel still hangs free in the air can leave it improperly seated. Skipping the crisscross tightening pattern can seat the wheel unevenly. Rotating directional tires side-

to-side puts the tread pattern running backward. Forgetting to re-torque wheel nuts after the first 50 to 100 miles of driving on a re-used wheel is another common miss.

Quick check

1. What pressure should every tire on your base-trim Fit hold, and what pressure does the spare need?
2. Why should you check tire pressure only when the tires are cold?
3. What torque should you use on your wheel nuts, and when in the rotation process should you apply it?
4. What is a directional tire, and how does rotating one differ from rotating a regular tire?
5. Where in this book should you look for the full flat-tire change procedure?

CHAPTER 27

Brakes — Inspection, Pads, Rotors, Fluid

What you will learn

- Why your Fit's front and rear brakes are not the same type
- How to inspect your front brake pads without removing the wheel
- How to replace front brake pads, and how to replace rear brake shoes
- How to bleed your brakes and what fluid to use
- Safety warnings that apply to every brake job in this chapter

The short version

Your 2008 Fit's front brakes are disc brakes, and its rear brakes are drum brakes, not disc. This matters because the rear job uses shoes and springs instead of pads and a caliper, and it is a different repair. Every brake job in this chapter needs jack stands, eye protection, and patience with springs under tension.

The full explanation

Front disc brakes, rear drum brakes

A **disc brake** uses a flat metal disc, called a **rotor**, that spins with the wheel. A **caliper** is a clamp that squeezes flat pads against both sides of the rotor to slow it down. A **drum brake** works differently. A hollow metal drum spins with the wheel, and curved brake **shoes** press outward against the inside of the drum to slow it down. Inside the drum, a small threaded part called the **self-**

adjuster (or just the **adjuster**) takes up slack as the shoes wear down. This keeps the brake working correctly without you adjusting it by hand every time.

Honda's own owner's manual states plainly that your Fit's front brakes are disc, and its rear brakes are drum. This applies across the 2008 Fit lineup, with no separate statement carving out a different design for base trim. Do not assume your rear brakes have pads and a caliper like the front; they do not.

Brake fluid

Your brake system, and your clutch system if you drive a manual, both use Honda Heavy Duty Brake Fluid DOT 3. A DOT 3 or DOT 4 fluid from a sealed container works as a temporary substitute. Never use DOT 5 brake fluid; it is not compatible with your Fit's braking system and can cause serious damage. On a manual-transmission Fit, the clutch pedal is operated by hydraulic fluid from the same specification, held in its own reservoir with a gray cap. Replace your brake fluid every 3 years, no matter what the Maintenance Minder display shows.

Torque and bolt sizes: what is confirmed, and what is not

Your wheel nuts, front or rear, torque to 80 lbf·ft (108 N·m) whenever a wheel comes off for brake work. Owners report a front caliper bracket-to-knuckle bolt torque of around 80 lbf·ft. This number comes from two community sources discussing a Fit Sport, not a confirmed base-trim source, so treat it as a starting point, not a certainty. No confirmed number exists yet for a rear caliper bracket bolt on this car. Since your Fit's rear brakes are drum, not disc, there is no rear caliper to begin with.

Step by step

Inspect the front brake pads through the wheel

Safety warnings, before you start: - Always let the brakes cool before you get close to the wheel if you just drove. The caliper and rotor can be very hot.
- Do not touch the wheel or brake parts with bare hands right after driving.

Steps: 1. Park on level ground and set the parking brake. 2. Turn the steering wheel toward you so you can see through the wheel spokes at that corner. 3. Shine the flashlight through the spokes at the caliper, the metal bracket that squeezes the pads onto the rotor. 4. Look for the pad material pressed against the flat, shiny rotor surface on each side of the caliper. 5. Estimate the pad's thickness; a healthy pad is clearly thicker than a thin metal backing plate. 6. Compare the inside and outside pad if you can see both; they should wear about evenly. 7. Repeat steps 2 through 6 on the other front wheel. 8. Listen for a metallic screeching sound the next time you brake gently; the manual says this warns you the pads need replacing.

How you know it worked: you can see both pads on each front wheel and estimate their thickness. Or, you learn you cannot see enough, and you need to remove the wheel or have a shop check them.

Replace the front brake pads

Safety warnings, before you start: - Always wear eye protection. Brake dust and springs under tension can hurt your eyes. - Do not breathe brake dust; never use compressed air or a dry brush on brake parts, per the manual's own maintenance-safety caution. Use brake-parts cleaner instead. - Do not work under a car held up only by a jack. Use jack stands. - Always work on one side at a time and leave the other wheel on the ground so you have a reference for how parts go back together. - Do not let the caliper hang by its brake hose; support or hang it so the hose has no weight on it. - Always pump the brake pedal several times before driving, until it feels firm, before you rely on the brakes again.

Steps: 1. Loosen the front wheel nuts, then raise and support the car on jack stands. 2. Remove the wheel and set it aside. 3. Look at how the old pads and springs sit before you remove anything, or take a photo. 4. Remove the caliper slide-pin bolts and lift the caliper off the bracket. 5. Support the caliper with a hook or bungee so it does not hang by the hose. 6. Pull the old pads out of the bracket. 7. Use the C-clamp against the old pad to push the caliper piston back into its bore. 8. Clean the bracket and slide pins with a wire brush and brake-parts cleaner. 9. Apply a thin coat of brake grease to the pad backing and slide pins. 10. Place the new pads into the bracket. 11. Slide the caliper back over the new pads and reinstall the slide-pin bolts. 12. Reinstall the wheel and snug the nuts by hand. 13. Lower the car, then torque the wheel nuts to 80 lbf·ft (108 N·m) in a crisscross pattern. 14. Repeat steps 1 through 13 on the other front wheel. 15. Pump the brake pedal several times with the engine off until it feels firm before driving. 16. Check the brake fluid reservoir level and top it off if needed. 17. Drive slowly at first and test the brakes gently before returning to normal driving.

How you know it worked: the brake pedal feels firm, not soft or spongy. The car stops evenly with no pulling to one side. There is no grinding or squealing under light braking.

Replace the rear brake shoes (drum brakes)

Safety warnings, before you start: - Always wear eye protection. Springs are under strong tension and can snap free unexpectedly. - Do not breathe brake dust; never use compressed air or a dry brush on brake parts. Use brake-parts cleaner instead. - Do not work under a car held up only by a jack. Use jack stands. - Always work on one side at a time, using the untouched side as your reference for how everything goes back together. - Do not get your fingers between a compressed spring and the backing plate. - Always keep grease off the brake shoe linings and the drum's inner surface; grease on the linings ruins braking. - Always press the brake pedal several times after reassembly, before driving, to seat the shoes and set the self-adjuster.

Steps: 1. Loosen the rear wheel nuts, then raise and support the car on jack stands. 2. Remove the wheel and set it aside. 3. Release the parking brake, then pull the drum straight off the hub (tap gently if it is stuck). 4. Take a photo of the spring and lever layout before touching anything. 5. Unhook the lower return spring with your spring tool or pliers. 6. Unhook the upper return spring the same way. 7. Remove the hold-down springs and pins that pin each shoe to the backing plate. 8. Lift out the old shoes together with the adjuster and parking-brake link still attached. 9. Transfer the adjuster, parking-brake lever, and springs onto the new shoes in the same layout as the photo. 10. Apply a thin dab of high-temperature grease to the backing plate's contact pads and the parking-brake lever pivot, keeping grease off the linings. 11. Set the new shoes onto the backing plate and reinstall the hold-down pins and springs. 12. Hook the lower return spring back into place. 13. Use your C-clamps or spring pliers to compress the shoes enough to hook the upper return spring. 14. Check that the shoes sit fully in their bosses (the raised mounting pads that hold each shoe in place) and the adjuster turns freely. 15. Slide the drum back onto the hub. 16. Reinstall the wheel and snug the nuts by hand. 17. Lower the car, then torque the wheel nuts to 80 lbf·ft (108 N·m) in a crisscross pattern. 18. Repeat steps 1 through 17 on the other rear wheel. 19. Press the brake pedal firmly several times with the engine off until the pedal feels normal. 20. Adjust the parking brake if it now feels loose. 21. Drive slowly at first and test the brakes gently before returning to normal driving.

How you know it worked: the brake pedal feels firm, and the parking brake holds the car on a hill. Both rear wheels turn freely with light drag when the car is off the ground. There is no grinding or dragging noise while driving.

Bleed the brakes

Safety warnings, before you start: - Always wear eye protection. Brake fluid can splash and it damages paint and eyes. - Do not let the brake fluid reservoir run dry during bleeding, or you will draw air back into the lines and have to start over. - Do not reuse fluid that comes out of the bleeder; only use fresh

fluid from a sealed container. - Always keep brake fluid off painted surfaces; wipe up any spill immediately. - Do not drive the car until the pedal feels firm and consistent, not soft or sinking.

Steps: 1. Remove old fluid from the reservoir with the baster, leaving a little so it does not run dry. 2. Fill the reservoir with fresh Honda Heavy Duty Brake Fluid DOT 3. 3. Slide clear tubing over the bleeder screw at the first wheel and route the other end into your catch bottle. 4. Have your helper press the brake pedal down and hold it. 5. Open the bleeder screw briefly to let fluid and air escape, then close it before the pedal is released. 6. Have your helper release the pedal only after the bleeder screw is closed. 7. Repeat steps 4 through 6 until no more air bubbles appear in the tubing. 8. Check the reservoir level and top it off before moving to the next wheel. 9. Move to the next wheel and repeat steps 3 through 8. 10. Recheck the reservoir level after all four wheels and fill to the MAX mark. 11. Press the brake pedal several times and confirm it feels firm and consistent, not soft.

How you know it worked: the pedal feels firm on the first press, not soft or sinking to the floor. No air bubbles appear in the tubing on a final check at each wheel.

What can go wrong

On front pads, pushing the piston back without first opening the brake fluid reservoir cap slightly can push fluid out or damage a seal. Forgetting to support the caliper can stretch or kink the brake hose. Skipping the pedal-pumping step before your first drive can mean the pedal goes to the floor instead of feeling firm.

On rear drum shoes, losing track of how the springs and adjuster were arranged is exactly why a before photo matters so much. Getting grease on the shoe linings can cause the brakes to grab or fade. Forcing the return spring on without a tool can make it snap loose unexpectedly. Not resetting the self-adjuster can leave the parking brake or pedal feeling loose. Several

independent do-it-yourself writeups single out the main return spring as the one step that trips up first-timers. If you are not comfortable with springs under tension, this is a reasonable job to hand to a shop.

On bleeding, letting the reservoir run empty partway through draws in fresh air, undoing your work. Opening the bleeder screw too far can let it come out, and over-tightening it can snap it. Skipping the final firm-pedal check and driving with air still in a line is the most dangerous mistake in this whole chapter.

Quick check

1. Are your Fit's rear brakes disc or drum? How does that change the parts and tools you need?
2. What brake fluid type does your Fit use, and which type must you never use?
3. How often should you replace your brake fluid, no matter what the Maintenance Minder shows?
4. Name two safety warnings that apply to both the front pad job and the rear shoe job.
5. How do you know your brake bleeding worked?

CHAPTER 28

Battery: Testing, Cleaning, Replacing, Jump Starting

What you will learn

- What size battery your 2008 Fit uses
- How to test your battery's voltage with a multimeter, a tool that reads electrical charge
- How to clean corrosion, the crusty buildup that forms on battery terminals
- How to replace the battery yourself, and why you need your radio code first
- Where to find the full jump-start steps if the battery goes dead on the road

The short version

Your Fit uses a 12-volt battery, size group 151R. You should check its condition every month. A cheap multimeter can tell you if the battery is healthy, half-charged, or failing. Corrosion on the terminals is easy to clean with baking soda and water, and replacing the whole battery takes basic tools and about 20 minutes. If the battery is already dead and you need to get moving right now, turn to Chapter 16. It covers jump starting in full.

The full explanation

Your battery, and why you check it monthly

A battery stores electrical power and delivers a big burst of it to start your engine. Your Fit's owner's manual tells you to check the battery's condition once a month. Most batteries have a small test indicator window on top. A

label on the battery explains what each color in that window means. The window's exact location can vary between battery brands, so check the label on your own battery.

The battery that came with your Fit is a group size 151R. A "group size" is a standard code for a battery's physical dimensions and terminal layout, so the right one fits your battery tray and connects correctly. Some owners report swapping in a group 51R battery instead, since it offers more cold-cranking amps, a rating of how much starting power a battery can deliver in cold weather. Owners report that a 51R is a noticeably tighter physical fit, and some say they had to modify the battery tray to make it work. For a straight-forward replacement with no modification, stick with the original 151R size.

A battery does one thing you should always respect: it gives off hydrogen gas, an explosive gas, during normal operation. This is true every time you work near the battery, for any reason. Keep flames, sparks, and cigarettes away from it, and always wear eye protection when you work close to it.

Testing the battery with a multimeter

A multimeter is a small handheld tool that reads electrical voltage, among other things. To test your battery, you touch its two probes to the battery's two posts and read the number on the screen. A rested battery, meaning the engine has been off at least an hour, should read about 12.6 volts or higher if it is fully charged. A reading near 12.2 volts means the battery is only about half charged, and anything below 12.0 volts usually points to a weak or failing battery. You can also have a helper crank the engine while you watch the meter; a healthy battery should not drop below about 9.6 volts while doing that. Full steps are below.

Cleaning corroded terminals

Corrosion looks like a white or yellowish, crusty powder around the battery posts and cable clamps. It forms slowly over time as the battery works, and it can weaken your electrical connection or even stop the car from starting.

Baking soda mixed with water reacts with this corrosion and breaks it apart, letting you rinse it away. A thin coat of grease afterward slows corrosion from coming right back.

Replacing the battery

Before you disconnect any cable, write down your radio's 5-digit anti-theft code. Many Honda audio systems lock themselves as soon as the battery loses power, and you need that code to unlock the radio again afterward. Always disconnect the negative cable first and reconnect it last; this order lowers the risk of a spark shorting something in the car's electrical system. The new battery needs to be the same group size, 151R, so it fits the tray and hold-down bracket, the metal piece that clamps the battery in place, without modification.

Jump starting: where to find it

This chapter does not repeat the jump-start steps. Chapter 16 already covers the full procedure, from checking your battery's condition to the exact order for connecting and disconnecting cables. If your battery is dead right now, go there.

Step by step

Test the battery with a multimeter

Safety warnings, before you touch a probe to the battery: - Always wear eye protection when working near the battery. It gives off explosive hydrogen gas during normal operation. - Do not smoke or bring sparks or flames near the battery while testing it. - Do not let the multimeter's metal probes touch both terminals, or any metal, at the same time by accident. This can cause a spark. - Always wash your hands after handling the battery or its terminals.

Steps: 1. Turn off the engine and every accessory, such as lights, radio, and fan, and let the car sit at least 1 hour. 2. Set the multimeter dial to 20 volts DC. 3. Touch the red probe to the positive (+) battery post. 4. Touch the black probe to the negative (-) battery post. 5. Read the number on the multimeter's screen. This is the resting voltage. 6. Compare the reading to this chart: 12.6 volts or higher means a full, healthy charge. 7. A reading near 12.2 volts means the battery is only about half charged. 8. A reading below 12.0 volts usually means a weak or failing battery. 9. Optional: have a helper crank the engine while you watch the meter. A healthy battery should not drop below about 9.6 volts while cranking. 10. If the reading is low, or the battery struggles to crank the engine, plan to charge or replace it.

How you know it worked: you have a clear resting-voltage number and, if you did the cranking check, a cranking-voltage number, and you know which category the battery falls into.

Replace the battery

Safety warnings, before you loosen a cable clamp: - Always wear eye protection and protective clothing, or have a skilled technician do this. The battery gives off explosive hydrogen gas. - Do not smoke or bring sparks or flames near the battery. - Always disconnect the negative (-) cable first, and reconnect it last, to reduce the risk of a spark shorting the vehicle's electrical system. - Always wash your hands after handling the battery, its posts, or its terminals. They contain lead and lead compounds. - Do not drop the battery. It is heavy and can crack, spilling acid.

Steps: 1. Turn off the engine and every accessory. 2. Write down or confirm you have your radio's 5-digit anti-theft code. 3. Loosen and remove the negative (-) cable clamp first, and set it aside so it cannot touch the terminal. 4. Loosen and remove the positive (+) cable clamp next. 5. Remove the bolt or clamp holding the battery hold-down bracket in place. 6. Lift the old battery straight out, using a strap or careful grip. It is heavy. 7. Set the new battery in the tray, in the same orientation as the old one. 8. Reinstall the hold-down

bracket and its bolt. 9. Reconnect the positive (+) cable clamp first this time, and tighten it. 10. Reconnect the negative (-) cable clamp last, and tighten it. 11. Start the engine to confirm it runs normally. 12. Turn on the radio and enter your 5-digit code if "ENTER CODE" appears. 13. Reset the clock if it lost its setting.

How you know it worked: the engine starts normally, the radio accepts your code, or never asked for one, and the clock is reset if needed.

Clean the battery terminals

Safety warnings, before you mix any solution: - Always wear eye protection and protective clothing, or have a skilled technician do this. The battery gives off explosive hydrogen gas. - Do not smoke or bring sparks or flames near the battery. - Always wash your hands after handling the battery or its terminals. They contain lead and lead compounds. - Always disconnect the negative (-) cable first if you need to remove a cable, and reconnect it last.

Steps: 1. Look at each terminal for a white or yellowish powder. This is corrosion. 2. Mix baking soda with a little water into a thick paste. 3. Cover the corroded terminal with the baking soda and water solution. 4. Watch it bubble up and turn brown as it reacts with the corrosion. 5. Once the bubbling stops, wash the terminal off with plain water. 6. Dry the battery and terminals with a cloth or paper towel. 7. Coat the clean terminals with a thin layer of grease to help prevent future corrosion. 8. Check that your radio still works. If it shows "ENTER CODE," enter your 5-digit code.

How you know it worked: the white or yellow powder is gone, the terminals look clean and shiny, and the cable clamps sit tight on the posts with no wobble.

What can go wrong

- **Testing right after driving, before the battery has rested.** This gives a falsely high reading. Wait at least an hour with the engine off.

- **Assuming a battery that reads 12.6 volts resting is automatically fine.** A battery can pass a resting-voltage test and still fail under starter load. The cranking check catches that.
- **Disconnecting the positive cable first.** This raises the risk of a short if a tool accidentally touches metal. Always take the negative cable off first.
- **Forgetting the radio code before disconnecting the battery.** You may have to dig through old paperwork afterward to find it.
- **Not securing the hold-down bracket.** A loose battery can shift while driving.
- **Buying a non-original battery size without checking the physical fit first.** A group 51R battery is a tighter squeeze than the factory 151R.
- **Skipping the grease coating after cleaning terminals.** Corrosion often comes right back within months.
- **Not washing off all the baking soda paste.** This leaves a residue behind on the terminal.

Quick check

1. What group size battery does your 2008 Fit use from the factory?
2. What voltage reading on a rested battery means a full, healthy charge?
3. Why do you always disconnect the negative cable first and reconnect it last?
4. What should you write down before you disconnect the battery, and why?
5. Where in this book do you go for the full jump-start procedure?

Spark Plugs

What you will learn

- What a spark plug does, and which plugs fit your 2008 Fit's engine
- The correct gap and torque for a new spark plug
- How to replace all four spark plugs yourself, step by step
- Why valve clearance, a small engine gap checked at the same visits, is worth knowing about
- What can go wrong during a plug swap, and how to avoid it

The short version

A spark plug ignites the fuel and air mixture inside your engine's cylinder. Your Fit takes an NGK IZFR6K-13 plug, or the equivalent Denso SKJ20DR-M13. The gap, the small space the spark jumps across, should be 1.2 to 1.3 millimeters (0.042 to 0.051 inches) on a new plug. Each plug tightens to about 13 pound-feet (18 newton-meters) of torque, a measure of twisting force. Replacing all four takes about 40 minutes the first time.

The full explanation

What spark plugs do, and when yours need changing

Every time a cylinder in your engine fires, a spark plug creates the spark that ignites the fuel and air inside it. Your Fit's owner's manual treats spark plug replacement as a Maintenance Minder sub item. This means the dashboard's Maintenance Minder computer decides when it is due, rather than a fixed mileage printed in the manual. No exact number is printed for your specific

car. Your Fit's plugs are marketed as iridium-style plugs, plugs with a hard metal tip that lasts longer than a standard plug. Some general guidance for this kind of plug points to roughly 100,000 miles. This figure is not specific to the 2008 Fit. We could not confirm it as an exact number for your car, so check your owner's manual and dashboard display instead.

Which plugs, what gap, what torque

Your Fit's Specifications table lists two matching plug part numbers: NGK IZFR6K-13 and Denso SKJ20DR-M13. Either one is correct; they are made to the same specification.

A new plug's gap, the small air space between the plug's electrodes where the spark jumps across, should measure 1.2 to 1.3 millimeters, or 0.042 to 0.051 inches. Most new plugs ship pre-gapped correctly from the factory. Use a gap gauge to check this measurement, but do not force-adjust a pre-gapped plug unless it is clearly wrong.

Torque means how tight a bolt or plug is turned, measured in force applied at a distance. Each spark plug tightens to about 18 newton-meters, which is 1.8 kilogram-force-meters or 13 pound-feet. A torque wrench, a tool that stops or signals once you reach a set tightness, keeps you from cracking the cylinder head's soft aluminum threads by overtightening.

Valve clearance: a related check at the same visits

Valve clearance is the small gap that lets your engine's valves open and close correctly as they heat up and cool down. Your Fit's manual has the technician check this clearance at the same Maintenance Minder visits, labeled A, B, 1, 2, or 3, where other service happens. It only gets adjusted if the engine sounds noisy; there is no separate fixed mileage number for adjusting it. Check this on a cold engine, meaning the cylinder head is below 100 degrees Fahrenheit (38 degrees Celsius). The correct clearance is 0.15 to 0.19 millimeters (0.006 to 0.007 inches) on the intake valves. The correct clearance is 0.26 to 0.30 millimeters (0.010 to 0.012 inches) on the exhaust valves. We could not

confirm a do-it-yourself procedure for adjusting valve clearance on this engine from this book's sources. This job is usually left to a technician, so check your owner's manual or a qualified shop if your engine is noisy.

Step by step

Replace the spark plugs

Safety warnings, before you touch a coil or a plug: - Always let the engine cool completely before starting. The cylinder head and exhaust stay hot long after the engine is off. - Do not drop anything into an open spark plug hole. A dropped bolt or tool can badly damage the engine. - Always work on one plug at a time so you never mix up which coil and boot go where. - Do not overtighten a plug. A torque wrench prevents cracking the cylinder head threads.

Steps: 1. Let the engine cool completely. 2. Open the hood and remove the plastic under-cowl or engine cover panel to reach the coils. 3. Unplug the electrical connector from the first ignition coil. 4. Remove the bolt holding that ignition coil, and lift the coil straight up and out. 5. Look inside the plug hole and clear away any loose dirt with a rag or brush. 6. Fit the spark plug socket over the old plug and turn it counterclockwise to remove it. 7. Inspect the old plug's tip for unusual color or heavy wear. 8. Check the new plug's gap against the specification on its box, but do not force-adjust a pre-gapped plug. 9. Apply a small amount of anti-seize compound, a paste that keeps metal threads from sticking together over time, to the new plug's threads. 10. Thread the new plug in by hand until it stops, to avoid cross-threading. 11. Tighten the plug the rest of the way with a torque wrench to about 13 pound-feet (18 newton-meters). 12. Reinstall that ignition coil, its bolt, and its electrical connector. 13. Repeat steps 3 through 12 for each remaining cylinder. 14. Reinstall the under-cowl or engine cover panel. 15. Start the engine and listen for smooth, even running with no misfire.

How you know it worked: the engine starts easily, idles smoothly with no shaking or misfire, and the check engine light does not come on after a short drive.

What can go wrong

- **Threading a plug in with the socket instead of starting it by hand.** This risks cross-threading the soft aluminum cylinder head.
- **Skipping the torque wrench and guessing "snug."** This can under-tighten or over-tighten the plug.
- **Mixing up which coil connector plugs into which coil.** Work on one cylinder at a time to avoid this.
- **Dropping a bolt into the plug hole while the coil is off.** Keep loose tools and hardware well clear of open plug holes.
- **Assuming your engine's plug interval matches general online guidance.** Your Fit's own Maintenance Minder display is the only source that actually tracks your car's driving.

Quick check

1. What two spark plug part numbers fit your 2008 Fit, and are they interchangeable?
2. What gap should a new spark plug have?
3. Why do you thread a new spark plug in by hand before using a torque wrench?
4. How tight should each spark plug be torqued to?
5. What is valve clearance, and when is it adjusted on your engine?

CHAPTER 30

Coolant, Transmission Fluid, Brake Fluid, Power Steering, and Washer Fluid

What you will learn

- What coolant type your 2008 Fit needs, and how much it holds
- How to check the coolant level and top it up, and how to do a full drain and fill
- What transmission fluid your car needs, and how to check and change it for your transmission type
- What brake and clutch fluid your Fit uses, and the one fixed rule that always applies
- Why this car has no power steering fluid to check at all
- What washer fluid to use, and what never to put in that reservoir

The short version

Your Fit uses five separate fluids under the hood, besides engine oil: coolant, transmission fluid, brake fluid, clutch fluid on manual-transmission cars, and washer fluid. There is no power steering fluid on this car at all, because your Fit steers with an electric motor instead of hydraulic pressure. Each fluid has its own correct type; using the wrong one can cause real damage, so this chapter tells you exactly which one goes where.

The full explanation

Engine coolant

Coolant is the liquid that flows through your engine and radiator to carry heat away and keep the engine from overheating. Your Fit's owner's manual calls for Honda Long-Life Antifreeze/Coolant Type 2, which comes pre-mixed at 50 percent antifreeze and 50 percent water. Never add straight, undiluted antifreeze or plain water on its own. If Honda's coolant is not available, another major-brand coolant made for aluminum engines, and free of silicates, can be used temporarily. Flush the system and refill it with Honda coolant as soon as you reasonably can.

Replacing the coolant entirely is a Maintenance Minder sub item, so the dashboard display decides when it is due; no fixed mileage number is printed. The system's capacity depends on which transmission your car has. After a coolant change, a manual-transmission Fit holds 3.8 liters (1.00 US gallon), and an automatic-transmission Fit holds 3.7 liters (0.98 US gallon). After a full engine overhaul, the total capacity is 5.4 liters (1.43 US gallons) for a manual, or 5.3 liters (1.40 US gallons) for an automatic. The separate reserve tank, the see-through plastic tank you check day to day, holds 0.4 liters (0.11 US gallon) on its own.

Transmission fluid

Transmission fluid keeps your gearbox's internal parts lubricated and, on an automatic, also carries the hydraulic pressure that shifts gears. Replacing it is a Maintenance Minder sub item on both transmission types, so the dashboard decides normal timing. Two fixed exceptions apply only to automatic-transmission Fits. If you regularly drive at low speed in mountains, change the fluid at 60,000 miles (100,000 kilometers), then every 30,000 miles (48,000 kilometers) after that. If you tow the car behind a motorhome, change it every 2 years or 30,000 miles (48,000 kilometers), whichever comes first.

If you have a manual transmission, your Fit uses Honda Manual Transmission Fluid (MTF). If it is not available, an SAE 10W-30 or 10W-40 motor oil carrying the API Certification Seal for gasoline engines works as a temporary substitute. Switch back to Honda MTF soon, since plain motor oil lacks the additives your gearbox needs and can make shifting stiffer over time. The **differential** is the gearbox part that splits engine power between your front wheels. A change takes about 1.6 US quarts (1.5 liters); the total amount, including the differential, is 1.7 US quarts (1.6 liters).

If you have an automatic transmission, your Fit uses Honda Genuine ATF-Z1 only. Mixing in another brand's automatic transmission fluid (ATF) can wear down your transmission, and Honda's own warranty does not cover damage caused that way. A drain-and-refill takes about 2.5 US quarts (2.4 liters); the total system capacity, only reached during a full teardown, is about 6.3 US quarts (6.0 liters). One note for buying fluid today: Honda's ATF-Z1 has since been discontinued across the industry and replaced by Honda ATF DW-1 for service fills. Ask your parts store for the replacement if the original is not on the shelf.

Brake fluid and clutch fluid

Brake fluid transmits the force from your foot on the brake pedal to the brakes at each wheel. Your Fit's manual calls for Honda Heavy Duty Brake Fluid DOT 3, or a DOT 3 or DOT 4 fluid from a sealed container as a temporary replacement. Never use fluid marked DOT 5; it is not compatible with this car's brake system and can cause serious damage.

One rule for brake fluid always applies, no matter what the Maintenance Minder display shows: replace it every 3 years. This is one of the small set of fixed-schedule items in your whole maintenance schedule, alongside the annual oil change and the 160,000-mile idle speed check covered in Chapter 21.

If your Fit has a manual transmission, it also uses a hydraulic clutch, meaning fluid pressure, not a cable, disengages the clutch when you press the pedal. This clutch fluid shares the exact same specification as the brake fluid above. Check that its level sits between the MIN and MAX marks on the reservoir, which has a gray cap. A low level can point to a leak in the clutch hydraulic system.

We could not confirm a step-by-step, do-it-yourself procedure for replacing brake or clutch fluid from this book's sources at this pass. Check Chapter 27 for more on your braking system, and Chapter 40 for a full brake job, or have a qualified shop replace this fluid on schedule.

Power steering: this car does not have hydraulic fluid to check

Power steering is the system that reduces how much effort it takes to turn your steering wheel. Many older cars use a hydraulic pump and a reservoir of power steering fluid for this. Your 2008 Fit does not. It uses Electric Power-Assisted Rack-and-Pinion Steering (EPS) instead, on every trim, which means an electric motor assists your steering rather than pressurized fluid. Your owner's manual's Fluid Locations diagram has no power steering fluid reservoir at all. It lists only engine oil, automatic transmission fluid, brake fluid, clutch fluid on manual-transmission cars, washer fluid, and the coolant reservoir. Your dashboard has its own dedicated electric power steering indicator instead, confirming this is an electrical system, not a hydraulic one. This also means there is no power steering fluid to check, top up, or ever change on your Fit.

Washer fluid

Washer fluid sprays onto your windshield when you use the washers, so the wipers have something to clean with instead of dry glass. Your Fit's manual calls for any commercially available windshield washer fluid. Never put engine antifreeze, or a vinegar-and-water solution, into this reservoir. We

could not confirm the exact reservoir capacity for the 2008 Fit from this book's sources. Check your owner's manual for that figure, and top the reservoir up when the level runs low.

Step by step

Check the coolant level and top it up

Safety warnings, before you touch the radiator cap: - Do not remove the radiator cap while the engine is hot. Hot coolant can spray out and scald you badly. Wait until the engine and radiator are cool. - Do not add plain water alone, or straight, undiluted antifreeze, to the reserve tank. - Do not add rust inhibitors or other additives. They may not be compatible with the coolant or engine parts.

Steps: 1. Park on level ground and let the engine and radiator cool completely. 2. Open the hood and find the coolant reserve tank, a translucent plastic tank, not the radiator itself. 3. Look at the coolant level against the MIN and MAX marks on the side of the tank. 4. If the level is at or below MIN, remove the reserve tank cap. 5. Pour Honda Long-Life Antifreeze/Coolant Type 2 in slowly, watching the level. 6. Fill until the level reaches the MAX mark. Do not overfill. 7. Put the reserve tank cap back on, and make sure it seats fully. 8. Wipe up any spilled coolant right away. It can damage engine bay parts. 9. Look around the reserve tank and radiator hoses for any obvious wet spots or drips that would explain a low level.

How you know it worked: the coolant level sits at or near the MAX mark on the reserve tank, the cap is seated, and you find no active leak underneath.

Drain and fill the coolant

Safety warnings, before you raise the car or open the radiator: - Do not remove the radiator cap while the engine is hot. Hot coolant can spray out and scald you. Wait until the engine and radiator are cool. - Do not work under a car held up only by a jack. Use jack stands or ramps rated for the weight. -

Always relieve pressure slowly. Turn the radiator cap counterclockwise to the first stop and let it hiss before pressing down and removing it. - Do not add plain water alone as a permanent fix, and do not mix Honda's concentrated coolant with plain water beyond what the label describes. - Always clean up spilled coolant immediately. It is toxic to pets and wildlife if left in a puddle.

Steps: 1. Let the engine and radiator cool completely. 2. Raise the front of the car and support it on jack stands, then remove the splash shield if one is fitted. 3. Relieve pressure at the radiator cap slowly, then remove the cap fully. 4. Place a drain pan under the radiator's drain plug, and loosen it to drain the radiator. 5. Place the drain pan under the drain bolt at the rear of the engine block, and remove it to drain the block. 6. Let all the coolant drain fully from both points. 7. Reinstall the block drain bolt with a new washer, and tighten it securely. 8. Tighten the radiator drain plug securely. 9. Remove, empty, and rinse the coolant reserve tank, then reinstall it. 10. Pour fresh Honda coolant into the radiator, up to the base of the filler neck. 11. Fill the reserve tank to the MAX mark. 12. Install the radiator cap loosely, not fully tight yet. 13. Start the engine and let it run until the radiator fan cycles on and off at least twice, to work air out of the system. 14. Turn off the engine, let it cool, and recheck the radiator and reserve tank levels. 15. Top off as needed, then put the radiator cap on fully and tightly. 16. Start the engine again and check underneath for leaks at both drain points. 17. Lower the car and remove the jack stands.

How you know it worked: the reserve tank sits at the MAX mark after a full warm-up-and-cool-down cycle, and there are no leaks at either drain point. The temperature gauge or warning light behaves normally on a test drive.

Check and change the manual transmission fluid

Safety warnings, before you raise the car: - Do not work under a car held up only by a jack. Use jack stands or ramps. - Always check and change the fluid with the transmission at normal operating temperature, and the car on level ground. - Do not overfill. Fluid should only reach the edge of the bolt hole, not spill out under pressure. - Always clean up any spilled fluid immediately.

Steps (level check only): 1. Drive the car until the transmission reaches normal operating temperature. 2. Park on level ground and shut off the engine. 3. Remove the transmission filler bolt. 4. Carefully feel inside the bolt hole with a clean finger for the fluid level. 5. If fluid does not reach the edge of the hole, add Honda MTF until it starts to run back out. 6. Put a new washer on the filler bolt, and reinstall it, tightened securely.

Steps (full drain and refill, continuing from a cool transmission): 7. Raise the car and support it on jack stands. 8. Place the drain pan under the transmission's drain bolt. 9. Remove the drain bolt and let the old fluid fully drain. 10. Reinstall the drain bolt with a new washer, tightened securely. 11. Lower the car back to level ground. 12. Remove the filler bolt and add fresh Honda MTF through a pump or long funnel, until it reaches the edge of the hole. 13. Reinstall the filler bolt with a new washer, tightened securely. 14. Check for leaks at both bolts after a short drive.

How you know it worked: fluid sits right at the edge of the filler bolt hole, neither dripping out under pressure nor clearly low. There are no leaks at either bolt after driving, and shifting feels normal, not stiff or grindy.

Check and change the automatic transmission fluid

Safety warnings, before you raise the car: - Do not work under a car held up only by a jack. Use jack stands or ramps. - Always check and fill the level with the engine at normal operating temperature, and the car on level ground. - Do not fill above the dipstick's upper mark. Overfilling can damage the transmission. - Always clean up any spilled fluid immediately.

Steps (level check only): 1. Park on level ground and start the engine, letting it run until the radiator fan comes on, then shut it off. 2. Wait about 60 seconds, but no longer than 90 seconds, before checking. 3. Remove the dipstick, the yellow-looped fluid gauge, and wipe it clean with a rag. 4. Insert the dipstick fully back into the transmission. 5. Remove it again, and read the fluid level against the upper and lower marks. 6. If the level is below the lower mark, add Honda ATF-Z1 through the dipstick tube until it is between the marks. 7. Reinsert the dipstick fully and securely.

Steps (full drain and refill, continuing after the level check): 8. Raise the car and support it on jack stands. 9. Place the drain pan under the transmission's drain plug. 10. Remove the drain plug and let the old fluid fully drain. 11. Reinstall the drain plug with a new washer, tightened securely. 12. Lower the car back to level ground. 13. Add fresh Honda ATF-Z1 through the dipstick tube using a pump or long funnel. 14. Recheck the level using steps 1 through 5 above, topping off as needed. 15. Check for leaks at the drain plug after a short drive.

How you know it worked: the dipstick reads between the upper and lower marks after the correct warm-up-and-wait routine, and there are no leaks at the drain plug. The car shifts smoothly, with no delay or harsh engagement.

What can go wrong

- **Topping off coolant with plain water instead of the correct pre-mixed type.** This dilutes the mixture and can reduce its protection against freezing or overheating.
- **Opening a hot radiator cap and getting scalded by escaping steam and coolant.** Always let the system cool first.
- **Not letting the engine run through a full warm-up cycle after a coolant drain and fill.** This can leave a trapped air pocket that causes overheating later.

- **Using plain motor oil in a manual transmission long-term instead of switching back to Honda MTF.** Motor oil lacks the additives your gearbox needs, and shifting can get stiffer over time.
- **Using any automatic transmission fluid other than Honda ATF-Z1, or its replacement, ATF DW-1.** Mixing fluids can wear down your transmission, and the damage may not be covered under warranty.
- **Checking automatic transmission fluid cold, or without waiting the 60- to 90-second window after shutdown.** This gives an inaccurate reading.
- **Assuming your Fit has power steering fluid to check because an older car you know did.** This car steers electrically. There is no fluid at all.
- **Skipping the 3-year brake fluid change because the dashboard display never mentioned it.** This is a fixed rule, separate from the Maintenance Minder.
- **Pouring antifreeze or a vinegar solution into the washer fluid reservoir.** Use only commercial washer fluid.

Quick check

1. What coolant type does your Fit require, and what should you never add to the system on its own?
2. Name the two fixed-mileage exceptions for automatic transmission fluid, and who they apply to.
3. What fluid does a manual-transmission Fit's clutch use, and where do you check its level?
4. Does your 2008 Fit have hydraulic power steering fluid? Why or why not?
5. What fixed rule applies to brake fluid no matter what the Maintenance Minder display says?

CHAPTER 31

Belts, Hoses, and the Timing Chain

What you will learn

- Why your 2008 Fit has a timing chain instead of a timing belt, and what that means for you
- What the drive belt is, and how your car's maintenance schedule handles it
- Why this book cannot give you a hose-replacement schedule for this chapter
- Why there is no do-it-yourself procedure in this chapter, and where to take the car instead

The short version

Your Fit's engine uses a timing chain, not a timing belt. A timing chain is a metal chain, not rubber, and Honda treats it as a lifetime part with no scheduled replacement. Your car also has one drive belt, which the Maintenance Minder tracks for inspection and adjustment on its own schedule. This chapter has no do-it-yourself procedure, because none of this book's sources include one for these parts; a technician checks them for you.

The full explanation

Timing chain, not timing belt

A timing belt is a rubber belt. It keeps the engine's valves and pistons moving together. Many cars need this belt replaced on a strict mileage schedule. A skipped or broken timing belt can be a serious, sudden problem. Your Fit's engine does not have one of these belts at all. It uses a timing chain instead, a metal chain that does the same job. Your Fit's own maintenance item table lists

two scheduled jobs: spark plug replacement and valve clearance inspection. It has no line at all for replacing a timing belt or timing chain. Honda treats the timing chain as a lifetime part under normal use. It has no scheduled replacement interval. It only needs service if a specific problem comes up. That problem could be unusual noise. It could be chain stretch, meaning the chain has worn and lengthened slightly. Or it could be the engine's timing actually skipping.

This is good news for your maintenance budget. Unlike a timing belt on many other cars, there is no expensive, scheduled chain replacement waiting for you at some fixed mileage. It is also a reason to pay attention if your engine ever sounds different or runs roughly. A timing chain problem is not something your maintenance schedule will warn you about ahead of time.

The drive belt

Your Fit has one drive belt. It spins accessories such as the alternator, the part that recharges your battery while the engine runs. Inspecting and adjusting this belt is a Maintenance Minder sub item. The dashboard's own maintenance computer decides when it is due. There is no fixed mileage printed in your manual for this job. This book's sources do not include a do-it-yourself procedure card for inspecting or adjusting this belt. A technician should check it during your regular Maintenance Minder service visits.

Hoses: what this chapter can honestly tell you

We could not confirm a specific inspection interval for coolant hoses from this book's sources at this pass. Your Fit's Maintenance Minder Item B inspection list does name some hoses. It lists brake hoses and lines, plus fuel lines and connections, as things a technician checks. It does not separately name engine coolant hoses in that list. Check your owner's manual for that item. Or ask your technician to look over your coolant hoses at your next service visit. This book does not have a confirmed, Fit-specific answer for that particular item.

Step by step

This chapter has no do-it-yourself procedure. Your Fit's engine does not have a timing belt to replace, so this book's research did not turn up a timing-belt-style replacement job. It also did not turn up a home procedure card for inspecting or adjusting the drive belt, or for checking coolant hoses. A technician handles all three of these jobs. This usually happens at your regular Maintenance Minder service visits, described in Chapter 21. If your engine ever sounds unusual, or you notice a belt squeal, have it looked at promptly. Do not guess at a fix yourself.

What can go wrong

- **Assuming your Fit needs a timing belt change at some mileage, the way many other cars do.** Your engine has a timing chain, not a belt, and Honda gives it no scheduled replacement interval.
- **Ignoring an unusual engine noise because "the timing chain doesn't need service."** A timing chain is only serviced when a problem shows up, such as noise, stretch, or a timing skip. Unusual noise is exactly the sign to have it checked.
- **Never having the drive belt inspected because the dashboard has not flagged it.** The Maintenance Minder decides the timing, but it still relies on your regular service visits to actually happen.
- **Assuming this chapter covers a hose-replacement schedule that this book could not confirm.** Check your owner's manual, or ask your technician, since this chapter is honest about that gap rather than guessing at a number.

Quick check

1. Does your 2008 Fit have a timing belt or a timing chain? What is the difference between the two?

2. Does your Fit's timing chain have a fixed replacement mileage? If not, when does it get serviced?
3. What does the drive belt do, and how does your Fit's maintenance schedule decide when to inspect it?
4. Why does this chapter not include a step-by-step procedure?

Lights and Bulbs, Wipers, Fuses

What you will learn

- Every bulb size and type in your 2008 Fit, from headlights to the dome light
- How to replace a headlight bulb, and tail, brake, or turn signal bulbs
- What size wiper blades your Fit uses, and how to replace them
- How to find and replace a blown fuse, and why you should never use the wrong amperage
- What can go wrong during each of these jobs

The short version

Your Fit uses a mix of halogen bulbs, meaning bulbs with a heated filament inside a gas-filled glass bulb. It also uses small wedge-style bulbs for everything from turn signals to the dome light. Replacing most exterior bulbs takes about 15 minutes and basic tools. Wiper blades should be checked every 6 months and replaced at the first sign of trouble. A blown fuse, a small safety device that breaks an electrical circuit, is one of the easiest and safest repairs in your whole car.

The full explanation

Every light on your Fit, and its bulb

Your Fit's owner's manual and its Specifications table describe the bulb for nearly every light on the car. The table below lists them.

Light	Bulb type	Fact ID
Headlight, high and low beam	HB2 (also cross-sold as 9003 or H4), halogen, 12 volt, 60/55 watt	F-WP09-048
Fog light (Sport model or accessory)	H11, halogen	F-WP09-049
Front turn signal	Amber, 12 volt, 21 watt	F-WP09-050
Rear turn signal	7443	F-WP09-051
Stop/taillight	7443	F-WP09-052
Back-up light	7443	F-WP09-053
License plate light	T10	F-WP09-054
High-mount brake light	UNVERIFIED trade number	F-WP09-055
Front side marker/parking light	T10	F-WP09-056
Rear side marker	T10	F-WP09-057
Dome/ceiling light	C5W, 31 millimeter festoon, a tube-shaped bulb with metal caps on each end	F-WP09-058
Cargo/luggage compartment light	T10	F-WP09-059

Your headlight bulb carries the trade name HB2 in Honda's own Specifications table. Parts stores commonly sell this same physical bulb under the name H4 or 9003 as well; confirm the cross-reference against your own bulb's markings before you buy a replacement. If your Fit has fog lights, either from the Sport trim or added as an accessory, they use an H11 halogen bulb.

Your front turn signals use an amber, 21-watt bulb. Several of your rear lights share the same 7443 trade number: the turn signal, the stop/taillight, and the back-up light. Each one still plugs into its own socket in the assembly. Your license plate light, front and rear side marker lights, and cargo compartment light all use a small T10 wedge bulb. We could not confirm the exact trade number for your high-mount brake light bulb from this book's sources. The

manual confirms it is a separate, owner-replaceable bulb, but does not print its part number. Check your owner's manual or a parts store for that specific number. Your dome light uses a festoon-style bulb, a small tube-shaped bulb with a metal cap on each end, listed as C5W in a 31-millimeter length.

Wiper blades

Check your wiper blades' condition at least every 6 months, per your owner's manual, and replace any blade at the first sign of cracking, hardening, or streaking. Your front wiper blades are two different sizes: 24 inches on the driver's side, and 16 inches on the passenger's side. Your rear wiper blade, on the tailgate, is 14 inches.

Fuses

A fuse is a small safety device that breaks an electrical circuit on purpose if too much current tries to flow through it. This protects the wiring behind it from damage or fire. Your Fit stores some spare fuses. Amperage rating, meaning the current level a fuse position is designed to carry, is printed on the fuse box's own lid diagram and in your owner's manual. Never substitute a fuse with a higher amperage rating than the one that was there originally. This raises the risk of damaging your electrical system.

Step by step

Replace a headlight bulb

Safety warnings, before you touch a bulb: - Do not touch the glass part of a halogen bulb with bare skin. Oil from your fingers can make it overheat and shatter when lit. Handle it by its base only. - Always clean the glass with denatured alcohol and a clean cloth before installing it, if you accidentally touch it. - Always let the engine cool if it was recently running, since you will reach in near the engine bay.

Steps: 1. To reach the driver's side bulb, start the engine, turn the wheel fully right, then turn the engine off. 2. To reach the passenger's side bulb, instead turn the wheel fully left. 3. Use a flat-tip screwdriver to remove the holding clip from the inner fender. 4. Pull the inner fender cover back to expose the back of the headlight. 5. Remove the electrical connector from the bulb by pulling it straight back. 6. Remove the rubber weather seal by pulling on its tab. 7. Unclip the end of the hold-down wire from its slot. 8. Pivot the hold-down wire out of the way and remove the old bulb. 9. Install the new bulb, holding it by its base, making sure its tabs sit in their slots. 10. Pivot the hold-down wire back into place and clip its end into the slot. 11. Check the wire's installation through the inspection window on the headlight assembly. 12. Install the rubber weather seal back over the back of the headlight assembly. 13. Push the electrical connector onto the new bulb until it is securely connected. 14. Reinstall the inner fender cover and the holding clip, pressing the clip's center to lock it. 15. Turn on the headlights to confirm the new bulb works.

How you know it worked: the headlight comes on and looks similarly bright and evenly aimed as the other side. The inner fender cover and clip are back in place.

Replace tail light and brake light bulbs

Safety warnings, before you pry the cover: - Do not force the trim tool or screwdriver against painted body panels without a cloth underneath. You can scratch the paint. - Always confirm which of the bulbs in the assembly is actually burned out before you disturb the others.

Steps for the rear light assembly, on the tailgate side: 1. Open the tailgate. 2. Place a cloth on the edge of the light assembly cover to protect the paint. 3. Pry the cover off carefully at the notch on its middle edge with a flat-tip screwdriver. 4. Identify which bulb is out: stop/taillight, back-up light, turn signal, or taillight. 5. Turn that bulb's socket one-quarter turn counterclockwise and remove it from the assembly. 6. Pull the burned-out bulb straight out

of its socket. 7. Push the new bulb straight into the socket until it bottoms out. 8. Turn on the lights, or press the brake pedal for a stop/taillight bulb, to confirm it works. 9. Turn the socket clockwise back into the light assembly until it locks. 10. Reinstall the light assembly cover.

Steps for the high-mount brake light, on the rear window: 11. Open the tailgate. 12. Remove the light assembly cover by pushing in the tabs on both sides and pulling it off. 13. Turn the socket one-quarter turn counterclockwise and remove it from the assembly. 14. Pull the old bulb straight out of its socket. 15. Push the new bulb straight into the socket until it bottoms out. 16. Put the socket back into the light assembly and turn it clockwise to lock it. 17. Press the brake pedal to confirm the new bulb lights up. 18. Push the cover back onto the light assembly until it locks in place.

How you know it worked: the specific light you replaced turns on when tested, and both covers are fully seated and locked.

Replace the wiper blades

Safety warnings, before you raise a wiper arm: - Do not open the hood while the wiper arms are raised off the glass. The hood can hit and damage the arms. - Always handle a raised wiper arm carefully. Do not let it snap down onto bare glass without a blade attached.

Steps for the front wiper blades: 1. Raise the wiper arms off the windshield, lifting the driver's side first, then the passenger's side. 2. Press and hold the lock tab on the blade assembly. 3. Slide the blade assembly toward the lock tab until it releases from the wiper arm. 4. Grasp the tabbed end of the old blade and pull firmly until it comes out of its holder. 5. If the new blade has no plastic or metal reinforcement strip, move the strip from the old blade into the new one's edge slots. 6. Make sure the new blade's rubber tabs fit into each notch of the reinforcement strip. 7. Slide the new blade into the wiper arm until it is engaged along its full length. 8. Lower the wiper arm back down onto the windshield. 9. Repeat steps 2 through 8 for the other front wiper arm.

Steps for the rear wiper blade, if equipped: 10. Raise the rear wiper arm off the rear glass. 11. Slide the old blade out of the wiper arm. 12. Move the reinforcement strip to the new blade if it does not already have one. 13. Slide the new blade assembly onto the wiper arm until it locks in place. 14. Lower the rear wiper arm back down against the glass.

How you know it worked: all wiper arms sit flat against the glass with the new blades locked in place. A test run of the wipers wipes cleanly, with no streaks, chatter, or missed spots.

Replace a fuse

Safety warnings, before you open a fuse box: - Always turn the ignition switch to LOCK (0) and turn off the headlights and every accessory before checking or changing fuses. - Do not install a fuse with a higher rating than the circuit calls for. This greatly increases the risk of damaging the electrical system. - Do not ignore a fuse that blows again right away with a same-rated replacement. That means there is a real electrical problem, and the vehicle should be checked by a qualified technician.

Steps: 1. Turn the ignition switch to the LOCK (0) position. 2. Make sure the headlights and all other accessories are off. 3. Figure out which box holds the fuse: primary under-hood, on the driver's side, interior, behind the coin tray, or secondary, on the battery. 4. Remove the cover from that fuse box. 5. Find the suspect fuse using the chart on the fuse box lid or in the manual. 6. Pull a small fuse straight out using the fuse puller from the interior box. 7. For a large fuse in an under-hood box, remove it with a Phillips screwdriver instead. 8. Look through the top of the fuse at the thin wire inside. A blown fuse shows a broken or melted wire. 9. If you are not sure which one is blown, check every fuse for that circuit, not only the first one. 10. Push in a new fuse of the exact same amperage rating. 11. Turn the ignition back on and test whether the device now works. 12. Reinstall the fuse box cover.

How you know it worked: the device that stopped working, such as a light, wiper, or power window, now works normally. The new fuse does not blow again right away.

What can go wrong

- **Buying a bulb without checking its cross-reference against your old bulb's markings.** A headlight sold as "H4" is often the same bulb as an HB2, but confirm this before you install it.
- **Touching a new halogen bulb's glass with bare fingers.** Oil from your skin can make it overheat and shatter when lit.
- **Guessing which rear bulb is out instead of testing each function first.** Test the brakes, turn signal, and reverse gear before you open any cover.
- **Prying a light assembly cover without a cloth underneath.** This can scratch the paint around it.
- **Forgetting to transfer the reinforcement strip into a wiper blade that ships without one.** The new blade will not sit correctly on the glass without it.
- **Letting a raised, unblocked wiper arm snap down onto bare glass.** Always support or lower an arm carefully.
- **Installing a higher-rated fuse to make it stop blowing.** This risks damaging the wiring or causing a fire.
- **Checking only one fuse and missing that a different one in the same circuit is the actual problem.** If your device still does not work, check every fuse on that circuit.
- **Forgetting that removing the radio fuse can lock the radio.** You may need your 5-digit code to unlock it again.

Quick check

1. What is the trade name for your Fit's headlight bulb, and what other name is it commonly sold under?
2. Name three of your Fit's rear lights that share the same 7443 bulb.
3. How often should you check your wiper blades, and what sizes does your Fit use, front and rear?
4. Why should you never install a fuse with a higher amperage rating than the original?
5. What should you do if a fuse blows again right away after you replace it?

CHAPTER 33

Keeping It Clean Inside and Out

What you will learn

- How to wash and wax your 2008 Fit's exterior without damaging the paint or wheels
- Why you should remove the rear antenna before a drive-through car wash
- How to clean your seat belts safely, and why they need to fully air-dry
- How to clean the interior and keep your floor mats from becoming a safety problem

The short version

Washing and waxing your Fit, plus a full interior clean, takes about 90 minutes the first time. Before a drive-through car wash, unscrew and remove the rear roof antenna by hand; the wash's spinning brushes can damage it. Never clean your seat belts with bleach, dye, or cleaning solvents, and always let them fully air-dry before you use the car again. Always re-anchor your floor mats into their factory anchor points once they are dry, since a loose mat can interfere with the pedals.

The full explanation

Washing and waxing the exterior

A clean, well-waxed exterior protects your Fit's paint from the sun, road salt, and everyday dirt. Wax, or a sealant, is a protective coating you apply on top of clean paint; it helps water bead up and roll off instead of sitting on the

surface. Your Fit's manual specifically warns against using a drive-through car wash without first removing the rear roof antenna by hand, since the wash's spinning brushes can damage it. Reinstall the antenna once you are done.

Your Fit's wheels are aluminum alloy, meaning a lightweight metal mix, and they have a protective clear-coat finish. Avoid harsh wheel cleaners or a stiff brush on them; these can damage that clear-coat. Use a car-wash soap made for vehicles, not dish soap, since dish soap can strip away wax.

Cleaning seat belts and the interior

Your seat belts should never be cleaned with bleach, dye, or cleaning solvents. If they get dirty, clean them with a soft brush and mild soap and water instead, and let them air-dry fully before you use the car again. A damp seat belt that gets rolled back up into its retractor can stay damp for a long time, so give it time to dry completely first.

When you clean the rest of the interior, vacuum the seats, carpet, and floor. Wipe hard surfaces, like the dashboard and door panels, with a mild cleaner and a soft cloth. Take out your floor mats and clean them separately. Once everything is completely dry, put the mats back and re-anchor them into their factory anchor points. A loose mat can slide forward and interfere with the pedals, or with a seat's weight sensor, which is a real safety issue, not only a tidiness one.

Step by step

Wash and wax the exterior, and clean the interior

Safety warnings, before you start: - Do not use a drive-through car wash without first unscrewing and removing the rear roof antenna by hand. The brushes can damage it. - Do not use bleach, dye, or cleaning solvents on the seat belts. - Always let seat belts air-dry fully before using the vehicle again. - Do not clean aluminum alloy wheels with harsh chemicals or a stiff brush. It can damage their protective clear-coat.

Steps for the exterior wash and wax: 1. If you plan to use a drive-through wash, unscrew and remove the rear antenna by hand first. 2. Rinse the whole car with plain water to knock off loose dirt. 3. Wash a small section at a time with car-wash soap and a wash mitt, then rinse that section. 4. Clean the wheels separately with a mild detergent and a soft brush or sponge, avoiding harsh wheel cleaners on alloy wheels. 5. Dry the car with a clean microfiber towel before water spots form. 6. Apply wax or sealant in small sections, following the product's instructions. 7. Buff each section with a clean, dry microfiber towel. 8. Reinstall the antenna if you removed it.

Steps for the interior: 9. Remove the floor mats and shake or vacuum them separately. 10. Vacuum the seats, carpet, and floor, using an attachment to reach corners. 11. Wipe the dashboard, door panels, and other hard surfaces with a mild cleaner and a soft cloth. 12. Wipe the insides of the seat belt loops with a cloth dampened in mild soap water or isopropyl alcohol. 13. If the seat belts themselves are dirty, clean them with a soft brush and mild soap and water, then let them air-dry fully. 14. Make sure carpeting and floor mats are completely dry before you put the floor mats back. 15. Re-anchor the floor mats onto their factory anchor points. A loose mat can interfere with the pedals or a seat sensor.

How you know it worked: the exterior is clean with an even, beaded look where waxed, and the wheels are clean without dulled or damaged clear-coat. The interior looks and smells clean, and every floor mat is dry and re-anchored in place.

What can go wrong

- **Washing the car in direct hot sun.** This makes soap and water dry into spots before you can rinse it off. Work in shade if you can.
- **Using a stiff brush or strong wheel cleaner on the aluminum alloy wheels.** This can damage their protective clear-coat finish.

- **Skipping the antenna removal before a drive-through wash.** The spinning brushes can bend or break it.
- **Cleaning seat belts with bleach or a solvent-based cleaner.** This can weaken the belt webbing itself.
- **Putting damp floor mats back down, or not re-anchoring them.** This can interfere with the pedals or a seat's weight sensor, which is a genuine safety concern.
- **Using dish soap instead of car-wash soap.** Dish soap can strip away any wax already on the paint.

Quick check

1. What should you do to the antenna before a drive-through car wash, and why?
2. Name two things you should never use to clean your seat belts.
3. Why do your wheels need a gentler cleaner and softer brush than the rest of the car?
4. Why is re-anchoring your floor mats a safety issue, not only a tidiness one?

PART VI

Diagnosing Problems

CHAPTER 34

How to Describe a Problem

What you will learn

- Why a clear description helps you find the real cause of a car problem
- How to describe *when* a noise, vibration, smell, or leak happens
- How to describe *where* you feel or notice it
- How your speed and engine speed give useful clues
- How color and smell narrow down a leak or an odor
- The difference between a "Fit-specific" finding and general automotive knowledge, a difference you will see again in Chapter 37

The short version

A good description has four parts: when it happens, where you notice it, how fast you were going, and what it looks or smells like. Chapters 35 through 37 sort information this same way, so a clear description lets you use those chapters fast. Write your notes down before you call a shop or look up a code.

The full explanation

Why this matters

"My car is making a weird noise" tells a mechanic almost nothing. It also will not help you use the tables in Chapter 37 or the flowchart in Chapter 36. The more specific you are, the faster anyone, including you, can narrow down the cause.

This chapter teaches you to notice four things about any problem: when it happens, where you notice it, your speed, and what it looks or smells like. These are the same four categories this book's research used to build the lookup tables in Chapter 37.

When does it happen?

Ask yourself when the noise, shake, or smell shows up. Common answers include: at idle, meaning while the engine runs but the car is not moving; while accelerating; while turning; while braking; or over bumps.

For example, a ticking sound at idle often points to a low oil level. It can also point to a problem in the VTEC system, the mechanism that switches the intake valves between a low-lift and a high-lift setting using engine oil pressure. A clunk that happens only when you shift into Drive or Reverse points to something different. It often means a worn engine or transmission mount, the rubber-and-metal part that holds the engine or transmission to the car's frame. Two very different sounds, both "at idle." But the timing detail, a constant tick versus a one-time clunk on shifting, points to two different systems.

Where do you notice it?

Location matters too. Do you feel a vibration through the steering wheel, or through the seat and floor? Do you hear a noise from the front of the car, the middle, or the back? Owners report that a clicking noise while turning often points to a worn CV joint. A CV joint, short for constant velocity joint, is a flexible joint in the axle that lets power reach the wheel while it turns. That same clicking sound, if it does not change while you turn the wheel, more likely points to a loose exhaust heat shield instead. A heat shield is a metal panel that protects nearby parts from a hot exhaust pipe.

How fast were you going?

Speed is one of the most useful clues for a vibration. A shimmy in the steering wheel that appears below about 20 mph, then goes away above that speed, often points to a tire that is out of round. It can also point to a bent wheel, or a stuck brake part dragging. A stronger shake that grows worse between roughly 45 and 65 mph often points to a wheel or tire that is out of balance. Out of balance means its weight is not spread evenly around the wheel. Writing down the exact speed range helps you or a mechanic skip straight to the likely cause.

Engine speed matters too, separately from road speed. A whine or grind from the front of the engine that gets louder as the engine revs, not as the car speeds up, points toward the alternator. The alternator is the part that charges the battery while the engine runs, and this noise can also come from the belt that spins it.

What does it look or smell like?

For a leak, note the color and exactly where under the car you see it. Amber or dark brown usually means engine oil. Red or pink usually means automatic transmission fluid on this car. The 2008 Fit uses electric power steering, not a hydraulic system, so there is no separate power-steering fluid to confuse it with. Green, yellow-green, orange, pink, or blue, with a sweet smell, points to coolant, the fluid that keeps the engine's temperature down.

For a smell, describe it in plain words: burning rubber, burning plastic, sweet and syrupy, rotten egg, raw gasoline, or hot oil. Each points toward a different system, which Chapter 37 explains in full.

Understanding urgency

Chapter 37's tables each give an urgency level: Low, Medium, Medium-high, or High. Low means the problem is usually only annoying, not dangerous. High means you should stop driving soon, sometimes right away, because the problem could hurt you, hurt the car badly, or both. When you describe a

problem to yourself or a shop, always mention any High-urgency sign. Examples include a burning smell, a grinding noise while braking, or a flashing warning light.

"Fit-specific" versus general knowledge

Some of what you will read in Chapters 35 through 37 comes from real 2007-2008 Honda Fit owners describing their own cars. This book marks those as "Owners report that..." Other information is true of almost any gasoline car and does not need a Fit-specific source, since it is standard automotive knowledge. Chapter 37 marks each row so you can tell which kind of information you are reading. Neither kind is more or less true. They are two different types of evidence.

Step by step

Use this short routine any time something feels or sounds wrong with your car.

1. Stop and think before you drive further, especially if the problem feels sudden or serious.
2. Note when it happens: at idle, accelerating, turning, braking, or over bumps.
3. Note where you notice it: steering wheel, seat and floor, front, middle, or back of the car.
4. Note your speed, or your engine speed, when it happens or gets worse.
5. Note the color of any fluid, and exactly where under the car it appears.
6. Note the smell, using plain words like burning, sweet, or rotten egg.
7. Write all of this down, or say it out loud to yourself, before you look anything up.
8. Use your notes with Chapter 35 (check engine light and codes), Chapter 36 (won't start), or Chapter 37 (lookup tables) to narrow down the cause.

What can go wrong

- **Being too vague.** "It sounds funny" describes nothing. "It ticks at idle, only when the engine is cold" describes something you can look up.
- **Skipping the "when."** A noise that only happens once, right after starting, is very different from a noise that happens constantly. Both matter, but they point to different causes.
- **Guessing the cause instead of describing the symptom.** Say what you heard, felt, saw, or smelled first. Save your guess about the cause for after you check the tables.
- **Ignoring a High-urgency sign because the car "still drives fine."** A car can still move while something inside it is failing badly. Treat a burning smell or a flashing light seriously right away.
- **Forgetting to check both speed and engine speed for a vibration or a whine.** Road speed and engine speed are not the same thing, and they point to different systems.

Quick check

1. What four things should you note about any car problem?
2. Why does the exact speed range matter for a vibration?
3. What is a CV joint, and what noise, at what time, points to a worn one?
4. What is the difference between a "Fit-specific" finding and general automotive knowledge?
5. Name one High-urgency sign you should never ignore, even if the car still seems to drive fine.

The Check Engine Light and OBD-II Codes

What you will learn

- What OBD-II means and what the check engine light is actually telling you
- Where to find the OBD-II port in your 2008 Fit
- How reading and clearing a code works, in general terms
- All 20 trouble codes this book found linked to this exact car, in plain English, with the first thing to check for each
- Why 2 of those 20 codes are less certain than the other 18, and why this book says so honestly

The short version

The check engine light, formally called the Malfunction Indicator Lamp (MIL), warns you that a system tied to emissions or engine control has a problem. OBD-II, short for On-Board Diagnostics, second generation, is the standard system every 1996-and-newer US gasoline car uses to store a trouble code describing that problem. A code reader plugged into the port under your dashboard shows you the code, and this chapter's table shows you what 20 common codes mean and where to start checking.

The full explanation

What OBD-II is, and what the check engine light means

OBD-II is a federal standard. Every gasoline car sold in the US since 1996 must have it, including your 2008 Fit. The system constantly watches the engine and emissions parts. When something goes outside its normal range, the car's computer, called the ECM (engine control module), saves a code describing the problem and turns on the check engine light.

Each stored code is called a DTC, short for Diagnostic Trouble Code. A DTC does not always mean a part is broken. It means the ECM saw a reading it does not expect, which could point to a broken part, a loose connector, a low fluid level, or something else entirely. That is why this chapter's table always lists a first thing to check, not only a part to replace.

Where the OBD-II port is

The OBD-II port, also called the data link connector (DLC), is a 16-pin plug you connect a code reader to. On your 2008 Fit, it sits under the dashboard on the driver's side, at or a little above the brake and gas pedals, close to the steering column. It usually has no label, and it may sit tucked up against the underside of the dash instead of hanging down in plain view. You will likely need to crouch down and feel for it with your hand, or put your head down near the pedals to see it.

This location is not random. A federal standard called SAE J1962 sets the rules. The port must sit between the driver's end of the dashboard and a point slightly past the center of the car. It must mount to the dashboard structure and be reachable from the driver's seat without tools. That rule is why "under the dash, driver's side, near the steering column" is the right first place to look on almost any 1996-or-newer US car. A source naming the Fit by name confirms this location specifically for your car.

Reading and clearing a code, in general terms

A code reader, sometimes called a scan tool, plugs into the OBD-II port. With the ignition turned on, it shows you any stored DTCs as a letter-and-number code, such as P0300. You then look up what that code means, which this chapter's table does for the 20 most common codes on this car.

Clearing a code turns off the check engine light, but it does not fix the problem that caused the code. If the underlying cause is still there, the code, and the light, will likely come back. This book could not confirm the exact menu steps for every code reader model, since readers vary by brand. Follow the instructions that come with your own reader.

Should you keep driving with the light on?

This depends on how the light behaves. A steady check engine light means a problem exists, but usually is not an emergency by itself. A flashing check engine light is more serious. It means an active, severe misfire, a condition where a cylinder is not firing correctly, which can send unburned fuel into the hot catalytic converter and destroy it. If your check engine light is flashing, pull over and stop driving. Do not continue on to your destination first.

The 20 codes this book found for the 2008 Fit

Every code below was checked against real reports from owners of 2007-2008 Honda Fits, not a generic list of what any car's computer can report. Most are backed by two independent owner reports describing this exact code on this exact generation of car. Two codes have only one such report each, and this book flags them honestly rather than presenting them as more certain than they are.

Key to the Source column: - **TSB** — Backed by an actual Honda Technical Service Bulletin, the strongest evidence in this list. - **2+** — Backed by two or more independent 2007-2008 Fit owner reports. - **1 only** — Backed by only one 2007-2008 Fit owner report found. Treat this as a lead, not a confirmed pattern.

Code	What it means	First thing to check	Source
P0300	Random or multiple cylinder misfire. More than one cylinder is skipping a spark.	Spark plugs first, then the ignition coil packs. Owners report that more than one coil pack can fail around the same time on this car.	2+
P0301	Cylinder 1 misfire.	Spark plug and ignition coil for cylinder 1. If that does not fix it, check the upstream oxygen sensor next.	2+
P0302	Cylinder 2 misfire.	Spark plug and ignition coil for cylinder 2. Owners report this code often appears alongside P0300.	2+
P0303	Cylinder 3 misfire.	Spark plug and ignition coil for cylinder 3. If plugs and coils don't fix it, worn fuel injectors are a real possibility on this engine.	2+
P0304	Cylinder 4 misfire.	Spark plug and ignition coil for cylinder 4.	2+
P0135	The heater inside the upstream oxygen sensor, the sensor that checks exhaust before the catalytic converter, is not drawing the current expected.	Check the 15-amp fuse for this sensor's heater circuit first. If the fuse is fine, check the sensor's wiring before replacing the sensor.	2+
P0141	Same heater problem as P0135, but on the downstream	Check the downstream sensor's connector and wiring	2+ (base-trim confirmed)

Code	What it means	First thing to check	Source
	oxygen sensor, located after the catalytic converter.	first. This exact code and repair has been directly confirmed on a base-trim 2007 Fit.	
P0420	The catalytic converter, the part that cleans exhaust gases, is not working as well as the computer expects.	Rule out an active misfire or a bad upstream oxygen sensor first. Both are cheaper to fix than a converter, and both can trigger this code without the converter actually being bad.	2+
P2646	Part of the VTEC system, which uses engine oil pressure to change the intake valve lift. This code means the switch that reports oil pressure to the computer is reading a lower voltage than expected.	Check engine oil level and condition first. Honda's own bulletin says this is the single most common actual cause.	TSB
P2647	The same VTEC oil pressure switch as P2646, but reading a higher voltage than expected.	Same as P2646: check engine oil level and condition first, before replacing the switch.	TSB
P2649	The VTEC oil control solenoid, the valve that actually moves oil pressure into the VTEC system. This is a different part from the switch in P2646/P2647.	Owners report a small, inexpensive O-ring or gasket at the solenoid as a common fix. This code often appears together with misfire codes on this car. Note: this code is not covered by the	2+ (owner reports only)

Code	What it means	First thing to check	Source
		same Honda bulletin as P2646/P2647, unlike an earlier version of this book's notes claimed.	
P0335	The crankshaft position sensor, which tracks engine speed and position for timing and fuel, is sending a signal the computer cannot use.	Check the sensor's electrical connector and wiring first. Owners report replacing the sensor and having the problem return right away, pointing to wiring more than the sensor itself.	2+
P0339	Same sensor and circuit as P0335, but the signal drops out only sometimes instead of failing completely.	Same as P0335: check the wiring and connector first.	2+
P0500	The vehicle speed sensor, which reports how fast the car is moving to the computer and speedometer, is not sending a usable signal.	Check the wiring at the transmission connector first. Owners on this car have twice found a broken wire, not the sensor itself, to be the actual cause.	2+
P0171	The fuel mixture keeps coming back leaner, meaning less fuel relative to air, than the target, and the computer is adding extra fuel to correct it.	Check for a vacuum or intake leak first, such as a loose or cracked hose. If that comes up clean, worn fuel injectors are a real second possibility on this car.	2+
P0172	The opposite of P0171: the fuel mix-	Check for a stuck-open EGR valve	2+

Code	What it means	First thing to check	Source
	ture keeps coming back richer, meaning more fuel than the target, than expected.	first, a part that recirculates exhaust gas into the intake. Then check fuel pressure before assuming a fuel injector problem.	
P2A00	The upstream air/fuel ratio sensor, Honda's more advanced version of an oxygen sensor, is reading outside its normal range.	Check the sensor's connector and wiring first. Watch out for a bad-fit after-market sensor if this code appears right after a do-it-yourself sensor replacement.	2+
P0451	The sensor that reports pressure inside the sealed EVAP system, which is the fuel tank, canister, and lines that trap gasoline vapor instead of releasing it into the air, is sending a signal the computer does not trust.	Check the sensor's electrical connector and its small vacuum hose fitting first, before replacing the sensor.	1 only
P0401	The EGR system, which feeds some exhaust gas back into the intake to lower combustion temperature, is not flowing as much as the computer expects.	Clean the EGR valve and its passage first. Carbon buildup restricting the valve is the most commonly cited cause.	2+
P0113	The intake air temperature sensor is sending a reading the computer considers too high, or	Check whether the code appeared right after disconnecting the battery, cleaning the throttle body, or	1 only

Code	What it means	First thing to check	Source
	possibly reading as an open circuit.	unplugging the sensor. A code caused this way can sometimes clear itself after normal driving.	

Being honest about the "1 only" codes

This book's research found only one 2007-2008 Fit-specific report for P0451 and for P0113, not the two independent reports found for every other code in this table. Both are still real, specific reports about this exact generation of car, not invented entries. But treat them as a lead worth checking, not a settled pattern the way the other 18 codes are.

What can go wrong

- **Assuming a code always means a broken part.** A DTC often points to a wiring problem, a loose connector, or a low fluid level instead. Always start with the "first thing to check" column, not a parts order.
- **Ignoring a flashing check engine light.** A flashing light means stop driving, not "get to it this week."
- **Clearing a code and assuming the problem is fixed.** Clearing only turns off the light. If the cause is still there, the code will likely return.
- **Treating P0451 or P0113 as a fully confirmed pattern.** Both only have one Fit-specific report behind them in this book's research.
- **Skipping the cheap check for P2646 or P2647.** Honda's own bulletin says checking the oil level and condition comes before replacing the switch.

Quick check

1. What does OBD-II stand for, and what does DTC stand for?

2. Where is the OBD-II port on your 2008 Fit, and why is it in that general area on almost any car built since 1996?
3. What does a flashing check engine light mean, and what should you do about it?
4. Which two codes in this chapter's table are backed by an actual Honda service bulletin, and which one code was previously, incorrectly, linked to that same bulletin?
5. Which two codes in the table have only one Fit-specific report, and how should you treat that information?

CHAPTER 36

Won't Start, Won't Crank, or Cranks But Won't Start

What you will learn

- The difference between "won't crank" and "cranks but won't start"
- A 9-step flowchart to work through when your Fit will not start
- What the immobilizer light tells you, and what to do if it blinks
- When the problem is likely the battery, and when it is likely something else
- When to try something yourself, and when to call for a tow

The short version

Cranking means the starter is spinning the engine over; starting means the engine actually fires and keeps running on its own. If the dashboard is completely dark, check the battery first. If the dash lights up but the engine will not crank, check your clutch pedal or shift position and the immobilizer light next. If the engine cranks normally but will not fire, think about flooding, fuel, or ignition.

The full explanation

Two different problems that feel similar

"My car won't start" can mean two very different things, and telling them apart is the fastest way to find the cause.

Won't crank means when you turn the key to START, nothing happens, or you hear only a click or a slow grinding sound. The starter, the motor that spins the engine to get it running, is not turning the engine over at a normal speed.

Cranks but won't start means the starter spins the engine at a normal speed, but the engine never fires up and keeps running on its own.

This chapter's flowchart below asks you a short question at each step, then tells you exactly where to go next depending on your answer. No step sends you down more than 3 possible paths.

The immobilizer

Your Fit has an immobilizer, an anti-theft system built into the key. The key has a chip inside it that the car's computer must recognize before the engine will start. A key-shaped light on the dashboard shows you what the immobilizer is doing. Step 3 of the flowchart below tells you what to watch for.

Step by step

The 9-step flowchart

Work through these steps in order. Each step tells you exactly which step to go to next, based on what you find.

Step 1. Turn the key to ON (the second position) without starting the engine. Do the dashboard warning lights and gauges light up normally? - If **yes**, go to Step 2. - If **no**, meaning the dashboard is completely dark, go to Step 8.

Step 2. Check your starting position. On a manual transmission, is the clutch pedal pressed all the way to the floor? On an automatic, is the shift lever fully in Park or Neutral, with the brake pedal pressed? The starter will not engage at

all unless this is correct. - If this is correct and the car still will not start, go to Step 3. - If it was **not** correct, fix it and try again. If it now starts, you are done.

Step 3. Turn the key to ON and watch the key-shaped immobilizer light. Does it flash, or stay lit, instead of going out normally? - If **yes**, go to Step 4. - If **no**, go to Step 5.

Step 4. Try your spare key. - If the spare key **starts** the car, your primary key or its chip is likely damaged. Stop here and have the primary key replaced by a dealer. - If the spare key **also fails** to start the car, the immobilizer system itself may be at fault. Stop here and see a dealer or a qualified shop.

Step 5. Turn the key to START. Does the engine crank, meaning turn over or spin, at a normal speed? - If **yes**, go to Step 6. - If it makes **no sound** and does not move at all, go to Step 8. - If you hear a **single click**, or a slow grinding sound and the engine barely turns, go to Step 8.

Step 6. The engine cranks normally but does not fire. Do you smell strong raw gasoline, or did you pump the gas pedal several times before trying to start? - If **yes**, this may be a flooded engine, meaning too much fuel has reached it. Wait a few minutes without touching the pedal, then try again without pumping the pedal. If it still will not fire, go to Step 7. - If **no**, go to Step 7.

Step 7. Check the fuel gauge. Is the tank empty or very close to empty? - If **yes**, add fuel and try again. - If **no**, this points to a fuel-delivery or ignition problem. It could be the fuel pump, a fuse, the spark plugs, or the ignition coils. This needs a scan tool and a shop visit, so stop here and arrange service.

Step 8. Check the battery terminals, the metal connection points on top of the battery. Are they clean, tight, and free of white or blue corrosion? - If **no**, clean and tighten them, then go back to Step 1. - If **yes**, go to Step 9.

Step 9. Jump-starting has real risks, including sparks near battery gases. Always follow every safety warning in Chapter 16 before you connect a single cable. Once you have read those warnings, jump-start the car using the

terminal order already covered in Chapter 16. Connect positive to positive on both batteries first. Then connect negative on the booster battery, then the unpainted metal ground point on your car. - Does the car **start**? If yes, drive for at least 20 to 30 minutes to recharge the battery, then have the battery and charging system tested soon. - If **no**, the starter, the alternator, or a deeper electrical fault is likely. Stop here and arrange a tow to a shop.

How you know it worked: the engine starts and keeps running on its own. The dashboard lights behave normally, and the immobilizer light goes out after its brief flash at key-ON.

What can go wrong

- **Skipping Step 2.** Many "won't start" calls turn out to be a clutch pedal not fully pressed, or a shifter not fully in Park. Always check this before assuming a bigger problem.
- **Missing the immobilizer light at Step 3.** A blinking key light means the chip in your key was not recognized. Trying to force a start without checking this wastes time.
- **Continuing to pump the gas pedal after a possible flooded engine.** This adds more fuel and makes flooding worse, not better.
- **Jumping straight to a jump-start before checking the battery terminals.** Dirty or loose terminals are checked at Step 8. That is an easy first fix that a jump-start alone would not solve.
- **Connecting jumper cables in the wrong order, or to the wrong point.** Follow the exact order in Step 9 and in Chapter 16 to avoid sparks near the battery.

Quick check

1. What is the difference between "won't crank" and "cranks but won't start"?

2. What two starting-position checks does Step 2 ask about, and why do they matter?
3. If the immobilizer light keeps blinking even with your spare key, what does that suggest, and what should you do?
4. What should you do first if you suspect a flooded engine?
5. In what order do you connect jumper cables at Step 9?

CHAPTER 37

Noises, Vibrations, Smells, and Leaks: Lookup Tables

What you will learn

- How to use four lookup tables to narrow down a noise, a vibration, a smell, or a leak
- What each table's urgency rating means for how soon you should act
- How to tell a "Fit-specific" finding from general automotive knowledge, at a glance, in each table
- Where this book's research found a gap, meaning no Fit-specific source at all

The short version

These four tables sort common car problems by when they happen, how fast you were going, what they smell like, or what color they are. Each row also names an urgency level, from Low to High, so you know how soon to act. A "Fit" tag means real 2007-2008 Fit owners reported this pattern; a "General" tag means the row is true of gasoline cars broadly.

The full explanation

How to read these tables

Every row below has a short **Source** tag: - **Fit** — Owners report that this specific pattern shows up on real 2007-2008 Honda Fits. - **General** — This is true of gasoline cars broadly, and did not need, or have, a Fit-specific report.

Use Chapter 34's method before you open these tables: note when it happens, where you notice it, your speed, and any color or smell. Then find the closest matching row below.

Urgency levels mean: - **Low** — Usually only annoying, not dangerous. Fix it when convenient. - **Medium** — Worth checking soon. It can get worse or more expensive if ignored. - **Medium-high** — Get it checked promptly. Waiting has a real cost. - **High** — This can hurt you, seriously damage the car, or both. Act now.

Table 1: Noises, by when they happen

When	Noise	Likely cause	Urgency	Source
At idle	Ticking or tapping from the top of the engine	Low oil level, or a fault in the VTEC oil pressure switch or solenoid. VTEC is the system that uses engine oil pressure to change the intake valve lift (see Chapter 35's P2646, P2647, and P2649)	Medium — check oil level today	Fit
At idle	Clunk when shifting into Drive or Reverse	A worn engine or transmission mount, the part that holds the engine or transmission to the car's frame	Medium — annoying now, but a worsening clunk can mean a fully failed mount	Fit
At idle, rough or unstable	Rough or hunting idle speed	A vacuum leak, a dirty throttle body, or a misfire code (see Chapter 35's P0300-series codes)	Medium — get it scanned for codes soon	Fit
Accelerating	Clicking that speeds up with engine speed but does not change with steering	An exhaust heat shield, a metal panel that protects nearby parts from a hot exhaust pipe, rattling loose	Low — annoying, but worth fixing before the shield falls off	Fit
Accelerating, especially under load	Misfire-related roughness, popping, or a flash-	Spark plugs, ignition coils, or fuel injectors (see Chapter	High if the check engine light is flashing, stop driving and arrange a	Fit

When	Noise	Likely cause	Urgency	Source
	ing check engine light	35's misfire codes)	tow; Medium if the light is steady	
Turning, especially sharp turns	Clicking that only happens while turning	A worn outer CV joint, the flexible joint in the axle that lets power reach the wheel while it turns	High — a fully failed CV joint can separate and cause loss of drive; get it inspected soon	Fit
Turning, low speed, such as in a parking lot	Rubbing or grinding	A brake pad wear indicator touching the rotor, or a worn CV joint or steering part	Medium-high — have brakes and the front end checked soon	Fit
Braking	High-pitched squeal	Worn brake pads, whose built-in wear tab is designed to squeal on purpose, or a missing anti-squeal shim on aftermarket pads	Medium-high — check pad thickness soon; worn-out pads can damage rotors	Fit
Braking	Grinding, metal-on-metal	Brake pads worn down to the metal backing plate	High — stop driving on it; rotors are likely being damaged and stopping distance is affected	Fit + General
Releasing the parking brake	Squeak only at that moment	Usually normal friction in the parking-brake mechanism	Low, if it only happens on release and brakes otherwise feel normal	Fit + General

When	Noise	Likely cause	Urgency	Source
Over bumps	Rattle from the dashboard or glovebox area	Loose interior trim clips, often the cowl panel under the windshield or a part behind the factory radio	Low — annoying, not a safety issue	Fit
Over bumps	Metallic clunk from the front end	Worn sway-bar end links or strut top-mount bushings, small parts in the front suspension	Medium — have the front suspension inspected; can worsen into a handling problem	Fit
Over bumps	Rattling from underneath, near the middle or rear of the car	A loose or cracked exhaust heat shield	Low-medium — usually only noise, but a shield that falls off can drag or become a fire risk near dry material	Fit
Any time, gets louder with engine speed, not road speed	Whining or grinding from the front of the engine	The alternator's bearing wearing out, or the serpentine belt and its tensioner	Medium-high — a failing alternator can leave you stranded with a dead battery; also check the Smells table below	Fit

Table 2: Vibrations, by speed range

Speed range	Vibration felt where	Likely cause	Urgency	Source
Idle or stopped	Through the seat or floor, not the steering wheel	A worn engine or transmission mount, or a rough idle from a misfire or vacuum leak code	Medium	Fit
Low speed, under about 20 mph, goes away above that	Shimmy in the steering wheel	A tire out of round, a bent wheel, or a stuck brake caliper part dragging	Medium	General
Mid-to-highway speed, roughly 45 to 65 mph, through the steering wheel	A shake that gets stronger with speed	A wheel or tire out of balance, meaning its weight is not spread evenly, or a bent rim	Medium-high — have wheels balanced and inspected soon; can speed up tire and bearing wear	Fit + General
Highway speed, roughly 55 to 65 mph, felt through the whole car, automatic transmission only	Shudder, jerk, or hesitation, not a steady vibration	Owners report a split opinion: it could be the automatic transmission itself, or worn ignition coils, valve clearance, or spark plugs presenting as a shudder	Medium-high — get it scanned for codes; do not assume it is only the transmission without a check, since the ignition-related cause is cheaper to fix	Fit
Any speed, a growl or vibration that grows with speed, sometimes louder in one turning direction	Vibration paired with a grinding or humming noise	A worn wheel bearing, the part that lets the wheel spin smoothly	Medium-high — a bearing that fails completely can cause a wheel to lock up or separate	General
				General

Speed range	Vibration felt where	Likely cause	Urgency	Source
Braking, any speed	Pulsation felt in the brake pedal	A warped or worn brake rotor	Medium — schedule a brake inspection; not usually an immediate emergency, but it will worsen	

Table 3: Smells

Smell	Likely cause	Urgency	Source
Burning rubber, especially after highway driving or with the heater on	The heater blower motor or its resistor overheating, or a drive belt slipping	Medium — have it checked soon; a slipping belt can eventually fail and strand you	Fit
Burning plastic or electrical	The alternator or another electrical part overheating; also check for a stuck power window switch, given this car's known switch recall	High — this can be a fire risk. Pull over, shut off the engine, and have it inspected before driving further	Fit
Sweet, syrupy smell, especially from the heater vents or under the hood	Leaking coolant, the fluid that keeps the engine's temperature down	Medium-high — check coolant level; do not keep driving if the temperature gauge or light also shows overheating	General + Fit
Rotten egg	A catalytic converter running too rich, or, in one specific report on this car, a faint exhaust smell only with the A/C on, while idling, after hard acceleration in hot weather	Medium — get it scanned for the fuel-trim, misfire, and catalyst codes covered in Chapter 35; a converter that keeps overheating from a rich mixture can be destroyed	General + Fit
Raw gasoline	A fuel leak, an EVAP system leak (the system that traps gasoline vapor), or an overfilled gas tank right after fueling	High if it is a steady, strong smell; do not start the engine near an open flame or smoking. Medium if it is a faint smell right after topping off the tank	General + Fit
Hot oil or a burning-oil smell	An oil leak dripping onto a hot exhaust part	Medium — check the oil level and look underneath for	General

Smell	Likely cause	Urgency	Source
		drips; oil pooling near hot exhaust for a long time can become a fire risk	
Clear, odorless puddle in humid or rainy weather	Normal air conditioning condensation	None — this is normal and not a leak to worry about	General

Table 4: Leaks, by color and location

Two things about this car matter for reading this table. First, the 2008 Fit uses electric power steering, not a hydraulic pump, so there is no separate red or pink power-steering fluid puddle to worry about. Any red fluid you see is almost certainly automatic transmission fluid instead, on automatic-transmission cars. Second, on a manual transmission, the clutch and the brakes share the same fluid reservoir and fluid type. A clear-to-amber leak near the driver's side firewall could point to either system.

Location	Color	Likely fluid or cause	Urgency	Source
Under the front of the car, roughly under the engine	Amber to dark brown, slightly oily	Engine oil	Medium-high — check the dipstick; do not run the engine low on oil	General
Under the front, closer to the center or passenger side, thin and slippery	Red or pink	Automatic transmission fluid; this car has no hydraulic power-steering fluid to confuse it with	Medium-high — have the transmission checked; low fluid can cause the highway shudder symptom in Table 2	General
Under the front, sweet-smelling	Green, yellow-green, orange, pink, or blue, depending on which coolant was last used	Coolant	Medium-high — do not keep driving if it is also overheating; top up and have the leak found and fixed	General + Fit
Under the middle of the car, near the driver's side firewall, clear to light amber, slippery, not sweet-smelling	Brake fluid or clutch fluid, since they share one reservoir on a manual-transmission Fit	High — brake fluid loss is a safety issue; stop driving and have it inspected right away if the pedal feels low or soft, or a brake warning light is on	General + Fit	
Under the middle-front area, thin, dark, smells strongly of gear oil	Manual transmission fluid	Medium — top up or have it checked; low fluid can cause hard shifting or gear noise	General	
Under the middle of the		None — this is normal	General	

Location	Color	Likely fluid or cause	Urgency	Source
car, clear, odorless, only in hot or humid weather with the A/C running	Air conditioning condensate water			
Under the rear of the car	Any fluid color	This book's research found no Fit-specific pattern for rear-of-car leaks; the fuel tank and rear brake and axle parts are the main candidates	High if it smells like gasoline; Medium otherwise, pending identification	General (gap: no Fit-specific source found)

What can go wrong

- **Treating any table as a final diagnosis.** These tables narrow things down. A shop visit or a code check may still be needed to confirm the exact cause.
- **Ignoring a High-urgency row because the car "seems fine."** Brake fluid loss, a burning-plastic smell, and a flashing check engine light are all High for a reason.
- **Mixing up red transmission fluid with power-steering fluid.** This car has no hydraulic power-steering fluid, so red or pink almost always means automatic transmission fluid instead.
- **Assuming the rear-of-car leak row is settled.** This book's research found no Fit-specific source for that row, and says so honestly rather than guessing.
- **Skipping Chapter 34's four questions before opening these tables.** When, where, at what speed, and what color or smell, make these tables far faster to use.

Quick check

1. What does a "Fit" tag mean in these tables, and what does a "General" tag mean?
2. What should you do if a row's urgency is listed as High?
3. Why is red or pink fluid under the front of this car almost always transmission fluid, not power-steering fluid?
4. Name one noise and one vibration that both point toward a worn CV joint or wheel bearing, and explain why.
5. Which row in the Leaks table does this book admit has no Fit-specific source, and what does that row cover instead?

CHAPTER 38

Known Weak Spots on the 2007-2008 Fit

What you will learn

- What "owners report that" means, and how it is different from a recall.
- Symptoms for 11 wear problems that first-generation Fit owners report.
- A rough shop repair cost for each problem, at today's prices.
- Whether each problem is a good DIY job, or one to leave to a shop.
- The one problem whose repair cost this book could not confirm for the 2008 Fit.

The short version

Owners of 2007-2008 Fits report 11 wear problems that show up again and again. None of these are recalls. They are patterns spotted by real drivers, not confirmed defects from Honda or the government. Every dollar figure below is a 2024-2026 shop price, not a 2008 price, so an actual 2008 repair bill would have been lower.

The full explanation

Why this is different from Chapter 18

Chapter 18 covered recalls and TSBs. A recall is a safety fix Honda must offer for free. A TSB is a repair instruction Honda sends to dealers. Both come from Honda or the government. This chapter is different. It covers patterns that owners on Fit forums have reported seeing again and again on their own cars. This book marks each one with the phrase "owners report that." That phrase

means the pattern comes from many individual drivers, not from an official investigation. A community-reported problem can still be real and common. It only means no official body has confirmed it.

A word about the prices in this chapter

Every dollar amount below comes from repair-cost websites and forum posts checked in 2026. These are today's shop prices, for cars generally 16 to 20 years old. They are not what a repair would have cost when your Fit was new, in 2007 or 2008. Prices also change by region, by shop, and by whether a part is new, aftermarket, or used. Treat every number here as a rough guide, not an exact quote.

The 11 weak spots

1. Motor mounts. Owners report that motor mounts wear out over the life of the car . A motor mount is a rubber-and-metal part that bolts the engine and transmission to the car's frame . When a mount wears out, you may feel a clunk when you shift into Drive or Reverse . You may also feel a clunk under hard acceleration . You may feel extra vibration at idle . Shop cost for one mount runs roughly \$495 to \$908, depending on which price page you check . New Honda mounts, called OEM parts, cost about \$130 each . OEM means Original Equipment Manufacturer, the same brand Honda used when it built your car . Generic aftermarket mounts run about \$30 to \$40 each . Aftermarket means a part made by a company other than Honda . This job is only partly DIY-able . DIY means "do it yourself," without paying a shop . You need a way to support the engine's weight while a mount is off . You also need jack stands and a torque wrench . A first-timer should have a helper or a shop manual close by .

2. Timing chain tensioner noise. Owners report that a ticking or rattling noise comes from the front of the engine . It is most noticeable at a cold startup, and it quiets down as the engine warms up . A timing chain tensioner keeps the timing chain tight, so the engine's valves open and close at the right

moment . This is the weakest-sourced item in this chapter . Researchers found only one usable online report of this exact problem on a 2008 Fit, not the two separate reports this book normally requires . Treat this as a possible pattern, not a settled fact . **We could not confirm this for the 2008 Fit.** For other small cars, this same kind of tensioner job can run from several hundred dollars to more than a thousand dollars . The job costs so much because a mechanic has to remove the front cover of the engine to reach the part . This is not a beginner job . A chain that slips out of time can bend the engine's valves . Take this repair to a shop .

3. Clutch master or slave cylinder wear (manual transmission only). Owners report that the clutch pedal can suddenly go spongy or soft . Sometimes the pedal drops to the floor with little resistance . Sometimes there is no fluid puddle under the car, because the leak happens inside the cylinder . A master cylinder and a slave cylinder work together to make the clutch disengage when you press the pedal . Shops usually replace both cylinders together . They do this because the second cylinder tends to fail soon after the first . A full job, with both cylinders and a fluid bleed, commonly runs \$700 to \$1,000 or more . This job is only partly DIY-able . Swapping the cylinders themselves is not hard . Bleeding air out of the hydraulic line afterward is easy to get wrong the first time .

4. Automatic transmission shudder (automatic transmission only). Owners report that the car shudders, jerks, or hesitates at about 55 to 65 miles per hour on flat ground . Forum owners disagree on the cause . Some blame the transmission itself . Others blame worn spark plugs or ignition coils instead . A torque converter is a part inside the automatic transmission that links the spinning engine to the transmission . If a worn torque converter turns out to be the cause, expect a bill around \$800 to \$1,600 . This is a general repair-shop price, not a number specific to the Fit . Diagnosing this shudder is not a DIY job . It needs real test time and, ideally, a scan tool that reads transmission codes . Changing the transmission fluid and filter first is a cheap, DIY-able step worth trying before paying for deeper diagnosis .

5. Air conditioning compressor and condenser problems. Owners report that two separate air conditioning complaints show up on this car . First, the condenser can get damaged by road debris . A condenser is a radiator-like part, mounted low at the front of the car, that cools the refrigerant . One 2008 Fit owner told the government this exact repair cost over \$800 . Second, some owners report a rattling compressor pulley when the air conditioning turns on . Others report air conditioning that cools for a minute and then stops . Shop cost for a full compressor replacement runs roughly \$1,100 to \$1,800, and can run higher . This is not a DIY job . Opening an air conditioning system needs a special license . That license is called EPA Section 609 certification . You also need equipment to safely remove the old refrigerant gas . A home garage does not have this equipment .

6. Front suspension clunk or knock. Owners report that a metallic clunk or knock comes from the front of the car over bumps . Forum owners most often trace this to one of two parts . The first is a worn sway bar end link . A sway bar end link is a small linking part that connects the sway bar to the suspension . The second is a worn strut top mount . A strut top mount is the rubber bushing where the top of a strut bolts to the car's body . Owner advice is to check the end links first, before paying for a full strut job . Sway bar end links cost roughly \$75 to \$200 to replace on one side, or \$200 to \$300 to do both sides at once . A full pair of struts, with new top mounts, runs roughly \$400 to \$900 . End links are a real beginner DIY job . The job needs only a floor jack, jack stands, and hand tools . Struts need a spring compressor tool and real caution . A released spring can badly hurt you .

7. Dashboard or glovebox rattle. Owners report that a rattle comes from the dashboard or glovebox area over bumps . It is most often traced to a loose plastic panel under the windshield . Sometimes it is a part behind the factory radio instead . The rattle tends to get worse in cold weather . There is no set price for this job, since it depends on finding the cause first . Shops typically bill it as one diagnostic fee visit . A diagnostic fee is the charge for the time a shop spends finding a problem before fixing it . That fee runs roughly \$50 to

\$150 at an independent shop . It can run \$100 to \$400 or more at a dealer . Once found, the actual fix often costs under \$20 in parts, such as a broken clip . This is a good DIY job for a patient beginner . It needs only basic plastic trim-removal tools .

8. Clearcoat peeling. Owners report that the clearcoat flakes or peels on flat, horizontal surfaces after several years . The roof and hood are the surfaces owners mention most . Clearcoat is the clear, protective top layer of a car's paint . Owners, and a paint-repair professional in one forum discussion, blame thin or improperly cured factory clearcoat . Fixing one panel, like only the roof, commonly costs \$500 to \$1,500 or more . A full repaint is sometimes needed once the damage has spread across several panels . A full repaint can run \$2,000 to \$5,000, and sometimes over \$10,000 . This is not a DIY job . Matching factory paint color takes real skill . Spraying an even clearcoat without runs also takes real skill and equipment .

9. Rust at the rear quarter panel and wheel well. Owners report that rust starts where the rear quarter panel meets the rear bumper cover . It also starts along a hidden seam near the rocker panel and wheel well . This spot collects road grit and moisture . Left alone, it can eat through to holes in the driver's rear wheel well . This mostly affects cars driven where roads get salted in winter . Cleaning and treating surface rust early costs roughly \$150 to \$900 . A real patch repair, once metal has to be cut and welded, runs roughly \$800 to \$1,800 . A full panel replacement can run up to about \$2,500 . Catching rust early is a real DIY job . It uses a rust converter product you can buy cheaply . Welding in a new patch panel needs skill and equipment most beginners do not have .

10. Battery drain. Owners report that the battery goes dead after sitting for two or three days . Some owners report the battery draining overnight instead . Reported causes include a faulty door switch, an electrical short, or an aftermarket add-on wired into the fuse box, such as an alarm . One technical forum post says a healthy 2008 Fit should draw only about 15 to 17 milliamps with everything off . A milliamp, written mA, is a small unit of electric

current . A draw above roughly 40 to 50 mA points to a real fault worth chasing . Basic diagnosis at a shop runs \$50 to \$100 . Tracing a harder short can run \$150 to \$400 or more . After that, a new battery costs \$50 to \$200 . A new alternator runs \$200 to \$600 . A wiring repair starts around \$100 and climbs with complexity . The diagnostic step is DIY-able with a simple multimeter and patience . Replacing a battery or a simple switch is beginner-easy . Chasing an intermittent wiring short is much harder .

11. Weak factory radio and speakers. Owners report that the factory speaker system sounds thin . Some trims are missing rear door speakers . Some head units lack a way to plug in an aftermarket amplifier . This is a comfort complaint, not a safety defect . A professional head-unit swap commonly costs \$150 to \$550 all-in for a basic unit . A unit with Bluetooth or a phone-mirroring screen costs more . Replacing all four speakers plus the head unit at a shop runs roughly \$600 to \$1,100 . This is one of the most beginner-friendly car projects there is . It needs only a screwdriver and a trim-panel tool . A wiring adapter is a good idea too, so you do not have to cut the factory wiring .

Putting it together

Most of these 11 items are wear-and-tear, not danger . None of them stop you from driving your car safely on their own . Still, always fix a real safety issue, like a shuddering transmission or a suspension clunk, before it gets worse . The next chapter shows you how to find a fair shop for any repair on this list that you cannot, or should not, do yourself.

Quick check

1. What does "owners report that" mean, and how is it different from a recall?
2. Name three of the 11 weak spots and their main symptom.
3. Which weak spot's repair cost could this book not confirm for the 2008 Fit, and why not?

4. Name one weak spot that is a good DIY job for a beginner, and one that is not.
5. Why are the dollar amounts in this chapter higher than what a 2008 Fit owner actually paid back in 2008?

CHAPTER 39

When to Go to a Shop, and How to Not Get Ripped Off

What you will learn

- The difference between an independent shop and a dealership, and why their prices differ.
- Typical hourly labor rates for each, at today's prices.
- Three situations where paying a dealer's higher rate still makes sense.
- How to read every line of a repair estimate before you agree to pay.
- Simple habits that protect you from a surprise bill.

The short version

Independent shops usually charge less per hour than dealerships. A dealer is still the right choice in a few specific cases. Every repair estimate has the same five parts: a diagnostic fee, labor, parts, shop supplies, and tax. Read every line before you agree to a repair, and compare the final bill to that estimate before you pay.

The full explanation

Independent shop versus dealership: labor rates

A labor rate is the hourly price a shop charges for a mechanic's time . It is separate from the cost of parts . An independent shop is a repair shop not owned by a car maker . Independent shops commonly charge \$80 to \$130 per hour . Dealerships commonly charge \$120 to \$180 per hour . A separate national study found that nearly half of all US repair shops price their labor

somewhere between \$120 and \$159 per hour . Dealerships tend to charge more for a few reasons . Their mechanics go through the car maker's own training programs . They must buy diagnostic tools the maker requires . They also carry higher costs from running a franchise and a larger building .

For a routine job you already understand, like brakes or a motor mount, an independent shop can often do equal-quality work for meaningfully less money . Look for a shop with mechanics who hold ASE certification, a widely recognized test for mechanics . A dealer becomes the better choice in three cases . First, the repair needs a special tool only the maker sells . Second, the repair is still covered by a warranty, a recall, or a TSB; see Chapter 18 for your car's own recall list . Third, the needed part is sold only through the dealer's own parts network .

How to read a repair estimate

Ask for a written estimate before any work starts . Do not settle for a spoken number only . A complete estimate has five parts .

Diagnostic fee. This is the charge for the time a shop spends figuring out what is wrong . It comes before any repair is agreed to . It commonly covers 60 to 90 minutes of labor . Many shops apply this fee toward your repair bill if you have the work done there . Ask this question before you say yes to the diagnosis .

Labor. This line shows the job name, a number of hours, and the shop's hourly rate . Hours are often written in tenths, so 3.5 means three and a half hours . You can check the math yourself by multiplying the hours by the rate .

Parts. Each parts line should say exactly what is being replaced . It should also say what kind of part it is . A part can be new OEM, meaning Original Equipment Manufacturer, the same brand Honda used . A part can be new aftermarket, made by a different company . A part can be remanufactured,

meaning a used part rebuilt to work like new . A part can also be plain used . A cheaper price on this line often means a lower-tier part, which is worth asking about .

Shop supplies. This line covers small extra costs, like rags, cleaning fluid, and small fasteners . It is normal to see this charge . It is usually about 5 to 10 percent of the labor total . A vague fee, with no clear dollar amount or percentage, is worth questioning .

Tax. Sales tax applies to parts in most states . Whether labor also gets taxed depends on your state . This line will vary by where you live .

Habits that protect you

Get every estimate in writing before work starts . Check that the paperwork lists your own car's VIN and mileage, not another car's . A VIN is your car's own 17-character serial number, covered in Chapter 18 . When the final bill arrives, compare it line by line against the original estimate . A bill that grew without a phone call explaining why is a warning sign .

Tying it back to Chapter 38

Chapter 38 told you which of the 11 weak spots a beginner can safely try at home. When a repair on that list is marked "no," or needs tools you do not have, use this chapter's habits before you pay. Ask for a written estimate. Read every line. Compare the final bill against it. Doing this each time protects you far more than trying to memorize a "fair price" for every possible repair.

Quick check

1. What is the usual hourly labor-rate range at an independent shop, and at a dealership?
2. Name three situations where paying a dealer's higher rate makes sense.
3. What are the five parts of a complete repair estimate?

4. What does OEM mean, and how is it different from aftermarket and re-manufactured?
5. What two things should you always compare before paying a repair bill?

PART VII

Repairs You Can Do at Home

Brake job, full

What you will learn

- The difference between a basic pad or shoe swap and a full brake job
- How to check a front brake rotor and replace it along with the pads
- How to check a rear brake drum and replace or resurface it
- The extra tools a full brake job needs beyond a basic pad or shoe job
- What can go wrong on these harder steps, and how to catch it

The short version

Chapter 27 already showed you how to swap front brake pads and rear brake shoes on their own. A full brake job goes one step further: you also check the front rotor and the rear drum, and replace either one if it is worn out. You do this work with the same jack stands, the same care, and mostly the same tools as Chapter 27, plus a few extras for the rotor and the drum.

The full explanation

A rotor is the flat metal disc that spins with your front wheel. The caliper squeezes your brake pads against it to slow the car. A drum is the metal cylinder that spins with your rear wheel. Your brake shoes press outward against the inside of the drum to slow the car.

Pads and shoes wear out on their own schedule, and so do rotors and drums. A rotor or drum can wear thin, get scored with deep grooves, or crack, even while the pads or shoes next to it still have life left. Both the rotor and the

drum carry a wear-limit number stamped into the metal. If your measurement goes past that number, or the metal is cracked or deeply grooved, you replace the part instead of reusing it.

This chapter does not repeat the pad and shoe steps from Chapter 27. It focuses only on what a full brake job adds: taking the rotor or drum off, checking it, and putting a new or resurfaced one back on.

Step by step

Front brakes: pads and rotors together

Do this job with the safety rules from Chapter 27 in mind. Always wear eye protection, since brake dust and rust flakes can hurt your eyes. Do not breathe brake dust, and never use compressed air or a dry brush on brake parts; use brake-parts cleaner instead. Never work under a car held up only by a jack; always use jack stands. Work on one wheel at a time and leave the other wheel on the ground, so you always have a reference for how the parts go back together. Never let the caliper hang by its brake hose; support it so the hose carries no weight. Never resurface or reuse a rotor that is cracked, deeply grooved, or thinner than its stamped minimum thickness; replace it instead. Always pump the brake pedal several times before you drive, until it feels firm again.

1. Loosen the front wheel nuts, then raise and support the car on jack stands, and remove the wheel.
2. Remove the caliper slide-pin bolts and lift the caliper off its bracket. Hang the caliper so it does not pull on the brake hose.
3. Remove the two caliper mounting-bracket bolts and lift the bracket off the steering knuckle, the metal part the wheel bearing sits in.
4. Check the rotor's edge for a minimum-thickness number stamped into the metal, and write it down before you remove the rotor.

5. If the rotor is stuck to the hub, tap it near its center with a rubber mallet. Never hit the braking surface.
6. Pull the old rotor off the hub studs.
7. Clean rust and debris off the hub face with a wire brush, then wipe it clean.
8. Apply a thin coat of anti-seize compound to the hub face only. Keep it off the braking surface.
9. Slide the new rotor onto the hub studs.
10. Clean the caliper bracket and slide pins with a wire brush and brake-parts cleaner.
11. Reinstall the caliper bracket. Owners report torquing its mounting bolts to 80 lbf-ft (108 N-m), but this number is not independently confirmed for this car.
12. Use a C-clamp against an old pad, or a block of wood, to push the caliper piston back into its bore.
13. Coat the new pads' metal backing and the slide pins with a thin layer of brake grease, then place the pads into the bracket.
14. Slide the caliper back over the new pads and reinstall the slide-pin bolts.
15. Reinstall the wheel, snug the nuts by hand, then lower the car and torque the wheel nuts to 80 lbf-ft (108 N-m) in a crisscross pattern.
16. Repeat steps 1 through 15 on the other front wheel.
17. Pump the brake pedal several times with the engine off, until it feels firm. Then check the brake fluid reservoir and top it off if needed.
18. Drive slowly at first and test the brakes gently in an empty area before you return to normal driving.

Rear brakes: drums

Chapter 27 already walked you through the rear brake shoe swap on this car, including the return spring, which is the hardest single step in that job. This section only adds the drum check. Do it in the same session as the shoe swap, since the drum has to come off to reach the shoes anyway.

Always wear eye protection when you inspect or clean a drum. Never reinstall a drum that is cracked, deeply scored, or worn past its stamped maximum diameter; replace it instead. Never work under a car held up only by a jack; always use jack stands.

1. Follow the first three steps of Chapter 27's rear brake job: raise the car, remove the wheel, and pull the drum off the hub.
2. Look for a maximum-diameter number cast or stamped into the drum's outer face.
3. Measure the drum's inside diameter with a micrometer or caliper, checking several points around the circle.
4. If any measurement goes past the stamped maximum diameter, or the surface has deep grooves or cracks, plan to replace both rear drums as a pair.
5. If the drum is within its limit but glazed or lightly scored, a machine shop can resurface it instead of replacing it. Ask the shop whether resurfacing costs less than a new drum.
6. If you are replacing the drum, do not transfer anything from the old drum itself. Only the shoes, springs, and adjuster carry over.
7. Finish the rest of Chapter 27's rear brake shoe steps, using the new or resurfaced drum in place of the old one.

What can go wrong

Getting anti-seize compound or grease on the rotor's braking surface, or on the pad linings, causes squealing and cuts your stopping power. Reusing a rotor or drum that is warped, cracked, or past its wear limit is the most common mistake in a full brake job. Always check the stamped number before you decide to reuse a part. Pushing a caliper piston back without first loosening the brake fluid reservoir cap can push fluid out or damage a seal. Letting a caliper hang by its hose can stretch or kink the hose. Replacing only one side's drum,

instead of both as a pair, can cause the car to pull or brake unevenly. And skipping the "pump the pedal first" step, on either job, can mean the pedal goes soft or all the way to the floor on your first drive.

Quick check

1. What two extra parts does a full brake job add on top of Chapter 27's basic pad and shoe swap?
2. How do you know a rotor or drum has to be replaced instead of reused?
3. Why should you never let a caliper hang by its brake hose?
4. Why should you check both rear drums, instead of only the one side that looks worn?
5. What should you do with the brake pedal before your first drive after the job, and why?

CHAPTER 41

Suspension: struts, shocks, sway bar links, bushings

What you will learn

- Why the front struts are a shop job, but the rear shocks are not
- What a shop will do when it replaces your front struts
- How to replace your rear shocks yourself, step by step
- How to replace your front sway bar end links and rear stabilizer bushing
- The difference between an end-link swap and a full sway bar replacement
- What can go wrong on each of these jobs

The short version

Your Fit's suspension holds the car up, controls its bounce, and keeps the tires planted on bumps. Some suspension parts are easy weekend jobs, and some are not. The front struts need a spring compressor and carry a real injury risk, so this book treats that one as a shop job. The rear shocks, the front sway bar end links, and the rear stabilizer bushing are all simpler jobs you can do yourself with basic tools and jack stands.

The full explanation

A strut and a shock absorber, often called a "shock" for short, do the same basic job: they slow down your suspension's bounce after a bump. A sway bar, also called a stabilizer bar, is a metal bar that connects the suspension on the car's left side to its right side. It reduces how much the body leans in turns. A

sway bar end link is a short linkage part that connects the end of the sway bar to the suspension. A bushing is a rubber or rubber-and-metal piece that cushions a joint and lets it flex slightly.

Your Fit's front suspension uses a MacPherson strut, a design where a coil spring wraps directly around the strut itself. That spring stores a large amount of energy while compressed. A tool called a spring compressor squeezes the spring so a mechanic can safely take the strut apart. If a compressor slips or is used the wrong way, it can release that stored energy suddenly and cause serious injury or death. This book could not confirm the exact top-nut torque number for this car's front strut from two independent sources. That gap is one more reason this book sends the front strut job to a shop, instead of guessing at that number.

Your Fit's rear suspension is different. The rear coil spring sits on its own in the rear axle beam, not wrapped around the shock. That means the rear shock job needs no spring compressor at all, which makes it a meaningfully easier job than the front strut.

This chapter also covers two smaller jobs that people often lump together under "sway bar links" and "bushings": swapping the front end links, and swapping the rear stabilizer bushing. Replacing the sway bar itself, the long bar rather than its end links, is a much bigger job on this car. It requires dropping the front subframe and the steering gearbox. This book treats a full sway bar swap as a shop job, and does not give full steps for it here.

What the shop will do: front struts

Even if a shop does this job, know what to ask about, before you read the summary below. Always treat a coil spring on a strut as if it stores enough energy to cause serious injury if it lets go suddenly and out of control. Never let anyone attempt this job with a homemade spring compressor, such as rope,

chain, or a bar clamp; a real job uses a purpose-built strut spring compressor. Always ask the shop to check, and if needed perform, a wheel alignment after the strut job.

A shop replacing your front struts will typically:

- Raise the car and remove the front wheel on each side.
- Disconnect the strut from the steering knuckle, then unbolt the brake hose bracket and any ABS sensor wiring clip from the strut.
- Remove the three top strut-mount nuts from inside the engine bay and lift the strut and spring out as one assembly.
- On a workbench, compress the spring with a proper spring compressor, then remove the top-mount nut while holding the shaft still with a hex wrench. Take the strut apart and rebuild it with the new part.
- Reinstall the new strut assembly in reverse order and torque every fastener to the specification in the shop's own service information.
- Perform a wheel alignment afterward. New struts can change ride height slightly. That can shift camber, how far the top of a tire tilts in or out, or toe, how far the tires point toward or away from each other.

Step by step

Rear shocks (both sides)

Never work under a car held up only by a jack; always use jack stands. Use a floor jack only to support the axle beam while you work, not as your only support. Always wear eye protection when working under the car. Never let the shock swing loose and strike the fuel tank, brake lines, or wiring; keep

control of its movement as you remove or install it. Always take care not to scratch the car's body while maneuvering the shock in and out of its mounting spot.

1. Raise the rear of the car, support it on jack stands, and remove both rear wheels.
2. Remove the wire guide grommet from both sides of the rear axle beam, the metal beam that the rear wheels attach to.
3. Position a floor jack under the lower spring seat on the axle beam. Raise it slightly, only enough for the suspension to begin compressing.
4. Remove the damper mounting bolt that connects the axle beam to the bottom of the shock.
5. Remove the access panel from the cargo area's side trim to reach the shock's top mount.
6. Hold the shock shaft still with a hex wrench, and remove the self-locking top nut.
7. Remove the top mounting washer and rubber mount.
8. Compress the shock by hand and lift it out of the car, taking care not to scratch the body.
9. Repeat steps 1 through 8 on the other side, if you are replacing both shocks.
10. Fit the rubber mount onto the new shock's top, and position the shock between the body and the axle beam.
11. Raise the floor jack slowly until the bottom mounting hole lines up. Loosely fit the new bottom mounting bolt. Honda's own procedure calls for a new bolt here, not the old one.
12. Raise the axle beam with the jack until the car's rear barely lifts off the jack stands, then torque the bottom mounting bolt to 40 lbf-ft (54 N-m).
13. Fit the top rubber mount, washer, and a new self-locking nut onto the shock shaft. Honda's own procedure calls for a new nut here too.

14. Hold the shaft still with the hex wrench and tighten the top nut. We could not confirm the exact torque number for this nut on this pass; check Appendix C or your new shock's instructions for it.
15. Reinstall the access panel, the wire guide grommets, and both rear wheels.
16. Lower the car and torque the wheel nuts to 80 lbf-ft (108 N-m) in a crisscross pattern.

Front sway bar end links (both sides)

Never work under a car held up only by a jack; always use jack stands. Always wear eye protection, especially when you use penetrating oil or a cutting tool on rusted fasteners. Never use a cutting wheel near brake lines, fuel lines, or wiring; check what is nearby before you cut a stuck fastener.

1. Raise the front of the car and support it on jack stands, with both front wheels off the ground.
2. Spray the end-link nuts with penetrating oil and let it sit for a few minutes.
3. Hold the end-link stud still with a hex key while loosening the nut with a wrench.
4. If the stud spins instead of the nut, grip the stud's shoulder with vice grips.
5. If a nut is fully seized, cut it off carefully with a cutting wheel rather than forcing it loose.
6. Remove the old end link from both its top connection and its bottom connection to the sway bar.
7. Fit the new end link, and start both nuts by hand.
8. Check the new part's instructions for the suspension position it wants during tightening: unloaded, with the wheel off the ground, or loaded to normal ride height.
9. Position the suspension the way the instructions call for, then tighten both nuts snug.
10. Repeat steps 2 through 9 on the other side.

11. Lower the car and check that both end links move freely as the suspension travels.

Rear stabilizer bushing

Never work under a car held up only by a jack; always use jack stands. Always cover nearby electrical connectors before using silicone spray, since silicone residue can interfere with switches and sensors.

1. Raise the rear of the car and support it on jack stands.
2. Remove the old stabilizer bushing from the axle beam.
3. Apply silicone spray to the new bushing to ease its installation.
4. Line up the new bushing's end face with the center of its mounting hole, and press it into the axle beam.
5. Make sure the stabilizer bar seats fully into the bushing's groove, and does not stick out past the end of the axle beam.

What can go wrong

Skipping the alignment check after a strut job can lead to uneven tire wear later. Reusing a worn or cracked top-mount bearing instead of replacing it along with the strut is a common shortcut that causes new noises. On the rear shocks, not supporting the axle beam with a floor jack before removing the bottom bolt can let the beam drop suddenly. Reusing the old self-locking top nut or bottom bolt, instead of the new ones Honda's procedure calls for, is another common mistake. On the sway bar end links, forcing a seized nut without penetrating oil first can snap the stud. Tightening the links with the suspension hanging free, instead of at normal ride height, can pre-load the rubber wrong. On the rear bushing, getting silicone spray on nearby electrical connectors can cause switch problems later. Finally, do not confuse a noisy or loose end link with a problem in the sway bar's own mounting bushings on the subframe. That bigger issue means the full bar has to come out, which is the shop-recommended job, not the end-link swap in this chapter.

Quick check

1. Why does this book send the front strut job to a shop but not the rear shock job?
2. What is a spring compressor for, and why is it dangerous if used the wrong way?
3. Name two parts Honda's own procedure says to replace with new ones every time during a rear shock job, rather than reusing the old ones.
4. What is the difference between replacing a sway bar end link and replacing the sway bar itself?
5. Why should you check silicone spray placement carefully during the rear stabilizer bushing job?

CHAPTER 42

Cooling system: thermostat, radiator, water pump

What you will learn

- What the thermostat, radiator, and water pump each do to keep your engine cool
- How to replace the thermostat
- How to replace the radiator
- How to replace the water pump, and why it is easier on this car than on many others
- The safety rules for working on a hot cooling system, and what can go wrong

The short version

Your engine's cooling system moves coolant, a special liquid, through the engine to carry heat away. The thermostat controls when that coolant starts flowing to the radiator. The radiator gets rid of the heat. The water pump pushes the coolant around the whole loop. All three jobs need the engine fully cool first. A hot system can spray scalding coolant when opened.

The full explanation

The thermostat is a small valve in the cooling system. It stays shut while the engine is cold, so the engine warms up faster. Once the engine reaches its normal operating temperature, the thermostat opens, and coolant can flow to the radiator. On your Fit, the thermostat sits inside a housing that Honda's own service information calls the "water outlet."

The radiator is a large, flat, finned unit at the front of the car. It takes hot coolant from the engine and lets outside air pass through it, cooling the coolant before it flows back to the engine. If your Fit has an automatic transmission, the radiator also carries a set of hose connections that cool the automatic transmission fluid. You need a radiator that matches your transmission type.

The water pump is the part that physically pushes coolant through the whole system. On many cars, the water pump is driven by the timing chain or timing belt deep inside the engine, which makes it a hard job. On your Fit, the water pump pulley is driven by the same external drive belt the Maintenance Minder already checks. You can reach it from the right front wheel well. That makes the water pump a meaningfully easier job on this car than on many others.

For all three jobs, check Appendix B for the correct coolant type and the amount your Fit needs after a refill. Coolant is toxic. Keep spills away from pets and children, since it can taste sweet to animals.

Step by step

Replace the thermostat

Always let the engine cool completely before you open the cooling system; a hot system can spray scalding coolant. Never open the radiator cap or the thermostat housing while the engine is warm or running. Always wear eye protection when you drain or refill coolant. Never let coolant sit on your skin for long, and never leave a spill where a pet or child could reach it. Never run the engine until you have refilled the system and bled the air out of it.

1. Make sure the engine is completely cool before you start.
2. Remove the air cleaner assembly to reach the thermostat housing area.
3. Drain the engine coolant into a clean drain pan.
4. Disconnect the upper radiator hose and the water bypass hose from the housing.

5. Unplug the coolant temperature sensor's connector. Then remove the housing from the cylinder head.
6. Lift the old thermostat out of the housing, and note which side faces the engine.
7. Remove all old gasket material from the housing's mating surfaces, its bolts, and the bolt holes, then clean and dry every surface.
8. Fit a new sealing ring to the coolant temperature sensor, and set the sensor aside for now.
9. Apply Honda liquid gasket, part number 08717-0004, 08718-0003, 08718-0004, or 08718-0009, to the housing's mating surface, following the bead pattern on the tube.
10. Install the housing quickly. Some Honda liquid gaskets start to skin over in as little as 4 to 5 minutes.
11. Plug the coolant temperature sensor's connector back in. Then reconnect the upper radiator hose and the water bypass hose.
12. Reinstall the air cleaner assembly.
13. Wait at least 30 minutes before you refill the system with coolant.
14. Do not run the engine for at least 3 hours after you install the housing, so the gasket can cure.
15. Refill the radiator with the correct coolant and bleed the air out, following your car's own fill procedure.
16. Clean up any spilled coolant right away.
17. Start the engine, let it reach normal operating temperature, and check for leaks around the housing.

Replace the radiator

Always let the engine cool completely before you drain coolant or disconnect hoses. Never open the radiator cap while the engine is warm or running. Always wear eye protection when draining or refilling coolant. Never work

under a car held up only by a jack; always use jack stands. Always keep your hands clear of the electric radiator fan; it can switch on even with the engine off, if the coolant is still hot.

1. Make sure the engine is completely cool before you start.
2. Raise the car and support it on jack stands, if you need to reach the lower hose and drain plug.
3. Drain the engine coolant into a clean drain pan.
4. Unplug the coolant temperature sensor connector near the radiator. Unclip its harness clamp.
5. If your car has air conditioning, unplug the A/C compressor clutch connector. Move its harness clamps out of the way.
6. Disconnect the lower radiator hose.
7. On an automatic transmission car, disconnect the automatic transmission fluid cooler hoses, and plug both the hose and the line to keep dirt out.
8. Lower the car if it is raised, and remove the coolant reservoir.
9. Unplug the radiator fan motor connector, and the air conditioning condenser fan motor connector if equipped. Unclip their harness clamps.
10. Disconnect the upper radiator hose. Then remove the radiator's upper mounting brackets.
11. Lift the radiator up and out. If your car has air conditioning, remove the condenser fan shroud first.
12. Remove the fan shroud, radiator cap, coolant temperature sensor, and drain plug from the old radiator. Reuse the ones that are still good, or transfer them by your new radiator's instructions.
13. Fit new O-rings and assemble the fan shroud, cap, sensor, and drain plug onto the new radiator.
14. Set the new radiator in place, making sure its lower rubber cushions seat firmly.

15. Reinstall the upper mounting brackets, then reconnect the upper radiator hose.
16. Plug in the fan motor connectors and their harness clamps. Then reinstall the coolant reservoir.
17. Raise the car again if needed, and reconnect the lower radiator hose.
18. On an automatic, remove the plugs and reconnect the transmission fluid cooler hoses.
19. Plug in the coolant temperature sensor connector, and the air conditioning compressor clutch connector if equipped. Reattach their harness clamps.
20. Lower the car. Refill the radiator with the correct coolant, and bleed the air out per your fill procedure.
21. Clean up any spilled coolant right away.
22. Start the engine, let it reach normal operating temperature, and check every connection for leaks.

Replace the water pump

Always let the engine cool completely before you drain coolant. Never open the cooling system while the engine is warm or running. Always wear eye protection when draining or refilling coolant. Never work under a car held up only by a jack; always use jack stands.

1. Make sure the engine is completely cool before you start.
2. Drain the engine coolant into a clean drain pan.
3. Raise the car and support it on jack stands, then remove the right front wheel.
4. Remove the splash shield, the plastic panel that covers the underside of the wheel well.
5. Loosen the water pump pulley's mounting bolts, then remove the drive belt.
6. Remove the water pump pulley's mounting bolts and the pulley itself.

7. Remove the water pump's mounting bolts, and lift the old pump away from the engine block.
8. Clean the old gasket material off the mounting surface.
9. Fit the new pump with a new gasket or O-ring, following its instructions.
10. Install the pump's mounting bolts and torque them per the new pump's instructions. We could not confirm an exact Honda torque number for this pass; check Appendix C for any later confirmation.
11. Install the water pump pulley, and loosely fit its mounting bolts.
12. Install the drive belt, then tighten the pulley's mounting bolts fully.
13. Install the splash shield and the right front wheel, then lower the car.
14. Refill the radiator with the correct coolant, and bleed the air out per your fill procedure.
15. Clean up any spilled coolant right away.
16. Start the engine, let it reach normal operating temperature, and check around the pump for leaks.

What can go wrong

Opening any part of the cooling system while the engine is still warm can spray scalding coolant on your skin. On the thermostat job, missing the liquid gasket's short install window lets it skin over before the housing is bolted down. Installing the thermostat backwards can keep the engine from warming up correctly, so check the old part's orientation before you throw it away. On the radiator job, forgetting to plug a disconnected transmission fluid cooler line lets dirt into the transmission's cooling circuit. Skipping the bleed-air step afterward can cause a false overheating reading on a later drive. On the water pump, do not mistake a small, normal amount of weeping at the pump's bleed holes for proof the old pump had failed. A healthy pump can show a trace of this. Judge a water pump by whether its pulley turns freely or by an active leak, not by a trace of weeping alone.

Quick check

1. What job does each of the thermostat, the radiator, and the water pump do in the cooling system?
2. Why must the engine be completely cool before you start any of these three jobs?
3. What makes the water pump on this car easier to reach than on many other cars?
4. What extra part must you disconnect and plug on an automatic transmission car during a radiator swap?
5. Why should you not run the engine right after installing a new thermostat housing?

CHAPTER 43

Starter, alternator, ignition switch

What you will learn

- What the starter, the alternator, and the ignition switch each do
- How to replace the starter yourself, step by step
- What a shop will do to replace the alternator, and why this book sends that job to a shop
- What a shop will do to replace the ignition switch, and why it is really two different jobs
- What can go wrong on each job, including battery-cable safety

The short version

The starter spins your engine to life when you turn the key or press the start button. The alternator charges your battery and powers your electrical system while the engine runs. The ignition switch is the part the key turns to control power to the starter and the rest of the car. The starter is a doable weekend job. The alternator and the ignition switch both go to a shop in this book, for two different reasons explained below.

The full explanation

On your Fit, the starter sits low on the driver's side of the engine. This book could not find a plain factory guide for removing and installing the starter on this car. The steps below come from real owners of this exact car instead. Reaching the starter's top bolt is widely reported as the hardest part of the job.

The alternator on your Fit sits in a hard spot. You must remove the intake manifold, the part that feeds air into the engine, just to reach it. That makes this a bigger job than a simple bolt-off swap. It also raises the chance of a vacuum leak or a bad gasket if it is not done with care. This book sends the alternator job to a shop for that reason.

The ignition switch is really two different jobs hiding under one name. One job replaces only the electrical switch behind the steering column covers. That is a moderate job for a confident home mechanic with basic tools. The other job replaces the whole lock cylinder, the mechanical part the key barrel turns inside. That job needs drilling out security shear bolts. These are bolts made to snap off flush once tightened, so a thief cannot unbolt the lock. On many Hondas, the job also needs reprogramming the immobilizer system to accept a new key. The immobilizer is the security system that only lets the engine start with a key it recognizes. Chapter 7 covers its dashboard light. A "hard to turn" key can be caused by either problem. Choosing the wrong repair wastes time and money. This book sends the whole ignition switch job to a shop so it can be diagnosed correctly first.

This car has no ignition-switch recall on record. This is wear-and-tear repair work, not a safety recall.

Step by step

Replace the starter

Always take off the negative battery terminal first, then the positive terminal, before you start this job. Always wear eye protection. Never let a wrench touch a positive battery or starter terminal and any grounded metal at the same time. This can cause a dangerous short and a burn. Never work under a car held up only by a jack. Always use jack stands.

1. Disconnect the negative battery terminal, then the positive terminal.

2. Remove the two 10 mm bolts holding the wiring harness to the intake manifold.
3. Remove the 10 mm bolt for the dipstick tube bracket.
4. Remove the 10 mm bolt on the starter that holds a nearby hose bracket.
5. Detach the radiator hose holder from the starter bracket.
6. Disconnect the electrical connectors from the starter's "S" (signal) and "B" (battery) terminals.
7. Remove the starter's mounting bolts. Use a long extension and a swivel joint on the hard-to-reach top bolt.
8. Guide the starter out from below the car, or through the engine bay, whichever your access allows.
9. Compare the new starter to the old one to confirm it matches before you install it.
10. Guide the new starter into place, and install its mounting bolts. We could not confirm an exact Honda torque number for this pass; follow your new starter's own instructions.
11. Reconnect the electrical connectors to the "S" and "B" terminals.
12. Reattach the radiator hose holder to the starter bracket.
13. Reinstall the dipstick tube bracket bolt and the two wiring harness bolts at the intake manifold.
14. Reconnect the positive battery terminal, then the negative terminal.
15. Start the car to confirm the new starter cranks the engine smoothly.

What the shop will do: alternator

Even if a shop does this job, know what to ask about first. Always disconnect the battery before this job. The alternator's output terminal stays electrically live even with the key off. Always wear eye protection if you help with any part of this job. Never skip a post-repair check of the charging voltage before you call the job done.

- Disconnect the battery, following the proper terminal disconnection order.
- Remove the drive belt, then remove the intake manifold to reach the alternator.
- Disconnect the alternator's electrical connector and its positive cable, then unbolt and remove the alternator.
- Install the new alternator, and torque its mounting bolts to 17 lbf-ft (24 N-m).
- Reconnect the alternator's connector and positive cable. Make sure the cable's ring terminal has its crimped side facing away from the alternator, and torque the cable's terminal bolt to 6 lbf-ft (8 N-m).
- Reinstall the intake manifold, using a new gasket if the old one is damaged, reinstall the drive belt, and reconnect the battery.
- Start the engine and confirm the charging system light goes out, and the battery charges normally.

What the shop will do: ignition switch

Even if a shop does this job, know what to ask about first. Always disconnect the battery before any steering-column work. The airbag system's wiring runs close to the column. Never let anyone bypass the immobilizer system instead

of properly reprogramming a new key. A bypassed immobilizer is a theft and safety risk. Always confirm with the shop that the airbag wiring near the column was not disturbed. If it was moved, confirm it was put back correctly.

- Diagnose whether the problem is the electrical switch, meaning the key turns but the car does not respond correctly. Or it could be the mechanical lock cylinder, meaning the key will not turn or turns roughly.
- If it is the electrical switch only: remove the plastic covers around the steering column, unplug and unbolt the old switch, and install the new one.
- If it is the lock cylinder or the full ignition lock housing: remove the steering column covers, and drill out the security shear bolts holding the housing. Remove the old housing and install a new assembly. This part of the job is reported at around 2 hours of labor.
- If a new lock cylinder or an immobilizer-linked part is installed, program the new key or keys to the immobilizer system so the engine will start.
- Test that the key turns smoothly through LOCK, ACCESSORY, ON, and START, and that the car starts reliably afterward.

What can go wrong

On the starter job, connecting the battery terminals in the wrong order is the most common mistake. Always take the negative terminal off first, and put it back on last. Forcing the starter's top bolt without a swivel joint can strip the bolt or its bracket. Mixing up the "S" and "B" connectors can keep the new starter from working right. On the alternator job, hooking up the positive cable with its ring terminal facing the wrong way is a common error. Skipping a post-job charging-voltage check is another. Either mistake can hide a problem until the battery goes flat later. On the ignition switch, paying for the wrong sub-repair wastes money. This can mean fixing the electrical switch instead of the lock cylinder, or the other way around, if the shop has not found the real problem first. Forgetting to reprogram the immobilizer for a new key can leave the car unable to start, even after a correct mechanical repair.

Quick check

1. What does each of the starter, the alternator, and the ignition switch do?
2. Why does reaching the starter's top mounting bolt matter so much on this job?
3. Why does this book send the alternator job to a shop instead of giving full DIY steps?
4. What are the two different jobs that can both be called "replacing the ignition switch"?
5. Why is disconnecting the battery in the correct order important before working on any of these three parts?

Exhaust and O2 sensors

What you will learn

- What an oxygen sensor does, and why your Fit has two of them
- How to replace an upstream or downstream oxygen sensor
- How to replace a rusted or broken section of exhaust pipe or muffler
- The warning signs that a job is too big for your driveway
- What to check before you call either repair finished

The short version

Your Fit's exhaust system carries hot gas away from the engine. Two small sensors along the way check the oxygen level in that gas. You can replace either sensor, or a rusted section of pipe or muffler, using basic tools. Always let the exhaust cool first, since these parts stay dangerously hot long after you shut off the engine.

The full explanation

What an oxygen sensor does

An oxygen sensor is a small part that threads into your exhaust pipe. It measures how much oxygen is left in the exhaust gas. Your Fit's engine has one exhaust bank, and two of these sensors sit along it.

The upstream sensor sits closer to the engine. It is also called the primary sensor, or the air-fuel-ratio sensor. It threads into the exhaust manifold, the metal pipe that collects gas straight from the engine's cylinders. The

downstream sensor sits farther back, on the catalytic converter. A catalytic converter is a part that cleans pollutants out of the exhaust gas. The downstream sensor watches how well that converter is working.

These two sensors connect to check-engine codes covered in Chapter 35. Code P0135 flags a problem with the upstream sensor's heater. Code P0141 flags the same kind of problem on the downstream sensor. A third code, P0420, means the computer thinks your catalytic converter is not cleaning the exhaust well enough. That code can also show up because of a bad upstream sensor or an engine misfire. Always rule those two causes out before you blame the converter itself.

Reaching the downstream sensor's electrical connector is harder than reaching the upstream one. On this car, you likely need to remove the center console from inside the car first. That is why this book rates the upstream sensor job easier than the downstream one, even though the two sensors look similar.

Exhaust pipes and mufflers

Your exhaust system is a chain of pipe sections and a muffler. These parts join at flanges, which are flat joints held together with bolts. Rubber hangers hold the whole system up under the car. Rust and road salt attack these flanges and hangers over the years. Eventually a section rusts through, or a joint seizes shut. This book could not find a Fit-specific, bolt-by-bolt guide for this exact car's exhaust system. The steps below use standard technique for this type of car and system. They are not a Honda-confirmed script written for the 2008 Fit specifically.

Step by step

Replacing an oxygen (O₂) sensor

What you need: - An oxygen-sensor socket (a socket with a slot cut for the sensor's wire), or a wrench that fits - Penetrating oil, for a rusted sensor - Anti-seize compound made for oxygen sensors — a paste that keeps metal parts

from rusting together - Jack and 2 jack stands, if you work from underneath - Basic interior trim tools, only if you replace the downstream sensor and need to remove the center console - A new sensor made for the exact position you are replacing. Upstream and downstream sensors are different parts. This book could not confirm a part number this pass. Ask a Honda parts counter, or check Appendix F.

Safety warnings: - Always let the exhaust system cool completely before you touch it. It stays dangerously hot long after the engine shuts off. - Always wear eye protection, especially when you use penetrating oil on a rusted sensor. - Do not work under a car held up only by a jack. Use jack stands. - Do not overtighten a new sensor. Overtightening can crack the sensor body.

Steps: 1. Make sure the exhaust system is completely cool before you start. 2. Decide whether you are replacing the upstream sensor or the downstream sensor. 3. Raise the car and support it on jack stands, if you need to work from underneath. 4. Disconnect the sensor's electrical connector. For the downstream sensor, you may need to remove the center console first. 5. Spray penetrating oil on the sensor's threads if it looks rusted. Let it sit a few minutes. 6. Use your oxygen-sensor socket or wrench to loosen and remove the old sensor. 7. Compare the new sensor's connector and threads to the old one, to confirm they match. 8. Thread the new sensor in by hand first, so you do not cross the threads. 9. Tighten it with the socket or wrench until snug. Do not overtighten it. 10. Reconnect the sensor's electrical connector. 11. Reinstall the center console, if you removed it for the downstream sensor. 12. Lower the car, if you raised it. 13. If you have a scan tool, clear any stored check-engine codes tied to the sensor. Then take a test drive.

How you know it worked: The check-engine light stays off for the code you were chasing, after a normal test drive. If you have a scan tool, the sensor's reading changes as the engine runs, instead of staying flat.

Replacing an exhaust section or muffler

What you need: - Jack and 2 jack stands - Penetrating oil - A wrench and socket set for the flange bolts or clamps - A cutting tool, such as a rotary tool or a hacksaw, for a seized joint - A rubber mallet, to help free a rusted flange - New gaskets (thin seals that stop exhaust leaks) and new self-locking nuts (nuts with a built-in insert that resists loosening on its own) for every flange joint you open. Honda's own guidance for this car calls for fresh ones, not reused ones. - New rubber hangers, for any that are cracked or torn - The replacement pipe, muffler, or section itself. This book could not confirm a part number this pass. Check with a Honda parts counter, a muffler shop, or Appendix F.

Safety warnings: - Always let the exhaust system cool completely before you touch it. - Always wear eye protection, especially when you use penetrating oil or a cutting tool on rusted fasteners. - Do not work under a car held up only by a jack. Use jack stands. - Do not cut near the fuel tank, fuel lines, or brake lines. Know what sits above and around a spot before you cut it. - Always support a heavy section, like the muffler, before you loosen its last hanger. It should not swing down and hit you or the ground.

Steps: 1. Make sure the exhaust system is completely cool before you start. 2. Raise the car and support it on jack stands. 3. Spray penetrating oil on every fastener and hanger rubber you plan to disturb. Let it sit. 4. Support the section you are removing with a jack, a strap, or a helper, before you loosen its last fastener. 5. Separate the flange joint at each end of the section. Use a rubber mallet if it is rusted tight. 6. If a clamp or bolt is fully seized, cut it carefully rather than force it and damage the pipe around it. 7. Remove the old section, working it free of its rubber hangers. 8. Compare the new section to the old one, to confirm it matches, before you install it. 9. Fit new rubber hangers if the old ones are cracked or torn. 10. Position the new section and hang it loosely on its rubber mounts before you tighten anything. 11. Fit new gaskets at each flange joint you opened. 12. Install new self-locking nuts at those

flange joints. 13. Tighten all the flange fasteners, with the system hanging naturally, not twisted or under strain. 14. Lower the car and start the engine. Listen and look for any exhaust leak or rattle.

How you know it worked: There is no hissing, ticking, or rattling from the exhaust while the engine idles. There is no visible leak at any joint. The exhaust hangs evenly, without touching the body or the ground.

What can go wrong

Oxygen sensor mistakes

- Confusing the upstream and downstream sensor, and ordering the wrong part.
- Overtightening the new sensor and cracking its body.
- Forcing a rusted sensor without penetrating oil, and snapping it off inside the exhaust manifold or converter.
- Not clearing the old fault code. The check-engine light can stay on even after a correct repair, until enough clean drive cycles pass.

Exhaust and muffler mistakes

- Reusing old gaskets or nuts at a flange joint instead of new ones, which can cause a leak later.
- Fully tightening one flange before the whole section is loosely hung, which can twist and stress the joints.
- Cutting a stuck fastener without checking what sits directly behind or above it first.
- Not supporting a heavy section before you remove its last hanger.

Quick check

1. What does an oxygen sensor measure, and where are your Fit's two sensors located?
2. Why should you rule out a bad upstream sensor or a misfire before you blame the catalytic converter for a P0420 code?
3. Name two safety warnings you must follow before working on either sensor or the exhaust pipe.
4. What should you do differently with an exhaust flange joint you open, compared to reusing the old parts?
5. How do you know an oxygen sensor swap or an exhaust repair actually worked?

CHAPTER 45

Interior fixes: door handles, window regulators, seat repair

What you will learn

- How to replace a door handle
- How to replace a window regulator, the mechanism that raises and lowers your window
- Why one detail in the window regulator job is not confirmed for your exact car, and what to do about it
- How to repair a torn or damaged seat cover
- What can go wrong in each of these three jobs

The short version

You can replace a door handle, a window regulator, or a torn seat cover with basic tools and some patience. Each job means taking a door panel or a seat apart, so work carefully and keep track of every clip and bolt. For the window regulator, one detail below comes from a different Honda Fit than yours, and this book says so plainly instead of guessing.

The full explanation

Door handles

A door handle connects to a metal rod inside the door. Pulling the handle pulls the rod, which opens the latch. The steps below come from a real repair guide for this car's rear door outer handle. The front door handle uses a very similar design. The door panel comes off, the rod disconnects, and the handle pries

out with a trim tool. A trim tool is a flat plastic tool made to remove panels without scratching them. Treat the exact bolt count as confirmed for the rear door, and only reasonably similar, not confirmed, for the front.

Window regulators, and an honest gap in this book's research

A window regulator is the mechanism inside your door that raises and lowers the glass. It usually has a small electric motor attached.

Here is where this book has to be honest with you. The most detailed, bolt-by-bolt window regulator guide this project could find was written for a 2011 Honda Fit. That is a later, different-generation Fit than yours, built between 2009 and 2013. Your 2008 Fit is a different car under the skin, even though it shares a name and looks similar. This book will not treat a detail from that different car as a confirmed fact about yours.

Here is the good news. A real 2007 Fit owner reports replacing a window regulator in their own driveway. That confirms the job itself works on your exact generation of car. Only one detail stays unconfirmed: the exact number of bolts holding the regulator in the door. The 2011-Fit guide reports roughly two bolts near the top and three near the bottom. Nobody has confirmed that exact count on a 2008 Fit. Count your own car's bolts as you go, rather than assuming this number in advance.

Seat repair

A torn seat cover is a general repair, not something unique to your Fit. No car-specific guide exists for patching a Fit seat, because the technique is the same on almost any car. The steps below use standard upholstery technique — the skill of repairing fabric, vinyl, or leather covers.

Step by step

Replacing a door handle

What you need: - A door or trim panel removal tool kit (plastic pry tools), so you do not scratch the door or break plastic clips - A clip remover tool, for the rod fastener - A basic screwdriver and socket set - Protective tape, to protect the paint near the handle while you pry - A new door handle. Confirm front vs. rear and driver vs. passenger side before you order one. This book could not confirm a part number this pass. Ask a Honda parts counter, or check Appendix F. - A replacement plastic access-cover cap, in case the old one cracks or tears during removal

Safety warnings: - Always wear gloves. Sharp metal edges and clips inside the door can cut your hands. - Do not pry directly against painted door metal without protective tape. A slipped trim tool can scratch the paint. - Always raise the window glass fully before you work inside the door, so it cannot drop while the panel is off.

Steps: 1. Remove the door panel to reach the inside of the door. 2. Raise the window glass fully before you continue. 3. If the door has power locks, disconnect the power door lock actuator connector (the small motor that locks and unlocks the door) and detach its harness clip. 4. Remove the rear plug cap and the plastic access cover, keeping them intact if you can. 5. Remove the maintenance caps that cover the handle's fasteners. 6. Detach the rod fastener with a clip remover, then disconnect the outer handle rod from the handle. 7. Remove the handle's mounting bolts. 8. Apply protective tape to the door around the handle, to protect the paint. 9. Pry the outer handle out with your trim tool, to release its hook, then remove it. 10. Compare the new handle to the old one, to confirm it matches, before you install it. 11. Fit the new handle into its opening and press it in until the hook engages. 12. Reinstall the mounting bolts. 13. Reconnect the outer handle rod and refit its clip fastener. 14. Reinstall the maintenance caps, the plastic access cover, and the rear plug cap. 15. If your door has power locks, reconnect the power door lock actuator

connector and its harness clip. 16. Reinstall the door panel. 17. Test the handle from outside. If your door has power locks, test that function too, before you call the job done.

How you know it worked: The door opens smoothly from outside with normal effort. The handle does not feel loose or gritty. Any power lock function on that door still works correctly.

Replacing a window regulator

What you need: - A door or trim panel removal tool kit (plastic pry tools) - A 10 mm socket wrench or nut driver. The exact bolt size and count are not independently confirmed for your car — see the note above. Confirm yours before you buy hardware. - A basic screwdriver set - A new window regulator, with or without a motor, depending on what failed, for the exact door and side you are fixing. This book could not confirm a part number this pass. Ask a Honda parts counter, or check Appendix F.

Safety warnings: - Always wear gloves. Broken window glass and sharp door-frame edges can cut your hands. - Do not remove the window glass while it is only partly supported. Make sure it cannot slide down and shatter while the door panel is off. - Always disconnect the window regulator motor's electrical connector before you handle the regulator. This avoids an accidental pinch if the motor still has power.

Steps: 1. Remove the door panel and the plastic watershield behind it. A watershield is a plastic sheet that keeps water out of the car's interior. 2. Move the window glass until its mounting fasteners are accessible. 3. Remove the glass's mounting fasteners, and carefully pull the glass out through the window slot. Support it the whole time. 4. Remove the bolts holding the old regulator to the door. A guide written for a later, different Fit generation reports roughly two bolts near the top and three near the bottom. Count your own car's bolts as you go, rather than assuming this number. 5. Disconnect the electrical connector for the window regulator motor. 6. Carry the old regulator out of the door. 7. Compare the new regulator to the old one, to confirm it

matches, before you install it. 8. Place the new regulator inside the door and reconnect the motor's electrical connector. 9. Line up the regulator to its original position. 10. Reinstall the mounting bolts, tightening them evenly. 11. Reinstall the window glass into its mounting fasteners. 12. Reinstall the watershield and the door panel. 13. Test the window's full up-and-down travel, and its auto up or down function if it has one, before you call the job done.

How you know it worked: The window moves smoothly through its full range, with no grinding, binding, or unusual noise. It stops properly at both the fully up and fully down positions.

Repairing a torn or damaged seat

What you need: - A ratchet or wrench for the seat mounting bolts, if you need to remove the seat for better access - Scissors, a heat gun on a low setting or a hair dryer, and a stiff-bristle brush, for vinyl or leather repair - A sewing kit with heavy-duty upholstery thread, for a small fabric tear - Clean rags and isopropyl alcohol, to clean the area before any repair - A vinyl or leather repair kit (color-matched filler and grain paper), for a torn vinyl or leather seat - Matching seat fabric, or a patch kit, for a torn cloth seat - Foam replacement or extra padding, if the seat's foam is also crushed or torn underneath the cover

Safety warnings: - Always disconnect the seat's electrical connectors before you remove a seat from the car. This includes wiring for a power seat, a seat heater, or a seat-belt pretensioner (a device that tightens the belt suddenly in a crash). - Do not cut into or disturb any wiring or part you cannot identify near the seat frame. Some seats route side-impact airbag wiring through them. - Always let a repaired area cure fully, per your repair kit's instructions, before you sit on it.

Steps: 1. Clean the damaged area with a rag and isopropyl alcohol, and let it dry. 2. Decide whether you can do the repair with the seat in the car, or need to remove the seat for full access. 3. If you remove the seat, disconnect its electrical connectors first, then remove its mounting bolts and lift it out with a helper. 4. For a small fabric tear, stitch the edges together from the underside

where you can, using heavy-duty thread. 5. Apply a fabric patch behind the tear for strength. 6. For a vinyl or leather tear, trim any loose or frayed edges. 7. Follow your repair kit's instructions to fill the tear with the color-matched compound. 8. Use the heat gun on a low setting, held well away from the surface, if your kit calls for heat-setting. 9. This step helps the vinyl or leather filler cure. 10. Press the grain paper from your kit onto the wet filler, to recreate the surface texture. 11. Peel it away once it cures, per the kit's instructions. 12. Blend the repaired area's color if needed, using your kit's color-matching guidance. 13. If you removed the seat, reinstall it and reconnect its electrical connectors and mounting bolts. 14. Let the repair cure fully before normal use, per your kit's stated time.

How you know it worked: The repaired area holds together under normal sitting and getting in and out. It does not visibly gap or peel once fully cured. Any seat electrical function still works normally if you removed the seat.

What can go wrong

Door handle mistakes

- Prying without protective tape, and scratching the paint around the handle.
- Losing or breaking a small plastic access cover or maintenance cap during removal.
- Forgetting to raise the window fully before you remove the door panel, risking a dropped or damaged glass.
- Not fully engaging the new handle's hook before you tighten the mounting bolts, leaving it loose.

Window regulator mistakes

- Letting the glass slide down and shatter while it is unsupported, with the door panel off.

- Reconnecting the motor while it still has power, risking a pinched finger if it activates unexpectedly.
- Not reinstalling the watershield correctly, letting water into the door and, eventually, into the car's interior.
- Assuming the bolt count from a different Fit generation's guide is correct, without checking your own door first.

Seat repair mistakes

- Skipping the cleaning step, so the repair compound does not bond well to a dirty or oily surface.
- Using too much heat too close to the vinyl, scorching it instead of curing the filler.
- Disturbing seat wiring without labeling the connectors first, then reconnecting something incorrectly.
- Sitting on the repair before it has fully cured.

Quick check

1. What does a window regulator do, and where is it located?
2. Why can't this book confirm the exact bolt count for your window regulator job?
3. What evidence confirms the window regulator job itself is doable on a 2008 Fit, even without a confirmed bolt count?
4. Name two safety steps you should follow before removing a car seat.
5. What tells you a door handle repair or a seat repair actually worked?

CHAPTER 46

Body: dents, rust, paint touch-up, headlight restoration

What you will learn

- How to pull out a small, shallow dent
- How to treat surface rust before it becomes a hole
- How to touch up a paint chip or scratch
- How to restore cloudy or yellowed headlight lenses
- When a body repair is too big for your driveway, and needs a shop instead

The short version

You can fix a small dent, treat early rust, touch up a paint chip, and clear up hazy headlights with basic tools and some patience. Rust is the one job with a hard line. Treat it early yourself, but once rust eats all the way through the metal, that spot needs a shop, not a driveway fix. Headlight haze is different. It comes from sunlight, not rust, and almost anyone can fix it in under an hour.

The full explanation

Dents, rust, and paint: one small job that grows

This book covers three small cosmetic body jobs together, because you often do them one after another on the same spot. First a dent, then any rust the dent exposed, then a paint touch-up to finish it.

Chapter 38 already flagged a specific rust spot on first-generation Fits like yours. It shows up where the rear quarter panel meets the rear bumper cover. It also shows up along a seam where a plastic cover hides the join between the

quarter panel, the rocker panel, and the underside of the car. That pocket collects road grit and moisture, and it is the first place rust tends to appear. This is more of a concern if you live somewhere that salts the roads in winter.

Surface rust is rust that sits only on top of the metal, not through it. Treating it is a beginner job: you clean it, treat it with a rust converter (a chemical that changes rust into a paintable surface), then prime and paint over it. A hole in the metal is a different story. Fixing a rust hole means cutting out the bad metal and welding in a new patch, which is not a beginner task. Covering a rust hole with filler alone might look fine for a while, but it does not stop the rust spreading underneath. A repair shop estimate for early surface or bubble rust treatment on this kind of spot typically runs \$150 to \$900. A real hole patch or panel replacement typically runs \$800 to \$2,500.

For paint, your touch-up color has to match your exact car. Your doorjamb certification label carries your car's paint code. This book could not independently confirm the exact paint code numbers for every 2008 Fit color. Treat any code you find online as a lead to check, not a confirmed fact. Always check it against your own doorjamb label first.

Headlight haze

Modern headlight covers are plastic, not glass, and plastic clouds over time from years of sunlight. Ultraviolet light, the part of sunlight that also causes sunburn, slowly breaks down the plastic's clear outer layer. This happens to nearly every car with plastic headlight covers, not only the Fit. So this job uses a general technique, rather than a Fit-specific guide.

Step by step

Fixing a small dent

What you need: - A plunger, or a dedicated dent-pulling tool, for a small, shallow dent in a flat panel

Safety warnings: - Always wear gloves and eye protection when you sand, wire-brush, or use chemical rust products, later in this same job. - Do not sand through to bare metal over a large area without protecting it with primer right away. Bare metal left exposed starts rusting again quickly.

Steps: 1. Clean the dented area with soap and water, then dry it. 2. For a shallow dent on a flat panel, use a plunger with some water on the rim, to create suction. 3. Pull outward in short, firm pulls. 4. For a dent a plunger cannot pull, a dedicated dent-puller kit with glue tabs and a slide hammer can help. 5. This kind of kit can fix many minor dents without damaging the paint. 6. Check the repaired area's paint for cracking. A cracked clear coat over a pulled dent may still need a touch-up.

Treating surface rust

What you need: - 220 to 600 grit sandpaper, to feather paint edges (sand them into a smooth, gradual taper instead of a hard ridge) and prep the rust spot - A wire brush, or a drill-mounted rust-removal wheel - Masking tape and paper, to protect the surrounding paint - A rust converter or inhibitor product, to treat the rust before you paint over it - Rust-inhibiting primer

Safety warnings: - Always wear gloves and eye protection when you sand, wire-brush, or use a chemical rust converter. - Do not use rust converter or primer in a closed, unventilated space. Work outdoors, or with good airflow. - Always let each layer — converter, primer, paint, and clear coat — dry or cure fully before you add the next one.

Steps: 1. Clean the rusted area with soap and water, then dry it fully. 2. Sand or wire-brush the rust down to a sound, unrusted surface. Feather the edges into the good paint around it. 3. Wipe away all the dust, then apply rust converter or inhibitor, following its instructions. 4. Once the converter has cured, apply rust-inhibiting primer over the treated area. 5. If the rust has already progressed to a hole in the metal, stop here. 6. Treat a rust hole as a job for a shop instead. See the note above.

Touching up paint

What you need: - Masking tape and paper - Fine sandpaper, 600 grit or finer - A small foam applicator or fine artist's brush - Touch-up paint matched to your car's paint code, found on your doorjamb certification label. Verify any code you find online against your own label first. - Clear coat, matched to your paint type, for the final layer over the touch-up paint

Safety warnings: - Always wear gloves and eye protection during sanding. - Do not skip letting each coat dry fully before you add the next one.

Steps: 1. Mask off the area around the chip, scratch, or repaired dent or rust spot with tape and paper. 2. If there is a rough edge, lightly sand the immediate area with fine sandpaper, then wipe it clean. 3. Apply touch-up paint in thin coats with a fine brush or applicator. Let each coat dry before the next, rather than applying one thick coat. 4. Once the paint has cured per its label, apply clear coat in thin coats over the repaired area. 5. Remove the masking tape and paper once everything has cured. 6. Wax the area once the repair is fully cured, blending it into the surrounding panel.

How you know it worked: The dent is flat enough that you cannot see it easily from a normal viewing distance. There is no more active rust bleeding through the paint at the treated spot. The touch-up paint is close enough in color and gloss that it does not stand out up close.

Restoring cloudy or yellowed headlights

What you need: - A headlight restoration kit. Most kits include sandpaper of several grits, a polishing compound, and applicator pads. - Masking tape, to protect the surrounding paint - A clean microfiber cloth - A spray bottle with water, to keep the sandpaper wet, if your kit calls for wet-sanding - A UV-protectant sealant, if your kit includes one

Safety warnings: - Do not attempt this job in direct, hot sunlight. The plastic, and any chemical products you use, can behave unpredictably in high heat. - Always mask off the paint around each headlight before you sand or polish, to avoid scratching it. - Do not use harsh solvents not meant for automotive plastic. They can cloud or crack the lens further.

Steps: 1. Wash both headlights with soap and water, then dry them. 2. Mask off the paint and trim around each headlight with tape. 3. Starting with your kit's coarsest sandpaper, wet-sand the lens in one direction, keeping the surface wet the whole time. 4. Rinse the lens and check your progress. The surface will look hazy white during sanding — this is normal and expected. 5. Move to the next-finer grit in your kit, and repeat the wet-sanding, rinsing between grits. 6. Continue through every grit your kit provides, finishing on the finest one. 7. Rinse and dry the lens fully. 8. Apply the polishing compound with an applicator pad, working it in small circles until the lens looks clear. 9. Wipe off any leftover polish residue with a clean microfiber cloth. 10. If your kit includes a UV-protectant sealant, apply it per the kit's instructions, to help slow future yellowing. 11. Remove the masking tape. 12. Repeat steps 2 through 11 on the other headlight.

How you know it worked: Both headlight lenses look clear instead of cloudy or yellowed. A nighttime check shows noticeably brighter, more evenly spread light on the road or a wall, compared to before.

What can go wrong

Dent, rust, and paint mistakes

- Painting directly over untreated rust. The rust keeps spreading invisibly under the new paint.
- Applying touch-up paint in one thick coat, which causes it to run or look lumpy.

- Skipping the clear coat step, leaving the touch-up looking dull or mismatched in gloss.
- Trying to patch a rust hole with filler alone, instead of cutting out the rusted metal and welding in new metal. Filler alone does not fix the underlying problem.

Headlight restoration mistakes

- Skipping a grit step in the sanding sequence, leaving fine scratches the polish cannot remove.
- Letting the sandpaper run dry, which can gouge the plastic instead of smoothing it.
- Forgetting to mask the surrounding paint, and scratching it with sandpaper or polish.
- Skipping the UV-protectant sealant step, if your kit includes one, and having the haze return sooner than expected.

Quick check

1. Where does Chapter 38 say rust tends to start first on a first-generation Fit like yours?
2. What is the difference between surface rust and a rust hole, and why does that difference matter for a DIY repair?
3. Where do you find your car's exact paint code, and why should you double-check any code you find online?
4. What causes headlight lenses to turn cloudy or yellow over time?
5. How do you know a headlight restoration job actually worked?

PART VIII

Souping It Up

Modding Philosophy — Goals, Budget, and the Law

What you will learn

- Why you should set a clear goal before you buy any part
- What federal law says about your car's emissions equipment
- What a CARB Executive Order (EO) is, and how to look one up
- How modifying your Fit can affect your insurance and your warranty
- What this book will never tell you to do

The short version

Before you spend money, decide what you want your Fit to do better, and set a budget you will actually keep. Federal law lets you modify your car, but it does not let you remove or defeat any part built to control exhaust emissions. Tell your insurance company about every modification you make, because an untold modification can leave you paying the full bill yourself.

The full explanation

Set a goal before you set a budget

Modding means changing your car from how the factory built it, usually to improve how it handles, stops, sounds, or looks. Before you buy a single part, decide what you actually want. A car built for **autocross** needs different parts than a car built to look sharp on the street. Autocross is a timed event held in a parking lot or on an airport lot. A car built for daily driving to school needs different parts than either one.

Write your goal down in one sentence. Then set a budget for the whole project, not only the first part. Mods add up fast. A **coilover** kit, an adjustable suspension upgrade covered fully in Chapter 50, is only useful if you can also afford an alignment afterward. Spend on baseline items first, covered in Chapter 48, before you spend on power or looks.

The law: what "tampering" means

The **Clean Air Act** is the federal law that sets emissions rules for every car sold in the United States, including your Fit. It makes two separate things illegal. First, it is illegal for anyone to manufacture, sell, or install a part built mainly to bypass, defeat, or disable a device that controls your car's exhaust emissions. That kind of part is called a **defeat device**, and this rule targets whoever makes, sells, or installs it, not only the driver.

Second, a separate part of the same law applies directly to you, the owner. It makes it illegal to knowingly remove or disable any emissions device on your own car after you bought it. This second rule is the one that applies to you if you do the work yourself.

Both rules apply everywhere in the United States, not only in states with strict smog laws. The federal government, through the EPA (Environmental Protection Agency), enforces them nationwide. **CARB**, the California Air Resources Board, adds its own state-level rules on top of the federal ones, mainly for cars titled or registered in California. Chapter 17 explains how to check your own state's inspection and emissions rules.

Here is the good news: EPA does not chase down every driver who installs an **aftermarket** intake or exhaust, meaning a part made by a company other than Honda. EPA uses what it calls "enforcement discretion," and it focuses on parts that actually raise emissions. EPA generally leaves a part alone if you can show a "reasonable basis" that it will not hurt emissions, proven one of three ways. The part can be identical in design and function to the one it

replaces. The modified car can still pass the same emissions tests it was certified with. Or the part can carry a CARB Executive Order that covers your exact vehicle.

The penalties for breaking these rules are real money, not a scare tactic. As of the date on EPA's own fact sheet, a manufacturer or dealer can be fined up to \$48,192 per violating vehicle or part. Any other person, meaning you, can be fined up to \$4,819 per violating vehicle or part. Faking or disabling emissions-related monitoring equipment, like sensors your check-engine light relies on, is a separate criminal violation on top of that fine.

CARB Executive Orders: the exemption a shop can check

A CARB **Executive Order (EO)** is a certificate California issues for one specific aftermarket part, after CARB's own engineers confirm the part does not raise emissions. An EO exempts that exact part, on that exact vehicle, from California's own anti-tampering law. A smog-check station or a CARB inspector can look up the EO number to confirm it is real.

Here is the part every reader of this book must remember. One EO does not automatically cover every car the part might physically bolt onto. An EO is tied to a specific engine family and a specific model year. An EO written for the second-generation 2009-2014 Fit, chassis code GE8, does not automatically cover your 2007-2008 Fit, chassis code GD3, even if the part looks identical on the shelf. Always confirm the EO you are reading names your exact model year before you trust it. Chapter 50 walks through this exact year-confusion trap using real suspension parts, so you can see how it plays out with actual products.

You can look up a real EO yourself at CARB's public tool, **<https://ww2.arb.ca.gov/aftermarket-parts-lookup>**. It offers five search categories, and "Performance & Add-on Parts" is the one you want for an intake, header, or exhaust. That category page leads into CARB's aftermarket-parts search

tool. Outside descriptions say you can search it by EO number, part number, or manufacturer, though this book could not fully confirm every field of that tool works as described.

An EO also matters outside California. EPA itself lists "a CARB EO covering the same part" and vehicle as one of the three accepted ways to pass its own federal reasonable-basis test. A correctly matched EO is useful evidence anywhere in the country, not only in California.

Warranties and insurance: what new modders forget

EPA's own fact sheet states plainly that tampering "can void manufacturer warranties and insurance agreements." That single sentence connects the legal rules above to two very practical costs.

A standard car insurance policy is priced around your car's factory configuration. A **turbocharger** is an engine power upgrade, covered fully in Chapter 54. Any change not made at the factory, whether a suspension part, a turbocharger, custom paint, or a stereo system, counts as a **modification** for insurance purposes. Insurance companies generally will not automatically pay full value for modified parts. To get that coverage, you typically need an added policy **endorsement**, meaning an add-on to your existing policy, sometimes called "custom parts and equipment coverage." That coverage only works on top of comprehensive and collision coverage you already carry.

If you add parts and never tell your insurance company, the company may refuse to cover those parts. You then become responsible for repairing or replacing them yourself, even though you pay for insurance every month. Call your insurer before you install anything, not after something breaks.

Not every modification raises your rate. Performance and appearance parts, like a bigger engine output or expensive aftermarket wheels, usually raise cost. They raise the car's value, and they often cost more to repair than factory

parts. Safety and security additions, like an alarm, a backup camera, or parking sensors, can sometimes earn a discount. This is not guaranteed, though, since some of that added technology is itself expensive to repair.

What this book will never tell you to do

This book never tells you to skip a safety step, and it never tells you to remove or disable emissions equipment for more performance. When a mod could hurt you or someone else, this book says so before it tells you how to do it. If a part looks too good to be true, like a free power gain with no downside, check its real fitment and its real legal status before you buy it. Chapter 50 teaches you exactly how to do that check yourself.

Quick check

1. What is a "defeat device," and who can the federal defeat-device rule punish?
2. Name the three ways EPA says an aftermarket part can pass its "reasonable basis" test.
3. Why must you confirm a CARB EO names your exact model year before you trust it?
4. What can happen if you install a modification and never tell your insurance company?
5. Give one example of a modification that can sometimes lower your insurance cost, and explain why it does not always work that way.

Stage 0 — Get Your Baseline Right First

What you will learn

- What "Stage 0" means, and why it comes before any performance part
- Why fresh fluids matter before you add stress to your Fit
- Why good tires and good brakes matter more than any bolt-on part
- Why this book cannot hand you an alignment procedure, and what to do instead

The short version

Before you buy a single performance part, make sure your Fit's basic maintenance is already current. Stage 0 means fresh oil, tires in good shape, and brakes that already work correctly. Stage 0 costs less than most single mods, and it protects every mod you add after it.

The full explanation

What "Stage 0" means

Most modding guides number their steps: Stage 1, Stage 2, and so on, from mild to wild. **Stage 0** is the step before all of those. It means bringing your Fit's existing maintenance up to date before you spend a dollar on power, handling, or looks. A stiffer suspension on old, worn shocks will not behave the way a review video shows. A grippier tire on a car with a slipping alignment will wear out fast and unevenly.

This chapter is short by design. Every Stage 0 item already has a full procedure card written in an earlier chapter of this book. This chapter tells you which chapter to open and why that item matters before you mod, instead of repeating steps you can already find there.

Fresh fluids

The first Stage 0 item is fresh engine oil and a fresh filter. Old oil has already picked up dirt and lost some of its ability to protect your engine. A car that will work harder, longer, or hotter than normal deserves oil that is not already tired. Chapter 24 covers the full oil-change procedure, including the tools, the drain-bolt torque spec, and every safety warning. Open Chapter 24 and follow its steps rather than guessing at your own order.

Good tires

The second Stage 0 item is good tires: correct, even air pressure, and no uneven wear from a rotation you skipped. This is a check, not a rebuild. Chapter 26 already covers how to check and set your tire pressure, and how to rotate your tires so they wear evenly. Read Chapter 26 and confirm both are already current before you spend money on new wheels or tires, which Chapter 49 covers next.

Stage 0 is not about buying new tires yet. It is about confirming the tires you already have are properly maintained, so any new tire or wheel choice starts from a clean baseline instead of hiding an existing problem.

Good brakes

The third Stage 0 item is good brakes. This means pads or shoes and fluid that are already in solid condition before you add track driving or spirited street driving to the mix. Your Fit's front brakes use pads and rotors, and its rear brakes use shoes inside a drum, which is a different design with different parts and a different repair. Chapter 27 covers inspecting the front pads through the

wheel, replacing the front pads, replacing the rear shoes, and bleeding the brake fluid. Work through Chapter 27's checks before you plan any brake upgrade in Chapter 51.

Alignment: the one gap in this book

Here is where this book has to be honest with you instead of guessing. **Alignment** is the process of adjusting your wheels' angles, camber, caster, and toe, so your car drives straight, handles predictably, and wears its tires evenly. This book does not include a step-by-step alignment procedure. We could not confirm a home alignment procedure for the 2008 Fit in this book's research. Check your owner's manual, and plan on a shop visit instead.

There is a practical reason for this gap. Alignment normally needs an **alignment rack**, a specialized shop machine that measures your wheel angles precisely and then adjusts them. This is not a typical home-garage job the way an oil change or a brake-pad swap is. For Stage 0, the right move is to have a shop check your alignment, especially if the car has ever been in a curb strike or a minor accident. Once you get to Chapter 50, you will also want a fresh alignment any time you change springs, coilovers, or camber parts.

A simple Stage 0 checklist

- Fresh engine oil and filter, per Chapter 24.
- Correct, even tire pressure and a current tire rotation, per Chapter 26.
- Healthy front brake pads, healthy rear brake shoes, and fresh brake fluid, per Chapter 27.
- An alignment check at a shop, especially after any curb strike, accident, or suspension work.

Quick check

1. What does "Stage 0" mean, and why does it come before power or handling mods?

2. Which chapter should you open for the exact oil-change steps at Stage 0?
3. Name the two tire-related checks that count toward Stage 0, and which chapter covers them.
4. Why does "good brakes" mean checking two different repair types on your Fit, not only one?
5. Why does this book send you to a shop for alignment instead of giving you home steps?

CHAPTER 49

Tires and Wheels — Sizes, Offset, and Weight

What you will learn

- What the numbers in a tire size and a wheel offset actually mean
- Which wheel and tire sizes owners report fitting a GD3 without rubbing
- Why a bigger wheel is not automatically a better wheel
- What autocross class a stock Fit runs in, and how tire choice connects to it

The short version

Fitting a tire and wheel to your Fit depends on diameter, width, and offset working together, not any single number. Owners of first-generation Fits report that a 195/50R15 tire on a 15x6.5-to-7-inch wheel fits without rubbing. Bigger wheels and wider tires get progressively tighter, and they also add weight your engine has to turn, which can work against your handling goal instead of helping it.

The full explanation

Reading a tire size, and what "offset" means

A tire size like 195/50R15 has three parts. The 195 is the tire's width in millimeters. The 50 is the **aspect ratio**, the tire's sidewall height as a percentage of that width. The R15 means it is a radial tire built for a 15-inch wheel diameter.

Offset is a separate number, measured in millimeters, that describes how far the wheel's mounting face sits from its true centerline. A higher offset number pulls the tire in toward the engine and suspension. A lower offset number pushes the tire out toward the fender. Offset, tire width, and wheel diameter all work together to decide whether a tire **rubs**. A tire rubs when it touches the fender, a suspension part, or a body panel while turning or going over a bump.

Sizes owners report fitting without rubbing

Owners report that 195/50R15 on a 15x6.5-to-7-inch wheel, at +40 to +42mm offset, is the combination that fits a stock-height GD3 without rubbing. A 205/45R16 tire on a 16x7-inch wheel is described as more marginal. It only works at a minimum offset of +42mm, and some owners report rear rubbing, especially with passengers in the back seat.

Owners also report on a 205/50R15 tire, one size wider than the 195/50R15 baseline. The consensus from those reports is that a 205/50R15 fits without severe rubbing. It runs tighter than the baseline size, though, and it may need the fender lip rolled back or a lower ride height to stay fully clear at full steering lock.

One owner who autocrossed a GD3 for four to five seasons reports a staggered setup. Up front sit 225/45R15 tires on 15x7.5-inch wheels, and the rear runs narrower 205/50R15 tires. A **staggered** setup means the front and rear tires are different sizes on purpose. This particular combination is reported by only one source, so treat it as one experienced owner's setup, not a confirmed community consensus.

Weight: why bigger is not always better

A wheel and tire are part of your car's **unsprung weight**, the mass that your springs do not support, since it sits below them and moves with the road surface. Heavier unsprung weight generally makes a suspension react slower and feel less precise, even if the tire itself grips well. This is one reason the

autocross setup above pairs a wider front tire with a narrower, lighter rear wheel. The wider front tire adds grip up front, without adding weight to the rear too.

Every size owners report above stays inside the 15-inch wheel diameter family. This matters because your Fit's Sport trim already used a 15-inch factory wheel. Staying at 15 inches keeps your speedometer and anti-lock brake system reading closer to correct than jumping straight to a 16-inch or 17-inch wheel would.

Autocross tire sizes and classing

Autocross is a timed event where you drive one car at a time through a course marked with cones, usually in a parking lot. A stock, unmodified 2008 Fit runs in **H Street (HS)** class under SCCA's national Solo rules, since the rulebook's "Fit" entry does not carry a model-year restriction. Once you add aftermarket wheels, tires, and other Street Touring-eligible parts, your Fit typically moves into **Street Touring FWD (STF)** class instead. STF allows a wider range of parts than H Street does.

Combining every source found for this chapter, two sizes come up most often for GD3 autocross: 195/50R15 and 205/50R15, both usually run on 15x7-to-7.5-inch wheels. Chapter 58 covers autocross classing and event prep in full detail; this chapter only connects your tire choice to that bigger picture.

A buying tip before you spend

Every size above is community-reported, meaning real owners shared it online, not a guarantee written by Honda or a tire manufacturer. Your own car's ride height, past accident history, and even factory tolerances can shift what actually clears. Before you buy a full set, ask the seller or a shop whether they have fit that exact size to a GD3 before. Check clearance on your own car at full steering lock before you commit.

Quick check

1. What do the three parts of a tire size like 195/50R15 tell you?
2. What is "offset," and why does changing it affect whether a tire rubs on the fender?
3. Which size do owners report as the safest baseline fit for a GD3, and which size is described as marginal?
4. Why can a bigger, heavier wheel work against your handling goal instead of helping it?
5. What SCCA class does a stock 2008 Fit run in, and what class do many modified GD3s move into?

Suspension — Springs, Coilovers, Sway Bars, and Alignment

What you will learn

- What a spring, a coilover, and a sway bar each actually do
- 9 real suspension products confirmed to fit your GD3, and how each was checked
- 7 real products that looked right but were rejected, and exactly why
- How to run that same fitment check yourself before you ever buy a part
- Street and track alignment targets, and why they are not the same numbers

The short version

A lowering spring drops your ride height and changes how stiff your suspension feels, and a coilover is a fully adjustable spring-and-shock unit. A sway bar reduces how much your car leans in a corner. This chapter lists 9 real products confirmed, from each product's own listing text, to fit your 2007-2008 GD3 Fit. It also lists 7 look-alike products this book rejected, because they were actually built for the different 2009-2014 GE8 Fit. Learning to spot that difference yourself is one of the most useful skills a Fit modder can build.

The full explanation

What these parts actually do

A **spring** is the coiled metal part that holds your car's weight up and compresses when the wheel hits a bump. A shorter, stiffer aftermarket spring usually lowers your ride height and reduces how much the car leans and dives. A **coilover** combines a spring around a shock absorber, called a **damper**, into one adjustable unit. Most coilovers let you set ride height independent of how hard the spring is preloaded, and many also let you adjust damping, meaning how quickly the shock compresses and rebounds.

A **sway bar**, also called an anti-roll bar, is a metal bar that connects the left and right suspension. It twists slightly in a corner to resist body lean. A thicker sway bar generally means a stiffer, less-leaning car.

Three more terms matter for the alignment section later in this chapter. **Camber** is how much a wheel tilts inward or outward at the top, viewed from the front of the car, measured in degrees. **Caster** is how much the steering axis tilts, viewed from the side. **Toe** is whether the wheels point slightly toward each other or away from each other, viewed from above.

The single most important rule in this chapter

Chapter 47 warned you that a part built for the 2009-2014 Fit, chassis code GE8, does not automatically fit your 2007-2008 Fit, chassis code GD3, even when it looks identical. This chapter puts that warning into practice with real parts, real listings, and real mistakes this book's own research caught.

Here is how you run the same check yourself. Read the product listing's own text, not only its headline. Look for the words "GD3," "2007-2008," or "07-08," or for two separate fitment lines, one for 2007 and one for 2008. A generic claim like "fits Honda Fit" is not enough on its own, since that same wording can quietly cover the GE8 too. Watch for three specific traps. A trusted brand can sell a near-identical part for the GE8 under a similar name. A

part number can look right while sitting on a different generation's page. And a Japanese-market production-date window can fail to line up with the US 2007-2008 model years, even when the chassis code looks correct.

9 real products confirmed for the GD3

Every product below states, in its own listing text, that it fits the 2007-2008 Fit or the GD3 chassis code by name. Confirm this yourself again at the time you buy, since listings change.

Product	Type	GD3 confirmation in the listing	Reported spec
BC Racing BR Series	Coilover	"2007-2008 Honda Fit FWD (GD3)," two separate year lines	5 kg/mm front and rear, ride height and damping adjustable
TEIN BASIC (DS-B36-LUAS2)	Coilover	"07+ GD3 Honda Fit FF"	Twin-tube, 22mm piston front and rear, ride height adjustable
Skunk2 Pro-C (541-05-6990)	Coilover	"06-08 Honda Fit" / "2006-2008 Honda Fit"	12-way adjustable, 8 kg/mm front, 6 kg/mm progressive rear, race-oriented
Fortune Auto 510 Series	Coilover	"Honda Fit (GD3) USDM 2007-2008," repeated on the page	Customer-selectable spring rates, 4k-16k kg/mm range
F2 Suspension Type 2	Coilover	"Honda Fit GD (2006-2008)"	Fitment confirmed; specific spring rates not confirmed this pass
TEIN S.Tech	Lowering spring	"2007-2008 Honda Fit FWD (GD3)," two separate year lines	Front 2.8 kg/mm, rear 6 kg/mm, drop about 1.8 in front / 1.2 in rear
H&R Sport Spring (51830)	Lowering spring	Fitment table names 2007 and 2008 Fit Base, and Sport, separately	Drop about 1.6 in front / 1.3 in rear
Eibach Pro-Kit (4069.140)	Lowering spring	Listing's own fitment list names only "2007 Honda Fit" and "2008 Honda Fit," no 2009+ line	Drop 1.6 in front / 1.3 in rear
Fortune Auto 500 Series	Coilover	"Honda Fit (GD3) USDM 2007-2008," same title pattern as the 510 Series	Entry-level version of the 510 Series above

That is 9 real spring and coilover products, across 6 different brands, each confirmed by its own listing text to fit the GD3. Notice that the H&R spring's own fitment table names your base trim directly, not only the Sport trim.

7 rejected products — the year-confusion trap, caught in the act

While researching the list above, this book also found 7 similar-looking products that failed the same check. Every one is recorded here so you can recognize the pattern, not only avoid these specific parts.

#	Category	Product	Why it was rejected
1	Lowering spring	Eibach Pro-Kit, part 4081.140	Listing says "2009-2013" / "09-12 Honda Fit," a GE8-only part with a different part number than the correct GD3 part above
2	Coilover	TEIN Street Basis Z	Listing title and URL say "2009-2014 Honda Fit FWD (GE8)"
3	Coilover	TEIN Street Advance Z	Listing tags itself "(GE8)," even though "2008" also appears in the title text
4	Camber bolt	Eibach Pro-Alignment 5.81260K	Listing says "Honda Fit (2009-2014)" (GE8)
5	Brake pad	EBC Yellowstuff	Listing says "Honda Fit (09-15)" (GE8/GK); no listing naming "2007-2008" or "GD3" was found
6	Camber bolt	SPC EZCam 81260, on a "09-14" page	Same physical part number as a confirmed GD3 fact below, but this specific page is tagged for the GE8, so it is not cited as a GD3 source
7	Coilover	TEIN FLEX Z	Listing does say "GD3," but its own Japanese production-date window, 2003.10 to 2007.09, ends be-

#	Category	Product	Why it was rejected
			fore the US 2008 model year begins

Row 3 is worth reading twice. A listing can say "2008" in its title and still not be for your car. That happens when the same listing also tags itself with a trim or region code, like "RS," that Honda never sold as a US 2008 base Fit. Row 7 shows an even sneakier version of the same trap: the chassis code is right, but the production-date window quietly excludes your car's actual build year. Read every date and code on a listing, not only the ones that jump out at you.

This chapter researched three categories: springs and coilovers, camber bolts, and brake pads. Across all three, 7 total candidates were found and rejected for not actually fitting the GD3, judged by their own listing text.

The same part, three different listings

One more case study drives the lesson home. The retailer Heeltote Automotive sells the identical SPC 81260 camber bolt on three separate product pages. One page is tagged "07-08 Honda Fit" for your GD3. Another is tagged "09-14 Honda Fit" for the GE8, and a third is tagged "15-20 Honda Fit" for the newer GK generation. A buyer who searches only by the bare part number, 81260, could click into the wrong page and never notice. Always confirm you are on the generation-matched page, not only the correct part number.

Sway bars for the rear

The Progress Technology 22mm rear sway bar, part 62.1060, is listed for "07-08 Honda Fit," and its own body copy adds "Compatible with GD Honda Fit." The Ultra Racing 16mm rear sway bar is listed for "Honda Fit (GD) 2001-2008," a thinner alternative to the Progress bar above.

One listing deserves a caution instead of a recommendation. A Whiteline sway bar link is sold under a web address that says "rear sway bar link." The same page's own title and body text, read directly, say "Front Sway Bar Link"

throughout instead. Because the page disagrees with itself about which end of the car this part is for, this book does not use it as a confirmed rear-bar part. As a bonus, a Cusco 24mm bar listed for "Honda Fit-Jazz GD1/GD3" was also found this pass. It is explicitly a front sway bar, not a rear one, so skip it if you are specifically shopping for a rear bar.

Camber and alignment adjustment parts

The SPC EZCam 14mm camber bolt, part 81260, gives roughly plus or minus 1.75 degrees of front camber adjustment when both bolts on a strut are replaced. It installs by swapping out one or both of your strut's existing pinch bolts for this eccentric, off-center bolt. Buy it from the page tagged "07-08 Honda Fit," not the "09-14" or "15-20" pages selling the identical part number.

Honed Developments, a shop focused specifically on the GD3, sells a strut-top part as part of its published GD3 suspension recipe. This part increases both front caster and front camber beyond what a camber bolt alone provides, though no numeric adjustment figure for it was confirmed this pass. Your Fit's rear suspension is a **torsion-beam axle**, a solid rear beam with no built-in camber or toe adjustment from the factory. Honed Developments also sells a rear alignment shim kit, listed for "GD & GK Jazz/Fit," that adds camber and toe adjustment where none exists from the factory.

Alignment specs: street versus track

These two alignment targets come from different sources and different goals, so this book labels each one clearly rather than blending them into a single number.

One GD3-focused suspension recipe publishes a **track or performance setup**. Up front: camber -3.5 to -4.5 degrees, caster +2 to +2.5 degrees, and toe 0 degrees. In the rear: camber -2 to -2.5 degrees, and toe +2 to 0 (in). This target came from a single source, so treat it as a strong starting point for a dedicated track car, not a universal rule.

GD3 owners report a gentler **street or daily-driving setup** instead: roughly -1.25 to -2.0 degrees of front camber, which balances comfortable driving with acceptable tire wear. Toe is kept as close to 0 as the front and rear allow. Owners report that rotating your tires regularly helps manage the extra wear that negative camber causes on a street-driven car. Notice how much less aggressive this is than the track numbers above; a track alignment on a daily driver chews through tires fast.

One number this book could not confirm is your Fit's original factory alignment spec. We could not confirm this for the 2008 Fit's factory alignment; check your owner's manual or a dealer. A source found this pass gave factory numbers alongside a tire fitment and pressure spec. That pressure spec conflicts with this book's own verified owner's-manual data elsewhere in this book. This book's research could not confirm which generation or market that source actually describes. Rather than guess, this book flags the conflict and sends you to your own door-jamb sticker and factory manual instead.

Quick check

1. Name the three alignment angles, camber, caster, and toe, and say what each one describes.
2. What is the difference between a lowering spring and a coilover?
3. List two ways a suspension listing can quietly reveal it is actually built for the GE8, not your GD3.
4. Why does the same SPC camber bolt appear on three separate product pages, and why does that matter to you as a buyer?
5. How do the track alignment numbers in this chapter differ from the street numbers, and why does that difference exist?

CHAPTER 51

Brakes for Mods — Pads, Fluid, Lines, Rotors, and Cooling

What you will learn

- Two brake pad compounds built for street use and track use, both confirmed for your GD3
- How to read a brake fluid boiling point, and why a bigger DOT number is not always the winner
- Three stainless steel brake line kits confirmed for your GD3
- Two upgrade areas, rotors and brake cooling, that this chapter could not confirm real GD3 products for

The short version

A track-oriented brake pad, a high-boiling-point brake fluid, and stainless steel brake lines all work together to give you a firmer, more consistent pedal under repeated hard braking. This chapter presents real manufacturer data on two pad compounds and two fluid types, all confirmed to fit or apply to your GD3. Fresh, correct fluid matters as much as any pad choice, since fluid that boils turns your brake pedal soft right when you need it most.

The full explanation

Two pad compounds, one confirmed fitment list

Hawk Performance sells two pad compounds worth knowing for a modified Fit. Hawk's own fitment database lists its HPS ("High Performance Street") front pad, part HB418F.646, for both the 2007 and 2008 Honda Fit, Base and Sport, by name. HPS is a street-oriented compound: lower noise and dust for daily driving, positioned below Hawk's track pad for outright heat capacity.

Hawk's HP+ ("High Performance Plus") front pad, part HB418N.646, appears on the identical fitment list, again naming both 2007 and 2008, Base and Sport, by name. HP+ is Hawk's higher-temperature track and autocross compound, generally preferred for repeated hard sessions, at the cost of more noise, more dust, and a longer bed-in period on the street. Both parts are confirmed for your GD3 by Hawk's own manufacturer fitment list naming your car and your base trim directly, not by a generic aftermarket claim.

Brake fluid: DOT numbers and boiling points

Brake fluid transfers the force from your foot on the pedal to the pads or shoes at each wheel, through a sealed hydraulic system. Brake fluid absorbs a small amount of water over time, and hot brakes can heat that fluid enough to boil it. Fluid that boils turns to gas bubbles, which compress instead of pushing the pads, and your pedal goes soft or sinks to the floor.

A **DOT rating** is a US safety standard that sets a minimum boiling point for a brake fluid. **Dry boiling point** is the boiling point of fresh, unused fluid. **Wet boiling point** is the boiling point after the fluid has absorbed a set amount of water, closer to how fluid behaves after real-world use.

Fluid	Dry boiling point	Wet boiling point
FMVSS 116 DOT 4, regulatory minimum	446°F (230°C)	311°F (155°C)
FMVSS 116 DOT 5.1 (non-silicone), regulatory minimum	500°F (260°C)	356°F (180°C)
Motul DOT 5.1, actual product data	522°F (272°C)	365°F (185°C)
Motul RBF 600 Factory Line, a racing DOT 4 fluid	594°F (312°C)	399°F (204°C)

Read that table carefully, because it teaches an important lesson. The racing-grade DOT 4 fluid at the bottom has a higher boiling point than the plain DOT 5.1 fluid above it. This is true even though 5.1 is technically the "higher" number. A DOT rating is a minimum standard, not a guarantee that a higher-numbered fluid always beats a lower-numbered one. When you shop for a track-capable fluid, compare actual boiling points on the data sheet, not only the DOT number printed on the bottle.

This chapter could not confirm which brake fluid your Fit's owner's manual specifies from the factory. We could not confirm this specific fact in this chapter's own research; Chapter 27 covers your factory-specified fluid type in full. Use this chapter's boiling-point table to compare upgrade options once you know your factory baseline from Chapter 27.

Stainless steel brake lines

A **stainless steel brake line** replaces the factory rubber brake hose with a braided steel outer layer over a sealed inner tube. Rubber hoses can swell slightly under hard, repeated braking, which can make the pedal feel soft even with good pads and fluid; a stainless line resists that swelling.

Three stainless line kits were confirmed for the GD3 this pass. A StopTech kit, part 950.40017, is listed as a front kit for "Honda Fit 2007-2008." The listing found this pass marks it discontinued at that retailer, so treat it as a lead to

chase down, not a guaranteed current purchase. A Fastline Performance kit carries the chassis code directly in its own part number, FLP-BLKIT-GD3, in addition to a "2007-08 Honda Fit" title. A Techna-Fit kit, part HN-2600, is listed for "2007-2008 Honda Fit" by two independent retailers.

Notice that the Fastline part number is the strongest fitment confirmation of the three. The manufacturer built the chassis code into the part number itself, not only into marketing text on a web page.

Rotors and brake cooling: an honest gap

Every product in this chapter so far came with real, checked fitment text. Rotors and dedicated brake-cooling parts, like brake ducts that route outside air to a hot rotor, did not turn up a confirmed GD3-specific product in this chapter's own research. We could not confirm specific rotor or brake-cooling products for the 2008 Fit in this pass. Check current listings yourself, and apply the same GD3-versus-GE8 check from Chapter 50 to anything you find.

A general point still applies even without a specific product to name. Any rotor you buy should be sized and drilled for your Fit's specific caliper and hub. Any brake-cooling duct should be checked for physical clearance against your specific bumper and fender, the same way a wheel or tire needs a clearance check in Chapter 49.

Quick check

1. Which two Hawk pad compounds are confirmed for your GD3, and which one suits daily street driving better?
2. What is the difference between a dry boiling point and a wet boiling point?
3. Why can a DOT 4 fluid sometimes have a higher boiling point than a DOT 5.1 fluid?
4. Name the three confirmed stainless brake line kits, and which one has the strongest fitment proof, and why.

5. Why did this chapter not name a specific rotor or brake-cooling product for your GD3?

Intake, Header, and Exhaust

What you will learn

- What an intake, a header, and an exhaust system each do on your Fit's engine.
- The real dyno numbers researchers found for each part on this exact car.
- Why a manufacturer's own claim is weaker proof than two separate independent tests.
- Why no clean, isolated dyno number exists for an exhaust system alone.
- How these parts perform together, instead of one at a time.

The short version

An intake, a header, and an exhaust each free up a small amount of power on your L15A1 engine . Every real dyno number found for these parts on this car is modest, in the single digits . No clean, isolated dyno test for an exhaust system alone was found for this car .

The full explanation

Words you need first

A dynamometer, or dyno, is a machine that measures how much power an engine makes . Some dynos measure power at the wheels . That number is written as whp . Wheel numbers are usually lower than crankshaft numbers . Power gets lost on the way to the wheels . A cold air intake replaces the factory air box and tubing . It feeds air into your engine . A header replaces the factory exhaust manifold . That is the metal piece that collects exhaust gas

right off the engine . A cat-back exhaust replaces the pipe and muffler after the catalytic converter . The catalytic converter is the part that cleans exhaust gas . An axle-back exhaust replaces only the section behind the rear axle . It is a smaller piece than a full cat-back . A test pipe, sometimes called a catless downpipe, is a straight pipe . It replaces the catalytic converter itself .

Intake gains

Researchers found dyno numbers for three intake products on the L15A1 . Injen's SP1511 cold air intake is built specifically for the 2007-2008 Fit . Injen advertises a gain of up to 7 hp, plus up to 8 lb-ft of extra torque . That same figure appears on a second retailer's page . Both pages trace back to the same original Injen claim . They are not two separately run tests . This book reports the number as Injen's own claim, not as two independent tests .

A different, independent video test used a Dynapack dyno on a car identified as a GD3 . An HPS Performance cold air intake gained a reported 5.9 whp at 5,685 rpm . It also gained 7.6 lb-ft of torque at 3,655 rpm . An HPS Performance short-ram intake pulls air from inside the engine bay instead of outside . It gained a reported 7.1 whp at 5,645 rpm and 6.4 lb-ft at 5,590 rpm . Three different intake products, tested two different ways, all landed in the same narrow range . Every reported gain is roughly 5 to 8 hp, and roughly 6 to 8 lb-ft . No source found claims a double-digit horsepower gain from an intake alone .

Header gains

Only one number-based dyno claim for a header alone was found . It came from a test on an otherwise completely stock GD3 Fit . A fitfreak.net forum thread reported a gain of "5+HP" . Owners report that figure; it was not independently re-tested by this research . That single number sits below this book's normal bar of two separate sources . Treat it as a rough hint, not a settled fact . A second header claim of "+9 hp" turned up during research . It named a header built for the 2009-2014 Fit . That is a different car, with a

different engine, so it does not count here . Header products built specifically for the 2007-2008 GD3 Fit are still sold today . No published dyno number is attached to that current product's own listing .

Exhaust gains, and the honest gap

Stop and remember Chapter 47 before you read this section. A test pipe removes your car's catalytic converter, the part that cleans exhaust gas. Removing it is exactly the kind of change the Clean Air Act calls a defeat device. Federal law makes that illegal on a car driven on public roads, with real fines attached. A test pipe belongs on a car built only for closed-course track or competition use. It does not belong on a car you also drive to school or work, unless a specific part carries a CARB EO that covers your exact car.

Community discussion describes a test pipe alone as "usually" adding about 3 to 5 whp . One owner reported a header, a test pipe, and a cat-back exhaust installed together . That combination raised the car from 92 whp to 100 whp . That is an 8 whp gain for all three parts combined, not the test pipe alone .

Here is the honest gap in this research. No clean, isolated dyno number for a cat-back or axle-back exhaust alone was found for this exact car . Every full-exhaust dyno result found was reported together with a header, a test pipe, or an engine tune . There is no way to know what the exhaust piece alone contributed . Rather than guess at a number, this book leaves it unconfirmed . Real cat-back and axle-back exhaust products built for the 2007-2008 Fit do exist today . A'PEXi and HKS both sell one . Neither brand's own product page publishes a specific dyno figure . Both use only general marketing language instead .

What happens when you combine all of them

Two community reports show what these parts can do together . Neither report isolates any single part's own share of the gain . One 2008 Fit owner started with a stock rating near 107 hp . He reported reaching about 115 whp . He got there by adding an intake, a header, a catless downpipe, a full exhaust, and an

engine tune together . A second build reported 115.46 whp from a header, a downpipe, an axle-back muffler, and a tune . A third build, with an intake, a header, and a downpipe, reached 101 whp with a different tune . These are real reports from real owners . Treat them as "several parts together," not proof of what any one part adds by itself .

Putting it together

An intake, a header, and an exhaust are each real, modest upgrades on the L15A1 . None of them is a magic power boost . Every isolated number found for these parts is small, in the single digits . The header claim rests on one source, not two, so treat it carefully . No isolated exhaust-only number exists at all . This book will not invent one . If you want a real, noticeable gain, expect it to come from combining several parts, plus a tune, not any single part alone .

Quick check

1. What is the difference between horsepower measured at the wheels and at the crankshaft?
2. Why does this book treat the Injen intake's "7 hp" claim as one piece of evidence, even though it appears on two web pages?
3. What single number could this book not confirm for the exhaust system, and why not?
4. Why was a "+9 hp" header claim left out of this chapter?
5. If a reported combo build gains power from an intake, a header, and a tune together, what can you NOT conclude from that number?

Engine Management

What you will learn

- What an ECU is, and what "tuning" or "reflashing" it actually means.
- The single most surprising fact in this whole book about aftermarket support: which Fit generation Hondadata actually supports.
- What Hondadata sells for your exact car, and what it costs.
- What does NOT exist for your car, so you do not waste money chasing it.
- Why a header dyno number "for a Honda Fit" you find online might not even be for your car.

The short version

A company called Hondadata sells a full engine-tuning system built specifically for your 2007-2008 Fit, right now, today . Hondadata has never sold anything like it for the newer 2009-2014 Fit, which is a different car with a different engine . If you assumed the newer Fit has better tuning support than yours, that assumption is backwards .

The full explanation

Words you need first

An ECU, short for Engine Control Unit, is the small computer that runs your engine . It decides things like fuel amount and spark timing . "Tuning" an ECU means changing those settings, usually to make more power or run differently . A "reflash" rewrites the software inside the factory ECU itself, through the car's OBD-II port . OBD-II is the standard data port under your

dashboard, used on all cars built after 1996 . A reflash tool does not need to physically cut any wires . A "piggyback" is a separate small box that sits between the ECU and its sensors . It changes some signals without touching the factory ECU's own software . A "standalone" system fully replaces the factory ECU with a new one . CARB is the California Air Resources Board, the state agency that sets emissions rules . A CARB EO, or Executive Order, is a specific approval number from that agency . It lets an aftermarket part be sold as legal for street use in every state, not only outside California .

The headline finding

Here is the single most important finding in this chapter, so read it twice . Hondata's FlashPro system is a real, full-reflash tuning product . Hondata's own product page lists it for "Vehicle: Fit, Year: 2007-2008, Market: US/Canada" . That is your exact car . This is not a guess or an inference . It is a direct statement from the manufacturer's own current catalog .

Now for the surprising part . A direct check of Hondata's own product catalog, filtered specifically to the Fit, shows only two products . Both of them are for the 2007-2008 car, your car . Zero products exist in that same catalog for the 2009-2014 Fit . Hondata's own customer forum has a thread from a 2009-2014 Fit owner asking "Still nothing for the L15A7?" . The L15A7 is the engine in the 2009-2014 Fit . A reader might expect the newer car to have more aftermarket support, since it is newer . For Hondata specifically, the exact opposite is true . Never assume the newer Fit has better tuning support without checking for yourself .

What Hondata sells for your car

Hondata sells two different FlashPro versions for the 2007-2008 Fit . The full version, product code FP-FIT-US, costs \$795 . It unlocks complete custom tuning and full datalogging . Datalogging means the system records engine data over time, so a tuner can study it later . It comes with 20 hours of on-board datalogging memory . The second version, product code FP-FIT-US-

CARB, costs \$595 . It offers datalogging plus a limited menu of preset tuning options, not fully open custom tuning . Hondadata states this CARB version carries CARB EO number D-742-1, which keeps it legal for street use in every state, including California .

Here is the real tradeoff . The \$595 CARB version keeps you street-legal everywhere, but limits you to Hondadata's own preset calibrations . The \$795 full version unlocks real custom tuning instead . That kind of tuning is what you need to get the most out of a header or other bolt-ons . Hondadata does not represent the full version as CARB-legal for street use, though . Choose based on whether legality or full tuning matters more to you .

Both versions connect through your car's OBD-II port . Neither one requires cutting into your factory ECU . Hondadata's own specification page lists the exact factory ECU part numbers this system supports . That range is RME-A11 through RME-A89 for US cars, and RME-C11 through RME-C89 for Canadian cars . An independent retailer's own product page reproduces that same exact part-number range, word for word . That second, separate confirmation makes this an unusually solid fact .

What does NOT exist for your car

Hondadata also makes older products called K-Pro and s300 . Neither one applies to any Honda Fit, in either generation . K-Pro is built only for Honda's K-series engines, used in cars like the Civic Si and Acura RSX . Your Fit does not have a K-series engine . The s300 line is built for much older, 1992-1999 OBD1-era Hondas . Your Fit is a newer OBD2 car . Very old-school Honda ECU-chipping tools, such as Uberdata, Neptune, or Crome, are also built only for those same 1992-1995 OBD1 Hondas . None of them can work on your car's ECU .

A plug-and-play standalone system, like an AEM Infinity harness, is a different kind of product than FlashPro . AEM sells this kind of harness for some older Honda engines . No dedicated, named harness for the Fit or the

L15A1 was found in this research . That does not prove one does not exist somewhere . It only means none was found . This is recorded as unverified, not as a confirmed "no" .

A real piggyback option, with real limits

At least one owner has used a piggyback controller, an Apexi VAFC, on a GD3 Fit . A piggyback, remember, sits between the ECU and its sensors rather than replacing the ECU . Owners report that the VAFC could adjust the air-fuel mixture . Owners also report that trying to use it to affect VTEC engagement triggered the check engine light . VTEC is Honda's variable valve-timing system . This is a single community report, so treat the specific limitation as reported, not fully confirmed .

A trap this research caught

While researching this chapter, a claim about a "+9 hp" header surfaced online . Tracing it back showed the header itself was built for the "2009-2012 Honda Fit," the newer car . The forum discussion about it also sat in a section for that newer car, not yours . This is exactly the kind of mix-up this book is built to catch . A number you find online for "a Honda Fit" might actually be for the newer, different car, not the one you own . Always check the model year before trusting a number .

Putting it together

Real, current, full-reflash ECU tuning exists for your exact car, sold today by a mainstream company, for \$595 or \$795 . This is not a niche or discontinued method . Hondata sells the same kind of product family for many other current Honda and Acura models . The newer 2009-2014 Fit, by contrast, has no equivalent Hondata product at all . Do not assume the newer car has better tuning support only because it is newer . For this brand and this exact question, your 2007-2008 car is the one Hondata actually built for .

Quick check

1. What is the difference between an ECU reflash and a piggyback controller?
2. What does Hondadata's own current product catalog show for the 2009-2014 Fit's engine, the L15A7?
3. What is the price difference between Hondadata's two FlashPro versions for your car, and what do you give up with the cheaper one?
4. Why does neither Hondadata's K-Pro nor its s300 line apply to any Honda Fit?
5. Why was a "+9 hp" header claim rejected from this chapter, even though it mentioned "a Honda Fit"?

Power Adders

What you will learn

- What a turbocharger and a supercharger are, and how they differ from a header or an exhaust.
- The three real, GD3-specific turbo or supercharger kits researchers found.
- Why all three of those kits are hard, or impossible, to buy new today.
- What real reliability problems owners and builders report.
- What else you would likely need to change if you add real boost: clutch, fuel system, and more.

The short version

Exactly three real, brand-named turbo or supercharger kits were ever built specifically for your GD3 Fit's L15A1 engine . All three are hard to buy new today; two are fully discontinued, and the third is small-batch and often out of stock . Adding real boost is reported to also demand a stronger clutch and a bigger fuel system, not only the turbo or supercharger kit itself .

The full explanation

Words you need first

A turbocharger is a small turbine that uses your engine's own exhaust gas to force extra air into the engine . A supercharger does the same job, but it is spun by a belt connected to the engine, not by exhaust gas . Both are ways to force more air in, so the engine can burn more fuel and make more power . This forced air is called boost, and it is measured in psi, or pounds per square

inch . An intercooler is a small radiator that cools the forced air down before it enters the engine . Cooler air is denser and safer for the engine . A CARB EO is a California approval number that lets a part be sold as legal for street use in every state . A clutch is the part that connects your engine to the transmission on a manual car . A flywheel is a heavy spinning disc the clutch grabs onto .

The three real kits

KraftWerks made a bolt-on supercharger kit built specifically for the manual-transmission 2007-2008 US Fit . It was also sold under the related brand name Jackson Racing . The kit used a Rotrex C15-60 supercharger, plus Hondata engine management and an intercooler, included in the box . It was never built for the automatic-transmission Fit . Neither KraftWerks' own current shop nor Jackson Racing's own current shop lists a Fit kit today . This kit is discontinued . Used kits can sometimes be found on sites like eBay instead .

Two different power figures were reported for this kit, from two different points in its life . An early write-up, at the kit's first announcement, reported 121 whp for the "base kit" . A later write-up described a "High-Boost" version, reporting 164 whp and 132 lb-ft of torque at 10 psi of boost . Both numbers are reported by enthusiast media, not independently re-tested by this research .

This kit's retail price was reported at \$4,495 plus shipping, from a single source . Reliability reports mostly cover the same Rotrex supercharger design used across other Honda models, not the Fit specifically . Owners of those other cars report belt and tensioner problems . One report describes a tensioner bolt snapping and the belt coming off during highway driving . Another describes belt edge wear from rubbing on the pulley . KraftWerks itself states it added an automatic belt tensioner specifically for the Fit kit, and reports no belt issues on it . That is the manufacturer's own claim, though, not an independent owner report .

HKS made a bolt-on turbo kit named specifically for "2007 and 2008 model year Honda Fits" with the 1.5L engine . This is confirmed directly by California's own CARB Executive Order record, D-186-27 . The kit used a

Garrett T25 turbocharger, a front-mount intercooler, and HKS's own F-Con IS engine management, which plugs into the OBD-II port . Maximum boost was set at 6.0 psi per that same official record . HKS described the kit as "legal in all 50 states," based on that CARB approval . HKS reported roughly a 30 percent gain in peak horsepower . That kit is also reported discontinued today, with only used or old-stock units left to find .

Unequal Fabrication, a small aftermarket parts maker, currently lists a "Honda Fit GD3 (07-08) Turbo Kit" at \$2,300 . The listing describes a bolt-on kit that keeps your air conditioning, heat, and fog lights working . The seller states this kit can reach "170hp on the stock internals," given a healthy clutch . That is a manufacturer claim, not an independent test . Separate tuning software, such as the Hondata FlashPro covered in Chapter 53, is required and not included in the price . The listing states the kit is "FOR OFF ROAD USE" . At the time of this research, the kit was shown as "Out of Stock," built to order with a variable wait time .

A warning about generic listings

Unbranded turbo kits also show up for sale on sites like eBay, labeled for the "GD3" or "L15A" . One such listing claims a huge 485 horsepower gain . That number is far above every other figure found in this research and is not backed up anywhere . This book does not use that number, and you should not trust it either . A GD3-specific community thread, from experienced builders, gives a more realistic picture instead . They describe the stock L15A1 as able to hold about 8 psi of boost safely . They suggest a realistic target of roughly 160 to 200 whp on stock internals . One named builder blew his engine at 220 whp on a custom turbo setup . He then rebuilt it with a larger bore to reach about 260 whp . This is a single community thread, so treat it as a reported estimate, not a lab-tested fact .

One well-written article about a turbocharged Fit was found and read in full, then set aside . The car in that article used the L15A7 engine, which is the newer 2009-2014 Fit's engine, not yours . It was excluded from this chapter entirely, so no number from it appears anywhere above .

What else you would likely need

Owners report that the stock clutch can slip once real boost is added . Exedy sells Sport Clutch kits built specifically for the 2007-2008 Fit in two strength levels . The Stage 1 kit is rated for 128 lb-ft of torque to the wheels, and costs about \$454 . The Stage 2 kit is rated for 160 lb-ft . For a daily-driven, non-boosted car, though, community advice says you should not need to upgrade the stock clutch at all . Some race-style clutches are reported to feel "too grabby" for comfortable daily driving . Save the clutch upgrade for if, and only if, you actually add boost .

Fuel system upgrades are also reported as necessary once you add real power . For the KraftWerks supercharger kit specifically, owners report upgrading to larger 410cc fuel injectors at 10 psi of boost . Changing injector size always requires a matching ECU retune, to avoid running dangerously lean . For custom turbo builds, owners report that aftermarket fuel pumps, such as a Walbro or Deatschwerks pump, are commonly used instead of the stock pump .

No source found in this research shows the stock axles failing specifically because of added horsepower . This is left honestly unverified, rather than guessed at .

Putting it together

Three real turbo or supercharger kits were built for your exact car, and all three are hard to buy new today . Reported power gains for these kits range from about 121 whp up to about 260 whp on a heavily built engine . Reliability problems are real and reported, mostly around belts, tensioners, and the

need for a stronger clutch and fuel system . Treat any huge, unsupported power claim you find online, like the 485 horsepower eBay listing, with real suspicion .

Quick check

1. What is the difference between how a turbocharger and a supercharger force air into the engine?
2. Name the three real, GD3-specific turbo or supercharger kits found in this research.
3. Why is a huge power claim on a generic eBay turbo-kit listing not trustworthy?
4. What two other systems do owners report needing to upgrade once they add real boost?
5. Why was a well-written turbo-Fit build article excluded from this chapter entirely?

Engine Swaps

What you will learn

- What an engine swap is, and why it is a much bigger job than a bolt-on part.
- The one documented engine family that has actually been swapped into a GD3 Fit.
- The real obstacles builders report: mounts, wiring, the ECU, the transmission, and more.
- Why this is one of the thinnest, smallest topics in this whole book.
- What was searched for and not found, so you know the honest limits of this research.

The short version

An engine swap means removing your car's factory engine and installing a different one . Exactly one engine family has a real, documented swap into a GD3 Fit: Honda's K-series engine . This is a genuinely small, niche corner of the Fit hobby; one community estimate puts the total number of K-swapped GD3 Fits worldwide at fewer than 50 .

The full explanation

Words you need first

An engine swap replaces a car's original engine with a different one, often from another model . Honda's K-series is a family of engines used in cars like the Civic Si and the Acura TSX . A wiring harness is the bundle of wires that

connects an engine and its ECU to the rest of the car . An immobilizer is an anti-theft system that stops the engine from starting without the correct key . A vehicle speed sensor, or VSS, tells the ECU how fast the car is moving . A limited slip differential, or LSD, is a transmission part that helps both drive wheels get power during hard cornering .

A legal reminder before you start

An engine swap is a much bigger emissions question than any bolt-on part in this book. Chapter 47 already covered the basic rule: federal law bans removing or defeating emissions equipment, and many states, especially California, apply extra rules specifically to swapped engines. This book's research did not confirm the exact swap-approval steps for this exact car, so this book will not guess at them. Check your own state's rules on engine swaps before you start, the same way Chapter 17 already told you to check your state's inspection and emissions rules.

The one documented swap family

Researchers found real, detailed build documentation for exactly one swapped-in engine family: Honda's K-series . Two specific K-series engines have documented builds: the K24A2, pulled from a wrecked Acura TSX, and the K20Z3, pulled from a Civic Si . No documented, completed swap of any other engine family was found for a GD3 chassis . That includes other Honda engine families, non-Honda engines, and electric motor conversions .

Two independent, named builders documented their own K-series swaps . One builder runs a website, pbomers.com, showing his own K24A2 swap into a GD3 Fit in detail . He says he has a running K24-swapped Fit . He also points to his brother's YouTube channel for video coverage of four completed swaps between the two of them . A second builder posted a 2014 forum thread about a K20Z3 swap into a 2007 GD Fit . He used an engine, ECU, and wiring harness pulled from a Civic Si . A different forum member replied that he had

"just went through this" the same way, about a month earlier . That second, independent reply shows this is a known method within the small community, not a one-off guess .

The real obstacles

Builders report real problems in several different car systems . On motor mounts: without stiffer aftermarket mounts, or reinforcing the stock ones, the engine is reported to rock and rub inside the tight engine bay . Hasport is the one mount brand specifically named by builders . On body clearance: the front support structure often has to be cut to fit the larger K24 intake manifold . A smaller K-series intake needs less cutting . The stock radiator and its fans can also foul the taller intake manifold, so some builders relocate the radiator and switch to slimmer fans .

Wiring is reported as one of the hardest parts of this swap . One builder said he contacted more than half a dozen harness companies and was ignored by all of them . His solution was to hand-build a harness, using an Acura TSX ECU, since the TSX shares similar wiring architecture with the Fit . The car's immobilizer has to be disabled or re-synced to the new ECU and key, at a Honda dealership . Builders also report that only certain donor transmissions work correctly . Honda changed its vehicle speed sensor design at some point . An older-design sensor will not read correctly on a newer ECU . Builders report using axles from a 2002-2004 Acura RSX, instead of the stock Fit axles, to make everything fit together . Because the Fit uses electric power steering, an idler arm borrowed from a Honda Civic Si is also reported as necessary . Even the exhaust header is reported as a genuinely tough part to source for this swap .

Cost and rarity

No total build-cost figure was found for a full K-series swap into a GD3 . This is left honestly unverified, rather than estimated or guessed at . One real, single component price point was found instead: a commercial Hasport-brand

engine mount kit for this exact swap sells for \$249 to \$285, depending on the retailer . That is only a small fraction of what a complete swap would actually cost . On rarity, one experienced forum member estimated that fewer than 50 K-swapped GD3 Fits exist worldwide . That is a rough personal guess, not an official count . A second, separate long-time forum member said he could not remember any currently active members with a K-swapped Fit . Both statements point the same direction: this swap is genuinely rare .

A real parts market, even so

Even though this swap is rare, a real commercial parts market exists to support it . Three separate retailers each independently sell the same Hasport-brand engine mount kit, specifically titled for the "07-08 Honda Fit" K-series swap . That does not confirm any one builder's total cost, but it does show real, ongoing commercial demand, from more than one seller .

Claims checked and set aside

One claim, that a professional shop called King Motorsports built a K24-swapped Fit for the SEMA car show, was searched for directly and could not be confirmed . Only an ordinary parts listing for a mount kit was found, no build article or show record . This claim is not used anywhere in this book .

A forum thread asking about swapping in a B-series or D-series Honda engine, a different engine family than K-series, was found . No completed, documented build of that kind into a GD3 chassis was found, though . This is recorded honestly as "few to no documented examples," not as a confirmed impossibility .

An electric motor, or EV, conversion was also searched for . One forum thread shows a GD3 owner asking questions about an EV conversion, which suggests at least planning-stage interest . No completed, documented GD3-specific EV conversion build was found . General EV conversion kits exist for other cars, but none found named the Fit specifically .

Putting it together

This is one of the smallest, most niche topics in this entire book . Exactly one engine family, Honda's K-series, has real documented swaps into a GD3 Fit . The obstacles are real and touch nearly every system in the car: mounts, wiring, the ECU, the transmission, the axles, even the steering . No total cost figure exists, and the swap is genuinely rare, at well under 50 known worldwide by one rough estimate . If you are considering this, expect a long, expensive, mostly self-taught project, not a weekend job .

Quick check

1. What is the one documented engine family that has been swapped into a GD3 Fit?
2. Name three different systems in the car that builders report as real obstacles for this swap.
3. Why does this book leave the total swap cost as unverified, instead of estimating it?
4. About how many K-swapped GD3 Fits does one community estimate suggest exist worldwide?
5. Why was a claim about a professional shop building one for a car show left out of this chapter?

Weight reduction and aerodynamics

What you will learn

- Which parts owners commonly remove from a base 2008 Fit to save weight
- How much weight the spare tire, its cover, and the jack system add up to
- How much weight the air conditioning system adds
- Why SCCA's Street category limits what you can remove
- Why aerodynamic parts are a thin topic for this specific car

The short version

Removing the spare tire, its cover, and the jack and tools together saves you roughly 30 to 35 pounds. Removing the air conditioning system saves about 23.46 pounds more. Aerodynamic parts like spoilers and splitters exist for this car, but they rarely help much at Fit speeds, and SCCA's Street category will not let you use them anyway.

The full explanation

Why weight matters

A lighter car turns, stops, and accelerates better with the same engine. Removing weight is one of the cheapest ways to make a stock Fit quicker at autocross or a track day. Autocross is a timed driving event where you drive one car at a time through a course marked with cones. SCCA stands for the Sports Car Club of America. It is a group that runs autocross and racing events and writes the rulebooks for them.

The spare tire system

In this book, the "spare tire system" means the spare tire itself, its cover, and the jack and tools that go with it.

Community-reported figures found for a first-generation Fit like yours put the spare tire itself at about 21 pounds (9,525 g). The fabric cover some Fits carry over the spare weighs about 7.76 pounds (3,520 g). The jack, wrench, tow hook, and mounting bracket together weigh about 4.91 pounds (2,227 g). Add those three together and you get roughly 30 to 35 pounds. These numbers come from a single online community source, so treat them as a good estimate, not an exact figure.

Removing the spare tire system is easy and reversible. You can put it back before you drive home. But driving without a spare means you have no backup if you get a flat tire on the way to or from the event. Plan around that risk.

The air conditioning system

Owners report that removing the entire air conditioning system — the condenser, compressor, refrigerant lines, brackets, and belt pulley — saves about 23.46 pounds (10,645 g). This job also frees up a small amount of engine power, since the compressor no longer drags on the engine's belt. This is a bigger job than pulling the spare tire, and it is usually a track-only or Street-Prepared-category modification, not something you do for a single event. **Always** have a licensed shop recover the refrigerant with the correct equipment before any A/C line is opened. Venting refrigerant straight into the air is illegal under federal law and harms the environment, so this is not a driveway job.

Sound deadening and rear seats

Owners report removing sound-deadening material — the tar-like padding under the carpet that quiets road noise — for further weight savings. One community source lists about 11 pounds of sound deadening found so far, but the source itself calls this tally incomplete. Sound deadening is a modest weight item on this car compared to the seats or the air conditioning system.

Rear seat weight is a different story. We could not confirm this for the 2008 Fit. Check your owner's manual. A number exists for the newer, second-generation Fit's rear seat, but that is a different car, so this book will not print it as if it applied to yours. The Fit's fold-flat Magic Seat design — Honda's name for rear seats that fold down flat to open up cargo space — makes removing the rear seat an easy, reversible job. That is likely why owners still do it, even without a confirmed weight figure.

The realistic number, and the extreme number

One community source describes a full "race trim" build that removes about 534.67 pounds (242,525 g) total. That number assumes gutting most of the interior, fitting lighter wheels and tires, a lightweight flywheel and battery, and a dedicated racing seat. That is not a realistic goal for a typical weekend autocrosser.

A more honest number keeps the car street-legal and Street-category-legal. Combining the spare tire system and the air conditioning system saves about 55 to 60 pounds. SCCA's Street category rule requires the car to run "as specified by the manufacturer with only standard equipment." That means a Street-classed Fit cannot legally remove its rear seats, sound deadening, or air conditioning at all. Owners who want to remove more have to move up into Street Prepared, Street Modified, Prepared, or Modified category first.

Aerodynamics: an honest, thin topic

Aerodynamic parts use the shape of moving air to push a car down onto the road, which can help it grip better in fast corners. For the base 2008 Fit, this is honestly a thin topic, for three reasons.

First, only a few parts are made specifically for this exact chassis. Rear spoilers are the most common part sold for a GD3 Fit, and most of them are small, styling-focused pieces rather than parts built to add real downforce. Front and side splitters — panels that redirect air under the front of the car — are also sold specifically for the GD3 chassis, by at least two retailers. Canards, small wing-like fins added to the front bumper, are the thinnest category. No canard was found molded specifically for this car's bumper shape; the parts sold are generic pieces trimmed to fit.

Second, SCCA's Street category will not let you use most of these parts anyway. The rulebook does not list splitters, canards, or non-factory wings as allowed Street-category modifications. You would need to move up to Street Prepared, Street Modified, Prepared, or Modified category before you could legally add them.

Third, aerodynamic parts work best at higher speeds, because the extra grip they create grows with the square of your speed. A Fit at typical street or track-day speeds is well below the speed range where a splitter or wing meaningfully outweighs the extra weight and drag it adds. Treat a rear spoiler as a styling choice, and a splitter as a niche option for a higher-category or track-only build, not a priority modification for this car.

A general checklist, beyond the specific numbers

Beyond the confirmed figures above, owners of small economy hatchbacks commonly remove other items too. These include the factory first-aid or warning-triangle kit, any aftermarket subwoofer or amplifier, and floor mats.

They also empty the trunk of anything not needed that day. These items do not have specific weight figures behind them in this book; they are general community practice, not measured facts.

Quick check

1. Roughly how much weight does removing the spare tire system save, and what risk do you take on by removing it before an event?
2. About how much does the air conditioning system weigh if fully removed?
3. Why can a Street-classed Fit not legally remove its rear seats or air conditioning?
4. Name the three reasons aerodynamics is described as a thin topic for this car.
5. What is a canard, and why are GD3-specific canards hard to find?

Seats, harnesses, roll bars

What you will learn

- What SCCA autocross rules require for seat belts and helmets in a stock Fit
- Why a stock Fit does not need a roll bar or harness for autocross
- What NASA requires for seat belts and roll bars at a track day
- Why NASA's roll bar rule depends on your car's body style, not your speed
- What happens once you move up to wheel-to-wheel racing

The short version

A stock, closed-body Fit does not need a roll bar or an aftermarket harness for autocross or a typical track day. Both SCCA and NASA only require a roll bar once a car is either heavily modified or has an open top, like a convertible. Neither rulebook ties the roll bar requirement to how fast you drive.

The full explanation

Two organizations, two kinds of event

This chapter covers rules from two organizations. SCCA, the Sports Car Club of America, runs autocross events, called Solo. NASA, the National Auto Sport Association, runs High Performance Driving Events, known as HPDE, along with Time Trial and full wheel-to-wheel racing. An HPDE is a track day where you drive your own car on a real racetrack. An instructor rides along at first, to help you build skill rather than to race other cars.

A roll bar is a metal hoop or frame built into the car to protect your head in a rollover crash. A roll cage is a fuller, more complete version of the same idea, with multiple bars protecting more of the cabin. A harness is a set of racing seat belts with more attachment points than a factory belt, usually four, five, six, or seven points instead of the factory's three.

SCCA autocross: what a stock Fit needs

For autocross, SCCA's current rulebook requires a lap belt in every car. A lap belt is the strap that crosses your hips; a shoulder belt is the strap that crosses your chest. Using a shoulder belt or a harness is strongly recommended, but it is not required in a closed car like your Fit. Your Fit's factory-installed belt system already meets this rule.

Here is a rule worth remembering carefully: SCCA only allows a non-factory harness in an open car. An open car means a convertible, or a car with its targa top or T-tops removed. Even then, the car also needs a roll structure that meets the rulebook's Appendix C standard. The top of that structure must sit at or above the top of your helmet. Your Fit is a closed hardtop with no factory open-top version, so this rule does not open a path to installing a racing harness in it. The rulebook does not let you swap a racing harness into a closed car at all.

Some autocrossers install a harness bar, a bar that lets you mount alternate belts, without a full roll cage. A harness bar is only a belt-mounting fixture. It is not a certified roll structure by itself. So installing a harness bar alone does not satisfy the roll-structure requirement above, and does not open the door to using a harness in a closed car either.

Helmets are a firmer rule. Every competitor, in every class, needs a helmet meeting one of a specific list of Snell, SFI, or FIA safety standards. Snell, SFI, and FIA are organizations that test and certify helmets for motorsport use, and none of them offer an exception based on speed or class. The chin strap must be fastened securely. Most cars with a normal-size windshield, like your Fit, do not need a full-face helmet, but open-wheel cars and karts do.

SCCA autocross: when a roll bar becomes mandatory

A roll bar or cage is not required for a closed car running in SCCA's Street, Street Touring, Street Prepared, or Street Modified categories. These are the categories a base, lightly-modified Fit would run in; see Chapter 58 for exactly which class. A roll bar only becomes mandatory in specific, heavily-modified classes: A Modified, B Modified, C Modified, F Modified, open cars in the Prepared category, D Modified, and E Modified. These "Modified" and "Prepared" classes are SCCA's most heavily-built, purpose-made-for-racing categories, well beyond a car with bolt-on street parts. Building a car into one of these classes is a major step beyond the stock or lightly-modified Fit this book focuses on.

If a roll bar is added to a car in one of the more modified categories, SCCA's Appendix C sets rules for it. The top of the bar cannot sit below the top of the driver's helmet. The rulebook also suggests it extend at least 3.0 inches (76.2 mm) above the helmet. It must use seamless steel tubing, not a bolted-together post design, and must attach to the car's structure at two points, forming a hoop shape.

NASA HPDE and Time Trial: what a stock Fit needs

For an HPDE or Time Trial day, NASA requires modern, good-condition factory seat belts, or a DOT-approved restraint system. DOT stands for the U.S. Department of Transportation, which sets safety standards for belts sold for street cars. A lap belt alone, with no shoulder strap, is not allowed. Your Fit's stock three-point belt already meets this rule, with no changes needed.

NASA also requires a helmet meeting Snell SA2010 or newer, or an equivalent standard, for both the driver and any passenger. Eye protection like goggles or a face shield is recommended, but is not a hard requirement the way the helmet is.

The roll bar rule that surprises a lot of people

Here is the most important, and most misunderstood, rule in this chapter. NASA's roll bar requirement for HPDE and Time Trial depends entirely on whether your car is open or closed. It does not depend on how fast you drive, or which run group you are in. NASA's rulebook states that open cars, meaning convertibles or open T-tops, should have a roll bar installed. Nothing in NASA's rules ties a roll bar requirement to a driver's lap time or top speed. Your base 2008 Fit is a closed hardtop with no factory open-top version. It does not need a roll bar for any NASA HPDE or Time Trial run group, no matter how fast you get.

Many drivers assume that going faster, or moving into a higher run group, eventually triggers a roll bar requirement. That assumption is wrong for both organizations covered in this chapter. The real trigger is always event type and car body style, never a specific speed or lap time.

There is one thing that does change your equipment needs on a closed car: adding a hard-shell, fixed-back racing seat. NASA's rulebook says stock belts should not be worn with that kind of seat, so a car with one needs a harness meeting a specific certification standard instead. This does not apply to your Fit's stock seats. It only matters if you later install an aftermarket racing seat.

Wheel-to-wheel racing: the one place equipment is always mandatory

Once you move beyond HPDE and Time Trial into wheel-to-wheel Competition — racing where more than one car is on the track at once, with finishing positions — the rules change completely. A full roll cage is required in every car, with no exception for car type or speed. The cage must meet detailed construction standards. It needs a continuous main hoop close to the roof, and a diagonal brace. Padding must cover anywhere your head or limbs could hit it. The cage also needs a minimum of six mounting points to the car's floor or rocker panels. A 5-, 6-, or 7-point harness, certified to a specific SFI or FIA standard, is also required, and that certification carries a hard expiration date with no exceptions.

Building a Fit to this level is a large step past autocross or an HPDE day, and beyond the scope of a first project car. If you ever consider it, treat this chapter as only a starting point, and work from the full current rulebook.

Quick check

1. Does a stock, closed 2008 Fit need a roll bar for SCCA autocross? Why or why not?
2. What is the real trigger for NASA's roll bar requirement at an HPDE or Time Trial day — speed, lap time, or something else?
3. Under SCCA's rules, when is an aftermarket harness legal, and why can you not install one in your Fit as it comes from the factory?
4. What is the difference between a roll bar and a roll cage?
5. What piece of aftermarket equipment can require a harness on a closed car, even without a roll bar rule forcing it?

PART IX

Getting Ready for the Track

CHAPTER 58

Autocross

What you will learn

- What autocross is and how it works
- What SCCA class a stock 2008 base Fit runs in
- What kinds of changes move your Fit into a faster class
- Whether an older Fit is still allowed to compete
- What to know before your first event

The short version

Autocross is a timed event where you drive one car at a time through a course marked out with cones, trying to beat the clock without hitting anything. A stock 2008 base Fit currently runs in SCCA's H Street class. Adding suspension, brake, or wheel and tire upgrades moves it up into a faster class, and adding race tires and engine tuning moves it up again.

The full explanation

What autocross actually is

SCCA, the Sports Car Club of America, calls its autocross program "Solo." In Solo, you drive one car at a time through a course laid out with cones on a large paved lot, usually a parking lot or an unused runway. You are timed, not racing side by side against another driver, so your only competition is the clock and everyone else's best run of the day. Cars are sorted into classes so that similar cars compete against each other, which is why knowing your class matters.

Your Fit's class: H Street

The current SCCA National Solo rulebook lists the Fit in H Street, written as HS. This classing was checked in two separate rulebook editions and is identical in both, so it has not changed recently. Street is SCCA's entry-level category. The rulebook itself calls it "the lowest barrier of entry." It allows only limited changes: mainly shocks, anti-roll bars — suspension parts that resist body lean in turns — and replacement tires. Beyond those changes, a Street-classed Fit is expected to stay close to how it left the factory.

Your 2008 model-year Fit is still fully eligible to compete. SCCA's own age rule allows a car to run in National events through the end of the 30th calendar year after its model year. For a 2008 car, that rule stays in effect through 2038, so age is not a concern for many years to come. Regional-only Solo events do not use this same age limit at all.

Moving up a class: Street Touring and Street Prepared

If you add the kinds of parts Street category does not allow, your Fit moves into a different class instead of staying in H Street. The next class up is E Street Touring, written EST. SCCA describes Street Touring as sitting between Street and the more heavily modified Street Prepared category. Moving into EST means adding aftermarket suspension parts, like shocks, springs, or sway bars. It can also mean a different differential — the part that lets your two drive wheels spin at different speeds through a turn — bolt-on engine parts, or a bigger brake kit. Wider or different-sized wheels and tires than Street category allows are permitted too. A car in this class is still expected to stay street-legal. It must keep its emissions equipment and OBD-II system — the car's onboard diagnostic computer, which tracks emissions and engine problems — working. Its factory anti-lock brakes must also keep working, though they may be turned off electronically.

Going further still moves your Fit into F Street Prepared, written FSP. Street Prepared builds on everything Street Touring allows, then adds a further step that is not really meant for driving on public roads. This includes race-

compatible tires and permanent bodywork changes, such as cutting or flaring fenders for tire clearance. It also includes front splitters, rear spoilers, and an aftermarket ECU — an aftermarket computer that controls engine tuning — for engine tuning. Emissions equipment can be removed, and the intake and exhaust become unrestricted. A Street Prepared car is also allowed further weight-reduction changes, like removing the air conditioning system, and more suspension upgrades than Street Touring allows.

Put together, the modification ladder for your Fit looks like this. Street or H Street keeps the car stock, aside from shocks, sway bars, and tires. Street Touring or EST adds bolt-on suspension, brake, and wheel upgrades while staying street-legal. Street Prepared or FSP adds race tires, tuning, and body changes, and is no longer necessarily street-legal.

Getting ready for your first event

Before your first event, confirm which class you plan to run. If your Fit is stock, or close to it, that class is H Street. Read Chapter 57 for what your Fit needs, and does not need, in terms of seat belts, harnesses, and helmets. Read Chapter 60 for a full checklist of what to bring and do the night before, the morning of, and between runs. Knowing your class ahead of time means you register correctly and compete against the right group of cars from your very first run.

Quick check

1. What class does a stock 2008 base Fit run in under SCCA's current rule-book?
2. What is the difference between Street, Street Touring, and Street Prepared, in terms of what each one allows you to change?
3. Through what year does SCCA's age rule keep a 2008 model-year car eligible for National events?

4. Name two changes that would move your Fit from H Street into E Street Touring.
5. Where in this book should you look for the safety-gear rules and the full event-day checklist?

HPDE / track days

What you will learn

- What HPDE means and how it differs from racing
- What NASA's tech inspection checks on the car itself
- What safety and general-condition items tech inspection checks
- What the actual paper tech form looks like
- The most common reasons cars fail tech inspection

The short version

HPDE stands for High Performance Driving Event, a track day where you drive your own car on a real track to build skill, not to race other drivers. Before you can drive, your car has to pass a tech inspection covering its mechanical condition and its safety equipment. A well-maintained, stock Fit with good tires and brakes should pass without trouble.

The full explanation

What tech inspection checks on the car itself

NASA's Club Codes and Regulations, the rulebook covering NASA's HPDE program, lists specific mechanical items every car must pass. Wheels and tires must show no cracked or damaged wheels and no exposed cords or bubbles on the tire, with every lug nut present and tight. Steering and suspension get checked for tightness, since very little or no play is allowed in the wheel bearings, suspension, or steering. The engine bay must show no significant fluid leaks from the engine, radiator, or hoses, and hoses must be in good

condition with no cracks. The radiator needs an overflow container holding at least one liter. Any oil breather or vent line must either return to the engine, or route to a catch tank of at least one liter. Brakes must work well enough to stop the car in a reasonable, safe distance, with adequate pedal pressure. Fluid must sit at or above the minimum level, and brake lines must be in good condition. The car's general appearance also gets checked against what the rulebook calls the "50/50 rule": it should look undamaged and straight at 50 mph from 50 feet away.

What tech inspection checks beyond the mechanicals

Tech inspection also covers safety items that are not part of the car's mechanical systems. Seatbelts and harnesses must be in good, undamaged, factory configuration, or use an approved aftermarket harness if the car has an aftermarket hard-shell seat. Roll bars are required on all open cars, but not on a closed car like your Fit. The battery must be securely fastened — a bungee cord or rubber cord alone is not acceptable — and its positive terminal must be covered with a non-conductive material. The fuel cap must seal well enough that fuel does not spill under hard driving. No exposed, live wiring is allowed inside the cabin, and seats must be securely fastened to the car's structure. Every loose object must come out of the passenger compartment before you go on track. At least one working rear-view mirror and at least two working brake lights, visible from 300 feet, are both required. Tow eyes — front and rear hooks a tow truck can safely pull from — are strongly recommended, though not always required.

On the process itself, you review the Technical Inspection Form before the event, then either have the car pre-teched at an authorized shop or teched at the track. Once the form is signed, you place it on the dash or windshield so a grid marshal can collect it and apply a tech sticker.

The real tech form, item by item

A real NASA HPDE/Time Trial Technical Inspection Form organizes its checklist into five boxes. Wheels and tires covers tire condition, all lug nuts present and tight, and no hubcaps or beauty rings left on. Steering and suspension covers front and rear wheel-bearing play and steering play. Engine covers fluid leaks from oil, fuel, or water, the radiator overflow and hose condition, and a battery secured with no bungee cords and covered terminals. Brakes covers pedal pressure, fluid level, line condition, and brake lights. Safety Equipment covers roll bars on open cars, seat belts or a harness, a Snell SA2005-or-newer helmet, and a covered battery if it sits inside the cockpit. Miscellaneous covers a secured gas cap, no exposed wires, and seats bolted down tight.

The form also reminds you to put contrasting-color numbers, at least 10 inches tall, on both sides of the car. It tells you to remove all loose objects, including the spare tire and floor mats. Finally, place your tech sticker at the top center of the windshield. The most common first-timer tech-inspection failures are worn brake pads, a loose battery, corded or cracked tires, leaking fluids, loose wheel bearings, and a missing tow hook. Inspect your car several days before the event, not the morning of, so you have time to fix or order parts for anything you find.

Quick check

1. What does HPDE stand for, and how is it different from wheel-to-wheel racing?
2. Name three mechanical items NASA's tech inspection checks under the car's hood.
3. Does a stock, closed Fit need a roll bar to pass tech inspection? Why or why not?
4. What are the five checklist boxes on a real NASA tech inspection form?

5. What are the most common reasons a first-timer's car fails tech inspection, and how can you avoid them?

CHAPTER 60

Track-day prep checklist

What you will learn

- What to do the night before a track day
- What to check and bring the morning of the event
- What to check between track sessions
- Why finishing car prep early matters more than doing it at the last minute

The short version

Finish your car prep several days before the event, not the night before, and pack your gear bag completely so the morning is simple. On the morning of, check your cold tire pressure and dress for the day's required clothing. Between sessions, recheck tire pressure and wheel torque, and take time to eat, drink, and rest.

The full explanation

The night before

Finish any car repairs or upgrades several days ahead of the event, not the night before. A last-minute repair that goes wrong, like a broken bolt or a surprise problem, is one of the most common ways a driver arrives at the track exhausted and unprepared. The night before, pack your car and gear bag completely, so the morning of the event stays simple. Pack your helmet and gloves, your completed or shop-signed tech inspection form, a pen, and a tire pressure gauge. Also pack a small tool bag with a jack, a wheel-lock key if your car uses one, and zip ties. Also pack a filled gas container if the track does not sell fuel, and a cooler with water or a sports drink and food. Bring

cash for tow-truck tips or tolls, and a toll transponder if you will drive toll roads to get there. Many event organizers hold check-in the evening before the event. Attending it avoids a morning-of line for waivers, the tech form, car numbers, and your event wristband or tech sticker.

The morning of

Check your tire pressures while the tires are still cold, right at the start of the day. Tire pressure rises once the tires warm up on track. Note your cold baseline before you leave, so you can adjust it correctly later. Confirm the car has no new fluid leaks and no dashboard warning lights before you leave home. Dress for the event's required clothing rather than showing up in shorts and sandals. Many organizations, including NASA, require long sleeves and pants, and closed-toe shoes. Bring sunscreen, comfortable weather-appropriate clothing, snacks, and plenty of water. A good rule of thumb is roughly one quart of fluid for every 50 pounds of body weight over a full track day. Heat and nervous energy both increase how much fluid you lose. Arrive with enough time before the mandatory drivers' meeting to unload, get through registration and tech, and not feel rushed.

Between sessions

After each session, recheck tire pressure, and tire temperature too if you have a temperature gun on hand. Track surface temperature and how hard you drive both change your tire pressure through the day. Recheck your wheel and lug torque periodically, since track driving can work fasteners loose faster than everyday street driving. Top off engine oil if your car is a known oil-consumer, or at least check the level, using oil and a funnel packed for exactly this purpose. Use the break between sessions to rehydrate, eat something, and rest in the shade, rather than only working on the car. Staying hydrated and fed directly affects how well, and how safely, you drive later in the day. Many organized programs hold an instructor debrief or driver meeting between

sessions. Taking a few notes on what changed between sessions is a widely repeated habit. A braking point, an odd noise, or a new tire pressure are all worth writing down, for tracking progress and problems across the day.

Quick check

1. Why should you finish car repairs several days before the event instead of the night before?
2. Name three things you should pack in your gear bag the night before.
3. Why do you check tire pressure while the tires are still cold in the morning?
4. What should you check on your wheels between sessions, and why does track driving make this more important than usual?
5. Besides working on the car, what should you do during the break between sessions?

CHAPTER 61

Driver skills

What you will learn

- Where to look with your eyes while driving on track
- Why smooth steering and pedal inputs make you faster, not just safer
- What trail braking means and why drivers use it
- How these skills connect to what you already do behind the wheel

The short version

Look far ahead, toward the path you want, not at a hazard or the track's edge. Smooth, gentle steering and pedal inputs keep more grip in your tires than sudden, jerky ones do. Trail braking means letting off the brake gradually as you steer into a corner, instead of coming off it all at once beforehand.

The full explanation

Where to look

A well-known professional driving coach's free instructional material teaches vision as one of the very first skills to learn, ahead of lines or braking technique. Train your eyes to look far down the track, not just at the pavement right in front of your car. Point your eyes toward the path you want the car to take. Avoid fixing your gaze on a hazard or the track's outer edge, since your eyes tend to pull the car toward wherever they are aimed. Turning your head to see into a corner, instead of keeping it pointed straight ahead, helps you spot that path earlier. Treat this as a constant scanning habit. Always pick up your next reference point further down the track, rather than glancing once

and then forgetting about it. Your overall awareness of the drive shapes how well you react. That means what you see, what you feel through the seat as the car grips or slides, and what you hear from the engine and tires.

Why smoothness matters

The same source teaches keeping your hands near the 9-and-3 o'clock position on the wheel, like the hands of a clock, through most corners on most tracks. A car that needs less wheel movement tends to be smoother and more predictable to drive. Keeping your hand movements small is directly tied to how smooth, and ultimately how fast, you can drive.

Here is why smoothness matters so much: weight transfer. Weight transfer is the way a car's weight shifts from one end, or one side, to the other as you brake, speed up, or turn. Braking shifts weight toward the front tires. Speeding up shifts it toward the rear. Turning shifts it toward the outside tires. A sudden, jerky input causes a bigger, more sudden weight shift than the same input made gradually. A tire's grip does not grow at the same rate as the extra load it carries. That is why a car actually loses more overall grip from a jerky input than a smooth one. That is the real reason smooth driving is fast driving, not only safer driving.

Trail braking, explained simply

Trail braking means gradually letting off the brake pedal with your foot as you steer into a corner, instead of coming completely off the brake before you begin turning. This trade swaps some of the brake's stopping force for turning force, right as the corner asks for it. Braking keeps extra weight, and therefore extra grip, on your front tires right when you are also asking them to change the car's direction. Coming off the brake gradually, instead of all at once, helps keep the front tires closer to their best grip as you turn in. Many quick drivers pay less attention to their braking point, and more attention to exactly when they finish coming off the brake. A smooth, late release keeps the car balanced, and keeps your eyes up on the corner instead of down at your feet.

Putting it together

These three skills work as a set, not on their own. Looking far ahead gives you the time you need to make your steering and pedal inputs smooth, instead of rushed. Smooth inputs keep your tires gripping their best, which is exactly what trail braking is trying to use as you enter a corner. None of these skills require a faster car. They only take practice, and they work just as well in a stock Fit as in any other car.

Quick check

1. Where should your eyes look while driving on track, and why does this matter?
2. What is weight transfer, and how does it connect braking, speeding up, and turning to your tires' grip?
3. Why is a smooth input often faster than a jerky one, not just safer?
4. What is trail braking, in your own words?
5. Why do many faster drivers pay more attention to how they finish releasing the brake than to where they start braking?

After the track

What you will learn

- What to do to your car right after you finish your last session
- Why you should avoid the parking brake right after hard driving
- What to check before you drive home
- What to check once you get home, for a fuller post-event inspection

The short version

Take a slow cool-down lap at the end of your last session, instead of braking hard and parking right away. Once the car has cooled, recheck your tire pressure, tread, and brake pedal feel before you drive home. At home, check for fresh leaks, fluid levels, pad wear, tire wear, and lug torque before you put the car back into everyday use.

The full explanation

Right after your last session

Drive one slower, gentle lap at the end of your last session, instead of braking hard and parking the car right away. This cool-down lap lets air move through the brakes, engine, and transmission, bringing their temperatures down gradually rather than all at once. Avoid using the parking brake, also called the hand brake, right after a hot session. A very hot brake rotor, clamped tight by the parking brake, can warp or damage the pad or rotor as it cools. Let the car cool with the parking brake off instead.

Before you drive home

Once the car and its tires have cooled, recheck your tire pressure and tread condition before you leave for home. Pressures measured right after a hot session read higher than your correct driving pressure, so wait until the tires cool before trusting the number. A hard session can also reveal cuts, exposed cords, or uneven wear that were not there that morning. Check your brakes too: the pedal should still feel firm, not soft or noticeably longer in travel than it felt that morning. Brake fluid can boil or fade under sustained hard track use, and a spongy or long pedal is a sign that has happened.

The fuller inspection, once you get home

Look under the car and around the engine bay, on clean, dry ground, for any fresh drips of oil, coolant, or brake fluid that were not there before the event. Check that your engine oil, coolant, and brake fluid are all still at the correct levels. A hard day of high-RPM driving and heavy braking uses up and stresses fluids faster than ordinary street driving does. Check the visible brake pad thickness at each wheel, and look for fresh cracks, heavy discoloration, or fluid residue near the calipers. Pad wear is one of the fastest-consumed items on a track day, so this check matters even if the car seemed fine on the drive home. Inspect all four tires for uneven wear, cuts, or a change in the tread surface. Re-torque every wheel's lug nuts once the car and wheels are fully cool. Track driving's repeated heating and cooling, combined with harder cornering loads, can work fasteners loose faster than normal driving does. Finally, walk around the car and listen for anything that came loose — in the engine bay, under the car, or inside. Check trim panels, undertray fasteners, and exhaust hangers before you put the car back into everyday use.

Quick check

1. Why should you drive a slower cool-down lap instead of stopping and parking right away?
2. Why should you avoid using the parking brake right after a hard session?

3. Why does tire pressure right after a session not tell you the correct driving pressure?
4. Name three fluids you should check the level of once you get home.
5. Why does track driving make it more important to re-torque your lug nuts than everyday driving does?

PART X

Appendices

APPENDIX A

Full specifications

This appendix lists every specification found for your 2008 Fit base trim: engine, transmission, dimensions, weight, cargo, fuel economy, crash ratings, and price. Most numbers below come straight from Honda's own 2008 press materials. Where a source disagreed with another, or where no reliable number could be found at all, that is written out honestly instead of leaving the row blank. A few terms are defined the first time they appear. **Redline** is the highest engine speed (in RPM, or revolutions per minute) the engine is designed to reach before the driver should shift or ease off. **Compression ratio** compares the volume inside a cylinder when the piston is at the bottom of its stroke to the volume when it is at the top; a higher ratio generally squeezes the air/fuel mixture more tightly before it burns. **Curb weight** is how much the car weighs by itself, with a full tank of fuel and all standard equipment, but no passengers or cargo. **Wheelbase** is the distance between the centers of the front and rear wheels. **MSRP** stands for manufacturer's suggested retail price — the sticker price Honda recommended before tax, license, registration, and destination charge.

Engine

Item	Value	Fact ID
Engine code	L15A1	F-WP01-001
Displacement	1.5 L (1,497 cc)	F-WP01-002
Horsepower	109 hp @ 5,800 rpm	F-WP01-003
Torque	105 lb-ft @ 4,800 rpm	F-WP01-004
Compression ratio	10.4:1	F-WP01-005
Bore x stroke	73.0 mm x 89.4 mm	F-WP01-006
Recommended fuel	Regular unleaded gasoline. Honda's own specification sheet and fueleconomy.gov both say "regular" fuel, but neither one prints an exact minimum octane number for this car. The commonly repeated "87 octane" figure is the standard US definition of "regular" gasoline in general, not a number confirmed for the Fit specifically, so it is not stated here as a verified Fit-specific fact.	F-WP01-007
Redline	6,500 rpm	F-WP01-008

Transmission

Item	Value	Fact ID
Transmission options	5-speed manual (standard) and 5-speed automatic (available), both offered on base trim, not only on Fit Sport	F-WP01-009
Manual gear ratios	1st: 3.462, 2nd: 1.870, 3rd: 1.321, 4th: 0.970, 5th: 0.757, Reverse: 3.231, Final drive: 4.29	F-WP01-010
Automatic gear ratios	1st: 2.996, 2nd: 1.679, 3rd: 1.067, 4th: 0.756, 5th: 0.550, Reverse: 1.957, Final drive: 4.56	F-WP01-011

Weight and dimensions

Item	Value	Fact ID
Curb weight, manual transmission	2,432 lb	F-WP01-012
Curb weight, automatic transmission	2,514 lb	F-WP01-013
Overall length	157.4 in	F-WP01-014
Overall width	66.2 in	F-WP01-015
Overall height	60.0 in	F-WP01-016
Wheelbase	96.5 in	F-WP01-017
Ground clearance	UNVERIFIED. Honda's own specification sheet does not list a ground-clearance figure at all, and a GD3-specific spec page found during research explicitly says the number is "market/trim dependent; verify locally" rather than giving one. The only number found anywhere — 5.9 in — came from a single blog that lumps the first-generation Fit (GD3, 2007-2008) and second-generation Fit (GE8, 2009-2014) together without separating them, and no second source could confirm it applies to your specific 2008 GD3 car. We could not confirm this for the 2008 Fit. Check your owner's manual, or measure it yourself from the ground to the lowest point of the frame or body.	F-WP01-018

Cargo and fuel

Item	Value	Fact ID
Cargo volume, rear seats up	21.3 cu ft	F-WP01-019
Cargo volume, rear seats folded down	56.8 cu ft, per Honda's own 2008 US press release. Two other numbers turned up during research and are recorded here rather than thrown away: a lower figure (41.9 cu ft) appeared only in a general web summary that did not clearly separate this car (GD3) from the later, different 2009+ Fit (GE8), so it was not trusted; and a higher figure (59.2 cu ft) came from a GD3-specific page that measures cargo volume by a different regional method than the one Honda uses for the US market. Honda's own US number, 56.8 cu ft, is used as the main value here because it is the one document that is both Tier 1 (from Honda itself) and explicitly for the 2008 US Fit.	F-WP01-020
Fuel tank capacity	10.8 US gal (41 L)	F-WP01-021

Fuel economy (EPA)

Item	Value	Fact ID
City MPG, manual trans- mission	28 mpg	F-WP01-022
Highway MPG, manual transmission	34 mpg	F-WP01-023
City MPG, automatic trans- mission	27 mpg	F-WP01-024
Highway MPG, automatic transmission	34 mpg	F-WP01-025

Tires and wheels

Item	Value	Fact ID
Factory tire size, base trim	P175/65R14 81S	F-WP01-028
Factory wheel size, base trim	14 in x 5.5J (14 x 5 1/2JJ), steel wheels with full covers	F-WP01-029

Appendix E has the full tire and wheel fitment table, including bolt pattern, center bore, offset, and tire pressures, which this WP-01 pass did not attempt to collect.

Crash test ratings

Item	Value	Fact ID
NHTSA frontal crash	Driver side: 5 stars. Passenger side: 5 stars. (NHTSA did not yet publish a single combined "overall" frontal star rating for a 2008-model-year car; that combined metric started with the 2011 ratings redesign, so only the separate driver/passenger scores exist for this car.)	F-WP01-036
NHTSA side crash	Driver side: 5 stars. Passenger side: 3 stars.	F-WP01-036
NHTSA rollover	4 stars (13.3% rollover risk)	F-WP01-036
IIHS moderate overlap front (original test)	Good	F-WP01-037
IIHS side (original test)	Good	F-WP01-037
IIHS head restraints and seats (whiplash protection)	Poor	F-WP01-037
IIHS small overlap front test	Not applicable. This test did not exist yet when your GD3 Fit was sold; IIHS introduced it starting with the 2012 model year.	F-WP01-037

IIHS states these ratings apply to both years the GD3 was sold in the US (2007 and 2008), even though the car IIHS physically crash-tested was a Fit Sport — IIHS says the safety-cage structure is the same across trims. NHTSA's rollover and crash record covers the "2008 Honda Fit 4-DR. w/ SAB" generally, without a separate breakdown for base versus Sport trim.

Price

Item	Value	Fact ID
Original MSRP, 5-speed manual	\$13,950, excluding tax, license, registration, and a \$670.00 destination charge; dealer prices may vary	F-WP01-038
Original MSRP, 5-speed automatic	\$14,750, excluding tax, license, registration, and a \$670.00 destination charge; dealer prices may vary	F-WP01-038

Production

Item	Value	Fact ID
Production plant and country	Suzuka, Mie Prefecture, Japan	F-WP01-035

APPENDIX B

Fluid capacities and types

This appendix lists every fluid in your 2008 Fit: the type or spec to buy, and how much your car holds. A few terms come up more than once. **Viscosity** describes how thick or thin a fluid flows, written as a code like 5W-20. **MTF** stands for manual transmission fluid, the fluid inside a manual gearbox. **ATF** stands for automatic transmission fluid, the fluid inside an automatic transmission. **DOT** stands for Department of Transportation. A DOT number on brake fluid is a US safety-grade rating, and a higher number is not automatically "better" for your car. Using the wrong one can cause damage. Numbers marked M/T are for cars with a manual transmission; numbers marked A/T are for cars with an automatic.

Engine oil

What	Value	Fact ID
Type	API Premium-grade 5W-20 detergent oil, with the API Certification Seal ("For Gasoline Engines"). Honda's own motor oil is the preferred brand. Synthetic oil is acceptable if it carries the same seal and viscosity. No oil additives are recommended.	F-WP09-016
Capacity, oil change including filter	3.8 US qt (3.6 L)	F-WP09-017
Capacity, oil change without filter	3.6 US qt (3.4 L)	F-WP09-018
Capacity, total (after a full engine overhaul or dry fill)	4.4 US qt (4.2 L)	F-WP09-019

Engine coolant

Coolant, also called antifreeze, is the liquid that flows through your engine and radiator to keep the engine from overheating.

What	Value	Fact ID
Type	Honda Long-Life Anti-freeze/Coolant Type 2 only, pre-mixed 50% antifreeze and 50% water. Never add straight antifreeze or plain water. If Honda coolant is not available, a major-brand non-silicate coolant rated for aluminum engines may be used temporarily, then the system should be flushed and refilled with Honda coolant as soon as possible.	F-WP09-020
Capacity after a coolant change, manual transmission	3.8 L (1.00 US gal)	F-WP09-021
Capacity after a coolant change, automatic transmission	3.7 L (0.98 US gal)	F-WP09-022
Capacity total, after engine overhaul, manual transmission	5.4 L (1.43 US gal)	F-WP09-023
Capacity total, after engine overhaul, automatic transmission	5.3 L (1.40 US gal)	F-WP09-024
Reserve tank capacity	0.4 L (0.11 US gal)	F-WP09-025

Manual transmission fluid (MTF)

What	Value	Fact ID
Type	Honda Manual Transmission Fluid (MTF) preferred. If unavailable, an SAE 10W-30 or 10W-40 motor oil with the API Certification Seal ("For Gasoline Engines") may be used as a temporary replacement, but motor oil is missing the additives a gearbox needs, and using it for a long time can leave your shifts feeling stiffer.	F-WP09-026
Capacity, fluid change	1.6 US qt (1.5 L)	F-WP09-027
Capacity, total (including the differential, after over-haul)	1.7 US qt (1.6 L)	F-WP09-028

The differential is the set of gears inside the transmission housing that lets your two front wheels spin at different speeds when you turn. It shares the same fluid as the manual transmission on your Fit.

Automatic transmission fluid (ATF)

What	Value	Fact ID
Type	Honda Genuine ATF-Z1 only. Do not mix it with other transmission fluids. Using a different fluid can cause the transmission to wear out faster, and damage from using the wrong fluid is not covered by Honda's new-vehicle warranty. Note for today's buyer: Honda ATF-Z1 has since been discontinued industry-wide and replaced by Honda ATF DW-1 for service fills on many older Hondas; ask your parts store for the current equivalent.	F-WP09-029
Capacity, fluid change	2.5 US qt (2.4 L)	F-WP09-030
Capacity, total (after over-haul)	6.3 US qt (6.0 L)	F-WP09-031

Brake fluid and clutch fluid

What	Value	Fact ID
Type	Honda Heavy Duty Brake Fluid DOT 3 preferred, or a DOT 3 or DOT 4 brake fluid from a sealed container as a temporary replacement. Brake fluid marked DOT 5 does not work with your Fit's braking system, and can cause extensive damage.	F-WP09-032
Clutch fluid (manual transmission only)	Manual-transmission Fits use a hydraulic clutch that shares the same fluid specification as the brake system above. Fluid level should sit between the MIN and MAX marks on the clutch fluid reservoir, which has a gray cap; a low level can point to a leak in the clutch hydraulic system.	F-WP09-033
Capacity	This work package's sources do not give a brake or clutch fluid capacity number for the 2008 Fit.	—

Power steering fluid

What	Value	Fact ID
Type / capacity	Not applicable. Your 2008 Fit uses Electric Power-Assisted Rack-and-Pinion Steering (EPS) on every trim. EPS uses an electric motor to help you turn the wheel, not a hydraulic pump and fluid reservoir, so there is no power steering fluid to check or change.	F-WP09-034

Windshield washer fluid

What	Value	Fact ID
Type	Any commercially available windshield washer fluid. Do not use engine antifreeze or a vinegar-and-water mix in the washer reservoir.	F-WP09-035
Capacity	We could not confirm this for the 2008 Fit. Check your owner's manual.	F-WP09-035

APPENDIX C

Torque specifications

Torque is a measure of twisting force — how hard you turn a bolt or nut with a wrench. A **torque wrench** is a tool that stops you (or clicks, or shows a reading) once you have reached a set amount of twisting force, so you do not over-tighten or under-tighten a fastener. Over-tightening can strip threads or crack parts; under-tightening can let a part loosen and fail while you drive. This appendix gives every torque number this project's research turned up for the 2008 Fit base trim, and it says so plainly whenever a number could not be confirmed. The service manual that mechanics normally use for this exact number (the Honda factory service manual) was not available as a free source during this research, so most values below come from the owner's manual (which prints a few) or from cross-checked online forum and repair-site sources (which print the rest, with lower confidence). The **confidence tier** column tells you how much to trust each number: Tier 1 means it came from Honda's own document; Tier 3 means a published repair reference or parts site; Tier 4 means an online forum or Q&A post. A "community-reported" value has not been checked against an official document — treat it as a helpful starting point, not a guarantee, and check it against your own shop's torque chart or the fastener's own markings before you rely on it.

Item	Torque value	Confidence / tier	Fact ID
Wheel lug nut (wheel nut)	80 lbf·ft (108 N·m, 11 kgf·m) — same value for a normal wheel or the compact spare	High / Tier 1 (owner's manual)	F-WP10-001
Oil drain plug (drain bolt)	29 lbf·ft (39 N·m, 4.0 kgf·m)	High / Tier 1 (owner's manual)	F-WP10-002
Oil filter	No numeric torque published by Honda. The owner's manual says to install the filter "according to the instructions that come with it" — hand-tighten per the filter maker's own directions (typically until the gasket touches the mounting surface, then turn an additional fraction of a turn as stated on the filter's box).	High confidence that no number exists / Tier 1 (owner's manual)	F-WP10-003
Spark plug	13 lbf·ft (18 N·m, 1.8 kgf·m)	Medium / Tier 3, corroborated by a Tier 4 source. Caution: the main source for this number files its page under a different generation of Fit than the 2007-2008 GD3 this book covers; it is used anyway because a second, independent source names a 2007 Fit specifically with the same number, and this engine fam-	F-WP10-004

Item	Torque value	Confidence / tier	Fact ID
		ily is not expected to change spark plug thread torque between adjacent generations — but that expectation was not independently confirmed against a document naming GD3 by name.	
Front brake caliper bracket (mounting bracket-to-knuckle) bolt	80 lbf·ft (community-reported)	Low / Tier 4. Both corroborating sources actually discuss a Fit Sport, not confirmed identical on base trim.	F-WP10-005
Rear brake caliper bracket bolt	UNVERIFIED. A figure of "41 ft-lbs" turned up repeatedly in general web searches, but every time it was traced back, it came from Honda Accord/Civic assembly-plant-specific threads, not the Fit. No Fit-specific source with this number was found. We could not confirm this for the 2008 Fit. Check your owner's manual, a Honda factory service manual, or ask a Honda parts counter or dealer before you rely on a number for this bolt.	Not applicable	F-WP10-006
Caliper slide pin (guide bolt), front	Approximately 25-26 lbf·ft (community-reported; the	Low / Tier 4, single independent source for the front value	F-WP10-007

Item	Torque value	Confidence / tier	Fact ID
	front number varies slightly between sources found, so it is given as a range rather than a single picked value)	naming this exact car	
Caliper slide pin (guide bolt), rear	Approximately 17 lbf·ft (community-reported)	Low / Tier 4	F-WP10-007
Strut top nut	UNVERIFIED. Only one online forum thread with specific numbers was found (front strut lower nut 25 lb-ft, upper nut 33 lb-ft, shock flange nuts 75 lb-ft, rear upper strut nut 22 lb-ft), and this project's rule requires two independent forum/community sources before using a number, not just one. A repair-reference excerpt confirms the general procedure and torque-spec diagrams exist for this car's front suspension, but its actual numbers were in an image that could not be read as text. We could not confirm this for the 2008 Fit. Check your owner's manual, a Honda factory service manual, or ask a Honda parts counter	Not applicable	F-WP10-008

Item	Torque value	Confidence / tier	Fact ID
	or dealer before you rely on a number for this nut.		
Battery hold-down	UNVERIFIED. The owner's manual describes cleaning, testing, and jump-starting the battery, but gives no clamp or bracket torque number. No source with a stated number for this car was found despite multiple searches. Small-car battery hold-down bolts are sometimes described generally as a "snug, do not overtighten" spec with no published torque number at all, but that general claim could not be confirmed as Honda's own guidance for this car either. We could not confirm this for the 2008 Fit. Check your owner's manual, a Honda factory service manual, or ask a Honda parts counter or dealer before you rely on a number for this bolt.	Not applicable	F-WP10-009-torque

APPENDIX D

Fuse and relay maps

Your 2008 Fit has three fuse boxes: one inside the car, and two in the engine compartment. This appendix lists every fuse position in all three boxes, with its amperage and the circuit it protects, straight from the owner's manual. It also tells you honestly where the manual does not give you a relay map, instead of making one up.

Fuse box locations

Fuse box	Location	How to open it	Fact ID
Interior (passenger compartment) fuse box	Behind the driver's coin tray, on the dashboard on the driver's side	Turn the coin tray's dial counterclockwise, then pull the tray toward you to remove it	F-WP21-002
Primary under-hood fuse box	Engine compartment, driver's side	Push the tabs on the fuse box cover to release the lid	F-WP21-003
Secondary fuse box	On the positive terminal of the battery, in the engine compartment	This box holds a single 80 A fuse that protects the battery circuit	F-WP21-004

Your Fit has three fuse boxes in total, not two. Many cars have only one box under the dash and one under the hood. Your Fit adds a third, very small fuse box mounted directly on the battery's positive terminal.

Fuse table — Interior (passenger compartment) fuse box

This box has 33 numbered positions. Six of them (No. 2, 5, 19, 20, 24, 30) are marked "Not Used" in the manual — they are empty slots with no fuse installed.

Fuse #	Amperage	Circuit(s) protected
1	10 A	Back Up Light
2	—	Not Used
3	10 A	METER
4	10 A	Turn Light
5	—	Not Used
6	30 A	Front Wipers
7	10 A	SRS
8	7.5 A	Daytime Running Light (Canadian models)
9	20 A	Rear Defogger
10	7.5 A	HAC (Hill-start Assist Control)
11	15 A	Fuel Pump
12	10 A	Rear Wiper
13	10 A	SRS
14	15 A	IGP (Ignition Power)
15	20 A	Left Rear Power Window
16	20 A	Right Rear Power Window
17	20 A	Right Front Power Window
18	7.5 A	TPMS — Tire Pressure Monitoring System (if equipped)
18	10 A	Daytime Running Light (Canadian models) — same slot, other equipment variant
19	—	Not Used
20	—	Not Used
21	20 A	Fog Light (if equipped)
22	10 A	Small Light (parking/side-marker lights)

Fuse #	Amperage	Circuit(s) protected
23	10 A	LAF (Linear Air/Fuel sensor)
24	—	Not Used
25	7.5 A	ABS
26	7.5 A	Radio
27	15 A	ACC Socket (accessory power socket / cigar lighter)
28	20 A	Power Door Lock (if equipped)
29	20 A	Driver's Power Window
30	—	Not Used
31	7.5 A	LAF (Linear Air/Fuel sensor)
32	15 A	DBW (Drive-By-Wire / electronic throttle)
33	15 A	Ignition Coil

Position No. 27 is your accessory power socket (cigar lighter) fuse — the one to check first if a phone charger or other plug-in accessory stops working. Position No. 18 is listed twice in the manual because it is an equipment-dependent slot: on a car with a Tire Pressure Monitoring System, it is a 7.5 A TPMS fuse; on a Canadian-market car, it is a 10 A daytime running light fuse instead. A US base-trim 2008 Fit without TPMS and without Canadian daytime running lights would not use either circuit at that position, so treat position 18 as reserved rather than assuming one fixed value.

Fuse table — Primary under-hood fuse box

This box has 16 numbered positions (No. 1-16), and all 16 are occupied — there are no "Not Used" slots in this box.

Fuse #	Amperage	Circuit(s) protected
1	80 A	Battery
2	60 A	EPS (Electric Power Steering)
3	50 A	Ignition
4	30 A	ABS
5	40 A	Blower Relay (HVAC blower motor)
6	40 A	Power Window
7	30 A	HAC Option (Hill-start Assist Control, if equipped)
8	10 A	Back Up
9	30 A	Small Light (parking/side-marker lights)
10	30 A	Cooling Fan
11	30 A	Condenser Fan, MG Clutch (A/C, if equipped)
12	20 A	Right Headlight
13	20 A	Left Headlight
14	10 A	Hazard (hazard warning flashers)
15	30 A	ABS F/S
16	15 A	Horn, Stop (stop/brake lights)

Fuse table — Secondary fuse box (on the battery)

This box holds exactly one fuse.

Fuse #	Amperage	Circuit(s) protected
1 (only fuse)	80 A	Battery

Checking and replacing fuses

Always replace a blown fuse with one of the exact same amperage rating shown in the tables above. Replacing a fuse with a higher-rated one greatly increases the chance of damaging your electrical system. If you do not have the correct-rated fuse on hand, the manual says to install a lower-rated one instead of a higher one, and to put in the correctly-rated fuse as soon as you can — a fuse with a lower rating may blow again on its own, which by itself does not mean something else is wrong.

Do not keep replacing a fuse that blows again in a short time with another fuse of the same rating. The manual says this points to a serious electrical problem. Leave the blown fuse in that circuit and have the car checked by a qualified technician instead.

To check whether a fuse is blown: (1) turn the ignition switch to LOCK (0) and make sure the headlights and other accessories are off; (2) remove the cover from the fuse box; (3) for the large fuses in the under-hood boxes, look through the top of the fuse at the wire inside; (4) for the smaller fuses in the primary under-hood box and all fuses in the interior box, pull each one out with the fuse puller; (5) look for a blown wire inside — a blown fuse's wire is broken or melted. Check the fuse listed for the affected device first, but check all the fuses before you decide a blown fuse is the cause.

A fuse puller is stored in the interior fuse box. Use it on the smaller fuses in the interior box and the smaller fuses in the primary under-hood box. The larger under-hood fuses are checked by looking through the top of the fuse, and removed with a Phillips-head screwdriver instead of the puller.

If you pull the radio fuse, or if the battery is disconnected, the audio system disables itself as an anti-theft feature. The next time you turn the radio on, "ENTER CODE" appears in the frequency display, and you must enter the audio system's 5-digit code using the radio's preset buttons to turn it back on. That code should be on a radio code card that came with the vehicle. You get

10 tries; after 10 failed attempts, you must leave the radio powered on for 1 hour before trying again. If the code card is lost, a dealer can look up the code using the audio system's serial number.

Relay map

We could not confirm this for the 2008 Fit. Check your owner's manual. The owner's manual documents fuses in detail, but it does not contain a separate relay location diagram, relay numbering, or a relay amperage/function chart anywhere in its "Taking Care of the Unexpected" chapter, its Specifications section, or its index. A physical relay box may still exist on the car, as it does on many similar cars, but no owner-facing map of it was found in this source.

APPENDIX E

Tire and wheel fitment table

This appendix gathers the numbers you need to buy tires or wheels that actually fit your 2008 Fit base trim. A few terms are defined here since they show up together for the first time. **Lug pattern** (also called bolt pattern or PCD) describes how many wheel bolts there are and the diameter of the circle they sit on — a wheel with the wrong lug pattern will not bolt onto your car at all. **Center bore** is the size of the round hole in the middle of the wheel that centers it on the hub; a wheel with too large a center bore needs a plastic ring (a hub-centric ring) to sit properly, and one with too small a bore will not fit over the hub at all. **Offset** describes whether the wheel's mounting surface sits toward the inside or outside of the wheel compared to its centerline, measured in millimeters; changing offset changes how far the tire sticks out from (or tucks into) the fender. **Cold tire pressure** means the pressure measured before you have driven the car, while the tires are still at outside-air temperature — driving heats up the air inside a tire and raises the reading, so checking "cold" gives you a consistent, accurate number.

Item	Value	Confidence / tier	Fact ID
Lug pattern (bolt pattern / PCD)	4x100 mm (4 lugs, bolt circle 100 mm across)	Medium / Tier 3, single source, no independent second source found	F-WP10-013
Center bore	56.1 mm	Medium / Tier 3, single source, no independent second source found	F-WP10-014
Factory wheel offset, base trim	ET45 (offset range given as 43-47 mm for the factory 5.5J x 14 wheel with the 175/65R14 tire)	Medium / Tier 3, single source; two other attempts to confirm this independently either failed to load or contradicted themselves and were not treated as a genuine competing claim	F-WP10-015
Factory tire size, base trim	P175/65R14 81S	High / Tier 1 (owner's manual), matches Appendix A	F-WP10-011
Factory wheel size, base trim	14 in x 5.5J (14 x 5 1/2JJ), steel wheel with full wheel covers	High / Tier 1 (owner's manual), matches Appendix A	F-WP10-012
Cold tire pressure, base trim, front and rear	32 psi (220 kPa, 2.2 kgf/cm ²) — the same number front and rear	High / Tier 1 (owner's manual)	F-WP10-010
Cold tire pressure, compact spare tire	60 psi (420 kPa, 4.2 kgf/cm ²)	High / Tier 1 (owner's manual)	F-WP10-010
Largest tire that fits without rubbing (community-reported)	195/50R15, on a 15x6.5 or 15x7 wheel with a +40 to +42 mm offset, is the size two independent online communities agree "fits without rubbing" at	Low / Tier 4, two independent community sources, both discussing this generation of Fit generally rather than confirming base trim	F-WP10-016

Item	Value	Confidence / tier	Fact ID
	stock ride height. A wider 205/45R16 (on a 16x7 wheel) is reported to work only with a high offset (+42 mm minimum), and even then some rubbing was reported at the rear, especially with passengers in the back seat — treat 205/45R16 as a borderline size, not a clean fit. One source also suggested a narrower 195/45R16 as a safer 16-inch alternative.	specifically against Fit Sport	

Note on the Fit Sport's numbers, for comparison only: the 2008 Fit Sport uses a different factory tire size, P195/55R15 84H, and a different factory wheel, 15 x 6JJ — both larger than the base trim's numbers above. These Sport-trim numbers are not this book's subject car but are included here so you do not confuse a Sport parts listing for a base-trim one.

APPENDIX F

Part numbers for common maintenance items

This appendix lists the Honda genuine part numbers, sizes, and specifications for the maintenance items you are most likely to buy for your 2008 Fit base trim: filters, spark plugs, the battery, wiper blades, and light bulbs. A **part number** is the unique code a manufacturer or parts store uses to identify one exact part, so you can order the correct one instead of guessing from a picture. A **trade number** is a standardized code (like "7443" or "T10") that many bulb makers use for the same size and shape of light bulb, so the same trade number generally works across different car brands. Where this project's research could not find a real part number, that is written out honestly below rather than left blank or guessed — do not buy a part based on a made-up number; ask a Honda parts counter or dealer, or bring your old part in for a match.

Filters

Item	Part number	Notes	Fact ID
Engine air filter	Honda 17220-PWA-505	Replaces older part number 17220-PWA-J10. Fitment confirmed for 2007-2008 Fit.	F-WP09-040
Cabin (dust and pollen) filter	Honda 80291-SAA-J01	Later superseded by part 80291-SAA-316. Fitment confirmed for 2007-2008 Fit. Present on models with air conditioning.	F-WP09-042
Oil filter	UNVERIFIED. No Tier 1 or Tier 2 source with a confirmed part number for this engine's oil filter was found during this project's research. Do not guess a number — ask your parts counter for "2008 Honda Fit 1.5L oil filter," or bring your old filter in for a match.	Not applicable	P-WP11-1

Spark plugs

Item	Part number	Notes	Fact ID
Spark plug (NGK brand)	NGK IZFR6K-13	Iridium-type plug per retailer listings; the owner's manual's own specifications table gives the bare part number without the word "iridium."	F-WP09-036
Spark plug (Denso brand)	DENSO SKJ20DR-M13	Genuine-equivalent to the NGK part above; either one is correct for this engine.	F-WP09-036
Spark plug gap	1.2-1.3 mm (0.042-0.051 in), standard/new	Confirm against the physical plug's own box or stamped specification before use; a second, independent web summary gave slightly different, overlapping numbers.	F-WP09-037

Battery

Item	Value	Notes	Fact ID
Battery group size	151R	This is the original-equipment size for the 2008 Fit. Group 51R is a commonly reported upgrade with more cold-cranking amps, but it is a noticeably tighter physical fit, and some owners report needing to modify the battery tray to use it. This number was not independently confirmed by a Honda document during this research pass, so treat it as medium confidence rather than certain.	F-WP09-046

Wiper blades

Item	Size	Fact ID
Front, driver's side	24 in	F-WP09-047
Front, passenger's side	16 in	F-WP09-047
Rear (hatchback wiper)	14 in	F-WP09-047

Note: the 2009-2014 Fit (a different car, chassis GE8, covered in this project's rule D8) uses different wiper sizes — 28 in / 14 in / 14 in — which confirms the sizes above are genuinely specific to your 2007-2008 first-generation Fit and not a general "Fit" number that also fits the later car.

Bulbs

Light	Bulb type / trade number	Fact ID
Headlight (high/low beam)	HB2 (the same physical bulb is also commonly sold under the names 9003 or H4 at parts stores; confirm the cross-reference against your own bulb's markings before buying), halogen, 12V, 60/55W	F-WP09-048
Fog light (Sport-model or accessory-equipped cars)	H11, halogen	F-WP09-049
Front turn signal	Amber, 12V, 21W (no trade number found; the owner's manual gives only the wattage)	F-WP09-050
Rear turn signal	7443 (aftermarket cross-reference; the owner's manual gives only the 21W wattage, not this trade number)	F-WP09-051
Stop/taillight	7443 (dual-filament, combined stop and tail function)	F-WP09-052
Back-up light	7443	F-WP09-053
License plate light	T10	F-WP09-054
High-mount brake light	UNVERIFIED as a trade number. The owner's manual confirms this is a separate, owner-replaceable bulb and gives its wattage as 5W, but no source found during this research prints a trade number for it (some other cars' high-mount brake lights are a sealed LED strip with no replaceable bulb at all, but that was not confirmed one way or the other for this exact car). We could not confirm a trade number for this bulb. Check your owner's manual, or bring the	F-WP09-055

Light	Bulb type / trade number	Fact ID
	old bulb to a parts counter for a match.	
Front side marker / parking light	T10	F-WP09-056
Rear side marker	T10	F-WP09-057
Dome/ceiling light	C5W, 31 mm festoon-type bulb	F-WP09-058
Cargo/luggage compartment light	T10	F-WP09-059

Other parts flagged as UNVERIFIED during this project's research

These items come up in maintenance procedures elsewhere in this book, but no confirmed part number was found for any of them. They are listed here so you know to ask, rather than assume no part exists:

Item	Status	Source
Oil drain plug crush washer	UNVERIFIED. No confirmed part number found. Buy from a Honda parts counter or dealer; this is a small, inexpensive washer that should be replaced every oil change.	P-WP11-1
Front brake pads, base trim	UNVERIFIED. No Tier 1 or Tier 2 source with a confirmed part number was found. Ask a Honda parts counter, or bring your old pads in for a match.	P-WP11-7
Rear brake shoes, base trim (drum brakes)	UNVERIFIED. No Tier 1 or Tier 2 source with a confirmed part number was found. Ask a Honda parts counter, or bring your old shoes in for a match.	P-WP11-8
Replacement wheel lug nuts	UNVERIFIED. No confirmed part number found. Ask a Honda parts counter for the correct lug nut if one is damaged or lost.	P-WP11-5

APPENDIX G

Glossary

This glossary defines every technical term used in this book, in one sentence each, at a level a new driver can follow. Terms are alphabetized. Where the dashboard or the owner's manual uses an abbreviation, such as "MIL" or "TPMS," the full name is given first.

ABS (Anti-lock Brake System): A safety system that pumps the brakes very fast on its own so your wheels don't lock up and skid when you brake hard.

Accessory position: The ignition key position that powers things like the radio and the power outlet without starting the engine.

Aftermarket: A part or accessory made by a company other than Honda, instead of by the car's original maker.

AKI (octane rating): The number on a gas pump's label that tells you how well that fuel resists engine knocking; the 2008 Fit needs at least 87 AKI.

Alignment: The process of adjusting a car's wheel angles — camber, caster, and toe — so it drives straight, handles predictably, and wears its tires evenly.

Alignment rack: A specialized shop machine that measures a car's wheel angles precisely and then adjusts them.

Alloy wheels: Wheels made from a lightweight metal blend instead of plain steel; the 2008 Fit's base trim has steel wheels with covers instead.

Aspect ratio: The tire's sidewall height, given as a percentage of the tire's width; it is the middle number in a tire size like 195/50R15.

Auto-down window: A power window feature where one firm push on the switch lowers the whole window by itself, without you holding the switch down.

Autocross: A timed driving event, usually held in a large parking lot or empty lot, where drivers race one at a time against the clock through a course marked out with cones.

Automatic transmission: A transmission that changes gears for you on its own, so there's no clutch pedal to press.

Auxiliary (aux) jack: A small audio plug on some car stereos that lets you plug in an outside music player; the US base 2008 Fit does not have one.

Axle-back exhaust: A replacement exhaust part that swaps out only the last section, from about the rear axle back to the tailpipe, usually just to change the sound.

Bill of sale: The dealer's or private seller's own sales contract for a car, showing the price, taxes, and fees.

Bleeder screw: A small screw at each brake that a wrench opens to push old fluid and trapped air out of the brake lines.

Bolt pattern (lug pattern): The number and spacing of the bolt holes on a wheel, which must match the studs on the car for the wheel to bolt on correctly.

Boost: Extra air pressure that a turbocharger or supercharger pushes into the engine, above normal air pressure, to help the engine make more power.

Bore and stroke: Two engine measurements — how wide each cylinder is (bore) and how far the piston travels inside it (stroke).

Brake fluid: The hydraulic fluid that transfers the force from a driver's foot on the pedal to the pads or shoes at each wheel.

Brake shoes: The curved parts inside a drum brake that press outward against the inside of the drum to slow the wheel.

Brake-parts cleaner: A solvent spray made for cleaning brake dust off brake parts during a pad or drum brake job.

Break-in period: The first several hundred miles of driving a new car, when the maker asks you to drive gently so engine parts can settle in properly.

Breaker bar: A plain, longer handle with no ratcheting action, used with a socket when a bolt is stuck and needs extra force.

Buyers Guide: A federally required window sticker on a used car at a dealer's lot that discloses the car's warranty status and major systems.

C-clamp: A tool that pushes a brake caliper's piston back into its housing to make room for new brake pads.

Cabin air filter: A filter, also called the dust and pollen filter, that cleans the air coming through a car's heater and air conditioner vents.

Caliper: A clamp that squeezes flat brake pads against both sides of a disc brake's rotor to slow the wheel.

Camber: How much a tire tilts inward or outward at the top when you look at the car from the front.

Campaign number: The specific ID number NHTSA assigns to one safety recall, used to look that exact recall up online.

Canard: A small, wing-shaped piece bolted to the front bumper that is meant to press the front of a fast-moving car down slightly.

CARB (California Air Resources Board): The California state agency that adds its own emissions rules on top of federal ones, mainly for cars titled or registered in California.

CARB EO number: A number the California Air Resources Board gives to an aftermarket part after testing, showing that part can legally be used on a car driven on public roads in California.

Cargo volume: How much space is available for luggage or gear, usually measured in cubic feet.

Caster: The tilt of the steering axis when viewed from the side of the car, which affects how the steering wheel returns to center and how stable the car feels driving straight.

Cat-back exhaust: A replacement exhaust system that starts right after the catalytic converter and runs back to the tailpipe, usually installed to change the exhaust sound or let gases flow out easier.

Catalytic converter: A part in the exhaust system that cleans harmful gases out of the exhaust before they leave the car.

Center bore: The hole in the middle of a wheel that fits around a matching part on the car's hub to help keep the wheel centered.

Certification label: A sticker on the driver's doorjamb that lists your VIN, along with the recommended tire sizes and tire pressures.

Charging system: The parts, mainly the alternator, that keep the car battery topped up while the engine is running.

Chassis code: A short factory code, like GD3, that identifies exactly which body and underlying platform a specific car was built on.

Check Fuel Cap message: A message that appears on the dashboard display when the gas cap is loose or missing.

Childproof lock: A small lever on a rear door that stops that door from being opened using the inside handle.

Clean Air Act The federal law that sets emissions rules for every car sold in the United States and makes it illegal to remove or disable a car's emissions-control devices.

Clutch (aftermarket): A stronger replacement clutch, built to grip harder than the factory clutch so it can handle more engine power without slipping.

Coilover: A shock absorber and spring built together as one adjustable unit, often letting an owner change the car's ride height and how stiff it feels.

Cold air intake: A replacement air-intake system meant to pull in cooler outside air instead of the warmer air already inside the engine bay, which can add a small amount of power.

Combination switch: The single lever on the steering column that also controls the turn signals and headlight high/low beams.

Compact spare tire: A small, lightweight temporary tire (sometimes called a "donut") meant only for short trips at reduced speed until the real tire is fixed.

Complaint (NHTSA): A report a vehicle owner files with the government describing a problem with their car, whether or not it turns into a recall.

Compression ratio: A number describing how much an engine squeezes the air-fuel mixture before it ignites it.

Consumer notice: The letter a manufacturer mails to a car's registered owner explaining a recall and what to do about it.

Cool-down lap: One slow, gentle lap driven at the end of a track session so the brakes, engine, and transmission cool down gradually instead of being parked while still very hot.

Cranks but won't start: A no-start problem where the starter spins the engine at a normal speed, but the engine never fires up and keeps running on its own.

Cruise control: A feature that holds the car at a steady speed on its own, so the driver doesn't need to keep a foot on the gas; not included on the 2008 Fit's base trim.

Curb weight: How much the empty car weighs with a full tank of gas but no passengers or cargo.

D3 (gear position): An automatic-transmission setting that works like Drive but keeps the transmission from shifting into 4th or 5th gear, useful for engine braking downhill.

Damper: The shock absorber inside a coilover, which controls how quickly the spring compresses and rebounds.

Daytime Running Lights (DRL): Lights that turn on by themselves whenever the engine is running, making the car easier to see in daylight; not fitted on the US 2008 Fit.

Defeat device: An aftermarket part built mainly to bypass, defeat, or disable a device that controls a car's exhaust emissions, which is illegal to manufacture, sell, or install.

Destination charge: A flat fee added to a new car's price to cover shipping it from the factory to the dealer.

Diagnostic fee: The charge a repair shop bills for the time it spends finding what is wrong with a car, before any repair is agreed to.

Diagnostic Trouble Code (DTC): A short code the car's computer stores when it detects a problem, which a mechanic reads with a scan tool.

Differential: The gearbox part that splits engine power between a car's drive wheels so they can spin at different speeds through a turn.

Directional tire: A tire with a tread pattern designed to roll only one way, marked with an arrow on its sidewall.

Disc brake: A brake that uses a flat metal disc, called a rotor, spinning with the wheel, squeezed by a caliper to slow it down.

Displacement: The total volume swept by all of an engine's cylinders, usually given in liters, which roughly shows how big the engine is.

Doorjamb: The frame area you see when a door is open; the certification label and VIN sticker are usually placed there.

DOT brake fluid rating: A number (like DOT 3, DOT 4, or DOT 5.1) that a federal safety standard uses to set a brake fluid's minimum boiling point; a higher DOT number does not always mean a higher boiling point for every specific product.

Downpipe (test pipe): The section of exhaust pipe right after the turbocharger or catalytic converter; a "test pipe" is a downpipe with the catalytic converter removed, usually meant only for track use.

Drum brake: A brake in which a hollow metal drum spins with the wheel while curved brake shoes press outward against the inside of the drum to slow it down.

Dry boiling point: The boiling point of fresh, unused brake fluid.

Dyno (dynamometer): A machine that measures how much power an engine or a car's wheels actually produce.

ECU (Engine Control Unit): The car's onboard computer that decides things like how much fuel to add and when to fire the spark plugs, based on readings from its sensors.

Electric Power Steering (EPS): A power-steering system that uses an electric motor, instead of hydraulic fluid pressure, to help you turn the steering wheel.

Electrolyte: The liquid inside a car battery that lets it store and release electricity; in very cold weather it can thicken or freeze.

Electronic Brake Distribution (EBD): A system that automatically balances how much braking force goes to the front wheels versus the rear wheels.

Emissions control system: The group of parts that work together to reduce pollution coming out of the engine and exhaust.

Endorsement (insurance): An add-on to an existing insurance policy, sometimes called "custom parts and equipment coverage," that covers modified parts a factory policy would not.

Engine air filter: A filter that sits in the engine bay and cleans the air the engine breathes before that air reaches the fuel injection system.

Engine swap: Removing a car's original engine and installing a different one, usually to get more power or better reliability.

EPA (Environmental Protection Agency): The US government agency that tests and publishes the official fuel-economy (MPG) numbers for new cars.

Fast idle: Running an engine a little faster than normal idle speed, often done to a booster vehicle's engine during a jump start.

FIA (Fédération Internationale de l'Automobile): An international motorsports organization whose safety standards, such as harness and helmet certifications, are often required at racing and track events.

Final drive ratio: An extra gear ratio built into the transmission that affects how the engine's speed relates to the wheels' speed in every gear.

Flat-tip (flathead) screwdriver: A screwdriver with a flat blade tip, used along with a Phillips screwdriver for the clips and screws on things like an air filter housing.

Flat-tow: Towing a car with all four wheels rolling on the ground behind another vehicle; the 2008 Fit's owner's manual does not support this method for either transmission.

Flatbed towing: Towing a car by loading all four wheels onto the back of a truck, so none of the car's wheels touch the road.

Floor jack: A larger tool on wheels, rated to hold much more weight than a trunk's scissor jack, used to lift a car for jack-stand work in the garage.

Flywheel: A heavy spinning disc attached to the engine that smooths out its power pulses; a lighter aftermarket flywheel lets the engine change speed faster.

Fog lights: Extra, low-mounted lights that help light the road edge in fog; standard on the Fit Sport trim only, not on base.

Friction point: The spot while slowly releasing a clutch pedal where the engine and transmission start to connect, and the car begins to move.

Fuel vapor recovery system: A system that catches gasoline vapor at the pump instead of letting it escape into the air; a fault in it can make the pump nozzle click off early and repeatedly.

Gear ratio: A number describing how many times the engine turns for every one turn of the wheels in a given gear; smaller numbers generally mean higher top speed, larger numbers mean more pulling power.

Ground clearance: The distance between the lowest point of the car's underside and the ground.

Ground point ("the stay"): A metal point in the engine bay, away from the battery itself, used as the final connection spot when jump-starting a car to reduce spark risk.

H Street (HS): An entry-level SCCA autocross class, meant for cars that are still mostly stock, with only limited changes like tires and shocks allowed.

Harness (racing harness): A multi-point seat belt system, with straps over both shoulders and between the legs, used in place of a factory 3-point belt for extra crash protection.

Harness bar: A bar bolted across the car behind the front seats, used to anchor a racing harness's shoulder straps.

Hazard warning lights: All four turn signals flashing together at once, used when parked in a dangerous spot or when a car is disabled.

Head restraint: The padded part above a seatback that supports your head and helps prevent whiplash in a crash; often just called a headrest.

Header: A set of pipes that connects each engine cylinder to the rest of the exhaust system; a performance header is shaped to let exhaust gases flow out more easily than the factory manifold does.

High beam: The brighter, longer-reaching headlight setting meant for dark roads with no oncoming traffic nearby.

Hood release: The handle inside the car, usually under the dashboard, that pops the hood partway open.

Horsepower: A measurement of how much power an engine can produce.

HPDE (High Performance Driving Event): A track day where drivers practice driving skills at speed with instruction, rather than racing wheel-to-wheel against other cars.

i-VTEC / VTEC: Honda's system for adjusting how and when engine valves open, which helps the engine make good power and stay efficient across a range of speeds.

IIHS (Insurance Institute for Highway Safety): An independent group, funded by insurance companies, that crash-tests vehicles and rates how well they protect the people inside.

Immobilizer: An anti-theft system that keeps the engine from starting unless it recognizes the electronic chip built into your key.

Intercooler: A small radiator-like part that cools down air after a turbocharger or supercharger has heated and pressurized it, before that air enters the engine.

Investigation (NHTSA): A formal government review into a possible safety defect, which may or may not lead to an official recall.

Jack stands: Metal supports slid under a lifted car so its weight rests on solid steel instead of on the jack itself.

Jacking point: A specific spot marked on the underside of a car's body where it is safe to place a jack.

Jump start: Starting a car with a dead battery by temporarily connecting jumper cables to a charged battery in another vehicle or a portable booster pack.

Keyless entry: A remote fob that lets you lock or unlock the doors by pressing a button instead of using the physical key; not included on the base 2008 US Fit.

Long Mode: One of the Fit's Magic Seat cargo layouts, used for sliding long items like lumber or a surfboard through to the front of the car.

Low beam: The normal, everyday headlight setting used whenever other traffic is nearby.

Lug wrench: A socket that fits a car's wheel nuts, paired with a breaker bar, used any time a wheel is taken off.

Magic Seat: Honda's name for the Fit's flexible 60/40-split rear seat, which can fold flat or flip up into several different cargo and passenger layouts.

Maintenance Minder: A dashboard system that tracks the car's condition over time and tells you, with a wrench-shaped light and a code, when a specific service is due.

Malfunction Indicator Lamp (MIL): The dashboard light — often called the "check engine light" — that comes on when the emissions-control system detects a problem.

Milliamp (mA): A small unit of electrical current, used to describe how much power a parked car's electrical system draws with everything off.

MMT: A manganese-based fuel additive that Honda recommends avoiding where possible because it can cause the check-engine light to come on.

Modding: Changing a car from how the factory built it, usually to improve how it handles, stops, sounds, or looks.

Moderate overlap front test: A crash test where only part of the front of the car strikes a barrier, similar to many real-world head-on crashes that aren't perfectly centered.

Modified category (SCCA): A group of SCCA autocross classes (labeled A through F Modified) built for cars taken far past a factory street car, closer to a dedicated race car; a roll bar or cage is mandatory in these classes.

Motor mount: A rubber-and-metal part that holds the engine and transmission in place and absorbs vibration; a worn one can cause a clunk when shifting gears.

MPG (miles per gallon): A measurement of how far a car can travel using one gallon of fuel.

MSRP (Manufacturer's Suggested Retail Price): The price a car's maker suggests dealers charge, before tax, fees, and any negotiating.

MTBE: A fuel additive some gasoline blends contain; the Fit's manual allows fuel with up to 15% MTBE by volume.

NHTSA (National Highway Traffic Safety Administration): The US government agency that manages vehicle recalls, safety ratings, and owner complaint records.

Occupant Detection System (ODS): Sensors that check the front passenger seat for a person, a child seat, or nothing at all, to help decide whether the airbag should go off.

Occupant Position Detection System (OPDS): Sensors that check exactly where a front passenger is sitting, so the airbag can be adjusted or turned off if someone is sitting too close to it.

Odometer: The gauge that shows the total distance a car has been driven over its whole life.

Odometer disclosure statement: The mileage reading at the time of a car's sale, signed by the seller, which federal law requires for most cars under 20 model years old.

OEM (Original Equipment Manufacturer) part: A replacement part made by, or to the same standard as, the part the car originally came with from the factory.

Oil filter wrench: A tool shaped to grip the round oil filter so it can be twisted loose.

Overlap crash test: A type of crash test that hits only part of a vehicle's front end, rather than the whole width, to better match many real-world crashes.

Paddle shifters: Levers mounted behind the steering wheel that let a driver manually bump an automatic transmission up or down a gear; found on the Fit Sport's automatic only, not on base.

Parasitic draw: A small amount of electrical current a parked, switched-off car still uses; too much of it drains the battery even when the car is left alone.

Parking brake: A separate brake, set by hand, that holds a parked car in place; also called an emergency brake.

Piggyback (tuning device): A small aftermarket box that changes the signals going to the factory ECU without replacing it, often used to adjust the fuel mixture.

Powertrain: The engine, transmission, and connected parts that work together to move the car.

Pre-purchase inspection (PPI): A check of a used car before you buy it, done by you or by a paid independent mechanic, covering the body, tires and brakes, engine bay and underside, interior and electronics, and a test drive.

Prepared category (SCCA): An SCCA autocross category for cars modified well past Street Prepared; an open car (no factory roof) in this category must have a roll bar.

Ratchet: A handle that turns a socket over a bolt head in one direction without lifting off between turns.

Readiness codes: Internal checks the emissions computer runs to confirm all its own monitoring systems have finished testing; some states check these during an emissions test.

Recall (safety recall): A free, manufacturer-run repair program ordered after a safety-related defect is found in a group of vehicles.

Redline: The highest engine speed considered safe to reach; going past it risks damaging the engine.

Refresh Mode: A Magic Seat layout where the front seats recline flat to rest, usable only while the car is parked.

Registration: The state's record that a car is legally allowed on the road.

Reverse lockout: A mechanism in a manual transmission that stops the driver from shifting straight from the highest gear into reverse without pausing.

Ride height: How high the bottom of the car sits above the ground; lowering it usually helps handling but reduces ground clearance.

Roll bar: A metal bar or hoop mounted behind the seats to help keep the roof from collapsing onto the people inside during a rollover crash.

Roll cage: A full frame of metal tubing built inside a car to protect the people inside during a serious crash; more extensive than a simple roll bar.

Rollover rating: A safety score estimating how likely a specific vehicle is to roll over in a single-vehicle crash.

Rotor: The flat metal disc that spins with the wheel in a disc brake and is squeezed by the caliper's pads to slow the car.

RPM (revolutions per minute): How fast the engine is spinning, shown on the tachometer.

Rub (tire rub): When a tire touches the fender, a suspension part, or a body panel while turning or going over a bump.

SAE Class S: A size and type classification for tire chains; the Fit's manual only allows this cable-type chain, and only on the front tires.

Salt-belt states: A group of US states where winter road salt is heavily used, which some recalls (like rust or corrosion-related ones) apply to instead of the whole country.

Security system: An anti-theft alarm system, standard on the Fit Sport trim only, not on the 2008 US base Fit.

Self-adjuster (adjuster): A small threaded part inside a drum brake that takes up slack as the brake shoes wear down, keeping the brake working correctly without manual adjustment.

SFI (SFI Foundation): A US organization that tests and certifies motorsports safety equipment, such as racing harnesses and helmets, to a set standard.

Shift interlock (shift lock): A safety mechanism that keeps an automatic transmission's shift lever locked in Park unless the brake pedal is pressed.

Shift lever: The lever a driver moves to choose a gear, in either a manual or an automatic transmission.

Side airbag: An airbag that deploys out of the seat or door area to protect against a side-impact crash.

Splitter: A flat panel added to the front or sides of a car's bumper, meant to help manage airflow underneath the car.

Spoiler: A wing-shaped piece mounted on the trunk or hatch; on a light, slower car this is mostly for looks, though on faster cars it can help push the rear tires down.

Spring (suspension): The coiled metal part that holds a car's weight up and compresses when a wheel hits a bump.

Spring rate: A number describing how stiff a spring is, based on how much force it takes to compress it a set distance; a higher number means a stiffer, firmer ride.

SRS (Supplemental Restraint System): The umbrella term for a car's airbags and their related crash sensors, working alongside the seat belts.

Stage 0: The step before any performance modification, meaning bringing a car's existing maintenance — fluids, tires, and brakes — up to date before spending money on power, handling, or looks.

Staggered (fitment): A wheel-and-tire setup where the front and rear tires are different sizes on purpose.

Stainless brake lines: Brake hoses wrapped in a woven stainless-steel sleeve instead of plain rubber, which resist expanding under hard, repeated braking.

Standalone engine management: A complete aftermarket computer that fully replaces the factory ECU, instead of working alongside it.

Street Modified: An SCCA autocross class, one step past Street Prepared, that allows major aerodynamic parts and other extensive changes on cars still built from a factory street car.

Street Prepared: An SCCA autocross class that allows bigger changes than Street Touring, such as race-only tires, engine tuning, and body modifications, and is no longer necessarily street-legal.

Street Touring: An SCCA autocross class that allows more modifications than Street classes, like aftermarket suspension, brakes, and wheels, while the car must still stay street-legal.

Supercharger: A device, driven by a belt from the engine, that forces extra air into the engine to help it make more power.

Sway bar: A suspension part connecting the left and right wheels that reduces how much the car's body leans in turns.

Synchronized transmission: A manual transmission design that lets gears mesh smoothly during a normal shift, without extra tricks like a double-clutch.

Tachometer: The gauge that shows how fast the engine is spinning, in RPM.

Takata airbag inflator: A now-recalled type of airbag part, made by the Takata company, that can explode and send metal fragments into the car after long exposure to heat and humidity.

Tall Mode: A Magic Seat layout where the rear seat cushions flip up, opening floor space for tall items.

Tech inspection: A safety check of a car, done before a track day or race, that looks at things like tires, brakes, fluids, and seat belts.

Technical Service Bulletin (TSB): A repair instruction a manufacturer sends to its own dealers about a known problem; it is not a safety recall and is not always a free repair.

Time Trial: A track event where each car is timed individually on an open track, instead of racing wheel-to-wheel against other cars at the same time.

Tire chain: A traction device wrapped around a tire to improve grip on snow or ice.

Title (vehicle title): The legal document that proves who owns a car; it should never be left in the glovebox.

Toe: How much a car's tires point inward or outward, viewed from above, compared to pointing straight ahead.

Torque: A twisting force; for a car this means both the pulling power an engine produces and the exact tightness a bolt or nut must be tightened to.

Torque wrench: A tool that tells you exactly how hard you are tightening a bolt, by feel or by a dial, measured in pound-feet or newton-meters.

Torsion beam: A simple, solid rear suspension design, used on the Fit, where a single beam connects the two rear wheels instead of letting them move fully independently.

TPMS (Tire Pressure Monitoring System): A system that watches tire pressure and warns the driver on the dashboard if a tire gets significantly low.

Trail braking: Easing off the brake pedal gradually while turning into a corner, instead of releasing the brake fully before starting to turn.

Tread wear indicator: A raised bar built into a tire's grooves that becomes visible once the tire is worn down and needs replacing.

Trim level: A specific version of a car model, such as "base" or "Sport," that comes with its own defined set of standard features.

Trip meter: A mileage counter, separate from the odometer, that a driver can reset to track the distance of a single trip.

Turbocharger: A device powered by the engine's own exhaust gases that forces extra air into the engine to help it make more power.

Unsprung weight: The mass of a car's wheels and tires that the springs do not support, since it sits below them and moves with the road surface.

Utility Mode: A Magic Seat layout where the rear seatback(s) fold down flat to make a larger, flat cargo floor.

VDS (Vehicle Descriptor Section): The middle part of a VIN that encodes details like the car's platform, engine, and body type.

VIN (Vehicle Identification Number): A unique 17-character code stamped on every car that identifies its maker, features, and production details.

Warranty extension: A manufacturer's decision to cover a specific repair for longer than the original warranty period, usually issued after a known problem turns up.

Wet boiling point: The boiling point of brake fluid after it has absorbed a set amount of water, closer to how the fluid behaves after real-world use.

Wheel chocks: Wedges placed against a tire that stays on the ground so the car cannot roll while another corner is lifted.

Wheel nut: One of the nuts that holds a wheel onto the car; these must be tightened to an exact torque spec, not just "as tight as possible" (also commonly called a lug nut).

Wheel offset: How far a wheel's mounting surface sits from the exact center-line of the wheel, which affects how far the tire sticks out from or tucks under the car.

Wheel-lift towing: Towing a car by lifting only its front (or rear) wheels off the ground on a tow truck's frame, while the other wheels stay on the road.

Wheelbase: The distance between the centers of the front and rear wheels.

Whp (wheel horsepower): Horsepower measured at the wheels, after some power is lost through the drivetrain, instead of measured at the engine itself.

Wing bolt: A bolt with flat, wing-shaped tabs that can be loosened or tightened by hand; used to hold the 2008 Fit's spare tire in its well.

WMI (World Manufacturer Identifier): The first three characters of a VIN, which identify the company and country that built the vehicle.

Won't crank: A no-start problem where turning the key to START produces nothing, or only a click or a slow grinding sound, because the starter is not turning the engine over at a normal speed.

Zone A/B/C recall: A way NHTSA and automakers roll out some large recalls in stages by region (zone) instead of notifying every owner nationwide at once, often based on local climate or humidity risk.

APPENDIX H

Resources

This appendix lists places you can actually go to look something up about your 2008 Fit: official Honda pages, government safety tools, owner communities, parts and fitment lookups, and track/autocross rulebooks. Each entry gives the resource's name, its URL, and one line on what it is good for.

Official Honda

Honda Owners site — <https://owners.honda.com/vehicle-information/information/2008/Fit/manuals> Official Honda site to download your 2008 Fit's owner's manual and look up other vehicle-specific documents.

Helm Inc. — factory service manual — https://www.helminc.com/helm/product2.asp?class_2=AHM&Sku=61SAA01&itemtype=N The official Honda factory service manual distributor; buy the real 2007-2008 Fit (KA) shop manual here if you want the same repair procedures Honda dealers use (paid).

Honda News — 2008 Fit specifications — <https://hondanews.com/en-US/releases/release-2911fc22bc993ed9e80deb004c34be5b-2008-honda-fit-specifications> Honda's own archived press release with the full factory spec sheet (engine, dimensions, weights, gear ratios) for the 2008 Fit and Fit Sport.

hfitinfo.com — <https://www.hfitinfo.com/> A free, page-by-page mirror of the full Honda Fit 2001-2008 (first-generation) owner's manual — useful if you don't want to deal with Honda's own PDF download flow.

ManualsLib — 2008 Fit Owner's Manual — <https://www.manualslib.com/manual/524850/Honda-2008-Fit.html> A free page-image/OCR viewer of the actual Honda-published 2008 Fit owner's manual, handy for searching a specific page or section.

Government / Safety

NHTSA vehicle lookup (recalls) — <https://www.nhtsa.gov/vehicle/2008/HONDA/FIT> Look up every official safety recall ever issued for your exact car by VIN or by year/make/model — always check this before assuming your car has had all its recall repairs done.

NHTSA vehicle lookup (complaints and investigations) — <https://www.nhtsa.gov/vehicle/2008/HONDA/FIT> The same NHTSA tool also shows owner-submitted complaints and any formal government investigations for the 2008 Fit, which can reveal problems before they become a recall.

IIHS ratings — 2008 Honda Fit — <https://www.iihs.org/ratings/vehicle/honda/fit-4-door-wagon/2008> Independent, insurance-industry-funded crash test ratings and results for the 2008 Fit 4-door wagon.

fuelconomy.gov — 2008 Honda Fit — <https://www.fueleconomy.gov/feg/Find.do?action=sbs&id=24920> The official US government fuel-economy page for the 2008 Fit, with EPA city/highway/combined MPG for both transmissions.

Community forums

FitFreak.net — 1st Generation (GD 01-08) forum — <https://www.fitfreak.net/forums/1st-generation-gd-01-08/> The largest dedicated first-generation Fit owner community; go here for real-world troubleshooting threads, "does anyone else have this problem" discussions, and DIY write-ups specific to the GD3 chassis. This is community-reported information only — always double-check anything safety-related against your own owner's manual.

Honda-Tech.com — Honda Fit forum — <https://honda-tech.com/forums/honda-fit-73/> A second large Honda-focused forum with its own active Fit section, useful as a second opinion when a FitFreak thread doesn't have enough replies to be confident in.

Parts and fitment

HondaPartsNow.com — https://www.hondapartsnow.com/2008-honda-fit--5dr_base-ka_5at-parts.html An authorized Honda dealer parts site where you can look up genuine part numbers and diagrams for your exact 2008 Fit trim and transmission.

Wheel-Size.com — **Honda Fit GD [2001-2008]** — <https://www.wheel-size.com/size/honda/fit/gd-2001-2008/> A quick-reference page for factory and compatible wheel/tire sizes, bolt pattern, and offset for the first-generation Fit — useful before buying replacement wheels or tires.

Haynes Repair Manual 42030 (Honda Fit 07-13) — <https://us.haynes.com/products/honda-fit-07-13-haynes-repair-manual> A widely available, affordable aftermarket repair manual covering the 2007-2013 Fit; it spans both the GD3 (2007-2008) and GE8 (2009-2013) generations, so check that any given page applies to your model year before following it (paid).

workshop-manuals.com — **Fit L4-1.5L fluid/spark plug/valve specs** — see the book's source records for the exact reachable URL A free reference page listing fluid capacities, spark plug part numbers, and valve clearance specs for the Fit's 1.5L engine — handy for a quick maintenance lookup without buying a full service manual.

Boslla — **2007-2018 Honda Fit bulb size guide** — <https://www.boslla.com/2007-2018-honda-fit-base-dx-ex-lx-sport-bulb-size-guide/> A free lookup table for every exterior and interior bulb size on the Fit, broken out by trim and model year.

AutoPadre — **Honda Fit battery and wiper blade sizes** — <https://www.autopadre.com/battery-size/honda-fit> and <https://www.autopadre.com/wiper-size/honda-fit> Quick-reference pages for the correct replacement battery group size and wiper blade lengths for the Fit across its model years.

Track / autocross organizations

SCCA — National Solo Rules — <https://www.scca.com/downloads> The Sports Car Club of America's official rulebook for autocross ("Solo") competition, including the class structure a stock 2008 Fit would compete in.

NASA — Club Codes and Regulations (CCR) — <https://nasa-assets.s3.amazonaws.com/document/document/282/2026.3.pdf> The National Auto Sport Association's official rulebook for HPDE (track day) and time-trial events, including safety-gear and tech-inspection requirements.

APPENDIX I

Recall and TSB list

This appendix is the complete, standalone reference for every NHTSA safety recall and Technical Service Bulletin (TSB) confirmed for the 2007-2008 Honda Fit. A **recall** is a free, manufacturer-run repair program ordered after a safety-related defect is found in a group of vehicles. A **TSB** is a repair instruction a manufacturer sends to its own dealers about a known problem; it is not a safety recall and is not always a free repair. Always check your own VIN at a recall lookup tool (see Appendix H), since coverage can depend on where a car was sold or registered, and on whether a part was already replaced under an earlier campaign.

All 8 confirmed NHTSA recalls at a glance

A direct pull of NHTSA's own recalls database for model year 2008 Honda Fit returns exactly 8 campaigns.

Campaign number	Recall began	Component	Fact ID
07V549000	December 12, 2007	Frontal airbag ODS/ OPDS wire harness corrosion (salt-belt states)	F-WP02-027
10V033000	March 19, 2010	Power window master switch water intrusion (1st campaign)	F-WP02-001
10V624000	January 24, 2011	Headlight low-beam wiring harness / combination switch	F-WP02-003
13V260000	June 24, 2013 (approx.)	Power window master switch water intrusion (re-recall)	F-WP02-002
16V344000	June 27, 2016	Passenger frontal airbag inflator, Takata, original-equipment and replacement, Zone A/ B/C	F-WP02-028
17V029000	March 2017	Passenger frontal airbag inflator, Takata, same defect family, later zone expansion	F-WP02-029
19V501000	August 12, 2019 (approx.)	Passenger frontal airbag inflator, Takata, remedy-part-only	F-WP02-004
20V770000	December 10, 2020 (approx.)	Drive shaft corrosion, manual transmission	F-WP02-005

Recall details

07V549000 — Frontal airbag ODS/OPDS wire harness corrosion

- **Recall began:** December 12, 2007
- **Applies to:** Certain 2007-2008 Fit vehicles originally sold, or currently registered, in 21 listed salt-belt states plus Washington, DC: Connecticut, Delaware, Illinois, Indiana, Iowa, Kentucky, Maine, Maryland, Massachusetts, Michigan, Minnesota, Missouri, New Hampshire, New Jersey, New York, Ohio, Pennsylvania, Rhode Island, Vermont, Virginia, West Virginia, and Wisconsin.
- **Component:** Frontal airbags — Occupant Detection System (ODS) / Occupant Position Detection System (OPDS) wire harness.
- **Defect:** The wire harness for the front passenger ODS/OPDS is routed underneath the carpet on the driver's side floorboard. In salt-belt areas, road salt tracked in on a driver's shoes can melt and seep through the carpet, reaching the harness. If the harness corrodes and a wire breaks, the SRS indicator light comes on, and the ODS/OPDS may fail to detect a child seat or an out-of-position passenger, so the passenger's frontal and side airbags will not be suppressed when they should be.
- **Consequence:** In a crash, a deploying front passenger airbag or side airbag increases the risk of injury to small or out-of-position occupants.
- **Remedy:** Dealers inspect and, if necessary, replace the wire harness, free of charge. Contact Honda at 1-800-999-1009.

10V033000 — Power window master switch water intrusion (1st campaign)

- **Recall began:** March 19, 2010 (Honda recall no. R28)
- **Applies to:** Certain model year 2007-2008 Fit, approximately 141,140 units sold in the US. This is an all-trims recall, not base-only.
- **Component:** Power window master switch, driver's door.

- **Defect:** Under severe conditions, rain water or spilled liquid entering through an open driver's window can reach the power window master switch and damage its printed circuit board. A damaged circuit board can short and overheat the switch connectors, which can melt and, in rare cases, produce smoke or fire.
- **Remedy:** Dealer inspects the switch and installs a cover around the switch housing to prevent water entry; if the switch is already damaged, it is replaced and a cover installed — free of charge. Contact Honda at 1-800-999-1009.

10V624000 — *Headlight low-beam wiring harness / combination switch*

- **Recall began:** January 24, 2011
- **Applies to:** 143,083 model year 2007-2008 Fit vehicles in the US. All-trims recall.
- **Component:** Exterior lighting — headlights, high/low beam dimmer (combination) switch and its wiring harness.
- **Defect:** An installation error in the wiring harness for the combination switch caused the low-beam circuit wires to be pulled tighter than intended. Repeated use of the headlight switch can wear the electrical connector terminal for the low-beam circuit, lowering conductivity and letting copper oxide build up at the contacts.
- **Consequence:** Low-beam headlights can become inoperative, reducing the driver's visibility and the vehicle's visibility to other traffic, increasing crash risk.
- **Remedy:** Dealer inspects the low-beam terminal and performs the necessary repair, free of charge. Contact Honda at 1-800-999-1009.

13V260000 — Power window master switch water intrusion (recall)

- **Recall began:** Approximately June 24, 2013; Honda's own recall notice states that by October 18, 2013, replacement parts were in stock in sufficient quantity to complete repairs on all affected units.
- **Applies to:** Certain model year 2007-2008 Fit. All-trims recall.
- **Component:** Power window master switch, driver's door — the same switch as 10V033000.
- **Defect:** Honda determined the 2010 remedy was not sufficient to prevent water intrusion into the switch. Under severe conditions, water can still reach the switch and impair its function, and a damaged switch can overheat and cause smoke, melting, or fire.
- **Remedy:** Dealer replaces the power window master switch, free of charge, using an updated part.

16V344000 — Passenger frontal airbag inflator, Takata, original-equipment and replacement, Zone A/B/C

- **Recall began:** June 27, 2016 (Honda numbers KA0, KA1, KA2, KA3)
- **Applies to:** Certain 2007-2008 Fit vehicles, among many other Honda/Acura models, in a geographic Zone A/B/C recall — the 2007-2008 Fit is named in every zone, though exact notification timing is zone-dependent. This campaign covers **both** original-equipment-installed and replacement passenger frontal airbag inflators.
- **Component:** Airbags, frontal, passenger side, inflator module.
- **Defect:** Passenger frontal airbag inflators, whether original equipment or replacement, may rupture due to propellant degradation after long-term exposure to absolute humidity and temperature cycling.
- **Consequence:** An inflator rupture may send metal fragments toward vehicle occupants, causing serious injury or death.

- **Remedy:** Dealers replace the passenger frontal airbag inflator, free of charge. Contact Honda customer service at 1-888-234-2138.

17V029000 — Passenger frontal airbag inflator, Takata, later zone expansion

- **Recall began:** March 2017 (Honda numbers KD9, KE0)
- **Applies to:** Certain 2007-2008 Fit vehicles, among other Honda/Acura models, in a later geographic Zone A/B/C expansion of the same defect family as 16V344000.
- **Component:** Airbags, frontal, passenger side, inflator module.
- **Defect:** Same as 16V344000 — passenger frontal airbag inflators, original equipment or replacement, may rupture due to propellant degradation from long-term exposure to absolute humidity and temperature cycling.
- **Consequence:** An inflator rupture may send metal fragments toward vehicle occupants, causing serious injury or death.
- **Remedy:** Dealers replace the front passenger airbag inflator, free of charge. Contact Honda customer service at 1-888-234-2138.

19V501000 — Passenger frontal airbag inflator, Takata, remedy-part-only

- **Recall began:** On or about August 12, 2019 (Honda internal numbers E5D and X5C)
- **Applies to:** Certain model year 2007-2008 Fit vehicles specifically equipped with a passenger frontal airbag inflator that was itself installed as a recall remedy or replacement service part — in other words, a Fit that already had its original passenger airbag inflator swapped under an earlier Takata campaign.
- **Component:** Passenger frontal airbag inflator.
- **Defect:** These replacement/remedy inflators may explode due to propellant (phase-stabilized ammonium nitrate) degradation from long-term exposure

to high absolute humidity, high temperature, and temperature cycling — the same underlying defect mechanism as the original mass recall, but affecting the second-round replacement parts.

- **Consequence:** An inflator explosion can send sharp metal fragments toward the driver or other occupants, risking serious injury or death.
- **Remedy:** Dealer replaces the passenger frontal airbag inflator again, free of charge. Contact Honda customer service at 1-888-234-2138.

20V770000 — Drive shaft corrosion, manual transmission

- **Recall began:** American Honda notified NHTSA on or about December 10, 2020.
- **Applies to:** 2007-2008 Fit vehicles with a **manual transmission** (nation-wide), plus 2009-2013 Fit vehicles with a manual transmission in specific salt-belt states or with a salt-belt registration history. Automatic-transmission 2008 Fits are not affected by this recall.
- **Component:** Powertrain — driveline — drive shaft.
- **Defect:** The protective coating on certain drive shafts may not have been applied properly during manufacturing, making the shaft more susceptible to corrosion from road salt or other contaminants; a sufficiently corroded shaft can break.
- **Consequence:** A broken drive shaft can cause a sudden loss of drive power.
- **Remedy:** Dealer inspects the drive shaft(s) and replaces them if necessary, free of charge.

Recall-sounding campaigns checked and confirmed to NOT apply to the 2008 Fit

It is easy to see "Honda Fit" in a recall name and assume your 2008 car is covered. These two campaigns were checked directly and do **not** apply to the 2008 model year:

Campaign number	Component	Why it does not apply
18V268000	Front passenger airbag module, incorrect installation	Names only 2007 and 2009-2013 Fit; 2008 is not listed.
19V182000	Driver frontal airbag inflator, Takata-related replacement part	Names only "2007 Fit" in its affected-vehicle list; 2008 is not present. (The 2008 Fit's passenger-side inflator is covered separately — see 19V501000, 16V344000, and 17V029000 above — but its driver-side inflator is not.)

Technical Service Bulletins (TSBs)

Two TSBs are confirmed applicable to the 2007-2008 Fit:

Bulletin number	Date	Title	Component
14-058	Approx. September 23, 2014	Warranty Extension: Electric Power Steering With DTC 32-09 Or 61-04	Electric power steering (EPS) system
13-021	Version 3, September 16, 2015	Rocker arm oil pressure switch causing Malfunction Indicator Lamp with DTC P2646/P2647/P2651/P2652	Rocker arm oil pressure switch

14-058 — Electric power steering warranty extension. Explicitly names the 2006-2009 Honda Civic and 2007-2008 Fit. American Honda extended the EPS warranty to 10 years from the original purchase date or 150,000 miles, whichever comes first, for vehicles with a stored diagnostic trouble code 32-09 (EPS current sensor) and/or 61-04 (open or short in the EPS motor harness). The bulletin describes the symptom as steering that feels heavier than normal or is hard to turn, particularly when the vehicle is not moving.

13-021 — Rocker arm oil pressure switch. The bulletin's "AFFECTED VEHICLES" table lists "2007-11 Fit — ALL — ALL" as one of several rows in a multi-model bulletin. It addresses an intermittently failing rocker arm oil pressure switch that causes the Malfunction Indicator Lamp to illuminate with DTC P2646/P2651 (circuit low voltage) or P2647/P2652 (circuit high voltage); the bulletin notes the symptom may be intermittent and may not be reproducible after clearing the code. The corrective action is to replace the rocker arm oil pressure switch (Honda part number 37250-PNE-G01) and its O-ring (part number 91319-PAA-A01).

Total TSB count. We could not confirm this for the 2008 Fit. Check your owner's manual, or ask a Honda dealer, for a full list of every bulletin issued for your specific car. Different aggregator sites reported conflicting totals (19 versus approximately 42), likely because they count bulletins differently, and neither count could be independently confirmed.

How to check for new recalls and TSBs yourself

Use the NHTSA vehicle lookup tool listed in Appendix H to search by your VIN or by year/make/model. This is the fastest way to see whether a new recall or TSB has been issued since this book was written, and it also shows owner-submitted complaints and formal government investigations, which can reveal problems before they turn into an official recall.